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(54) **MINIMAL NEPHRIN PROMOTER**

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(57) **ABSTRACT**

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Jun. 30, 2021 (WO) PCT/GB2021/051668

A promoter comprising (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17 and wherein the promoter has a length of about 1.1 kb or less.

Specification includes a Sequence Listing.

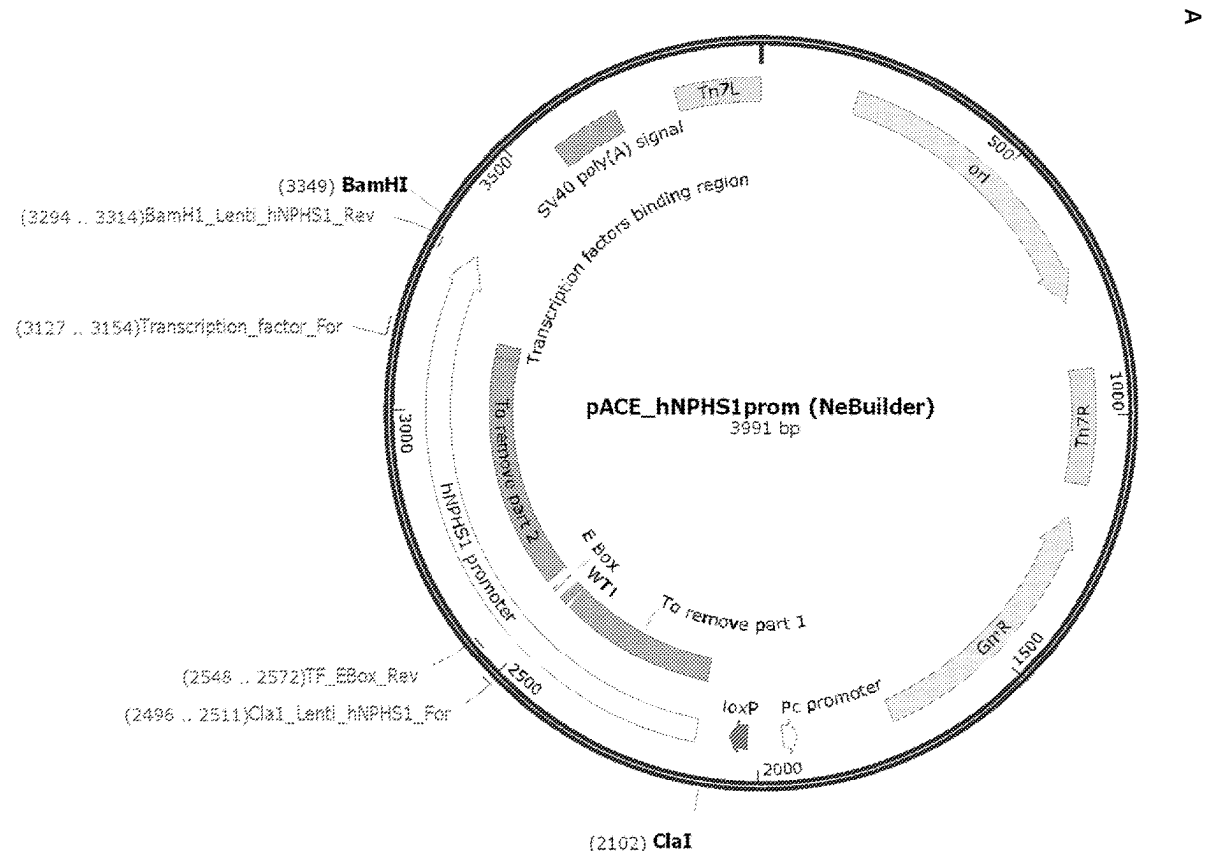


FIGURE 1

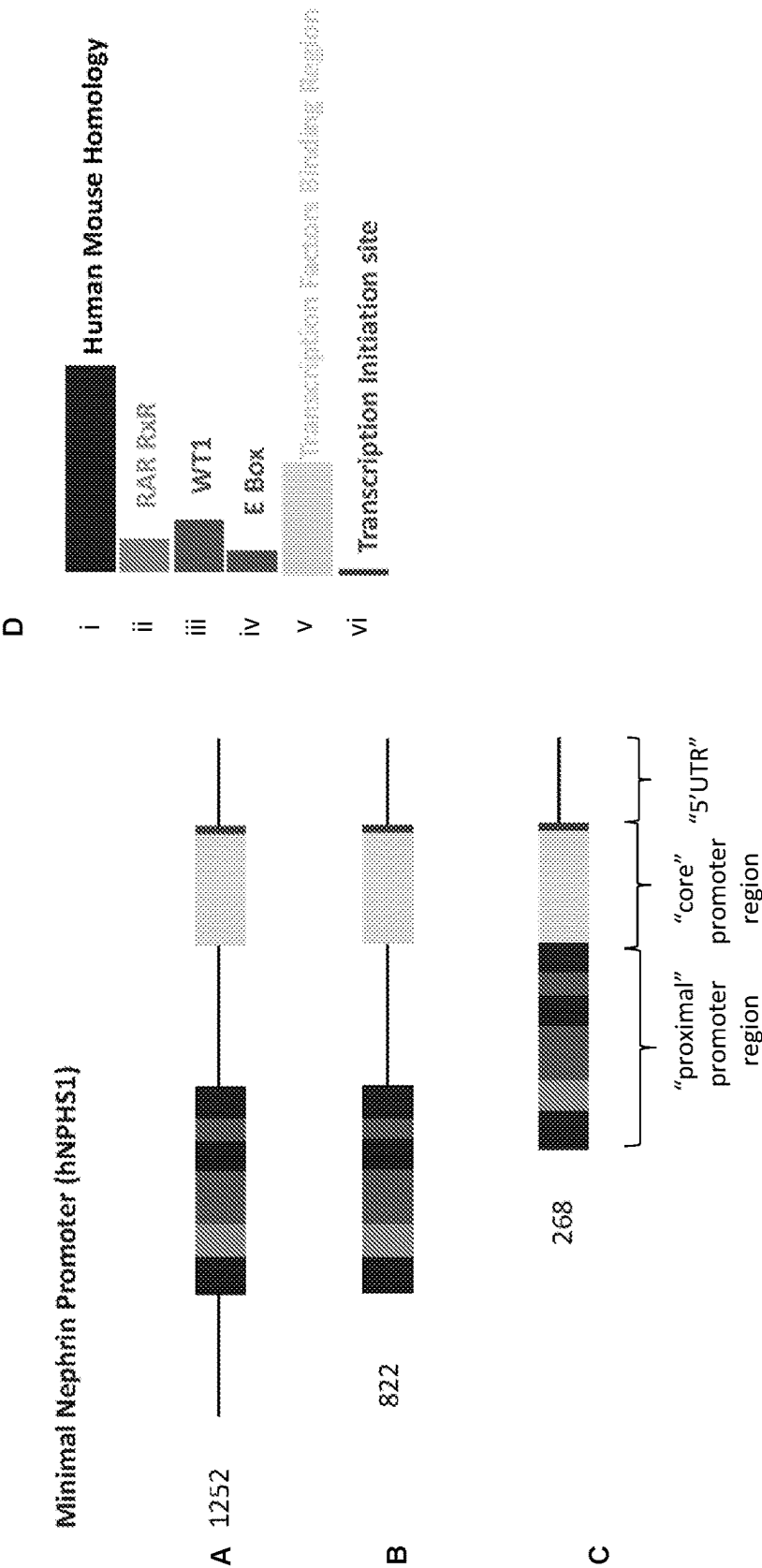


FIGURE 2

A

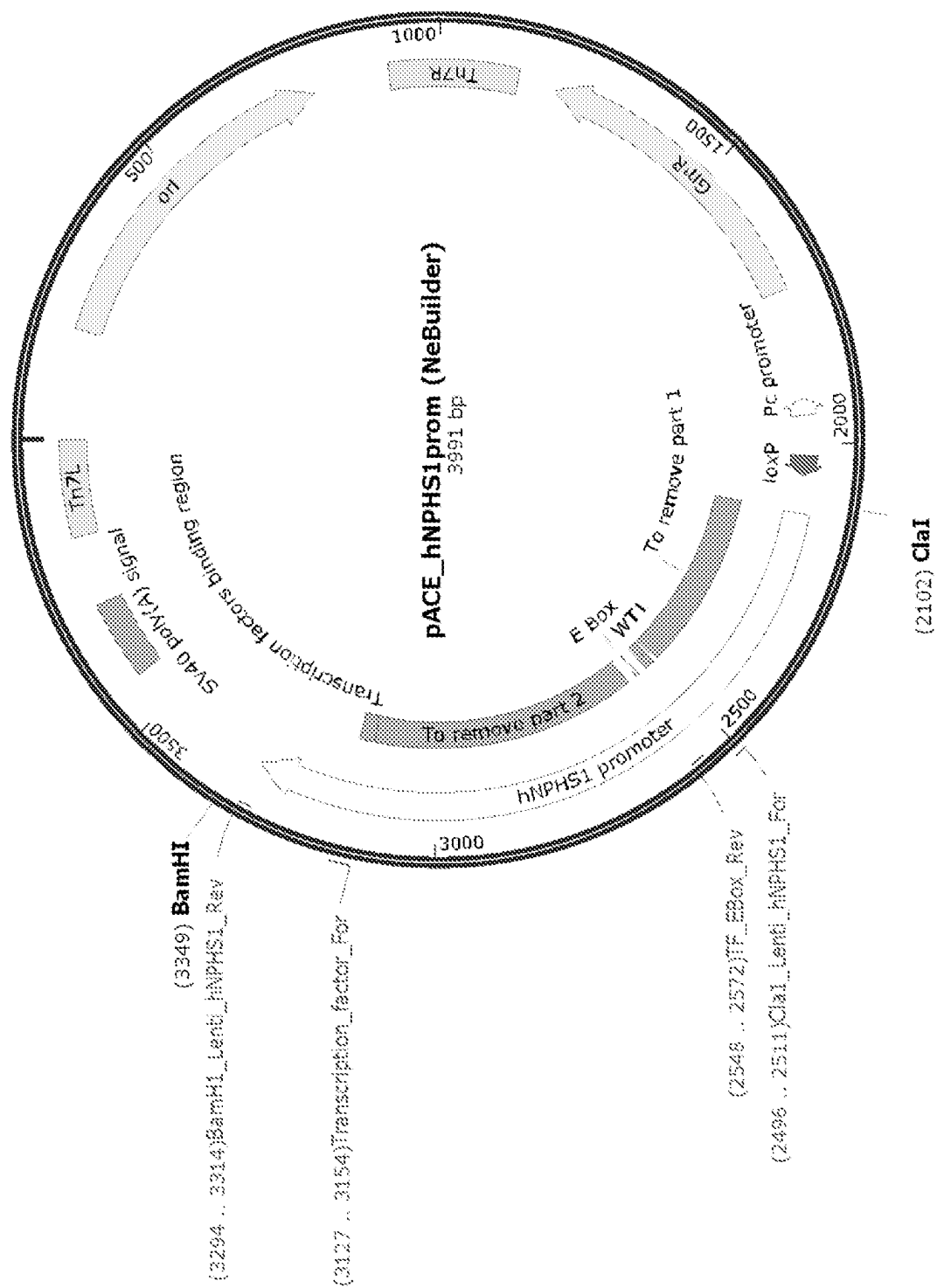


FIGURE 3

A

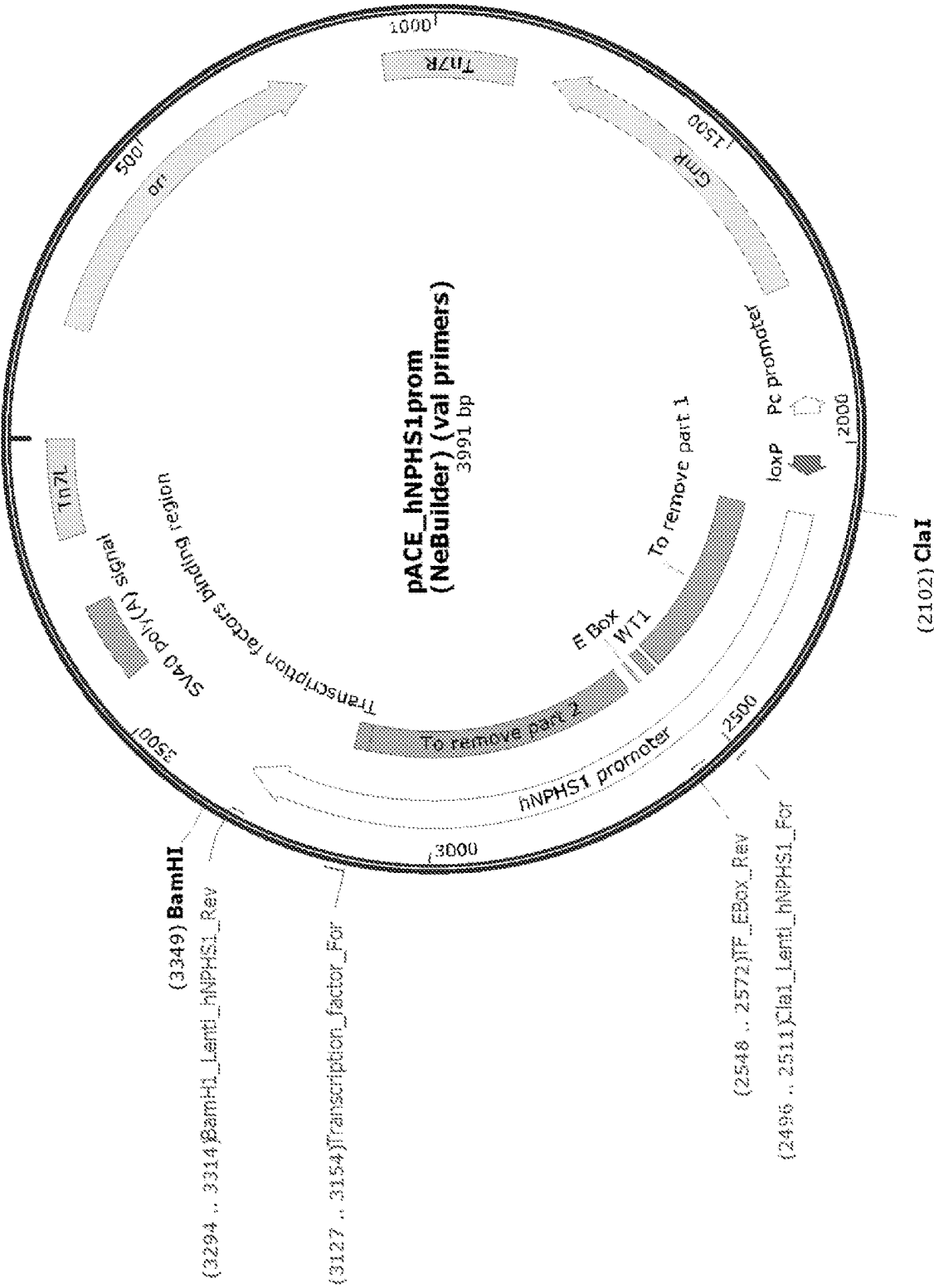


FIGURE 3 (CONTINUED)

B

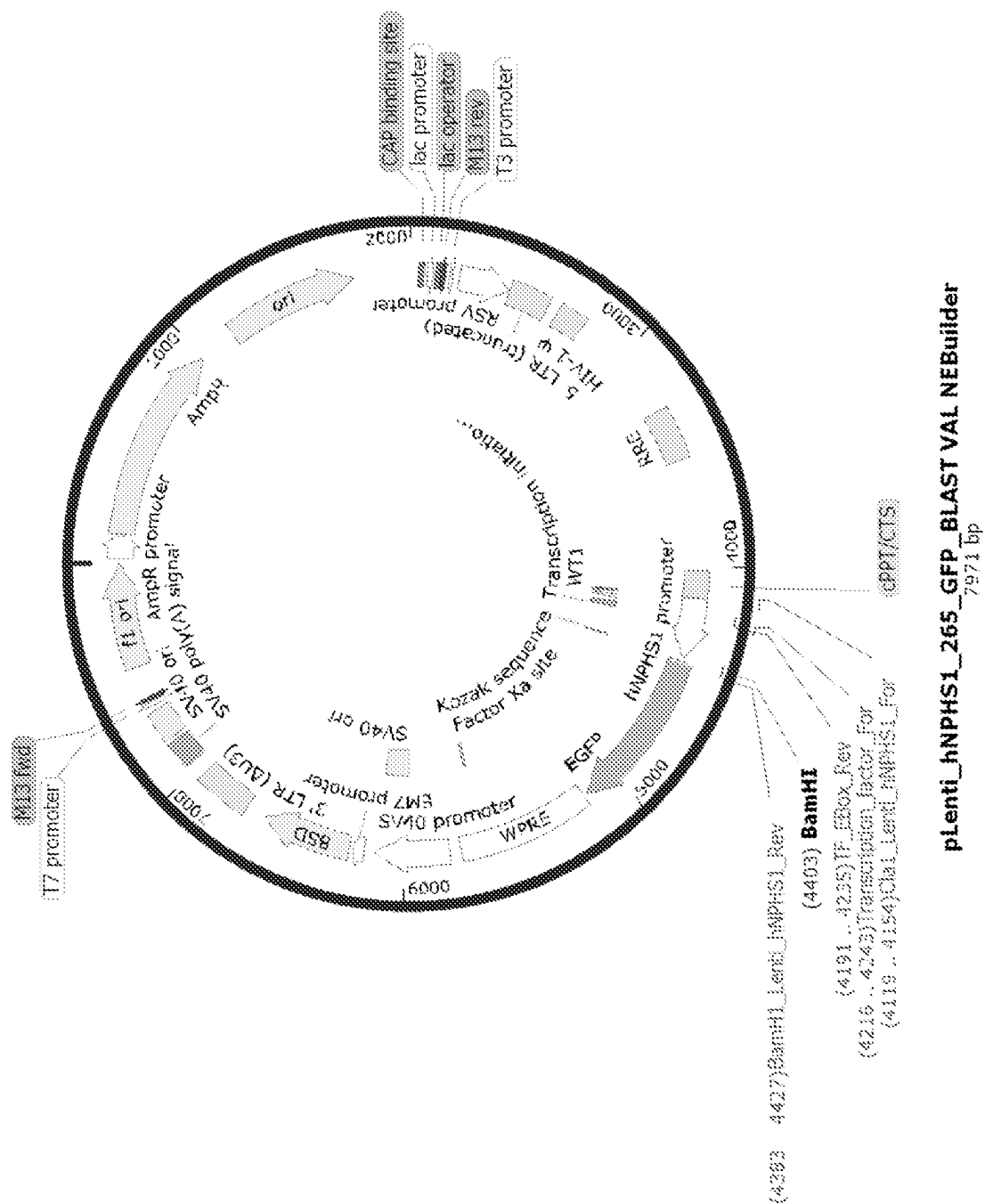
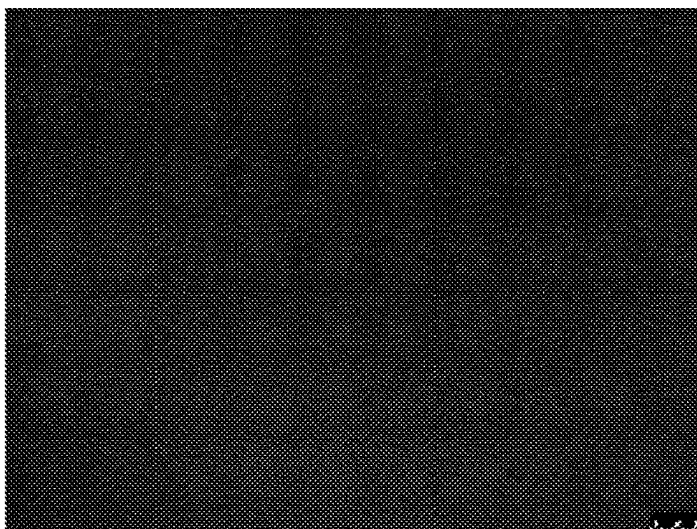


FIGURE 4

A
Untransduced



B
mini hNPSH1 GFP



C
FL hNPSH1 GFP

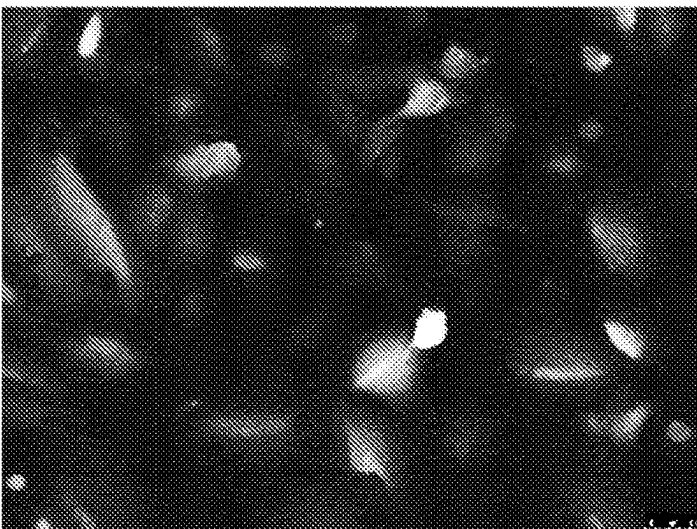


FIGURE 5

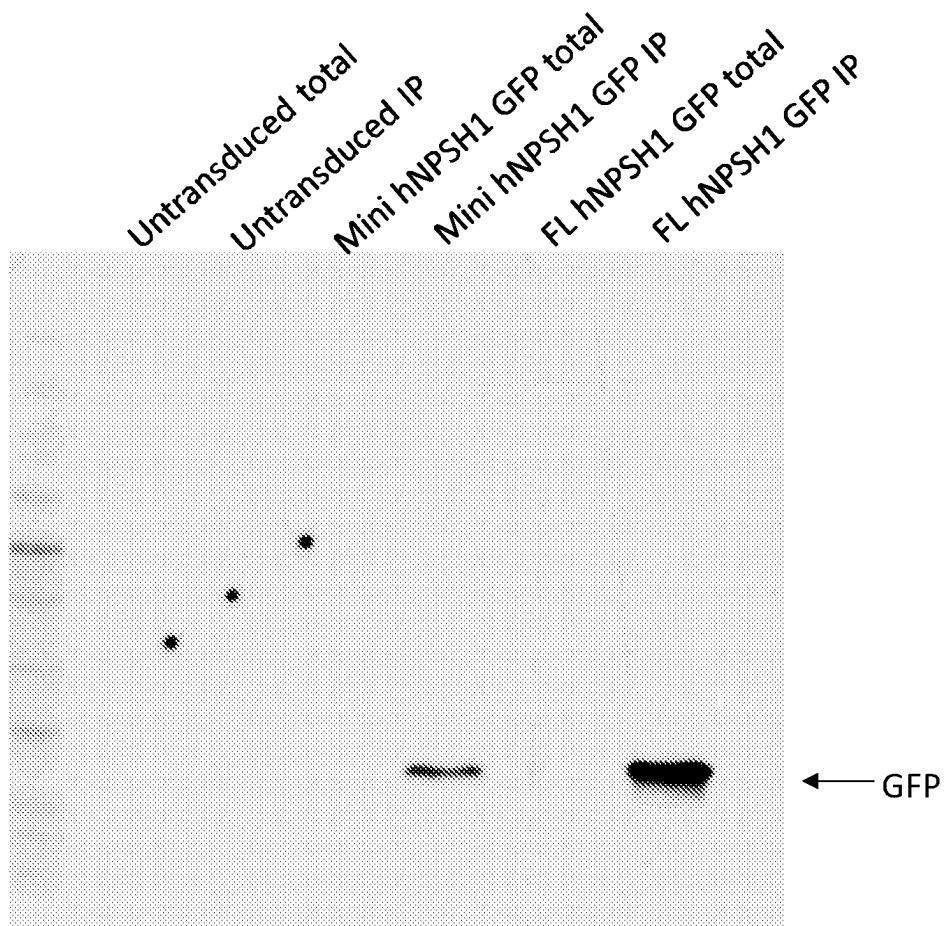


FIGURE 6

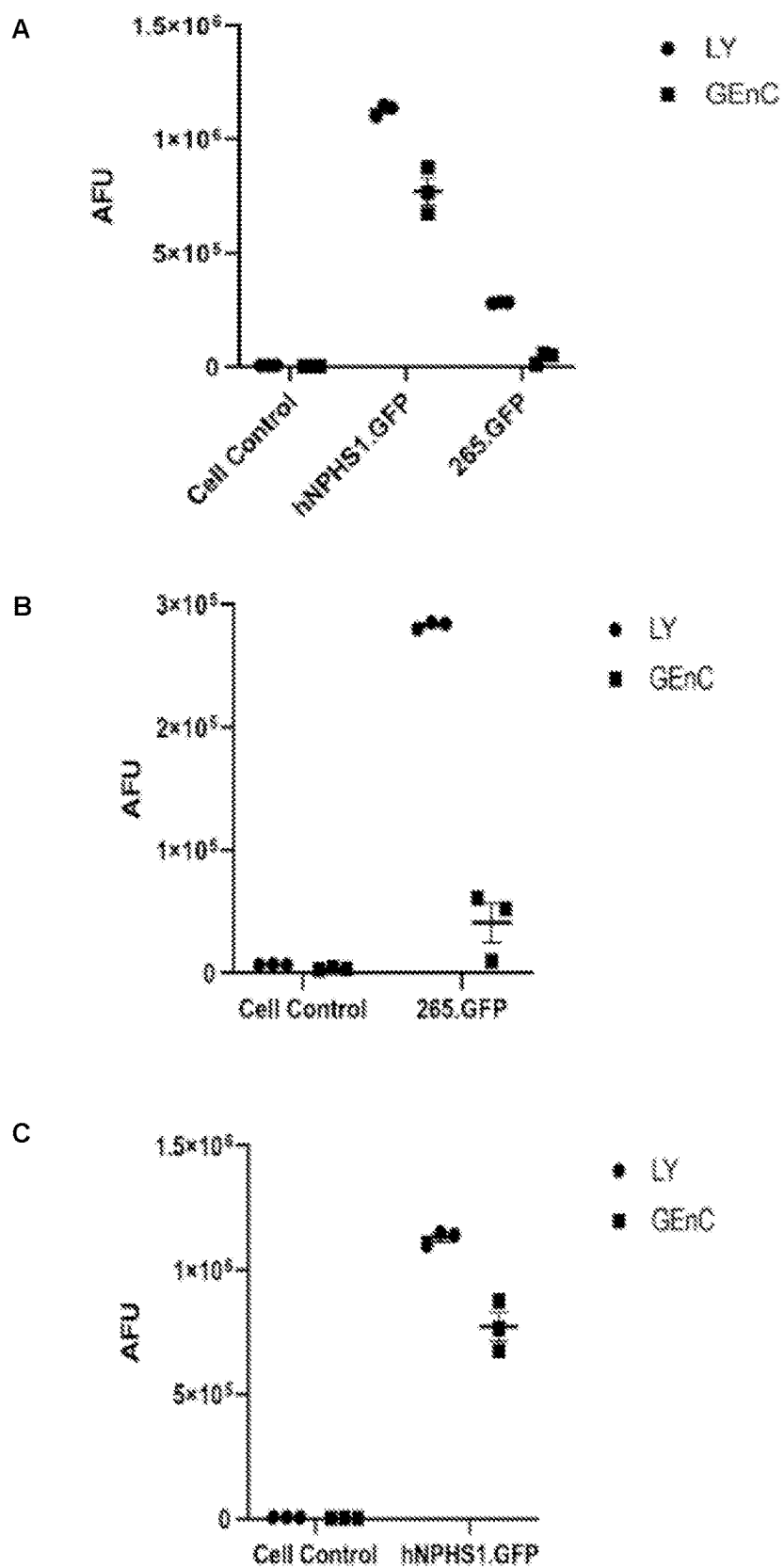
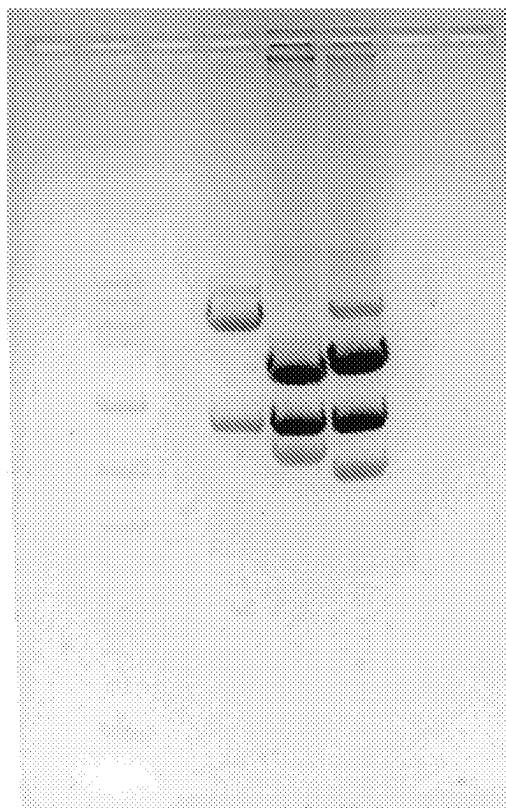


FIGURE 7 (CONTINUED)

D



E

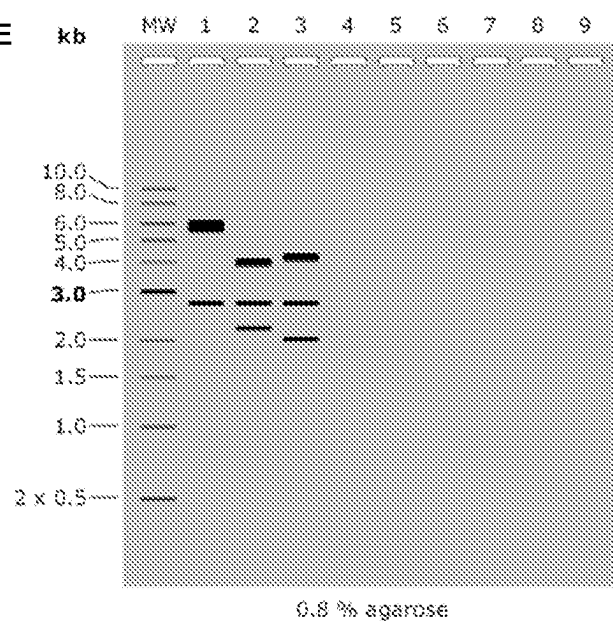


FIGURE 8

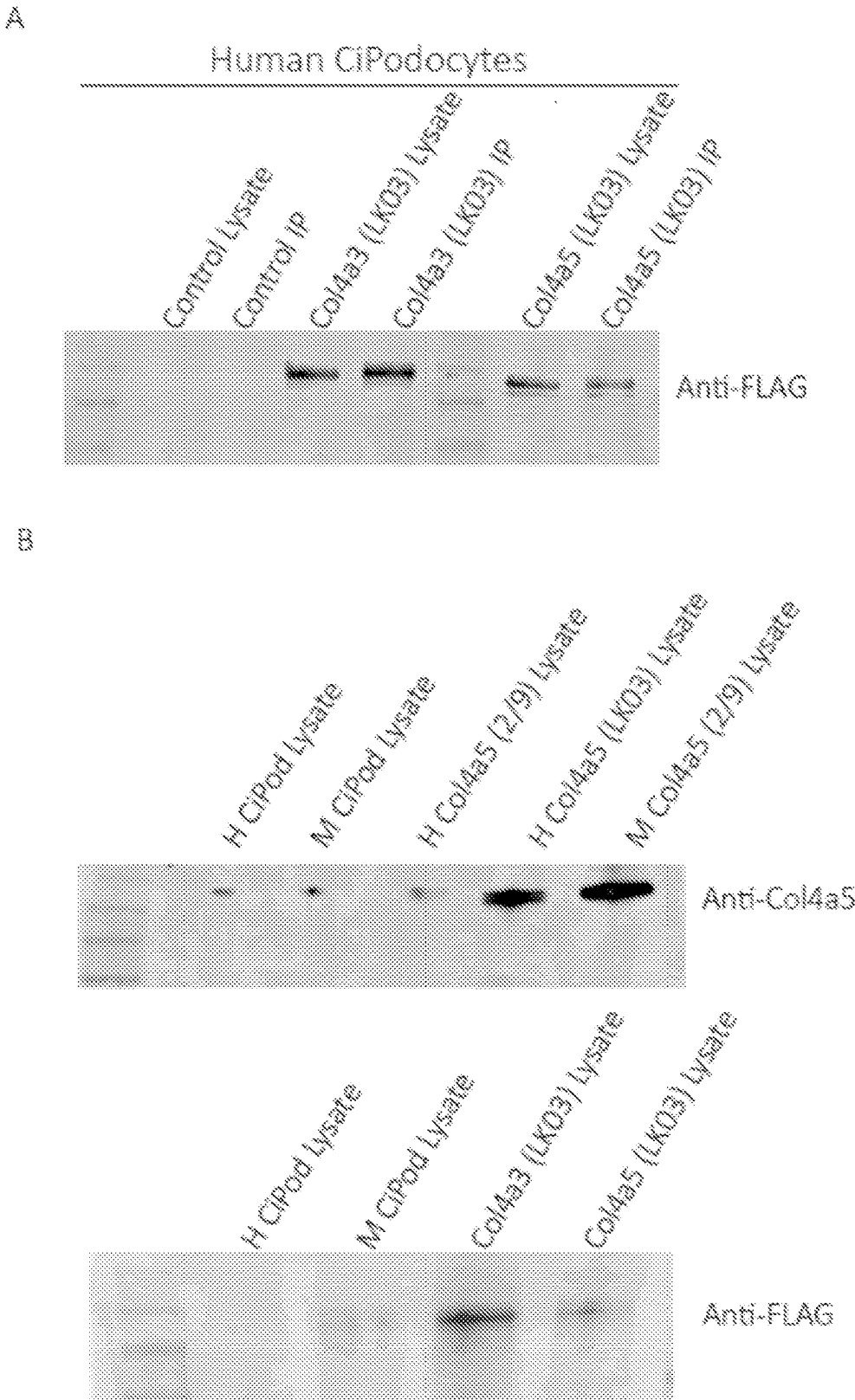


FIGURE 8 (CONTINUED)

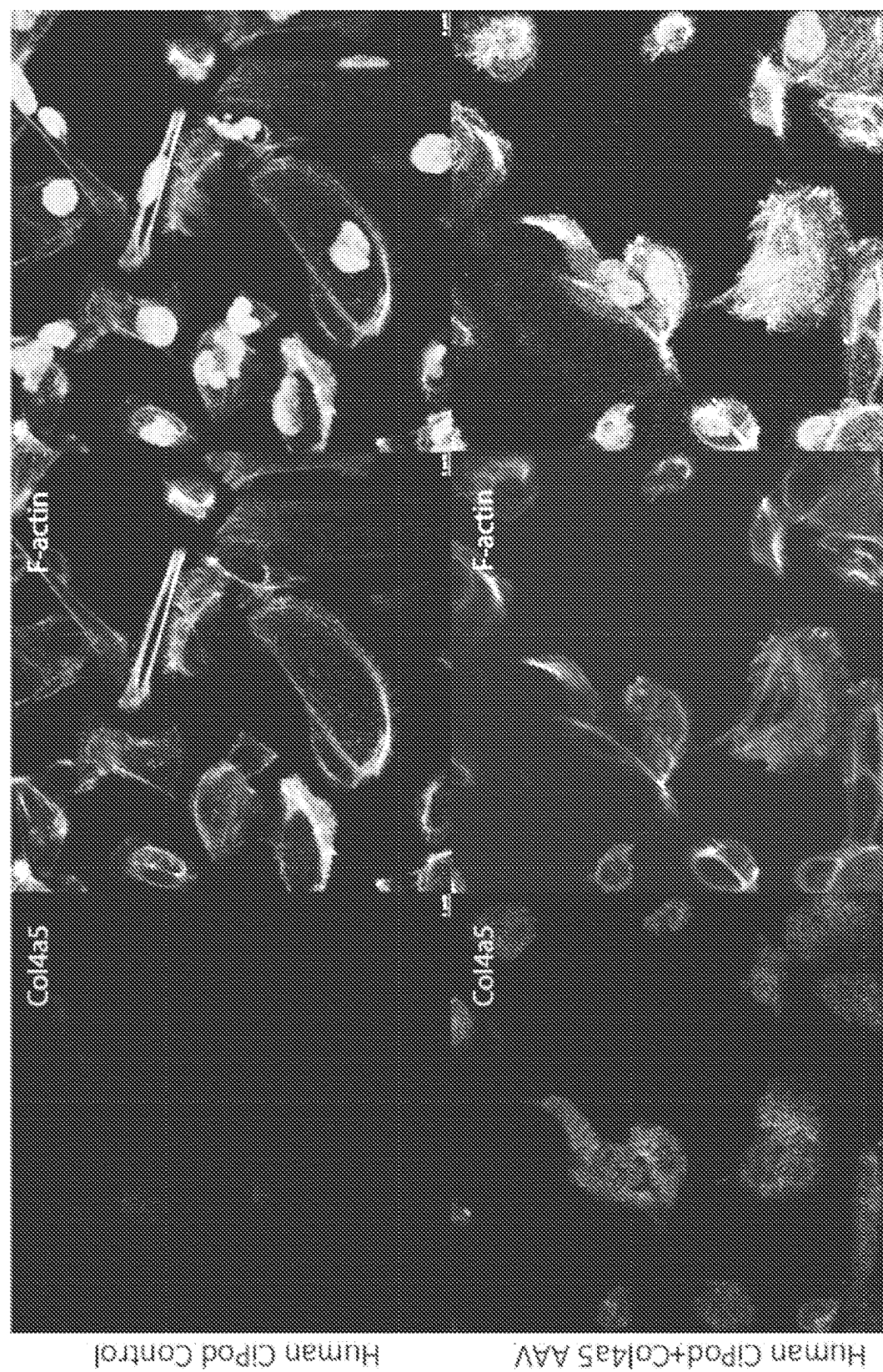


FIGURE 9

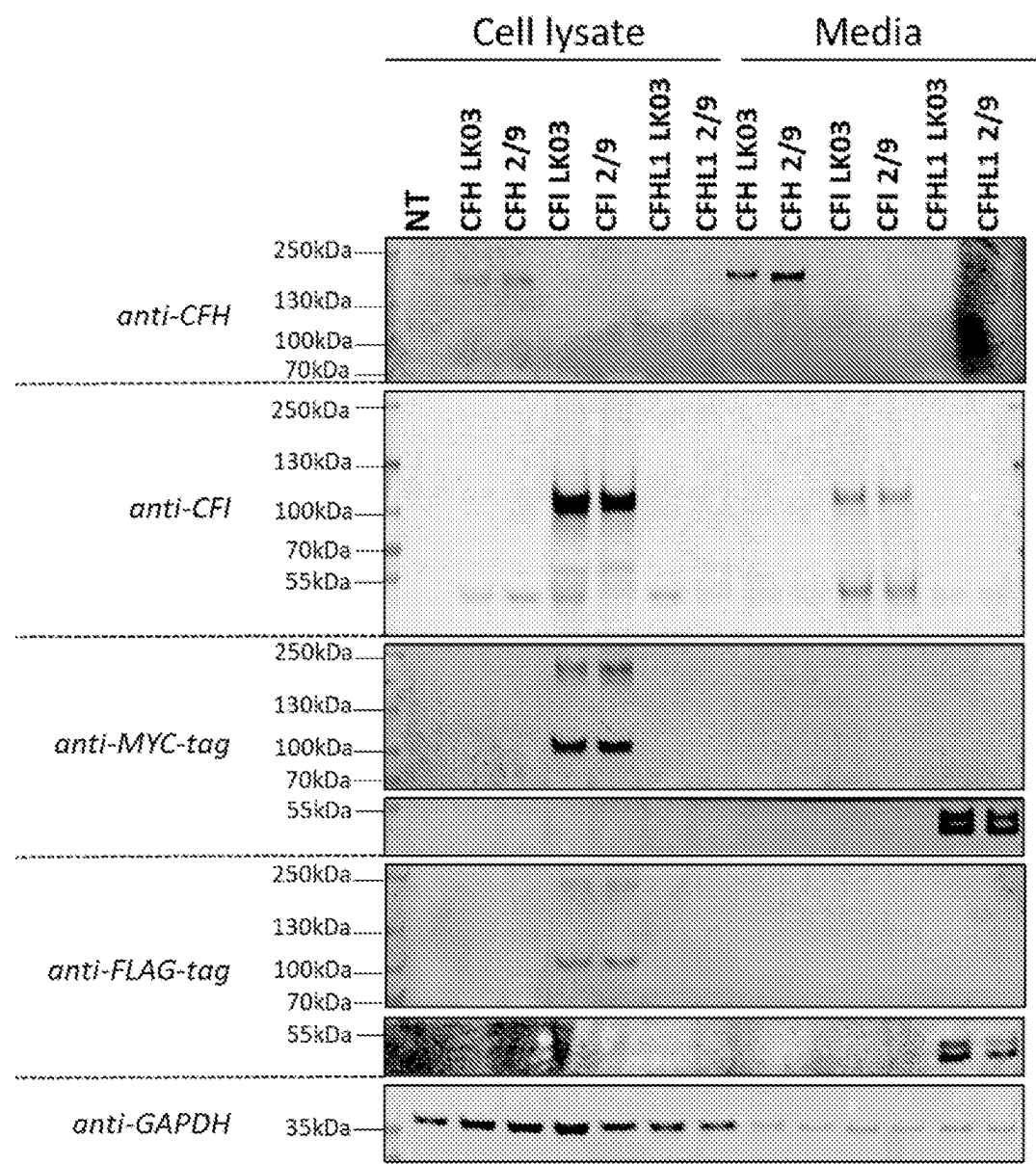
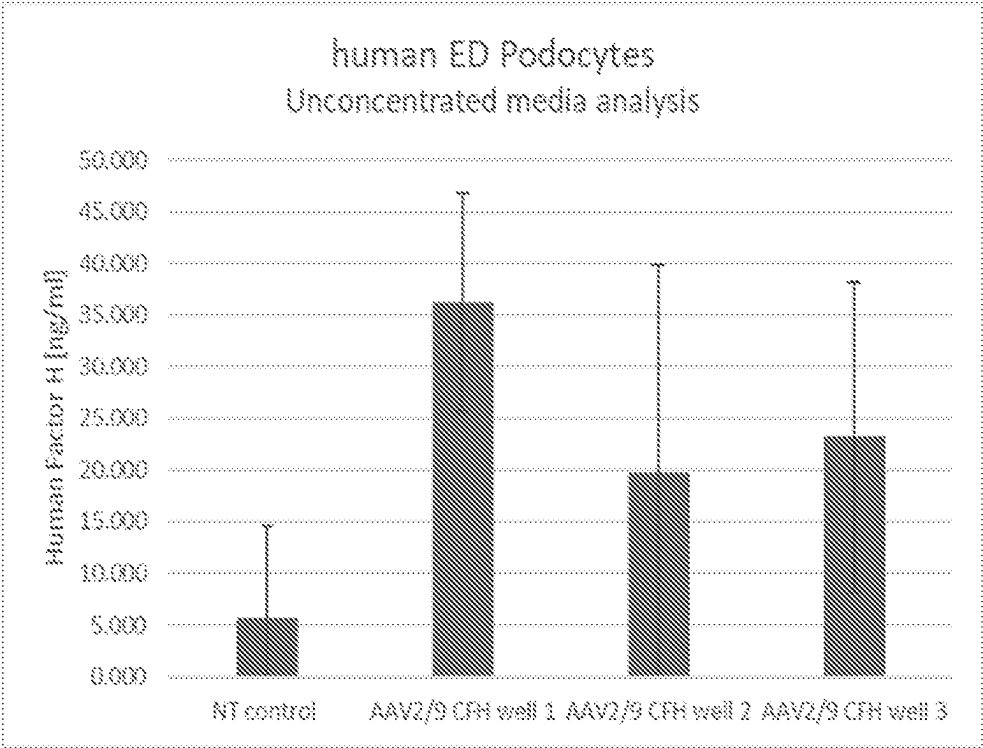


FIGURE 10

A



B

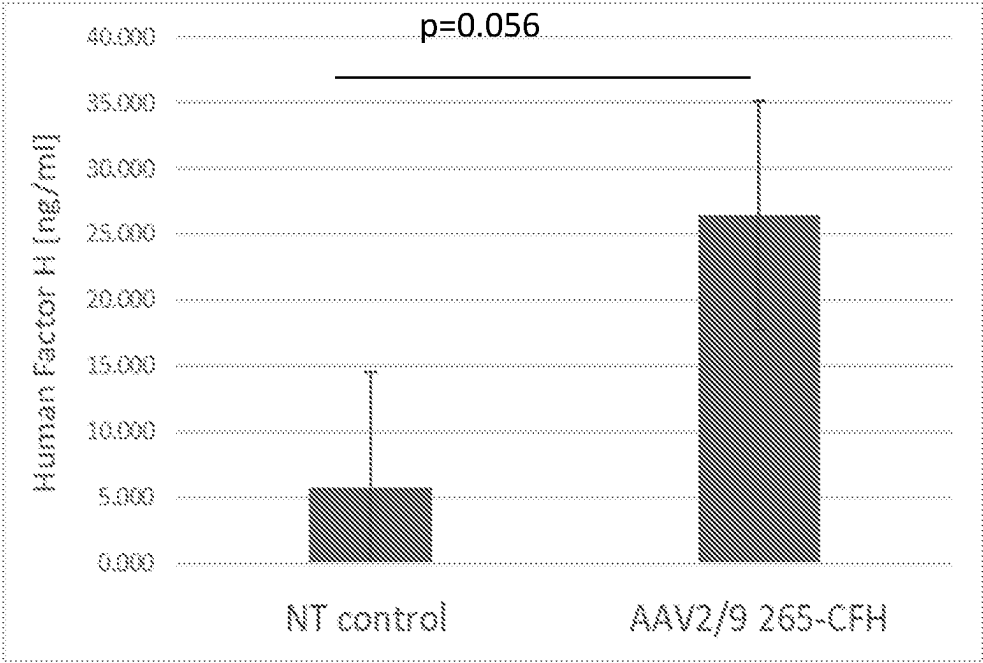


FIGURE 10 (CONTINUED)

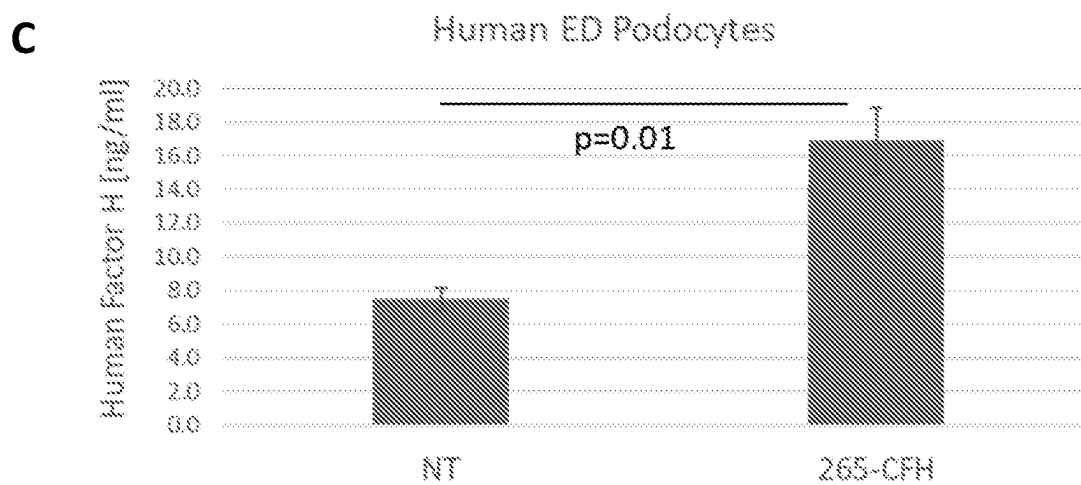


FIGURE 11

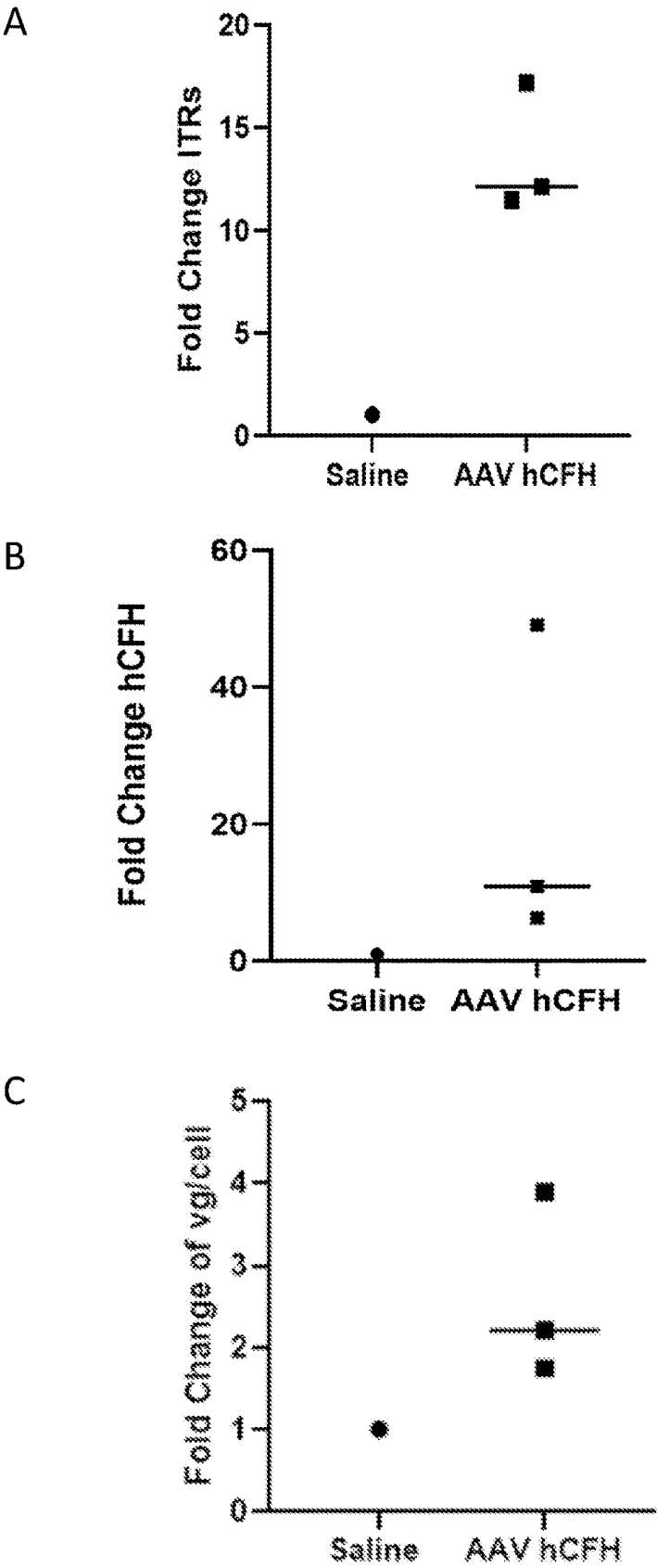


FIGURE 11 (CONTINUED)

D

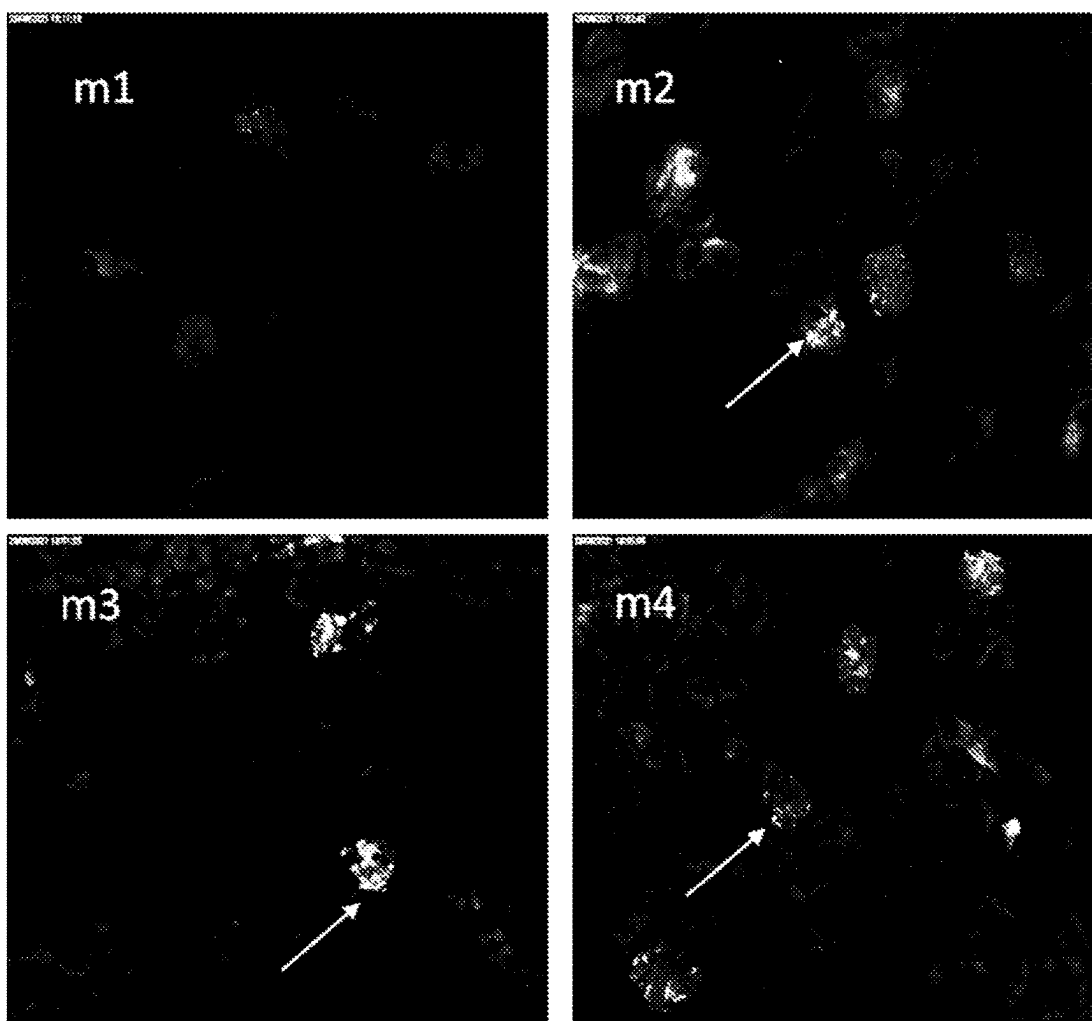


FIGURE 12

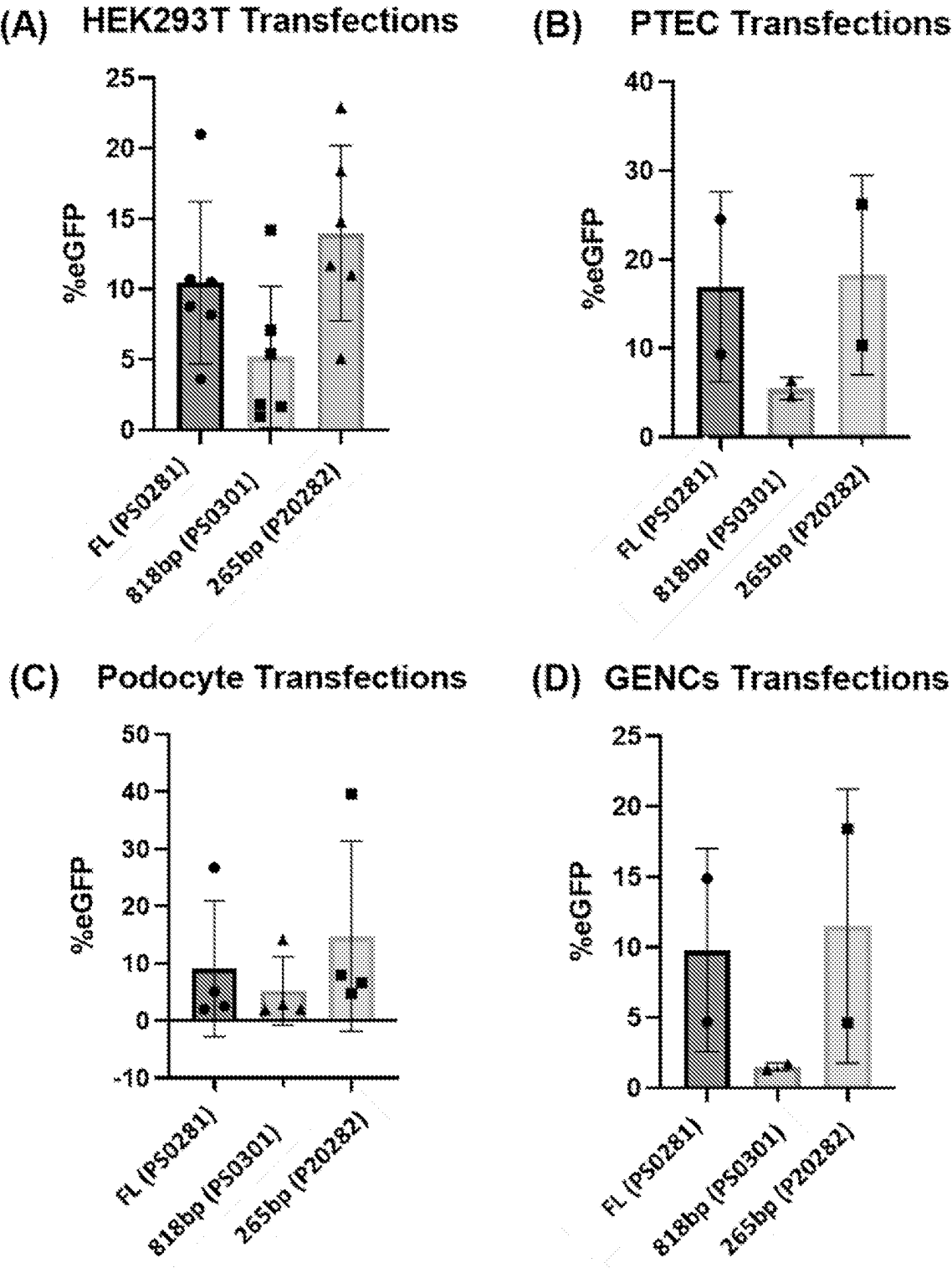


FIGURE 13

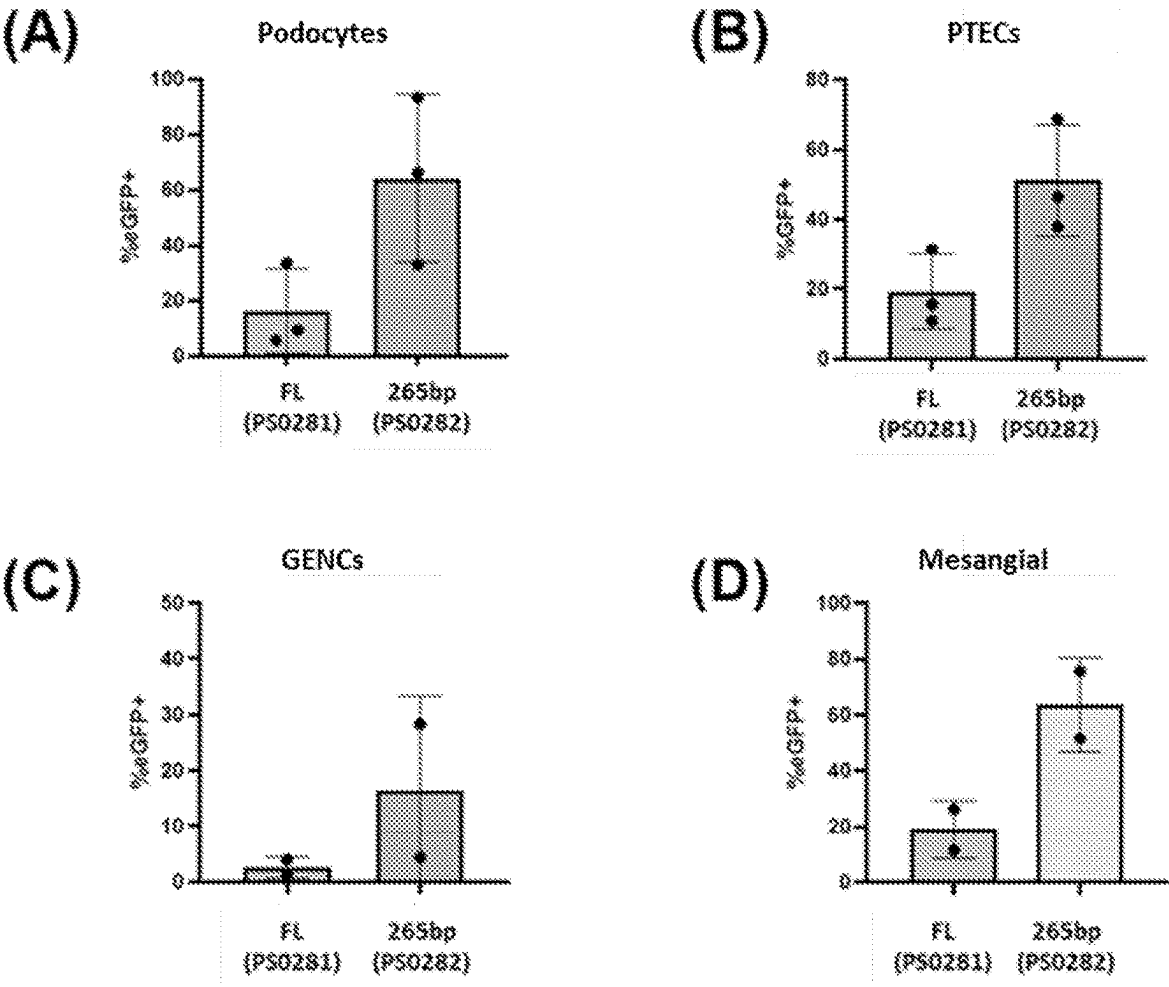


FIGURE 14

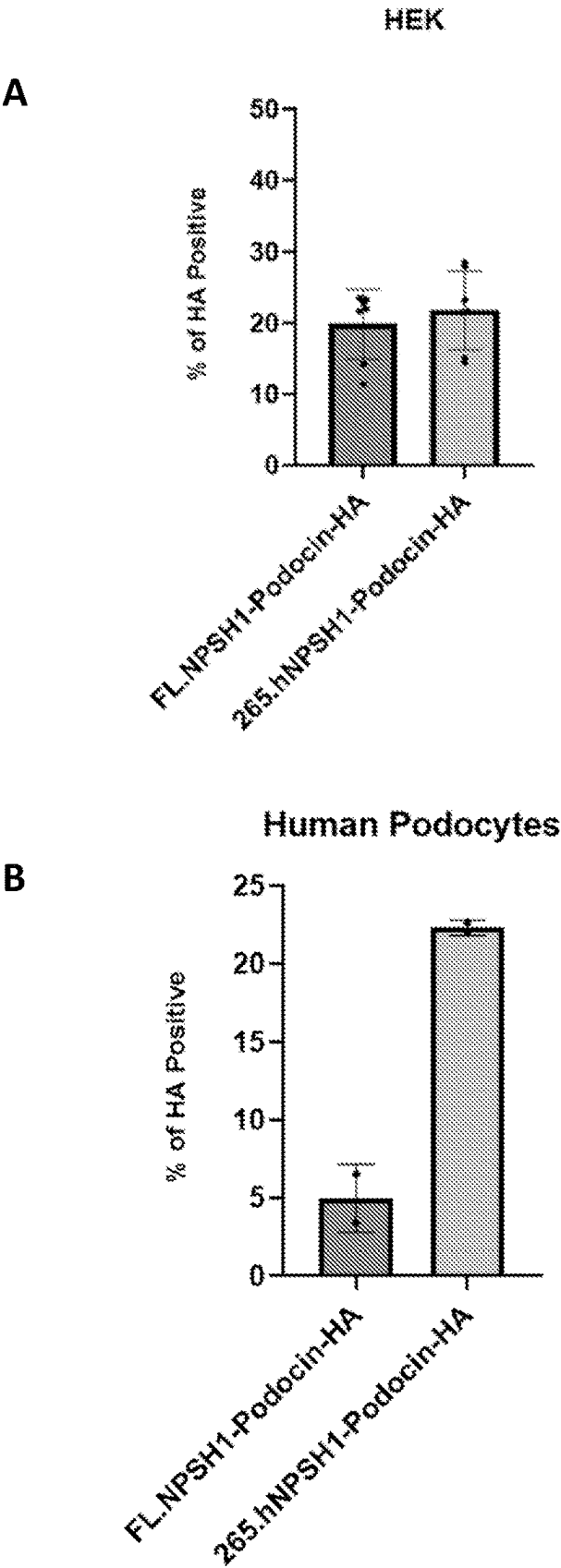
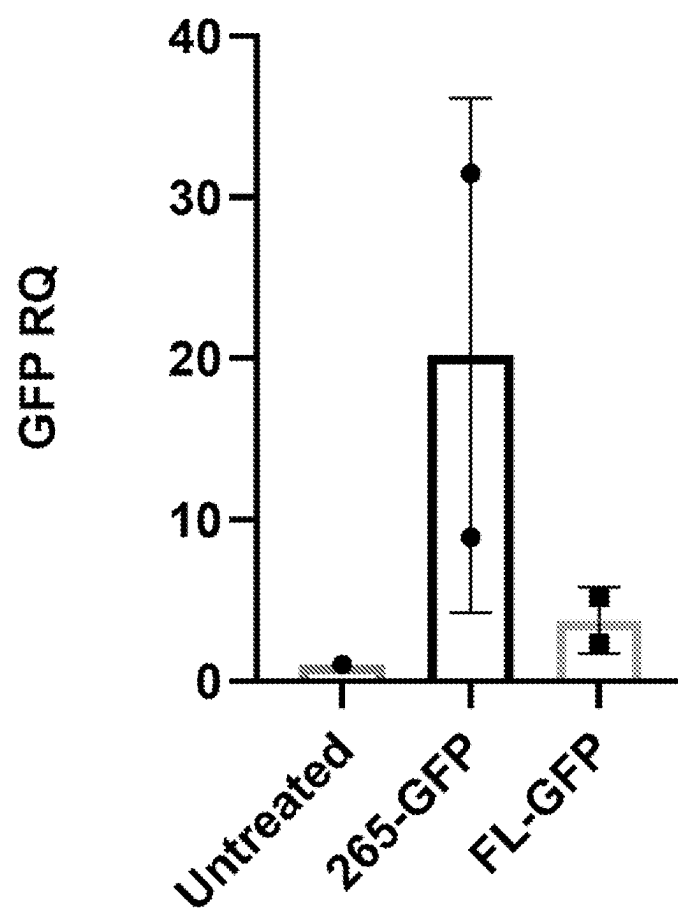


FIGURE 15



MINIMAL NEPHRIN PROMOTER**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] This application is a U.S. National Phase of International Patent Application No. PCT/GB2022/050649, filed on Mar. 14, 2022; which claims priority to International Patent Application No. PCT/GB2021/051668, filed on Jun. 30, 2021, and International Patent Application No. PCT/GB2021/050633, filed on Mar. 12, 2021, and Application No. 2103470.7, filed on Mar. 12, 2021 in the United Kingdom.

INCORPORATION BY REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

[0002] A Sequence Listing is incorporated herein by reference as part of the disclosure. The sequence listing has been submitted as a text file named "59411_SeqListing.txt", which was created on Mar. 26, 2024, and is 146,453 bytes in size.

FIELD OF THE INVENTION

[0003] The present invention relates to promoters which are capable of driving transgene expression in podocytes. The present invention also relates to polynucleotides and vectors comprising said promoters.

BACKGROUND TO THE INVENTION

[0004] There are many diseases which affect kidney function by attacking the glomerulus. The glomerulus filters approximately 180 litres of plasma each day, and the healthy glomerular filtration barrier has an astonishing ability to retain about 99.9% of large proteins including albumin over our lifetimes without clogging. The glomerular filtration barrier (GFB) comprises 3 main layers: the glomerular endothelial cell, the glomerular basement membrane (GBM) and the podocyte.

[0005] The GBM is made of a highly crosslinked macromolecular meshwork of type IV collagen, proteoglycans, and laminin. Genetic forms of glomerular disease can be caused by genetic defects in these molecular structures. For example, Alport syndrome is caused by pathogenic variants in the COL4A3, COL4A4 and COL4A5 genes, which result in abnormalities of the collagen IV α 345 network of basement membranes. Alport syndrome affects approximately 1 in 5,000-10,000 of all individuals in continental Europe and the USA. The condition usually presents during childhood and is associated with a spectrum of phenotypes that include a progressive loss of kidney function, and can also include hearing loss and eye abnormalities. Other GBM-associated diseases include Pierson syndrome and Nail-patella syndrome (Chiang, C. K. and Inagi, R., 2010. Nature Reviews Nephrology, 6 (9), p.539).

[0006] The podocyte has also been implicated as a key cell in the progression of glomerular disease. Podocytes are mesodermally derived cells that are highly specialized and found only in the renal glomerulus. They exhibit unique characteristics such as foot processes and slit diaphragms, which are critical for glomerular filtration. Podocyte-associated genetic glomerular diseases include Nephrotic Syndrome, Frasier syndrome and Denys-Drash syndrome, Schimke immuno-osseous dysplasia, and Epstein and Fench-

ner syndrome. (Chiang, C. K. and Inagi, R., 2010. Nature Reviews Nephrology, 6 (9), p.539).

[0007] Accordingly, glomerular cells, such as podocytes, represent a potential target for gene therapy approaches.

[0008] In order to maximise gene therapy potential, promoter sequences which can drive transgene expression in glomerular cells, such as podocytes, are required. Wong et al. (American Journal of Physiology Renal Physiology; 2000; 279 (6); F1027-32) described a 1.25-kb DNA fragment from the human nephrin promoter and 5'-flanking region that is capable of directing podocyte-specific expression. This promoter has been used to achieve kidney-specific expression of GFP using an AAV9 vector (Picconi et al.; 2014; Molecular Therapy-Methods & Clinical Development; 1, 14014).

[0009] At present a significant amount of vector cargo capacity (e.g. AAV cargo capacity) is taken up with non-coding elements such as the nephrin or podocin promoters, WPRE elements and polyadenylation sequence. Thus, there is a need in the art for minimal promoters which can drive transgene expression in glomerular cells, such as podocytes, in order to provide maximal flexibility for gene therapy approaches for glomerular diseases.

SUMMARY OF THE INVENTION

[0010] The present invention is based on the inventors surprising provision of a minimal promoter which is capable of driving transgene expression in kidney cells, in particular glomerular cells, such as podocytes.

[0011] In one aspect, the present invention provides a promoter comprising (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 4 or 17 and wherein the promoter has a length of about 1.1 kb or less.

[0012] In some embodiments, the present invention provides a promoter which comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, but wherein:

[0013] (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 1 to 430; and/or

[0014] (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061;

[0015] or a nucleotide sequence with at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity thereto.

[0016] In some embodiments:

[0017] (iii) the position corresponding to position 493 of SEQ ID NO: 1 is G, the position corresponding to position 1080 of SEQ ID NO: 1 is T, the position corresponding to position 1169 of SEQ ID NO: 1 is C, and the position corresponding to position 1249 of SEQ ID NO: 1 is G.

[0018] In one aspect, the present invention provides a promoter which comprises or consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to a nucleotide sequence according to SEQ ID NO: 1 but wherein:

[0019] (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 100 to 430; and/or

[0020] (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061.

[0021] In some embodiments:

[0022] (iii) position 493 of SEQ ID NO: 1 is substituted with G, position 1080 of SEQ ID NO: 1 is substituted with T, position 1169 of SEQ ID NO: 1 is substituted with C, and position 1249 of SEQ ID NO: 1 is substituted with G.

[0023] The promoter may comprise (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 4 or 17 and/or the promoter may have a length of about 1.1 kb or less.

[0024] The promoter may be operable in a kidney cell. The promoter may be capable of driving transgene expression in kidney. The promoter may be a kidney-specific promoter. The promoter may be operable in a podocyte cell. The promoter may be capable of driving transgene expression in podocytes. The promoter may be a podocyte-specific promoter.

[0025] Suitably, the promoter further comprises (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 7 or 20. In some embodiments, the promoter further comprises (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 6 or 19.

[0026] Suitably, the promoter further comprises (iii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 8, or one or more fragments thereof.

[0027] Suitably, the promoter according to any preceding claim, wherein the promoter comprises from 5' to 3': (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 4 or 17; (iii) optionally a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 8, or one or more fragments thereof; and (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 7 or 20.

[0028] In some embodiments, the promoter comprises from 5' to 3': (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 4; and (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 7. In some embodiments, the promoter consists of from 5' to 3': (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 4; and (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 7, and wherein the promoter has a length of about 1.1 kb or less.

[0029] In some embodiments, the promoter consists of from 5' to 3': (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 17; and (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 20. In some embodiments, the promoter consists of from 5' to 3': (i) the nucleotide sequence of SEQ ID NO: 17; and (ii) the nucleotide sequence of SEQ ID NO: 20.

[0030] In some embodiments, the promoter comprises from 5' to 3': (i) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 17; and (ii) a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 20, and wherein the promoter has a length of about 1.1 kb or less. In some embodiments, the promoter comprises from 5' to 3': (i) the nucleotide sequence of SEQ ID NO: 17; and (ii) the nucleotide sequence of SEQ ID NO: 20, and wherein the promoter has a length of about 1.1 kb or less.

[0031] The promoter may have a length of about 1.0 kb or less, about 0.9 kb or less, about 0.8 kb or less, about 0.7 kb or less, about 0.6 kb or less, about 0.5 kb or less, about 0.4 kb or less, or about 0.3 kb or less. The promoter may have a length of from 0.265 kb to 1.0 kb, from 0.265 kb to 0.9 kb, from 0.265 kb to 0.8 kb, from 0.265 kb to 0.7 kb, from 0.265 kb to 0.6 kb, from 0.265 kb to 0.5 kb, from 0.265 kb to 0.4 kb, or from 0.265 kb to 0.3 kb.

[0032] The promoter may comprise: (a) a retinoic acid receptor binding site; (b) a WT1 binding site; (c) an enhancer box; (d) a transcription factor binding region; and/or (e) a transcription initiation site. Suitably, the retinoic acid receptor binding site comprises or consists of a nucleotide sequence shown as SEQ ID NO: 10 or a nucleotide sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 10. Suitably, the WT1 binding site comprises or consists of a nucleotide sequence shown as SEQ ID NO: 11 or a nucleotide sequence having one, two or three substitutions, deletions, or insertions compared to SEQ ID NO: 11. Suitably, the enhancer box comprises or consists of a nucleotide sequence shown as SEQ ID NO: 12 or a nucleotide sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 12. Suitably, the transcription factor binding region comprises or consists of a nucleotide sequence shown as SEQ ID NO: 13 or 43 or a nucleotide sequence having one, two, three, four or five substitutions, deletions, or insertions compared to SEQ ID NO: 13 or 43. Suitably, the transcription initiation site comprises or consists of an "AG" dinucleotide. Suitably, the transcription factor binding site is operably linked to the transcription initiation site, optionally wherein the transcription factor binding site is directly upstream of the transcription initiation site.

[0033] The nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 4 or 17 may comprise one or more of: (a) a retinoic acid receptor binding site; (b) a WT1 binding site; and (c) an enhancer box. One or more of the following nucleotide sequences may be present in the nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 4 or 17: (a) GGGGTCA at a position corresponding to position 7 to

position 13 of SEQ ID NO: 4 or 17; (b) CGGAGGCTGGG-GAGGCA at a position corresponding to position 14 to position 30 of SEQ ID NO: 4 or 17; and (c) ATGTG at a position corresponding to position 49 to position 53 of SEQ ID NO: 4 or 17.

[0034] In some embodiments, the promoter comprises or consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 2 or SEQ ID NO: 3.

[0035] In some embodiments, the promoter comprises or consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 15, SEQ ID NO: 59, or SEQ ID NO: 47.

[0036] In preferred embodiments, the promoter comprises or consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 47. In more preferred embodiments, the promoter comprises or consists of the nucleotide sequence of SEQ ID NO: 47.

[0037] In one aspect, the present invention provides a promoter which consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 2.

[0038] In one aspect, the present invention provides a promoter which consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 3.

[0039] In one aspect, the present invention provides a promoter which consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 15.

[0040] In one aspect, the present invention provides a promoter which consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% identity to SEQ ID NO: 59.

[0041] In one aspect, the present invention provides a promoter which consists of a nucleotide sequence having at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to SEQ ID NO: 47.

[0042] In one aspect, the present invention provides a promoter which consists of the nucleotide sequence of SEQ ID NO: 47.

[0043] In one aspect, the present invention provides a polynucleotide comprising the promoter according to the present invention.

[0044] The promoter may be operably linked to a protein coding sequence. In some embodiments, the protein coding sequence encodes NPHS2 or a fragment and/or variant thereof; a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof; or CFI, CFH or FHL-1, or a fragment and/or variant thereof.

[0045] In some embodiments, the protein coding sequence encodes NPHS2, or a fragment and/or variant thereof. In some embodiments, the protein coding sequence encodes an amino acid sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 55 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 56 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at

least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 57 or a fragment thereof.

[0046] In some embodiments, the protein coding sequence encodes a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof. In some embodiments, the protein coding sequence encodes an amino acid sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to one of SEQ ID NOs: 21-23 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to one of SEQ ID NOs: 24-26 or a fragment thereof.

[0047] In some embodiments, the protein coding sequence encodes CFI, or a fragment and/or variant thereof. In some embodiments, the protein coding sequence encodes an amino acid sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 49 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 50 or a fragment thereof.

[0048] In some embodiments, the protein coding sequence encodes CFH, or a fragment and/or variant thereof. In some embodiments, the protein coding sequence encodes an amino acid sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 51 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 52 or a fragment thereof.

[0049] In some embodiments, the protein coding sequence encodes FHL-1, or a fragment and/or variant thereof. In some embodiments, the protein coding sequence encodes an amino acid sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 53 or a fragment thereof. In some embodiments, the protein coding sequence comprises or consists of a nucleotide sequence which is at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or at least 99% identical to SEQ ID NO: 54 or a fragment thereof.

[0050] In some embodiments, the protein coding sequence is operably linked to one or more further regulatory elements, such as a post-transcriptional regulatory element and/or a polyadenylation sequence.

[0051] In one aspect, the present invention provides a vector comprising the polynucleotide according to the present invention. Suitably, the vector is capable of transducing podocytes, optionally wherein the vector is capable of specifically transducing podocytes.

[0052] Suitably, the vector is a viral vector, such as an adeno-associated virus (AAV) vector, a lentiviral vector, a retroviral vector, an adenoviral vector, an adeno-associated viral vector, a herpes simplex viral vector, an alphaviral vector, a flaviviral vector, a rhabdoviral vector, a measles viral vector, a Newcastle disease viral vector, a poxviral vector, and a picornaviral vector, preferably wherein the vector is an AAV vector. Suitably, the viral vector is in the form of a viral vector particle, preferably wherein the viral vector is in the form of an AAV vector particle.

[0053] In some embodiments, the vector is the form of an AAV vector particle encapsidated by AAV3B, LK03, or AAV9 capsid proteins.

[0054] In one aspect, the present invention provides a cell comprising the polynucleotide according to the present invention, or the vector according to the present invention.

[0055] In one aspect, the present invention provides a pharmaceutical composition comprising the polynucleotide according to the present invention, the vector according to the present invention, or the cell according to the present invention.

[0056] In one aspect, the present invention provides a polynucleotide according to the present invention, vector according to the present invention, or cell according to the present invention for use in medicine.

[0057] In one aspect, the present invention provides for use of the promoter according to the present invention to drive expression of a coding sequence. The expression may be kidney-specific. Suitably, the expression is podocyte-specific.

DESCRIPTION OF DRAWINGS

[0058] FIG. 1—Schematic illustration of minimal nephrin promoters

[0059] (A) An exemplary full-length nephrin promoter may be 1249 bp in length (excluding the start codon), hereafter referred to as the “FL” nephrin promoter. (B) An exemplary minimal nephrin promoter with the 5' region deleted may be 819 bp in length (excluding the start codon), hereafter referred to as the “midi” nephrin promoter. (C) An exemplary minimal nephrin promoter with the 5' region deleted and central region deleted may be 265 bp in length (excluding the start codon), hereafter referred to as the “mini” nephrin promoter. (D) The following regions of the nephrin promoter are indicated: (i) human mouse homology region, (ii) retinoic acid receptor (RAR) binding site, (iii) WT1 binding site, (iv) Enhancer box, (v) transcription factors binding region, and (vi) transcription initiation site.

[0060] FIG. 2—Schematic of a lentiviral vector comprising GFP operably coupled to midi nephrin promoter

[0061] (A) pACE_hNPHS1 promoter was used as a template, to introduce BamHI and ClaI restriction sites. (B) Final construct vector comprising GFP operably coupled to midi nephrin promoter.

[0062] FIG. 3—Schematic of a lentiviral vector comprising GFP operably coupled to mini nephrin promoter

[0063] (A) pACE_hNPHS1 promoter was used as a template to PCR and gel extract the two sections of the promoter. (B) Final construct vector comprising GFP operably coupled to mini nephrin promoter.

[0064] FIG. 4—Expression of GFP in CiPodocytes following transduction with lentiviral vectors

[0065] Human CiPodocytes stably expressing GFP-tagged nephrin promoters were generated using the lentiviral approach. GFP expression was observed by fluorescence microscopy. (A) Untransduced CiPodocytes. (B) CiPodocytes stably expressing the GFP-tagged mini nephrin promoter. (C) CiPodocytes stably expressing the GFP-tagged FL nephrin promoter.

[0066] FIG. 5—Expression of GFP in differentiated ciPodocytes following transduction with lentiviral vectors

[0067] Lentiviral vectors comprising GFP operably linked to nephrin promoters were transduced into differentiated

conditionally immortalised podocytes (ciPodocytes). Immunoprecipitation (IP) was used to detect GFP expression.

[0068] FIG. 6—Human glomerular cells transduced with Lentivirus—GFP.nephrin promoter (FL or 265)

[0069] FACS analysis displaying median GFP fluorescence (AFU) of all live singlets of conditionally immortalised human podocytes (LY) and glomerular endothelial cells (GEnC) using a Novocyte Analyser. Untransduced cells (Cell Control) were compared with those transduced with lentivirus constructs harbouring a GFP expression cassette controlled by a full length human nephrin promoter (hNPHS1.GFP) or a mini human nephrin promoter (265. GFP). All cells were differentiated for 10 days, trypsinised (100 uL) and diluted in PBS, 2% FBS, 1:1000 DRAQ7 (150 uL). Data and error bars represent 3 technical repeats (100 uL, >2500 cells)+SEM.

[0070] FIG. 7—AAV transfer plasmids comprising COL4A3, COL4A4, and COL4A5 coupled to mini nephrin promoter

[0071] (A) Schematic for pAAV.265.Col4a3.3flag.sv40, an AAV plasmid comprising COL4A3 coupled to mini nephrin promoter. SmaI sites are shown and the following fragments are expected following restriction with SmaI: 1. 6238 bp, 2. 2753 bp, 3. 56 bp, 4. 11 bp, 5. 11 bp. (B) Schematic for pAAV.265.Col4a4.3flag.sv40, an AAV plasmid comprising COL4A4 coupled to mini nephrin promoter. SmaI sites are shown and the following fragments are expected following restriction with SmaI: 1. 4052 bp, 2. 2753 bp, 3. 2224 bp, 4. 56 bp, 5. 11 bp, 6. 11 bp. (C) Schematic for pAAV.265.Col4a5.3flag.sv40, an AAV plasmid comprising COL4A5 coupled to mini nephrin promoter. SmaI sites are shown and the following fragments are expected following restriction with SmaI: 1. 4272 bp, 2. 2753 bp, 3. 2032 bp, 4. 56 bp, 5. 11 bp, 6. 11 bp. (D) Restriction digest with SmaI. MW=1 Kb DNA ladder, Lanes 1, 2 and 3 correspond to the digests for plasmids shown in (A), (B), and (C), respectively. (E) Schematic showing restriction digest.

[0072] FIG. 8—Podocytes transduced with AAV.COL4.nephrin265.Sv40 virus

[0073] (A) Immunoprecipitation experiments of the full-length FLAG-tagged Col4a3 (LK03) or Col4a5 (LK03) in human differentiated CiPodocytes (conditionally immortalised) pulled down with the anti-FLAG antibody. Anti-FLAG antibody precipitated both Col4a3 and Col4a5. Human-FLAG IgG was used as a control. (B) Western blots of protein lysates showing the expression levels of Col4a3 (LK03 capsid serotype), Col4a5 (LK03) and Col4a5 (2/9 capsid serotype) in human or mouse differentiated CiPodocytes. Non-infected human and mouse CiPodocytes were used as controls. (C) Confocal images showing immunofluorescence staining of transduced Col4a5 in Human wild-type CiPodocytes/Col4a5 3xFlag AAV CiPodocytes with F-Actin. Col4a5 is present at the cytosolic level in the human differentiated podocytes infected with Col4a5 3xFlag AAV virus in comparison to the wild-type counterpart.

[0074] FIG. 9—Expression of CFH, CFI and CFHL1 in HEK cells

[0075] Protein expression analysis by western blot in 293T Human Embryonic Kidney cells. Cells were transfected with pAAV-265-CFH, pAAV-FL-CFI or pAAV-FL-CFHL1. NT (non-transfected 293T HEK cells). Expression of each of the transgenes is demonstrated in the cell lysate and/or media

using protein-specific antibodies or anti-FLAG-tag or anti-MYC-tag antibodies. HEK293T cells transfected with plasmid expressing the CFH transgene under the control of the 265 bp minimal nephrin promoter expressed and secreted human Factor H.

[0076] FIG. 10—Transduction of Factor H mutated podocytes (“Human early disease (ED) podocytes”) with AAV2/9 265-CFH or transfection with plasmid encoding CFH

[0077] (A) Analysis of human Factor H concentration using an ELISA assay. Podocytes transduced with AAV2/9 virus containing the CFH transgene demonstrated higher concentrations of human Factor H in the culture media than the non-transduced control. (B) The average human Factor H concentration from (A). (C) Analysis of human Factor H concentration using an ELISA assay. Podocytes transfected with plasmid expressing the CFH transgene under the control of the 265 bp minimal nephrin promoter demonstrated higher concentrations of human Factor H than the non-transfected control.

[0078] FIG. 11—Expression of CFH in the kidney in WT mice

[0079] Gene expression profile demonstrating an increase in (A) viral ITRs, (B) human CFH cDNA, and (C) viral particles in wild-type mice injected with pAAV.NPHS1 (265).hCFH.WPRE.bGH in comparison to mice injected with the saline control. (D) Immunofluorescence imaging of kidney sections derived from wild-type C57BL6 mice injected with the saline control (m1) or the AAV gene therapy product (pAAV.NPHS1 (265).hCFH.WPRE.bGH, n=3) (m2/m3/m4). Arrows indicate the co-localization of nephrin and CFH in the mouse glomerulus. Zoom=20×.

[0080] FIG. 12—Expression of GFP using 265 bp v 818 bp v FL minimal nephrin promoters (plasmid transfection) in kidney cells

[0081] Expression of eGFP in cells transfected with plasmids containing Full Length (PS0281), 818 bp (PS0301) and 265 bp (PS0282) minimal nephrin promoters driving eGFP. (A) Transfection results in HEK293T cells. (B) Transfection results in Proximal Tube Epithelial cells (PTECs). (C) Transfection results in human Podocytes. (D) Transfection results in Glomerular Endothelial cells (GENCs).

[0082] FIG. 13—Expression of GFP using 265 bp v FL minimal promoters (AAV transduction) in kidney cells

[0083] Expression of eGFP in cells transduced with AAV particles incorporating transgene cassettes containing Full Length (PS0281) and 265 bp (PS0282) minimal nephrin promoters driving eGFP. (A) Transduction results in human Podocytes. (B) Transduction results in Proximal Tube Epithelial cells (PTECs). (C) Transduction results in Glomerular Endothelial cells (GENCs). (D) Transduction results in Mesangial cells.

[0084] FIG. 14—Expression of human podocin using 265 bp v FL minimal nephrin promoters

[0085] (A) Protein expression analysis by FACs in Lenti-X 293T Human Embryonic Kidney cells. Cells were transfected with 3 µg of plasmid with PEI at ratio of 1:3. Each plasmid expresses human Podocin-HA driven by either full length (FL.NPSH1-Podocin-HA) or 265 bp (265.hNPSH1-Podocin-HA) minimal nephrin promoter. Expression of human Podocin is demonstrated by anti-HA antibody on Day 2 post transfection. N=3 data expressed as mean values±SD. (B) Protein expression analysis by FACs in Human Podocytes cells. Cells were transduced with AAV particles, incorporating human Podocin-HA transgene cas-

sette driven by either full length (FL.NPSH1-Podocin-HA) or 265 bp (265.hNPSH1-Podocin-HA) minimal nephrin promoter. Expression of human Podocin is demonstrated by anti-HA antibody on Day 9 post transduction and differentiation.

[0086] FIG. 15—In vivo expression of GFP using 265 bp v FL minimal nephrin promoters via AAV transduction

[0087] Evaluation of 265- and FL-nephrin promoter in vivo in WT mice (C57/BI6). AAV2/9 particles were generated expressing GFP under the control of either full-length (FL-GFP) or 265 bp (265-GFP) minimal nephrin promoters. Tail vein injection of ~1×10¹³ AAV2/9 per mouse (2× mice per treatment group and 1× untreated mouse). Tissue harvest of kidney occurred 4 weeks post injection. qPCR analysis was performed and GFP copy number normalised to 18S (housekeeping gene).

DETAILED DESCRIPTION

[0088] Various preferred features and embodiments of the present invention will now be described by way of non-limiting examples.

[0089] It must be noted that as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

[0090] The terms “comprising”, “comprises” and “comprised of” as used herein are synonymous with “including”, “includes” or “containing”, “contains”, and are inclusive or open-ended and do not exclude additional, non-recited members, elements or method steps. The terms “comprising”, “comprises” and “comprised of” also include the term “consisting of”.

[0091] The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that such publications constitute prior art to the claims appended hereto.

[0092] This disclosure is not limited by the exemplary methods and materials disclosed herein, and any methods and materials similar or equivalent to those described herein can be used in the practice or testing of embodiments of this disclosure. Numeric ranges are inclusive of the numbers defining the range. Unless otherwise indicated, any nucleic acid sequences are written left to right in 5' to 3' orientation; amino acid sequences are written left to right in amino to carboxy orientation, respectively.

[0093] The practice of the present invention will employ, unless otherwise indicated, conventional techniques of chemistry, biochemistry, molecular biology, microbiology and immunology, which are within the capabilities of a person of ordinary skill in the art. Such techniques are explained in the literature. See, for example, Sambrook, J., Fritsch, E. F. and Maniatis, T. (1989) *Molecular Cloning: A Laboratory Manual*, 2nd Edition, Cold Spring Harbor Laboratory Press; Ausubel, F. M. et al. (1995 and periodic supplements) *Current Protocols in Molecular Biology*, Ch. 9, 13 and 16, John Wiley & Sons; Roe, B., Crabtree, J. and Kahn, A. (1996) *DNA Isolation and Sequencing: Essential Techniques*, John Wiley & Sons; Polak, J. M. and McGee, J. O'D. (1990) *In Situ Hybridization: Principles and Practice*, Oxford University Press; Gait, M. J. (1984) *Oligonucleotide Synthesis: A Practical Approach*, IRL Press; and Lilley, D. M. and Dahlberg, J. E. (1992) *Methods in Enzymology*:

DNA Structures Part A: Synthesis and Physical Analysis of DNA, Academic Press. Each of these general texts is herein incorporated by reference.

Promoters

[0094] The present invention provides a promoter for glomerular gene therapy.

[0095] A “promoter” is used herein according to its typical meaning to refer to a sequence of DNA which regulates and/or initiates transcription of a RNA from a DNA sequence.

[0096] Suitably, the promoter is operable in mammalian cells, e.g. human cells. The promoter may be capable of driving transgene expression in mammalian cells, e.g. human cells. The promoter may be a mammalian promoter, e.g. a human promoter.

[0097] The promoter may be operable in kidney cells. The promoter of the invention may be capable of driving transgene expression in the kidney. Examples of kidney cells in which the promoter may be operable include, but are not limited to glomerular cells.

[0098] The promoter may be operable in glomerular cells. The promoter of the invention may be capable of driving transgene expression in the glomerulus. The mature glomerulus contains four cell types: Parietal epithelial cells that form Bowman’s capsule, podocytes that cover the outermost layer of the glomerular filtration barrier, glycocalyx-coated fenestrated endothelial cells that are in direct contact with blood, and mesangial cells that sit between the capillary loops (Vaughan, M. R. and Quaggin, S. E., 2008. *Journal of the American Society of Nephrology*, 19 (1), pp. 24-33).

[0099] The promoter may be operable in a podocyte cell. The promoter of the invention may be capable of driving transgene expression in podocytes.

[0100] The promoter of the invention may be a tissue-specific promoter. As used herein, a “tissue-specific promoter” is a promoter which preferentially facilitates expression of a transgene in a specific type of cells or tissue. Suitably, a tissue-specific promoter may facilitate higher expression of a transgene in one cell-type as compared to other cell-types. For example, a tissue-specific promoter may be a promoter which facilitates transgene expression levels at least 10% higher, at least 20% higher, at least 30% higher, at least 40% higher, at least 50% higher, at least 100% higher, at least 200% higher, at least 300% higher, at least 400% higher, at least 500% higher, or at least 1000% higher in one cell-type as compared to expression levels in other cell-types.

[0101] Suitably, the promoter is a kidney-specific promoter. In some embodiments, the promoter is a glomerular-specific promoter. In some embodiments, the promoter is a podocyte-specific promoter.

[0102] Transgene expression may be measured by any suitable method known in the art. For example by measuring the expression of a reporter transgene, e.g. Green fluorescent protein (GFP), operably linked to the promoter, wherein expression of the reporter transgene correlates with the ability of the promoter to facilitate expression of a gene. Expression of the reporter transgene, e.g. GFP may be determined by any suitable method e.g. FACS. For example, a kidney-specific promoter may facilitate higher expression of a reporter transgene in kidney cells compared to other cell types (e.g. CNS, retina, lung, pancreas, heart, or muscle cells). For example, a podocyte-specific promoter may

facilitate higher expression of a reporter transgene in conditionally immortalised podocytes compared to other cell-types. Suitable podocyte cell lines will be well known to those of skill in the art, for example CIHP-1. Methods to generate immortalized podocytes will be well known to those of skill in the art. Suitable methods are described in Ni, L., et al., 2012. *Nephrology*, 17 (6), pp. 525-531.

[0103] Suitably, the promoter is a minimal kidney-specific promoter. In some embodiments, the promoter is a minimal glomerular-specific promoter. In some embodiments, the promoter is a minimal podocyte-specific promoter.

[0104] The promoter may have a length of about 1.2 kb or less. Suitably, the promoter has a length of about 1.18 kb or less, about 1.17 kb or less, about 1.16 kb or less, about 1.15 kb or less, about 1.14 kb or less, about 1.13 kb or less, about 1.12 kb or less, about 1.11 kb or less, or about 1.10 kb or less. Suitably, the promoter has a length of about 1.15 kb or less.

[0105] The promoter may have a length of about 1.1 kb or less. In some embodiments, the promoter has a length of about 1.1 kb or less, 1.0 kb or less, about 0.9 kb or less, about 0.8 kb or less, about 0.7 kb or less, about 0.6 kb or less, about 0.5 kb or less, about 0.4 kb or less, or about 0.3 kb or less.

[0106] In some embodiments, the promoter has a length of about 0.8 kb or less, about 0.7 kb or less, about 0.6 kb or less, about 0.5 kb or less, about 0.4 kb or less, or about 0.3 kb or less. In some embodiments, the promoter has a length of 818 bp or less. In some embodiments, the promoter has a length of 800 bp or less.

[0107] In some embodiments, the promoter has a length of about 0.5 kb or less, about 0.4 kb or less, or about 0.3 kb or less. In some embodiments, the promoter has a length of about 0.3 kb or less.

[0108] The promoter may have a length of about 250 bp or more. In some embodiments, the promoter has a length of about 250-1100 bp, 250-1000 bp, 250-900 bp, 250-800 bp, 250-700 bp, 250-600 bp, 250-500 bp, 250-400 bp, 250-300 bp.

[0109] The promoter may have a length of about 265 bp or more. In some embodiments, the promoter has a length of about 265-1100 bp, 265-1000 bp, 265-900 bp, 265-800 bp, 265-700 bp, 265-600 bp, 265-500 bp, 265-400 bp, 265-300 bp.

[0110] In one embodiment, the promoter has a length of 250-300 bp, 250-280 bp, 255-275 bp, 260-270 bp, or about 265 bp. In one embodiment, the promoter has a length of 800-850 bp, 800-840 bp, 810-830 bp, 815-825 bp, about 819 bp, or about 818 bp.

Nephrin Promoters

[0111] The promoter may be a synthetic promoter. As used herein, a “synthetic promoter” is a promoter which does not occur naturally. Typically, a synthetic promoter will be produced by genetic modification of a pre-existing promoter. Thus, a synthetic promoter may be derived from a pre-existing promoter, i.e. identical to a pre-existing promoter, except comprising one or more genetic modifications (e.g. substitutions, deletions, or insertions).

[0112] Suitably, the promoter may be derived from the nephrin (NPHS1) promoter. Suitably, the promoter may be a minimal nephrin promoter. The NPHS1 gene encodes nephrin, which is selectively expressed in podocytes.

[0113] A human NPHS1 promoter has been described in Moeller et al. 2002 J Am Soc Nephrol, 13 (6): 1561-7 and Wong M A et al. 2000 Am J Physiol Renal Physiol, 279 (6): F1027-32. This NPHS1 promoter is a 1.2 kb fragment and appears to be podocyte-specific. The 1.2 kb promoter region lacks a TATA box, but has recognition motifs for other

transcription factors e.g. PAX-2 binding element, E-box and GATA consensus sequences.

[0114] Exemplary nephrin promoters are shown as SEQ ID NO: 1, SEQ ID NO: 14, SEQ ID NO: 44, and SEQ ID NO: 45.

Exemplary nephrin promoter-1,249 bp

(SEQ ID NO: 1)

CCTGCAGGGCCCACTAGTCTGTAATCCCAGCATTTTGGGAGGCTGAGGCAGATGGATCACCTGAGGTC
AGGAGTTCGAGACCAGCCTGGCCAACATGATGAAACCCGTCTCTAGTAAAAATACAAAAATTAGCCA
GGCATGGTGTCTATATACCTGTAGTACCAGCTACTTGGGAGACAGAGGTGGGAGAATTACTTGAACCTG
GGAGGTTCAAGCCATGGGAGGTGGAAGTTGCAGTGAGCCGAGATGCCACTGCACTCCAGCCTGAGCAA
CAGAGCAAGACTATCTCAAGAAAAAAGAAAGAAAGAAAGGAGTTCAGGTCATGTATCAGGGC
AAGGAAGAGCTGGGGGCCAGCTGGCTGCTCCCTGCTGAGCTGGGAGACCACCTTGATCTGACTTCT
CCCATCTTCCAGCCTAAGCCAGGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAGGAACGCGCTG
GCATGTGCTGACAGGGAATTTTATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCA
GAGCTGGGGGCTCTGGAAGGTACCTGGGGAGGTTGCAGCTGTGAGAATGAGCTCAAGCTGGGTGAGAG
AGCAGGGCTGACTCTGCCAGTGCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCCTGCTGGACT
CTGGGCTGCAGGTCTTCTTGAAAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGG
AGAGAAGATAGAGATTGAGAGAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGA
GAGAGAACGGCTTAGACAAGGAGAGAAAGATGGAAAGATAAAGAGACTGGGCGCAGTGCTCAGCCCT
GTAATCCCAACACTTGGGAGGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTG
GCCAACATAGTGAGACCCGTCTCTAAAAAAGAAAAAAGAAAAAAGAAAAAAGAAAAAAGTTT
TTTTAAAGAGACAGAGAAAGAGACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGACTCACAG
GGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGTAACGGAAGAGATAAAAAAGAAAGCAGGT
GGCAGAGACACAGAGAGAGGGACCCAGAGAAAGCCAGACAGACGAGGTGGCTGGCAGCGGGCGCTGT
GGGGGTACAGTAGGGGACCTGTC

Exemplary variant nephrin promoter-1,192 bp

(SEQ ID NO: 14)

CACCTGAGGTCAGGAGTTCGAGACCAGCGTGGCCAACATGATGAAACCCGTCTCTAGTAAAAATACA
AAAATTAGCCAGGCATGGTGTCTATATACCTGTAGCACCAGCTACTTGGGAGACAGAGTGGGAGAATT
ACTTGAACCTGGGAGGTTCAGCCATGGGAGGTGGAAGTTGCAGTGAGCCGAGATGCCACTGCACTCC
AGCCTGAGCAACAGAGCAAGACTATCTCAAGAAAAGAAAGAAAGAAAGAGACTTCCAGGTCA
TGTATCAGGGCAAGGAAGAGCTGGGGGCCAGCTGGCTGCTCCCTGCTGAGCTGGGAGACCACCTTG
ATCTGACTTCTCCCATCTTCCAGCCTAAGCCAGGCCCTGGGGTCACGGAGGCTGGGAGGCACCGAG
GAACGCGCTGGCATGTGCTGACAGGGGATTTTATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTG
GGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTACCTGGGGAGGTTGCAGTGTGAGAATGAGCTCAAG
CTGGGTGAGAGAGCAGGGCTGACTCTGCCAGTGCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGG
CCTGCTGGACTCTGGGCTGCAGGTCTTCTTGAAAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGA
GGGGTGGCAGGAGAGAAGATAGAGATTGAGAGAGAGAGAGAGAGAGACAGAGAGAGAGGAAGAGAC
AGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGAGAGAAAGATGGAAGATAAAGAGACTGGGCGCA
GTGGCTCACGCTGTAATCCCAACACTTGGGAGGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTC
TGAGATCAACCTGGCCAACATAGTGAGACCCGTCTCTAAAAAAGAAAAAAGAAAAAAGAAAAAGA

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AAAAAAGTTTTTTTAAAGAGACAGAGAAAGAGACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGA
 GATACTCAGAGGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGTAACGGAAAGAGATAAAAAA
 GAAAAGCAGGTGGCAGAGACACACAGAGAGGGACCCAGAGAAAGCCAGACAGACGCAGGTGGCTGGCA
 GCGGGCGCTGTGGGGGTACAGTAGGGGGACCTGTG

Exemplary variant nephrin promoter-1,192 bp

(SEQ ID NO: 44)

CACCTGAGGT CAGGAGTTCGAGACCAGCGTGGCCACATGATGAAACCCCGTCTCTAGTAAAAATACA
 AAAATTAGCCAGGCATGGTGCTATATACCTGTAGCACCAGCTACTTGGGAGACAGAGGTGGGAGAATT
 ACTTGAACCTGGGAGGTTCAAGCCATGGGAGGTGGAAGTTGCAGTGAGCCGAGATGCCACTGCACTCC
 AGCCTGAGCAACAGAGCAAGACTATCTCAAGAAAAGAAAGAAAGAAAGAGACTTGCCAAGGTCA
 TGTATCAGGGCAAGGAAGAGCTGGGGGCCAGCTGGCTGCTCCCTGCTGAGCTGGGAGACCACCTTG
 ATCTGACTTCTCCCATCTTCCAGCCTAAGCCAGGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAG
 GAACGCGCTGGCATGTGCTGACAGGGAATTTATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTG
 GGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTACCTGGGGGAGGTTGCACTGTGAGAATGAGCTCAAG
 CTGGGT CAGAGAGCAGGGCTGACTCTGCCAGTGCCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGG
 CCTGCTGACTCTGGGCTGCAGGTCCTTCTTGAAAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGA
 GGGGTGGCAGGAGAGAAGATAGAGATTGAGAGAGAGAGAGAGAGAGACAGAGAGAGAGGAAGAGAC
 AGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGAGAGAAAGATGGAAGATAAAGAGACTGGGCGCA
 GTGGCTCAGCCTGTAAATCCCAACTTGGGGAGGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTC
 TGAGATCAACCTGGCCAACATAGTGAGACCCCGTCTCTAAAAAAGAAAAAAGAAAAAAGAAAAAGA
 AAAAAAGTTTTTTTAAAGAGACAGAGAAAGAGACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGA
 GACACTCAGAGGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGTAACGGAAAGAGATAAAAAA
 GAAAAGCAGGTGGCAGAGACACAGAGAGAGGGACCCAGAGAAAGCCAGACAGACGCAGGTGGCTGGCA
 GCGGGCGCTGTGGGGGTACAGTAGGGGGACCTGTG

Exemplary variant nephrin promoter-1,192 bp

(SEQ ID NO: 45)

CACCTGAGGT CAGGAGTTCGAGACCAGCGTGGCCACATGATGAAACCCCGTCTCTAGTAAAAATACA
 AAAATTAGCCAGGCATGGTGCTATATACCTGTAGCACCAGCTACTTGGGAGACAGAGGTGGGAGAATT
 ACTTGAACCTGGGAGGTTCAAGCCATGGGAGGTGGAAGTTGCAGTGAGCCGAGATGCCACTGCACTCC
 AGCCTGAGCAACAGAGCAAGACTATCTCAAGAAAAGAAAGAAAGAAAGAGACTTGCCAAGGTCA
 TGTATCAGGGCAAGGAAGAGCTGGGGGCCAGCTGGCTGCTCCCTGCTGAGCTGGGAGACCACCTTG
 ATCTGACTTCTCCCATCTTCCAGCCTAAGCCAGGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAG
 GAACGCGCTGGCATGTGCTGACAGGGGATTTATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTG
 GGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTACCTGGGGGAGGTTGCACTGTGAGAATGAGCTCAAG
 CTGGGT CAGAGAGCAGGGCTGACTCTGCCAGTGCCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGG
 CCTGCTGACTCTGGGCTGCAGGTCCTTCTTGAAAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGA
 GGGGTGGCAGGAGAGAAGATAGAGATTGAGAGAGAGAGAGAGAGAGACAGAGAGAGAGGAAGAGAC
 AGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGAGAGAAAGATGGAAGATAAAGAGACTGGGCGCA
 GTGGCTCAGCCTGTAAATCCCAACTTGGGGAGGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTC
 TGAGATCAACCTGGCCAACATAGTGAGACCCCGTCTCTAAAAAAGAAAAAAGAAAAAAGAAAAAGA

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AAAAAAGTTTTTTTAAAGAGACAGAGAAAGAGACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGA

GATACTCAGAGGGAAGAGGGGAAGAGGAAAACGAGAAAGGGGAGGAGTAACGGAAAGAGATAAAAAA

GAAAAGCAGGTGGCAGAGACACAGAGAGAGGGACCCAGAGAAAGCCAGACAGACGCAGGTGGCTGGCA

CGGGGCGCTGTGGGGGTACAGTAGGGGGACCTGTCT

[0115] Suitably, the promoter of the present invention is derived from SEQ ID NO: 1 or a variant that has at least 70% identity to SEQ ID NO: 1. Suitably, the promoter of the present invention has one or more deletions compared to SEQ ID NO: 1 or a variant that has at least 70% identity to SEQ ID NO: 1. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 1. An exemplary variant is SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45.

[0116] Suitably, the variant of SEQ ID NO: 1 comprises one or more of the following substitutions: A493G, C1080T, G1169C, and C1249G. Suitably, the variant of SEQ ID NO: 1 comprises each of the substitutions A493G, C1080T, G1169C, and C1249G (see SEQ ID NO: 14).

[0117] Suitably, the promoter of the present invention is derived from SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45, or a variant that has at least 70% identity to SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45. Suitably, the promoter of the present invention has one or more deletions compared to SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45, or a variant that has at least 70% identity to SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 14, SEQ ID NO: 44, or SEQ ID NO: 45.

[0118] Suitably, the promoter of the present invention is derived from SEQ ID NO: 14, or a variant that has at least 70% identity to SEQ ID NO: 14. Suitably, the promoter of the present invention has one or more deletions compared to SEQ ID NO: 14, or a variant that has at least 70% identity to SEQ ID NO: 14. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 14.

[0119] Suitably, the promoter of the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and having one or more deletions, e.g. one or two deletions, or nucleotide sequence with at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity thereto. Suitably, the promoter of the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and having two or more deletions, or nucleotide sequence with at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity thereto. The deletions may be any size. Suitably, the deletions are each at least 50 bp, at least 100 bp, at least 150 bp, at least 200 bp, at least 250 bp, at least 300 bp, at least 350 bp, or at least 400 bp in size. Suitably, the deletions are each from 50 to 500 bp, from 100 to 500 bp, from 150 to 500 bp, from 200 to 500 bp, from 250 to 500 bp, from 300 to 500 bp, from 350 to 500 bp, or from 400 to 500 bp in size.

[0120] In some embodiments, the promoter of the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, but wherein:

[0121] (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 1 to 430; and/or

[0122] (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061;

[0123] or a nucleotide sequence with at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity thereto.

[0124] In some embodiments:

[0125] (iii) the position corresponding to position 493 of SEQ ID NO: 1 is G, the position corresponding to position 1080 of SEQ ID NO: 1 is T, the position corresponding to position 1169 of SEQ ID NO: 1 is C, and/or the position corresponding to position 1249 of SEQ ID NO: 1 is G.

[0126] In preferred embodiments:

[0127] (iii) the position corresponding to position 493 of SEQ ID NO: 1 is G, the position corresponding to position 1080 of SEQ ID NO: 1 is T, the position corresponding to position 1169 of SEQ ID NO: 1 is C, and the position corresponding to position 1249 of SEQ ID NO: 1 is G.

[0128] Although the nucleotide positions above are referred to by reference to a position in SEQ ID NO: 1, the skilled person would be able to readily identify the corresponding nucleotide location in variants thereof, for example by aligning SEQ ID NO: 1 and the variant sequence. For example, position 493 of SEQ ID NO: 1 corresponds to position 436 of SEQ ID NO: 14; position 1080 of SEQ ID NO: 1 corresponds to position 1023 of SEQ ID NO: 14; position 1169 of SEQ ID NO: 1 corresponds to position 1112 of SEQ ID NO: 14; and position 1249 of SEQ ID NO: 1 corresponds to position 1192 of SEQ ID NO: 14.

[0129] For example, the promoter of the present invention may comprise or consist of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% identity to a nucleotide sequence according to SEQ ID NO: 1, but wherein:

[0130] (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 1 to 430; and/or

[0131] (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061.

[0132] In some embodiments:

[0133] (iii) position 493 of SEQ ID NO: 1 is substituted with G, position 1080 of SEQ ID NO: 1 is substituted with T, position 1169 of SEQ ID NO: 1 is substituted with C, and/or position 1249 of SEQ ID NO: 1 is substituted with G.

[0134] In preferred embodiments:

[0135] (iii) position 493 of SEQ ID NO: 1 is substituted with G, position 1080 of SEQ ID NO: 1 is substituted with T, position 1169 of SEQ ID NO: 1 is substituted with C, and position 1249 of SEQ ID NO: 1 is substituted with G.

[0136] Suitably, n1 is an integer from 50 to 430, from 100 to 430, from 150 to 430, from 200 to 430, from 250 to 430, from 300 to 430, from 350 to 430, or from 400 to 430. In some embodiments, n1 is an integer from 100 to 430. In some embodiments, n1=430, i.e. position 1 to position 430 of SEQ ID NO: 1 is deleted in the promoter of the invention.

[0137] The difference between n3 and n2 specifies the size of the deletion. Suitably, $n3 \geq n2 + 49$, $n3 \geq n2 + 99$, $n3 \geq n2 + 149$, $n3 \geq n2 + 199$, $n3 \geq n2 + 249$, $n3 \geq n2 + 299$, $n3 \geq n2 + 349$, $n3 \geq n2 + 399$, $n3 \geq n2 + 449$, $n3 \geq n2 + 499$, or $n3 \geq n2 + 549$. In some embodiments, $n3 \geq n2 + 49$.

[0138] The values that n2 and n3 take determine where the deletion is. Suitably, n2 and n3 are each integers from 550 to 1050, n2 and n3 are each integers from 600 to 1000, n2 and n3 are each integers from 650 to 950, n2 and n3 are each integers from 700 to 900, n2 and n3 are integers from 750 to 850.

[0139] In some embodiments, n2=508 and n3=1061, i.e. position 508 to 1061 of SEQ ID NO: 1 is deleted in the promoter of the invention.

Promoter Regions

[0140] The present inventors have determined the regions of the nephrin promoter which drive transgene expression.

[0141] A promoter typically comprises a “core” and a “proximal” region. The “core promoter region” may comprise a transcription start site, a RNA polymerase binding sites and a general transcription factor binding site. The “proximal promoter region” may comprise primary regulatory elements and specific transcription factor binding sites which are required, for example, to facilitate effective and controllable transcription. The size and components of both the core and proximal promoter regions typically vary in a gene specific manner. A promoter may also comprise a 5' untranslated region (5' UTR) (also known as a leader sequence) downstream of the core promoter region and upstream from the initiation codon. (See e.g. FIG. 1).

[0142] The promoter may be a hybrid promoter. As used herein, a “hybrid promoter” comprises a combination of elements derived from different promoters. For example, a hybrid promoter may comprise a proximal promoter region derived from one pre-existing promoter and a core promoter from another pre-existing promoter to achieve the desired transgene expression. A muscle hybrid promoter is described in Piekarczyk, K., et al. (2019). Methods & clinical development, 15, 157-169.

[0143] In some embodiments, the promoter of the present invention comprises (i) the nucleotide sequence shown as SEQ ID NO: 4, or a variant which is at least 70% identical to SEQ ID NO: 4. Without wishing to be bound by theory, it is considered that a nucleotide sequence having at least about 70% identity to SEQ ID NO: 4 may provide a proximal promoter region.

Exemplary proximal promoter region

(SEQ ID NO: 4)
GGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAGGAACGCGCCTGGCAT
GTGCTGACAGGGAATTTTATGCTCCAG

[0144] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 4. The promoter may comprise a variant of SEQ ID NO: 4 shown as SEQ ID NO: 17. Suitably, a variant of SEQ ID NO: 4 comprises the substitution A63G (see SEQ ID NO: 17).

[0145] In some embodiments, the promoter of the present invention comprises (i) the nucleotide sequence shown as SEQ ID NO: 17, or a variant which is at least 70% identical to SEQ ID NO: 17. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 17.

Exemplary variant proximal promoter region

(SEQ ID NO: 17)
GGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAGGAACGCGCCTGGCAT
GTGCTGACAGGGGATTTTATGCTCCAG

[0146] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 5 or 18, or a variant which is at least 70% identical to SEQ ID NO: 5 or 18; the nucleotide sequence shown as SEQ ID NO: 6 or 19, or a variant which is at least 70% identical to SEQ ID NO: 6 or 19; and/or the nucleotide sequence shown as SEQ ID NO: 7 or 20, or a variant which is at least 70% identical to SEQ ID NO: 7 or 20.

[0147] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 5, or a variant which is at least 70% identical to SEQ ID NO: 5. Without wishing to be bound by theory, it is considered that a nucleotide sequence having at least about 70% identity to SEQ ID NO: 5 may provide a core promoter region.

Exemplary core promoter region

(SEQ ID NO: 5)
GAGCAAGACAGAGAGAGACACTCACAGGGAAG

[0148] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 5. The promoter may comprise a variant of SEQ ID NO: 5 shown as SEQ ID NO: 18. Suitably, a variant of SEQ ID NO: 5 comprises the substitution C19T (see SEQ ID NO: 18).

[0149] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 18, or a variant which is at least 70% identical to SEQ ID NO: 18. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 18.

Exemplary variant core promoter region
(SEQ ID NO: 18)
GAGCAAGACAGAGAGAGATACTCACAGGGAAG

[0150] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 6, or a variant which is at least 70% identical to SEQ ID NO: 6. Without wishing to be bound by theory, it is considered that a nucleotide sequence having at least about 70% identity to SEQ ID NO: 6 may provide a 5'UTR.

Exemplary 5'UTR
(SEQ ID NO: 6)
AGGGGAAGAGGAAAACGAGAAAGGGAGGAGAGTAACGGAAGAGATAAAA
AAGAAAAGCAGGTGGCAGAGACACAGAGAGAGGGACCCAGAGAAAGCCAG
ACAGACGCAGGTGGCTGGCAGCGGGCGCTGTGGGGGTCACAGTAGGGGGA
CCTGTC

[0151] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 6. The promoter may comprise a variant of SEQ ID NO: 6 shown as SEQ ID NO: 19. Suitably, a variant of SEQ ID NO: 6 comprises the substitutions G76C and/or C156G (see SEQ ID NO: 19).

[0152] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 19, or a variant which is at least 70% identical to SEQ ID NO: 19. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 19.

Exemplary variant 5'UTR
(SEQ ID NO: 19)
AGGGGAAGAGGAAAACGAGAAAGGGAGGAGAGTAACGGAAGAGATAAAA
AAGAAAAGCAGGTGGCAGAGACACACAGAGAGGGACCCAGAGAAAGCCAG
ACAGACGCAGGTGGCTGGCAGCGGGCGCTGTGGGGGTCACAGTAGGGGGA
CCTGTG

[0153] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 7, or a variant which is at least 70% identical to SEQ ID NO: 7. Without wishing to be bound by theory, it is considered that a nucleotide sequence having at least about 70% identity to SEQ ID NO: 7 may provide a core promoter region and a 5'UTR.

Exemplary core promoter region and 5'UTR
(SEQ ID NO: 7)
GAGCAAGACAGAGAGAGACACTCACAGGAAGAGGGGAAGAGGAAAACGA
GAAAGGGAGGAGAGTAACGGAAGAGATAAAAAAGAAAGCAGGTGGCAG
AGACACAGAGAGAGGGACCCAGAGAAAGCCAGACAGACGCAGGTGGCTGG
CAGCGGGCGCTGTGGGGTCACAGTAGGGGACCTGTC

[0154] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 7. The promoter may comprise a variant of SEQ ID NO:

7 shown as SEQ ID NO: 20. Suitably, a variant of SEQ ID NO: 7 comprises the substitutions C19T, G108C and/or C188G (see SEQ ID NO: 20). Suitably, a variant of SEQ ID NO: 7 comprises the substitutions C19T, G108C and C188G (see SEQ ID NO: 20).

[0155] In some embodiments, the promoter of the present invention comprises (ii) the nucleotide sequence shown as SEQ ID NO: 20, or a variant which is at least 70% identical to SEQ ID NO: 20. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 20.

Exemplary variant core promoter region and 5'UTR
(SEQ ID NO: 20)
GAGCAAGACAGAGAGAGATACTCACAGGGAAGAGGGGAAGAGGAAAACGA
GAAAGGGAGGAGAGTAACGGAAGAGATAAAAAAGAAAGCAGGTGGCAG
AGACACACAGAGAGGGACCCAGAGAAAGCCAGACAGACGCAGGTGGCTGG
CAGCGGGCGCTGTGGGGTCACAGTAGGGGGACCTGTC

[0156] In some embodiments, the promoter of the present invention comprises (iii) a nucleotide sequence having at least 70% identity to SEQ ID NO: 8, or one or more fragments thereof. Suitably, the promoter comprises a nucleotide sequence having at least about 70% identity to SEQ ID NO: 8, or one or more fragments thereof, immediately downstream of the proximal promoter region and/or immediately upstream of the core promoter region.

[0157] The promoter may comprise a nucleotide sequence that has at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to SEQ ID NO: 8, or one or more fragments thereof. The promoter may comprise the nucleotide sequence of SEQ ID NO: 8, or one or more fragments thereof.

Exemplary optional promoter region
(SEQ ID NO: 8)
CTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGGCTC
TGGAAGGTACCTGGGGAGGTTGCACTGTGAGATGAGCTCAAGCTGGGT
CAGAGAGCAGGGCTGACTCTGCCAGTGCCTGCATCAGCCTCATCGCTCTC
CTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCTCTTTGAAAGGC
TGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGAT
AGAGATTGAGAGAGAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAG
ACAAAAGGAGAGAGAACGGCTTAGACAAGGAGAGAAAGATGGAAGATAA
AGAGACTGGGCGCAGTGGCTCACGCCGTGAATCCCAACACTTGGGGAGGC
CAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAA
CATAGTGAGACCCCTCTCTAAAAAAGAGAGAGAGAGAGAGAGAGAGAGAG
GAAAAAAGTTTTTTAAAGAGACAGAGAAAGAGACTCAGAGATTGAGA
CTGA

[0158] Suitably, the one or more fragments are (a) a 5' terminal fragment; and/or (b) a 3' terminal fragment. Suitably, the 5' terminal fragment may be immediately downstream of the proximal promoter region. Suitably, the 3'

terminal fragment may be immediately upstream of the core promoter region. For example, the promoter of the present invention may comprise:

[0159] (a) a nucleotide sequence having at least 70% to positions 1 to x of SEQ ID NO: 8; and/or

[0160] (b) a nucleotide sequence having at least 70% identity to positions y to 554 of SEQ ID NO: 8;

[0161] wherein x and y are integers, and $y > x$.

[0162] The fragment(s) of SEQ ID NO: 8 may be any length. Suitably, the fragment(s) may have a length of about 500 bp or less, 450 bp or less, 400 bp or less, 350 bp or less, 300 bp or less, 250 bp or less, 200 bp or less, 150 bp or less, 100 bp or less, 50 bp or less, 40 bp or less, 30 bp or less, 20 bp or less, or 10 bp or less.

[0163] In some embodiments, the promoter does not comprise SEQ ID NO: 8.

[0164] In some embodiments, the promoter of the present invention comprises a nucleotide sequence having at least 70% identity to SEQ ID NO: 9, or a fragment thereof. Suitably, the promoter comprises a nucleotide sequence having at least about 70% identity to SEQ ID NO: 9, or a fragment thereof, immediately upstream of the proximal promoter region.

[0165] The promoter may comprise a nucleotide sequence that has at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to SEQ ID NO: 9, or a fragment thereof. The promoter may comprise the nucleotide sequence of SEQ ID NO: 9, or one or more fragments thereof.

Exemplary optional upstream promoter region
(SEQ ID NO: 9)
CCTGCAGGGCCCACTAGTCTGTAATCCAGCATTTTGGGAGCTGAGGCA
GATGGATCACCTGAGGTGAGGAGTTCGAGACCAGCCTGGCCAACATGATG
AAACCCCGTCTCTAGTAAAAATACAAAATTAGCCAGGCATGGTCTCTATA
TACCTGTAGTACCAGCTACTTGGGAGACAGAGGTGGGAGAATTACTTGAA
CCTGGGAGGTTCAAGCCATGGGAGGTGGAAGTTGCAGTGAGCCGAGATGC
CACTGCACCTCCAGCCTGAGCAACAGAGCAAGACTATCTCAAGAAAAAAA
GAAAGAAAGAAAGGGACTTGCCAAGGTGATGTATCAGGGCAAGGAAGAGC
TGGGGGCCAGCTGGCTGCTCCCTGCTGAGCTGGGAGACCACCTTGATC
TGACTTCTCCCATCTTCCAGCCTAAGCCA

[0166] Suitably, the fragment is a 3' terminal fragment. For example, the promoter of the present invention may comprise a nucleotide sequence having at least 70% identity to positions z to 430 of SEQ ID NO: 9, wherein z is an integer.

[0167] The fragment of SEQ ID NO: 9 may be any length. Suitably, the fragment may have a length of about 400 bp or less, 350 bp or less, 300 bp or less, 250 bp or less, 200 bp or less, 150 bp or less, 100 bp or less, 50 bp or less, 40 bp or less, 30 bp or less, 20 bp or less, or 10 bp or less.

[0168] In some embodiments, the promoter does not comprise SEQ ID NO: 9.

[0169] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0170] (i) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 4 or 17;

[0171] (iii) optionally a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 8, or one or more fragments thereof; and

[0172] (ii) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 7 or 20.

[0173] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0174] (i) a nucleotide sequence having at least 90% identity to SEQ ID NO: 4 or 17;

[0175] (iii) optionally a nucleotide sequence having at least 90% identity to SEQ ID NO: 8, or one or more fragments thereof; and

[0176] (ii) a nucleotide sequence having at least 90% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 90% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 90% identity to SEQ ID NO: 7 or 20.

[0177] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0178] (i) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 4 or 17;

[0179] (iii) optionally (a) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to a 5' terminal fragment of SEQ ID NO: 8; and/or (b) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to a 3' terminal fragment of SEQ ID NO: 8; and

[0180] (ii) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 7 or 20.

[0181] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0182] (i) a nucleotide sequence having at least 90% identity to SEQ ID NO: 4 or 17;

[0183] (iii) optionally (a) a nucleotide sequence having at least 90% identity to a 5' terminal fragment of SEQ ID NO: 8; and/or (b) a nucleotide sequence having at least 90% identity to a 3' terminal fragment of SEQ ID NO: 8; and

[0184] (ii) a nucleotide sequence having at least 90% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 90% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 90% identity to SEQ ID NO: 7 or 20.

Promoter Elements

[0185] The present inventors have determined the functional elements of the nephrin promoter which drive transgene expression.

[0186] The promoter of the invention may comprise one or more of the following elements: (a) a retinoic acid receptor binding site; (b) a WT1 binding site; (c) an enhancer box; (d) a transcription factor binding region; and (e) a transcription initiation site.

[0187] Suitably, the promoter of the invention comprises all of the following elements: (a) a retinoic acid receptor binding site; (b) a WT1 binding site; (c) an enhancer box; (d) a transcription factor binding region; and (e) a transcription initiation site.

[0188] A retinoic acid receptor (RAR) binding site refers to a polynucleotide sequence which is capable of binding RAR alpha, RAR beta, and/or RAR gamma. The RAR binding site may comprise or consist of a nucleotide sequence shown as SEQ ID NO: 10, or a nucleotide sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 10. The substitutions, deletions, or insertions may be any substitution, deletion, or insertion of a single nucleotide such that the RAR binding site retains at least one of its endogenous functions.

Exemplary RAR binding site
(SEQ ID NO: 10)
GGGGTCA

[0189] A WT1 binding site refers to a polynucleotide sequence which is capable of binding the zinc finger polypeptide encoded by the Wilms' tumor suppressor gene, WT1. The WT1 binding site may comprise or consist of a nucleotide sequence shown as SEQ ID NO: 11, or a nucleotide sequence having one, two or three substitutions, deletions, or insertions compared to SEQ ID NO: 11. The substitutions, deletions, or insertions may be any substitution, deletion, or insertion of a single nucleotide such that the WT1 binding region retains at least one of its endogenous functions.

Exemplary WT1 binding site
(SEQ ID NO: 11)
CGGAGGCTGGGGAGGCA

[0190] An enhancer box refers to a DNA response element found in some eukaryotes that acts as a protein-binding site. The enhancer box may comprise or consist of a nucleotide sequence shown as SEQ ID NO: 12, or a nucleotide

sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 12. The substitutions, deletions, or insertions may be any substitution, deletion, or insertion of a single nucleotide such that the enhancer box retains at least one of its endogenous functions.

Exemplary enhancer box
(SEQ ID NO: 12)
ATGTG

[0191] One or more of (a) a retinoic acid receptor binding site; (b) a WT1 binding site; and (c) an enhancer box may be present in the proximal promoter region. Suitably, each of (a) a retinoic acid receptor binding site; (b) a WT1 binding site; and (c) an enhancer box are present in the proximal promoter region

[0192] In some embodiments, one or more of the following elements is present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17: (a) a RAR binding site at a position corresponding approximately to position 7 to position 13 of SEQ ID NO: 4 or 17; (b) a WT1 binding site at a position corresponding approximately to position 14 to position 30 of SEQ ID NO: 4 or 17; and (c) an enhancer box at a position corresponding approximately to position 49 to position 53 of SEQ ID NO: 4 or 17. In some embodiments, each of the elements are present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17.

[0193] In some embodiments, one or more of the following nucleotide sequences is present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17: (a) GGGGTCA at a position corresponding to position 7 to position 13 of SEQ ID NO: 4 or 17; (b) CGGAGGCTGGGGAGGCA at a position corresponding to position 14 to position 30 of SEQ ID NO: 4 or 17; and (c) ATGTG at a position corresponding to position 49 to position 53 of SEQ ID NO: 4 or 17. In some embodiments, each of the nucleotide sequences are present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17.

[0194] Suitably, the promoter may comprise a transcription factor binding region comprising or consisting of a nucleotide sequence shown as SEQ ID NO: 13 or a nucleotide sequence having one, two, three, four or five substitutions, deletions, or insertions compared to SEQ ID NO: 13. The substitutions, deletions, or insertions may be any substitution, deletion, or insertion of a single nucleotide such that the transcription factor binding region retains at least one of its endogenous functions. An exemplary variant is shown in SEQ ID NO: 43. Suitably, a variant of SEQ ID NO: 13 comprises the substitution C19T (see SEQ ID NO: 43).

Exemplary transcription factor binding region
(SEQ ID NO: 13)
GAGCAAGACAGAGAGAGACTCACAGGGA

[0195] Suitably, the promoter may comprise a transcription factor binding region comprising or consisting of a nucleotide sequence shown as SEQ ID NO: 43 or a nucleotide sequence having one, two, three, four or five substitutions, deletions, or insertions compared to SEQ ID NO: 43. The substitutions, deletions, or insertions may be any substitution, deletion, or insertion of a single nucleotide such

that the transcription factor binding region retains at least one of its endogenous functions.

Exemplary transcription factor binding region
(SEQ ID NO: 43)
GAGCAAGACAGAGAGAGATACTCACAGGGA

[0196] Other suitable transcription factor binding regions will be well known to those of skill in the art. For example, other suitable transcription factor binding regions include TACGAT (SEQ ID NO: 37), TATAAT (SEQ ID NO: 38), GATACT (SEQ ID NO: 39), TATGAT (SEQ ID NO: 40), and TATGTT (SEQ ID NO: 41).

[0197] Suitably, the promoter may comprise a transcription initiation site which comprises or consists of an “AG” dinucleotide.

[0198] Suitably, the transcription factor binding site is operably linked to the transcription initiation site. Suitably, the transcription factor binding site may be directly upstream of the transcription initiation site. Without wishing to be bound by theory, it is considered that the transcription factor binding site and the transcription initiation site may provide a core promoter region.

[0199] Suitably the promoter may comprise a 5' untranslated region. The 5' untranslated region may comprise or consist of a nucleotide sequence having at least about 70%, 80%, 90%, 95% or 99% sequence identity to SEQ ID NO: 6 or 19. The 5' untranslated region may comprise or consist of SEQ ID NO: 6 or 19.

[0200] Suitably, the 5' untranslated region is operably linked to the transcription initiation site. Suitably, the 5' untranslated region may be directly downstream of the transcription initiation site.

[0201] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0202] (i) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 4 or 17, wherein each of the following elements is present: (a) a RAR binding site at a position corresponding approximately to position 7 to position 13 of SEQ ID NO: 4 or 17; (b) a WT1 binding site at a position corresponding approximately to position 14 to position 30 of SEQ ID NO: 4 or 17; and (c) an enhancer box at a position corresponding approximately to position 49 to position 53 of SEQ ID NO: 4 or 17;

[0203] (iii) optionally a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at

least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 8, or one or more fragments thereof; and

[0204] (ii) a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 6 or 19, or a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 7 or 20.

[0205] In some embodiments, the promoter of the present invention comprises or consists of from 5' to 3':

[0206] (i) a nucleotide sequence having at least 90% identity to SEQ ID NO: 4 or 17, wherein each of the following elements is present: (a) a RAR binding site at a position corresponding approximately to position 7 to position 13 of SEQ ID NO: 4 or 17; (b) a WT1 binding site at a position corresponding approximately to position 14 to position 30 of SEQ ID NO: 4 or 17; and (c) an enhancer box at a position corresponding approximately to position 49 to position 53 of SEQ ID NO: 4 or 17;

[0207] (iii) optionally a nucleotide sequence having at least 90% identity to SEQ ID NO: 8, or one or more fragments thereof; and

[0208] (ii) a nucleotide sequence having at least 90% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 90% identity to SEQ ID NO: 6 or 19, or a nucleotide sequence having at least 90% identity to SEQ ID NO: 7 or 20.

Exemplary Promoters

[0209] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 2, or a variant which is at least 70% identical to SEQ ID NO: 2.

[0210] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 2, or a variant which is at least 70% identical to SEQ ID NO: 2 and wherein the promoter has a length of about 1.1 kb or less.

[0211] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 2, or a variant which is at least 70% identical to SEQ ID NO: 2.

Exemplary minimal nephrin promoter-819 bp

(SEQ ID NO: 2)

GGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAGGAACGCGCCTGGCATGTGCTGACAGGGAATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCIGGGGGCTCTGGAAGGTA
CCTGGGGGAGGTTGCACTGTGAGAATGAGCTCAAGCTGGGTCAGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGA
GAGAAAGATGGAAGATAAAGAGACTGGGCGCAGTGGCTCACGCCCTGTAATCCCACTTGGGGAGG
CCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAACATAGTGAGACCCCGTC
TCTAAAAAAAAAAGAAAAAAAAAGAAAAAAAAAGTTTTTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGACTCACAGGGAAGAGGGGAAGAGGAAAACG
AGAAAGGGAGGAGAGTAACGGAAAGAGATAAAAAAGAAAGCAGGIGGCAGAGACACAGAGAGGGGA
CCCAGAGAAAGCCAGACAGACGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTACAGTAGGGGGACCT
GTC

[0212] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 2. The promoter may comprise or consist of a variant of SEQ ID NO: 2 shown as SEQ ID NO: 15 or SEQ ID NO: 46.

[0213] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 15, or a variant which is at least 70% identical to SEQ ID NO: 15.

[0214] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ

ID NO: 15, or a variant which is at least 70% identical to SEQ ID NO: 15 and wherein the promoter has a length of about 1.1 kb or less.

[0215] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 15, or a variant which is at least 70% identical to SEQ ID NO: 15.

[0216] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 15.

Exemplary minimal nephrin promoter variant-819 bp

(SEQ ID NO: 15)

GGCCCTGGGGTCACGGAGGCTGGGGAGGCACCGAGGAACGCGCCTGGCATGTGCTGACAGGGGATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGCTCTGGAAGGTA
CCTGGGGGAGGTTGCACTGIGAGAATGAGCTCAAGCTGGGTCAGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGAGAGAGAACGGCTTAGACAAG
GAGAGAAAGATGGAAGATAAAGAGACTGGGCGCAGTGGCTCACGCCCTGTAATCCCACTTGGGGGA
GGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAACATAGTGAGACCCCG
TCTCTAAAAAAAAAAGAAAAAAAAAGAAAAAAAAAGTTTTTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGATACTCACAGGGAAGAGGGGAAGAGGAAAACG
AGAAAGGGAGGAGAGTAACGGAAAGAGATAAAAAAGAAAGCAGGIGGCAGAGACACACAGAGAGGGGA
CCCAGAGAAAGCCAGACAGACGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTACAGTAGGGGGACCT
GTG

[0217] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 46, or a variant which is at least 70% identical to SEQ ID NO: 46.

[0218] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 46, or a variant which is at least 70% identical to SEQ ID NO: 46 and wherein the promoter has a length of about 1.1 kb or less.

[0219] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 46, or a variant which is at least 70% identical to SEQ ID NO: 46.

[0220] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 46.

Exemplary minimal nephrin promoter variant-819 bp (SEQ ID NO: 46)

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GGCCCTGGGGTACGGAGGCTGGGGAGGCACCGAGGAACGCGCTGGCATGTGCTGACAGGGGATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTA
CCTGGGGGAGGITGCACTGIGAGAATGAGCTCAAGCTGGGTCAGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGA
GAGAAAGATGGAAGATAAAGAGACTGGGCGCAGTGGCTCACGCTGTAATCCCAACACTTGGGGAGG
CCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAACATAGTGAGACCCCGTC
TCTAAAAAAAAAAGAAAAAAAAAAGAAAAAAAAAAGTTTTTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGATACTCACAGGGAAGAGGGGAAGAGGAAAACG
AGAAAGGGAGGAGAGTAACGGAAGAGATAAAAAAGAAAAGCAGGTGGCAGAGACACAGAGAGAGGGA
CCCGAGAGAAAGCCAGACAGACGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTACAGTAGGGGGACCT
GTC

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[0221] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 58, or a variant which is at least 70% identical to SEQ ID NO: 58.

[0222] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ

ID NO: 58, or a variant which is at least 70% identical to SEQ ID NO: 58 and wherein the promoter has a length of about 1.1 kb or less.

[0223] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 58, or a variant which is at least 70% identical to SEQ ID NO: 58.

Exemplary minimal nephrin promoter-818 bp (SEQ ID NO: 58)

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GGCCCTGGGGTACGGAGGCTGGGGAGGCACCGAGGAACGCGCTGGCATGTGCTGACAGGGAATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTA
CCTGGGGGAGGITGCACTGIGAGAATGAGCTCAAGCTGGGTCAGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGAGAGAGAACGGCTTAGACAAGGA
GAGAAAGATGGAAGATAAAGAGACTGGGCGCAGTGGCTCACGCTGTAATCCCAACACTTGGGGAGG

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CCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAACATAGTGAGACCCCGTC
TCTAAAAAAAAAAGAAAAAAAAAGAAAAAAAAAGTTTTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGACACTCACAGGGAAGAGGGAAGAGGAAAACG
AGAAAGGGAGGAGAGTAACGGAAAGAGATAAAAAAGAAAGCAGGTGGCAGAGACACAGAGAGAGGGA
CCCAGAGAAAGCCAGACAGACGCAGGTGGCTGGCAGCGGGCGCTGTGGGGGTCACAGTAGGGGGACCT
GT

[0224] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 58. The promoter may comprise or consist of a variant of SEQ ID NO: 58 shown as SEQ ID NO: 59 or SEQ ID NO: 60.

[0225] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 59, or a variant which is at least 70% identical to SEQ ID NO: 59.

[0226] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ

ID NO: 59, or a variant which is at least 70% identical to SEQ ID NO: 59 and wherein the promoter has a length of about 1.1 kb or less.

[0227] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 59, or a variant which is at least 70% identical to SEQ ID NO: 59.

[0228] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 59.

Exemplary minimal nephrin promoter variant-818 bp

(SEQ ID NO: 59)

GGCCCTGGGGTCACGGAGGCTGGGAGGCACCGAGGAACGCGCCTGGCATGTGCTGACAGGGGATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTA
CCTGGGGGAGGTTGCACTGTGAGAATGAGCTCAAGCTGGGTGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCCTAGGCTCCTGGCTGCTGGACTCTGGGCTGCAGTCTCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGTGAGGGGTGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGGAGAGAGAACGGCTTAGACAAG
GAGAGAAAGATGGAAGATAAGAGACTGGGCGCAGTGGCTCACGCCTGTAATCCCAACACTTGGGGA
GGCCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAACATAGTGAGACCCCG
TCTCTAAAAAAAAAAGAAAAAAAAAGAAAAAAAAAGTTTTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGATACTCACAGGGAAGAGGGAAGAGGAAAACG
AGAAAGGGAGGAGAGTAACGGAAAGAGATAAAAAAGAAAGCAGGTGGCAGAGACACACAGAGAGGGA
CCCAGAGAAAGCCAGACAGACGCAGGIGGCTGGCAGCGGGCGCTGTGGGGGTCACAGTAGGGGGACCT

GT

[0229] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 60, or a variant which is at least 70% identical to SEQ ID NO: 60.

[0230] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 60, or a variant which is at least 70% identical to SEQ ID NO: 60 and wherein the promoter has a length of about 1.1 kb or less.

[0231] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 60, or a variant which is at least 70% identical to SEQ ID NO: 60.

[0232] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 60.

Exemplary minimal nephrin promoter variant-818 bp

(SEQ ID NO: 60)

GGCCCTGGGGTACGGAGGCGTGGGAGGCACCGAGGAACGCGCTGGCATGTGCTGACAGGGGATTTT
ATGCTCCAGCTGGGCCAGCTGGGAGGAGCCTGCTGGGCAGAGGCCAGAGCTGGGGGCTCTGGAAGGTA
CCTGGGGGAGGTTGCACTGIGAGAATGAGCTCAAGCTGGGTCAGAGAGCAGGGCTGACTCTGCCAGTG
CCTGCATCAGCCTCATCGCTCTCTAGGCTCCTGGCCTGCTGGACTCTGGGCTGCAGGTCCTTCTTGA
AAGGCTGTGAGTAGTGAGACAAGGAGCAGGAGIGAGGGGIGGCAGGAGAGAAGATAGAGATTGAGAGA
GAGAGAGAGAGACAGAGAGAGAGGAAGAGACAGAGACAAAAGAGAGAGAACGGCTTAGACAAGGA
GAGAAAGATGGAAGATAAAGAGACTGGGCGCAGTGGCTCACGCTGTAATCCCAACTTGGGGAGG
CCAAGGTGGGAGGATGGCTTGAAGGAAAGAGTCTGAGATCAACCTGGCCAAACATAGTGAGACCCCGTC
TCTAAAAAAGAGAAAAAAGAAAAAAGAAAAAAGTTTAAAGAGACAGAGAAAGAG
ACTCAGAGATTGAGACTGAGAGCAAGACAGAGAGAGATACTCACAGGGAAGAGGGGAAGAGGAAAACG
AGAAAGGGAGGAGTAGTACGGAAGAGATAAAAAAGAAAGCAGGIGGCAGAGACAGAGAGAGGGA
CCCAGAGAAAGCCAGACAGACGCGAGGIGGCTGGCAGCGGGCGCTGTGGGGGTACAGTAGGGGGACCT
GT

[0233] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 3, or a variant which is at least 70% identical to SEQ ID NO: 3.

[0234] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 3, or a variant which is at least 70% identical to SEQ ID NO: 3 and wherein the promoter has a length of about 1.1 kb or less.

[0235] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 3, or a variant which is at least 70% identical to SEQ ID NO: 3.

Exemplary minimal nephrin promoter-265 bp

(SEQ ID NO: 3)

GGCCCTGGGGTACGGAGGCTGGGAGGCACCGAGGAACGCGCTGGCAT
GTGCTGACAGGAATTTATGCTCCAGGAGCAAGACAGAGAGACACTC

-continued

ACAGGGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGTAACGGAAA
GAGATAAAAAAGAAAAGCAGGTGGCAGAGACACAGAGAGAGGGACCCAGA
GAAAGCCAGACAGACGCGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTGACA
GTAGGGGGACCTGTC

[0236] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 3. Suitably, a variant of SEQ ID NO: 3 comprises one or more of the substitutions A63G, C96T, G185C, and C265G. Suitably, a variant of SEQ ID NO: 3 comprises the substitutions A63G, C96T, G185C, and C265G (cf SEQ ID NOs: 17-20). The promoter may comprise or consist of a variant of SEQ ID NO: 3 shown as SEQ ID NO: 16 or SEQ ID NO: 47.

[0237] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 16, or a variant which is at least 70% identical to SEQ ID NO: 16.

[0238] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 16, or a variant which is at least 70% identical to SEQ ID NO: 16 and wherein the promoter has a length of about 1.1 kb or less.

[0239] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 16, or a variant which is at least 70% identical to SEQ ID NO: 16.

[0240] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 16.

Exemplary minimal nephrin promoter variant-265 bp
(SEQ ID NO: 16)
GGCCCTGGGGTACCGAGGCTGGGGAGGCACCGAGAACGCGCTGGCAT
GTGCTGACAGGGGATTTATGCTCCAGGAGCAAGACAGAGAGAGATACTC
ACAGGGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGAGTAACGGAAA
GAGATAAAAAAGAAAAGCAGGTGGCAGAGACACAGAGAGAGGGACCCAGA
GAAAGCCAGACAGACGCGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTACA
GTAGGGGGACCTGTC

[0241] In some embodiments, the promoter of the present invention comprises or consists of the nucleotide sequence shown as SEQ ID NO: 47, or a variant which is at least 70% identical to SEQ ID NO: 47.

[0242] In some embodiments, the promoter of the present invention comprises the nucleotide sequence shown as SEQ ID NO: 47, or a variant which is at least 70% identical to SEQ ID NO: 47 and wherein the promoter has a length of about 1.1 kb or less.

[0243] In some embodiments, the promoter of the present invention consists of the nucleotide sequence shown as SEQ ID NO: 47, or a variant which is at least 70% identical to SEQ ID NO: 47.

[0244] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 47.

Exemplary minimal nephrin promoter variant-265 bp
(SEQ ID NO: 47)
GGCCCTGGGGTACCGAGGCTGGGGAGGCACCGAGAACGCGCTGGCAT
GTGCTGACAGGGGATTTATGCTCCAGGAGCAAGACAGAGAGAGATACTC
ACAGGGAAGAGGGGAAGAGGAAAACGAGAAAGGGAGGAGAGTAACGGAAA
GAGATAAAAAAGAAAAGCAGGTGGCAGAGACACACAGAGAGGGACCCAGA
GAAAGCCAGACAGACGCGAGGTGGCTGGCAGCGGGCGCTGTGGGGGTACA
GTAGGGGGACCTGTG

Polynucleotides

[0245] The present invention provides a polynucleotide comprising the promoter of the present invention.

[0246] Polynucleotides of the invention may comprise DNA or RNA, preferably DNA. They may be single-stranded or double-stranded. Polynucleotides such as DNA polynucleotides may be produced recombinantly, synthetically or by any means available to those of skill in the art. They may also be cloned by standard techniques. The polynucleotide may be an isolated polynucleotide.

[0247] Longer polynucleotides will generally be produced using recombinant means, for example using polymerase chain reaction (PCR) cloning techniques. This will involve making a pair of primers (e.g. of about 15 to 30 nucleotides) flanking the target sequence which it is desired to clone, bringing the primers into contact with mRNA or cDNA obtained from an animal or human cell, performing a polymerase chain reaction under conditions which bring about amplification of the desired region, isolating the amplified fragment (e.g. by purifying the reaction mixture with an agarose gel) and recovering the amplified DNA. The

primers may be designed to contain suitable restriction enzyme recognition sites so that the amplified DNA can be cloned into a suitable vector.

[0248] The polynucleotides may be modified by any method available in the art. Such modifications may be carried out in order to enhance the in vivo activity or lifespan of the polynucleotides of the invention.

Protein-Coding Sequence

[0249] The promoter may be operably linked to one or more protein-coding sequences.

[0250] The protein-coding sequence may encode any polypeptide of interest. For example, the protein-coding sequence can encode any polypeptide associated with a glomerular disease. The protein-coding sequence may encode a polypeptide involved in with a GBM-associated genetic glomerular disease, such as Alport Syndrome. The protein-coding sequence may encode a polypeptide involved in podocyte-associated genetic glomerular disease.

[0251] Suitably, the protein-coding sequence may encode a COL4A3, COL4A4, COL4A5, NPHS2, CFH, CFL, CFHL1, C1INH, C4BP, MASP2, C3, C5aR1, C5, C5a, CD55, CD35, CD46, CD59, vitronectin, clusterin, ADCK4, ALG1, ARHGAP24, ARGHDIA, CD151, CD2AP, COQ2, COQ6, DGKE, E2F3, EMP2, KANK2, LAGE3, LMNA, LMX1B, MAF B, NUP85, NUP93, NXF5, OSGEP, PAX2, PDSS2, PMM2, PODXL, SCARB2, SGPL1, Smad7, TP53RK, TPRKB, VDR, WDR73, WT1, ZMPSTE24, APOL1, NPHS1, TRPC6, NUP107, NUP133, NUP160, ACTN4, INF2, ANKFY1, ANLN, CRB2, ITGA3, KANK1, KANK4, MAGI2, MYO1E, OCRL, PTPRO, SMARCAL1, SYNPO, TBC1D8B, XPO5, TNS2, NLRP3, or VEGFC polypeptide.

[0252] Suitably, the protein-coding sequence encodes a polypeptide which has a length of 1450 amino acids or more, 1500 amino acids or more, 1550 amino acids or more, 1600 amino acids or more, or 1650 amino acids or more.

[0253] For a protein-coding polynucleotide, it will be understood by a skilled person that numerous different polynucleotides can encode the same polypeptide as a result of the degeneracy of the genetic code. In addition, it is to be understood that skilled persons may, using routine techniques, make nucleotide substitutions that do not affect the polypeptide sequence encoded by the polynucleotides of the invention to reflect the codon usage of any particular host organism in which the polypeptides of the invention are to be expressed.

[0254] The protein-coding sequence may be codon-optimised. Different cells differ in their usage of particular codons. This codon bias corresponds to a bias in the relative abundance of particular tRNAs in the cell type. By altering the codons in the sequence so that they are tailored to match with the relative abundance of corresponding tRNAs, it is possible to increase expression. By the same token, it is possible to decrease expression by deliberately choosing codons for which the corresponding tRNAs are known to be rare in the particular cell type. Thus, an additional degree of translational control is available. Codon usage tables are known in the art for mammalian cells (e.g. humans), as well as for a variety of other organisms.

[0255] The protein coding nucleotide sequences disclosed herein may comprise or lack stop codons at their 3' end. Thus, the present disclosure encompasses the SEQ ID NOS

disclosed herein with the stop codons present or absent. COL4A3, COL4A4 and COL4A5 polypeptides

[0256] The protein-coding sequence may encode a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof.

[0257] COL4A3, COL4A4 and COL4A5 proteins are approximately 170-185 kDa homologous polypeptides containing collagenous Gly-X-Y repeat sequences frequently interrupted by non-collagenous sequences and forming a triple helix repeat. Each polypeptide also contains a large globular non-collagenous domain at the carboxyl-terminal end.

[0258] Alport syndrome (AS) is caused by pathogenic variants in the COL4A3, COL4A4 and COL4A5 genes, which result in abnormalities of the collagen IV α 345 network of basement membranes.

[0259] The COL4A3, COL4A4 or COL4A5 polypeptide or a fragment or derivative thereof may be capable of forming a collagen IV α 345 network.

[0260] Approximately 200-300 amino acids may be removed from each of the COL4A3, COL4A4 and COL4A5 polypeptides to produce a truncated transgene suitable for a mini-gene approach. The amino acids may be removed from the triple helix repeat. Preferably the amino acids are not removed from the non-collagenous region.

[0261] In some embodiments the COL4A3, COL4A4 and COL4A5 polypeptides are full-length polypeptides.

[0262] Preferably, the COL4A3, COL4A4 or COL4A5 polypeptide is human. An example human COL4A3 is the COL4A3 having the UniProtKB accession number Q01955. An example human COL4A4 is the COL4A4 having the UniProtKB accession number P53420. An example human COL4A5 is the COL4A5 having the UniProtKB accession number P29400.

[0263] Suitably, the COL4A3 peptide may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 21, or a variant which is at least 70% identical to SEQ ID NO: 21. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 21.

[0264] Suitably, the COL4A4 peptide may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 22, or a variant which is at least 70% identical to SEQ ID NO: 22. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 22.

[0265] Suitably, the COL4A5 peptide may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 23, or a variant which is at least 70% identical to SEQ ID NO: 23. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 23.

Exemplary COL4A3 amino acid sequence-Uniprot reference Q01955
(SEQ ID NO: 21)
MSARTAPRPQVLLPLLLVLLAAPAASKGCVCKDKGQCFCDAKGEKGEKGFPGPGSPGQKGFTGP
EGLPGPQGPKGFPGLPGLTSGKVGRIISGLPGFSGSPGLPGTPGNTGPYGLVGVPGCSGSKGEQGFPG
LPGLTGYPGIPGAAGLKGQKGAPEKEEDIELDAKGDPLPGAPGPQGLPGPPGFPVGPVGGPPGPFPG
FPGAMGPRGPKGHMGERVIGHKGERGVKGLTGPPGPPGTIVITLTGPDNRDLKGEKDKGAMGEPGP
PGPSGLPGESYSGEKGAPGDPGLQKPKGDVPGFPGSEGVKGNRGFPLMGEDGIGKQKGDIGPPGF
RGPTYYDYQEKDEGTPGPPGPRGARGPQGPSGPPGVPGSPGSSRPGLRGAPGWPLKGSKGERGR
PGKDAMGTPGSPGAGSPGLPGSPGPPGPGDIVFRKGPPGDHGLPGYLGSPGIPGVDPKGEPLGLC
TQCPYIPGPPGLPGLPGLHGVKGIPIGRQAAGLKGSPGSPGNTGLPGFPFGPAQGDPLKGEKGETL
QPEGQVGVGPDPLRGQPKRGKLDGIPGTPGVKGLPGPKGELALSGEKGDGPPGDPGSPGSPGAPG
AGPPGYGPQGEPLQGTQGVPGAPGPPGEAGPRGELSVSTPVPGPPGPPGPHGPGPQPPGIPGSLG
KCGDPGLPGPDGEPGIPGIGFPPGPPGPKGDQGFPGTKGSLGCPGKMGEPGLPGKPLPGAKGEPAVAM
PGGPGTPGFPGERGNSGEHGEIGLPLGLPGTPGNEGLDGRGDPGPGPPGEQGPGRGICIEGPRGA
QGLPLNGLKGQQGRGKTGPKGDPGIPGLDRSGFPGETGSPGIPGHQGEMLGQGRGYPGNPGIILGP
PGEDGVIGMMGFPGAIGPPGPPGNPGTPGQRGSPGIPGVKQGRGTPGAKGEQDGKGNPGPSEISHVIG
DKGEPGLKGFAGNPGKGNRGVPMPLKGLKGLPGAPGPPGPRDGLSTGNPGEPGLRGIPIGSMGNM
GMPGSKGKRGTGLFPGRAGRPLPGIHLQGDKGEPGYSEGTRPGPPGPTGDPGLPGDMGKKGEMGQP
GPPGHLGPAGPEGAGSPGSPGLPGKPGPHGDLGFKIKGLLGGPPGIRGPPGLPGFPSPGPMGIRGD
QGRDGIPGPAGEKGETGLLRAPPGRGNPGAQGAKGDRGAPGFPGLPGRKGAMGDAGPRGPTGIEGFP
GPPGLPGAIIPGQTNRGPPGSRGSPGAPGPPGPPGSHVIGIKGDKGSMGHPGKPGPPGTAGDMGPPG
RLGAPGTPGLPGRGDPGFQGFPGVKGEKGNPGFLGSGPPGPIGPKGPPGVRGDPGLTKIISLPGSP

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GPPGTPGEPGMQGEPPGPPGNLGPCGPRGKPGKDGPPTPGPAGEKGNKGSKEPAGSDGLPGL
KKGKRGDSGSPATWTTTRGFVFRHSQTTAIPSCPEGTVPVLYSGFSFLFVQGNQRAHQDLGLTGLSCLQR
FTTMPPLFCNVNDVCFNFSRNDYSYWLSTPALMPMMAPITGRALEPYISRCTVCEGPAIAIAVHSQT
TDIPPCPHGWISLWKGFSFIMFTSAGSEGTGQALASPGSCLEEFRAFPFLECHGRGTCNYYSNSYSFW
LASLNPFRMFRKPIPISTVKAGELEKIIISRCQVCMKKRH

Exemplary COL4A4 amino acid sequence-Uniprot reference P53420
(SEQ ID NO: 22)

MWSLHIVLMRCSFRLTKSLATGPWSLILILFSVQYVYSGKKYIGPCGGRDCSVCHCVPEKGSRGPPG
PPGPQGPGLGAPGPIGLSGEKGMRGDRGPPGAAGDKDKGPTGVPGFPGLDGIPGHPGPPGPRGKP
GMSGHNGSRGDPGPPGGRGALGPGGPLGHPGEGKEKGNVFI LGAVKGIQGDGRDPLPLPGSWGAG
GPAGPTGYPGEPGLVGPQPGRPGLKGNPGVGVKGQMGDPGEVQOQSGPGPTLLVEPPDFCLYKGEK
GIKGIPIGMVGLPGPPRGKGESGIGAKGEKGIPIGFPGRDPGSGYSGPFGPLKGLGLVGDPLGLGLI
GPKGDPGNRGHPGPPGVLTPLPLKGPDPGDPGPRYGETGDVGPFGPPGLLGRPGEACAGMIGPPG
PQGFPLPLPGEAGIPGRPDSAPGKPGKPGSPGLPGAPGLQGLPGSSVIYCSVGNPGPQGIKGVGP
PGRGPKGEKGNELCACPEPMGPPGPPGLPGRQSGKDLGLPGWLTKGDPGPPGAEGPPGLPGKH
GASGPPGNKGAKGMVSVRVKGHKGGERGPDGPPGPGQPGSHGRDGHAGEKGDPPGDHEDATPGGK
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FDGPPGKGFPPGQAGPLSGSDGHKGRPGTPTAEIPGPPGFRGMDGDPGFGGEKGS SPVGPFGPPG
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PGRKGDIGEDGYPGGPPGPIGDGPKGFGPGYLGGLLVLSQTDQEPTCPLGMPRLWTGYSLLYL
EGQEKAHNQDLGLAGSCLPVESTLPFAYCNIHQVCHYAQRNDRSYWLASAAPLPMPLSEEAIRPYVS
RCAVCEAPAQAVAVHSQDQSI PPCPQTWRSLWIGYSFLMHTGAGDQGGGQALMSGSCLED FRAAPFL
ECQGRQGTCHFFANKYSFWLT TVKADLQFSSAPAPDTLKESQAQRQKISRCQVCVKYS

Exemplary COL4A5 amino acid sequence-Uniprot reference P29400
(SEQ ID NO: 23)

MKLRGVSLAAGLFLALSLWGQPAEAAACYGCSPGSKDCSGIKGEKGERGFPGLGHPGLPGFPPE
GPPGPRGQKGGDGI PGPPGPKGIRGPPGLPGFPPTPLPGMPGHDGAPGPQGI PGCNGTKGERGPPGS
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GIGVQGPFPPIGPIGQPLHGIPEKGDGPPGLDVPGPGERGSPGIPGAPGPIGPPGSPGLPG
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LPGPSGQSIIKGDAGPPGIPGQPLKGLPGPQGPQGLPGTPGPPGDPGRNGLPGFDGAGGRKGDPLG
PQPGTRGLDGPDPGLQGPFGPPGTSSVAHGFLITRHSQTTDAPQCPQGLQVYEGFSLLYVQGNK
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AVCEAPAVVIAVHSQTIQIPHCPQGWDSLWIGYSFMMHTSAGAEGSGQALASPGSCLEEFRSAPFIEC
HGRGTCNYIANSYSFWLATVDVSDMFSKPQSETLKAGDLRTRISRCQVCMKRT

[0266] The protein-coding sequence may comprise or consist of a COL4A3, COL4A4 or COL4A5 transgene.

[0267] An example transgene encoding COL4A3 is provided in NM_000091.5. An example transgene encoding COL4A4 is provided in NM_000092.5. An example transgene encoding COL4A5 is provided in NM_000495.5.

[0268] Suitably, the COL4A3 transgene may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 24, or a variant which is at least 70% identical to SEQ ID NO: 24. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 24.

[0269] Suitably, the COL4A4 transgene may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 25, or a variant which is at least 70% identical to SEQ ID NO: 25. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 25.

[0270] Suitably, the COL4A5 transgene may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 26, or a variant which is at least 70% identical to SEQ ID NO: 26. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 26.

Exemplary COL4A3 transgene sequence

(SEQ ID NO: 24)

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Exemplary COL4A4 transgene sequence

(SEQ ID NO: 25)

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Exemplary COL4A5 transgene sequence

(SEQ ID NO: 26)

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ctggttacctcgtgaaccgggaagggtgggtgaaaagggccaaaagggtgacactggccccacctgga

cctcctggacttgtgaatccttagacctgggactgggtataaactataggagaaaaaggaaacattggggtg
gcctgggttgcttgagaaaaaggagagcaggatttcttggaaatcacagggtccacctggccttctg
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ggatgatgaaggaccacctggaatttccattcctggacctcctggactgacggacagacctggggctcc
tgggtctccagggtcctcctggccctgctggccctcacattcctcctagtgatgagatatgtgaaccag
gccctccaggccccccaggatctccagggtgataaaggactccaaggagaaacaggagtgaagggtgac
aaagggtgacacttgcttcaactgcattggaactggatttccagggtcctcagggtcaacctgggttgc
agggtctcccagggtcctccaggatctcttgggttctcctggacagaaaagggtgaaaaaggacaagctgggt
caactgggtcccaaaggattaccaggcattccaggagctccagggtgctccagggttctcctggatctaaa
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aaggagagcctgggtggaattacttttaagggtgaaaagggtccccctgggaaccagggttaccagggt
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catacaagggtgtggcaggaaaatccaggccagccaggaaataccagggtcctaaagggtatccagggtcaga
ctataaccaggccggggaagcctgggttgctcctggtaaccaggcagagatgggtgatgtagggtctcca
gggtgacctggacttccagggtcaaccagggttgcagggtgatacctggtagcaaggagagaaccagggtat
ccctggaattgggtctcctggaccacctgggtcccaaaggcttctcctggaattccaggacctccaggag
cacctgggacacctggaagaattgggtctagaaggccctcctggggccaccgggttccaggaccaaaag
ggatgaaccaggatttgcatctacctgggccacctgggtccaccaggacttccagggttcaaaggagcact
tgggtccaaaagggtgatcgtgggttccaggacctccgggtcctccaggagcgcactgggttagatgggc
tccctggaccaaaagggtgatgttgaccaaaatggacaacctggaccaatgggacctcctgggtgcca
ggaatagggtgtcagggtaccaccagggtaccaccagggtatcctgggtccaatagggtcaacctgggttaca
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gaggcagtcagggtatccccggagcacctgggtcctataggacctccagggtaccagggttccagggt
aaagcagggtcctctggatttccagggtaccaaaagggtgaaatgggtatgatgggacctccagggtccacc
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agccaggacttctggccctacaggagaaaaaggtagttaaaggagagcctggccttccagggtcctcct
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tggagttcagggtccaaaagggttatcagggttgcctggagaccagggtcaacctggactgagtggaac
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gtgatcctgggtatttcaagcattgggtctccagggtcttctgggtccaaagggtgagcctgggtctgcct
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cctgggtcaaccagggtcctcagggtgaaaaagggtcaaacccgggtcaagatgggtatcctgggtaccaggtg
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ccaggctttcacggtttccctgggtgtgcagggtcccccaggccctcctgggtctccgggtccagctct
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gtcctccagggtcctcctggaaatggaggtattaaaggagagaagggaatccaggccaacctgggcta
cctgggttgctgggtttgaaaggagatcaaggaccaccaggactccagggtaatcctggcggcggg
tctcaatggaatgaaaggagatcctgggtcctcctgggtgtccaggattccaggcatgaaaggacca
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gagatcctggacgcaatggactccttggttgatgggtgcaggagggcgaaaggagaccagggtctg
ccaggacagccagggtaccctgggttggtgggtccctcctgggtccagatggattgcaagggtcccccagg
tccccctggaacctcctctgttgacatggatttcttattacacgccacagccagacaacggatgcac
cacaatgcccacagggaacacttcagggtctatgaaggcttttctcctgtatgtacaaggaaataaa
agagccacaggtcaagacttggggacggctggcagctgccttcgtcgtcttagtaccatgcctttcat
gttctgcaacatcaataatgtttgcaactttgcttcaagaaatgactattcttactggtctctaccc
cagagcccatgccaatgagcatgcaaccctaaaggccagagcatccagccattcattagtcgatgt
gcagtatgtgaagctccagctgtggtgatcgagttcacagtcagacgatccagattccccattgtcc
tcagggtatgggattctctgtggattggttattccttcctgatgcatacaagtgcaggggcagaaggct
cagggtcaagccctagcctccctgggttcctgcttggaaagagtttcgttcagctcccttcacgaatgt
catggggaggggtacctgtaactactatgccaaactcctacagcttttggtggcgaactgtagatgtgtc
agacatgttcagtaaacctcagtcagaaacgctgaaagcaggagacttgaggacacgaattagccgat
gtcaagtgtgcatgaaggacataa

[0271] The COL4A3, COL4A4 or COL4A5 transgene may comprise an intron or intronic sequences, which can be used to improve gene expression. The COL4A3, COL4A4 or COL4A5 transgene may comprise a protein tag, such as a hemagglutinin (HA) tag. HA can be used as an epitope tag and has been shown not to interfere with bioactivity or biodistribution of proteins to which it has been added. The protein tag can facilitate detection, isolation, and purification of the transgene. Other suitable protein tags may include Myc tags, polyhistidine tags and flag tags.

[0272] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof.

[0273] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof.

Nephrotic Syndrome (NS)-Associated Transgene

[0274] The protein-coding sequence may comprise or consist of a NS-associated transgene.

[0275] Nephrotic syndrome (NS) is a chronic kidney disease characterized by significant proteinuria, hypoalbumin-

emia, oedema and hyperlipidemia. The NS-associated transgene may be a gene associated with a monogenic form of NS and expressed in podocytes.

[0276] Suitable NS-associated transgenes include NPHS2, ADCK4, ALG1, ARHGAP24, ARGHDIA, CD151, CD2AP, COQ2, COQ6, DGKE, E2F3, EMP2, KANK2, LAGE3, LMNA, LMX1B, MAFB, NUP85, NUP93, NXF5, OSGEP, PAX2, PDSS2, PMM2, PODXL, SCARB2, SGPL1, Smad7, TP53RK, TPRKB, VDR, WDR73, WT1, ZMPSTE24, APOL1, NPHS1, TRPC6, NUP107, NUP133, NUP160, ACTN4, INF2, ANKFY1, ANLN, CRB2, ITGA3, KANK1, KANK4, MAGI2, MYO1E, OCRL, PTPRO, SMARCAL1, SYNPO, TBC1D8B, XPO5, TNS2 and NLRP3.

[0277] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the NS-associated transgene.

[0278] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the NS-associated transgene.

NPHS2

[0279] The protein-coding sequence may encode NPHS2, or a fragment and/or variant thereof.

[0280] "NPHS2" is the abbreviated name of the polypeptide encoded by the NPHS2 gene and is also known as

podocin. NPHS2 is a 42 kDa hairpin like membrane-associated podocyte-specific protein that is a key component of the protein complex at the slit diaphragm; the cell-cell junction between adjacent podocyte foot processes. It localises to lipid rafts and interacts with other important slit diaphragm proteins like nephrin, CD2AP and TRPC6. It is essential in the maintenance of the slit diaphragm, and consequently the integrity of the glomerular filtration barrier.

[0281] A fragment and/or variant of NPHS2 may retain NPHS2 activity. For example, a fragment and/or variant of podocin may regulate glomerular permeability. Suitably, a fragment and/or variant of NPHS2 may have the same or similar activity to NPHS2, e.g. may have at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the activity of NPHS2.

[0282] A person skilled in the art would be able to generate fragments and/or variants using conservative substitutions, based on the known structural and functional features of NPHS2 (see e.g. Tabassum, A., et al., 2014. Interdisciplinary

Sciences: Computational Life Sciences, 6 (1), pp. 32-39), and/or based on known variants (see e.g. NCBI Gene ID: 7827 and NCBI HomoloGene: 22826). Suitably, a fragment and/or variant of NPHS2 comprises a transmembrane domain, with two cytoplasmic domains at the N- and C-terminus.

[0283] The NPHS2 gene is conserved in chimpanzee, Rhesus monkey, dog, cow, mouse, and rat. The NPHS2 may be a human NPHS2. Suitably, the NPHS2 may comprise or consist of a polypeptide sequence of UniProtKB accession Q9NP85, or a fragment and/or variant thereof.

[0284] In some embodiments, the NPHS2 comprises or consists of an amino acid sequence which is at least 70% identical to SEQ ID NO: 55 or a fragment thereof. Suitably, the NPHS2 comprises or consists of an amino acid sequence which is at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 55 or a fragment thereof.

[0285] In some embodiments, the NPHS2 comprises or consists of SEQ ID NO: 55 or a fragment thereof.

(SEQ ID NO: 55)

MERRARSSRESRGRGRTPHKENKRAKERSGGGRGRQEAGPEPSGSRAGTGPGEPRAPATVVDVE
VRGSGEETEVVALLESERPEEGTKSSGLGACEWLLVLISLLFIIMTFPFSIWFCVKVQYERVIIFR
LGHLPLGRAKGPFLLPCLDITYHKVDLRLQTLFHEIVTKDMFIMEIDAI CYYRMENASLLLSLL
AHVSKAVQFLVQTTMKRLLAHRSLEILLERKSIAQDAKVALDSVTCTIWKIKVERIEIKDVRPLAGLQH
SLAVEAEAQRQAKVRMIAAEAEKAASESLRMAAEILSGTPAAVQLRYLHTLQSLSTKEPSTVVLPPLPFD
LLNCLSSPSNRTQGSPLFPSPSKPVEPLNPKKKDSPML
Exemplary NPHS2 amino acid sequence

[0286] In some embodiments, the NPHS2 transgene comprises or consists of a nucleotide sequence which is at least 70% identical to SEQ ID NO: 56 or a fragment thereof. Suitably, the NPHS2 transgene comprises or consists of a nucleotide sequence which is at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 56 or a fragment thereof.

[0287] In some embodiments, the NPHS2 transgene, comprises or consists of the nucleotide sequence SEQ ID NO: 56 or a fragment thereof.

(SEQ ID NO: 56)

ATGGAGAGGAGGGCGCGAGCTCCTCCAGGGAGTCCC GCGGGCGAGGCGGCAGGACTCCGCACAAGGAG
AACAAGAGGGCAAAGGCCGAGAGGAGCGGGGGGGCCGCGGGCGCCAGGAGGCTGGGCCCGAGCCGTCG
GGCTCCGACGGGCGGGGACCCCGGGGAGCCCCGAGCGCCCGCCGACGGTGGTGGACGTGGATGAG
GTCCGAGGCTCCGGCGAGGAGGACCCGAGGTGGTGGCGCTGTGGAGAGCGAGCGGCCGAGGAAGGT
ACCAAACTCTCCGGCTTAGGGCCCTGTGAGTGGCTTCTGTCTCATTTCCCTGCTTTCATCATCATG
ACCTTCCCTTTTTCATCTGGTTCTGCGTAAAGGTTGTACAAGAGTATGAAAGAGTAATATATATCCGA
CTGGGACATCTGCTTCTGGAAGAGCCAAAGGCCCTGGTCTTTCTTTTTTTTGGCCCTGCTGGATACC
TACCACAAGGTTGACCTTCGTCTCCAACTCTGGAGATACCTTTTCATGAGATCGTGACCAAGACATG
TTTATAATGGAGATAGATGCCATTGTCTACTACCGAATGGAAAATGCCTCTCTTCTCCTAAGCAGTCTT

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GCTCATGTATCTAAAGCTGTGCAATTCCTTGTGCAAACCACTATGAAGCGTCTCCTAGCACATCGATCC
CTCACTGAAATTCCTCTAGAGAGGAAGAGCATCGCCCAAGATGCAAAGTTGCCTTGGATTTCAGTGACC
TGTATTGGGGAATCAAAGTGAGAGAGATAAGAAATTAAGATGTGAGGTTGCCAGCTGGGCTTCAGCAC
TCACTGGCTGTGGAGGCTGAAGCGCAAAGACAAGCCAAAGTGCAGGATGATTGCTGCAGAAGCGGAAAAG
GCTGCTTCTGAGTCCCTGAGGATGGCAGCTGAGATTCTGTGAGGCACCCCTGCTGCTTTCAGCTTCGA
TACCTCCACACCCCTTCAGTCTCTGTCCACAGAGAAGCCTTCCACTGTGGTTTACCTTTGCCATTGAC
CTACTGAATTGCCTGTCTTCTCCAGCAACAGAACTCAGGGAAGCCTCCCTTCCCAAGTCCTTCCAAA
CCTGTTGAGCCACTAAATCCTAAAAAGAAAGACTCTCCCATGTTA
Exemplary NPHS2 transgene sequence

[0288] In some embodiments, the NPHS2 transgene comprises or consists of a nucleotide sequence which is at least 70% identical to SEQ ID NO: 57 or a fragment thereof. Suitably, the NPHS2 transgene comprises or consists of a nucleotide sequence which is at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 57 or a fragment thereof.

[0289] In some embodiments, the NPHS2 transgene, comprises or consists of the nucleotide sequence SEQ ID NO: 57 or a fragment thereof.

[0290] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding NPHS2, or a fragment and/or variant thereof.

[0291] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding NPHS2, or a fragment and/or variant thereof.

(SEQ ID NO: 57)

ATGGAGAGGAGGGCGCGAGCTCCTCCAGGGAGTCCGCGGGCGAGGCGGAGGACTCCGCACAAGGAG
AACAAGAGGGCAAAGGCGAGAGGAGCGGCGAGGCCGCGGGCGCCAGGAGGCTGGGCCGAGCCGTCG
GGCTCCGACGGGCGGGGACCCCGGGGAGCCCCGAGCGCCCGCCACGGTGGTGGACGTGGATGAG
GTCCGAGGCTCCGCGAGGAGGGCACCGAGGTGGTGGCGCTGTTGAGAGCGAGCGGCCGAGGAAGGT
ACCAATCCTCCGGCTTAGGGCCTGTGAGTGGCTTCTGTCTCATTTCCCTGCTCTTCATCATCATG
ACCTTCCCTTTTCCATCTGGTCTGCGTAAAGGTTGTACAAGAGTATGAAAGAGTAATTATATCCGA
CTGGGACATCTGCTTCTGGAAGAGCCAAAGGCCCTGGTCTTTTCTTTTTTTTGCCCTGCCTGGATACC
TACCACAAGGTTGACCTTCGTCTCCAACTCTGGAGATACCTTTTCATGAGATCGTGACCAAGACATG
TTTATAATGGAGATAGATGCCATTTGCTACTACCGAATGGAAAATGCCTCTCTTCTCCTAAGCAGTCTT
GCTCATGTATCTAAAGCTGTGCAATTCCTTGTGCAAACCACTATGAAGCGTCTCCTAGCACATCGATCC
CTCACTGAAATTCCTCTAGAGAGGAAGAGCATCGCCCAAGATGCAAAGTTGCCTTGGATTTCAGTGACC
TGTATTGGGGAATCAAAGTGAGAGAGATAAGAAATTAAGATGTGAGGTTGCCAGCTGGGCTTCAGCAC
TCACTGGCTGTGGAGGCTGAAGCGCAAAGACAAGCCAAAGTGCAGGATGATTGCTGCAGAAGCGGAAAAG
GCTGCTTCTGAGTCCCTGAGGATGGCAGCTGAGATTCTGTGAGGCACCCCTGCTGCCGTTTCAGCTTCGA
TACCTCCACACCCCTTCAGTCTCTGTCCACAGAGAAGCCTTCCACTGTGGTTTACCTTTGCCATTGAC
CTACTGAATTGCCTGTCTTCTCCAGCAACAGAACTCAGGGAAGCCTCCCTTCCCAAGTCCTTCCAAA
CCTGTTGAGCCACTAAATCCTAAAAAGAAAGACTCTCCCATGTTATAG
Exemplary NPHS2 transgene sequence

Vascular Endothelial Growth Factor (VEGF) C Transgene

[0292] The protein-coding sequence may comprise or consist of a vascular endothelial growth factor (VEGF) C transgene.

[0293] VEGFC is a lymphangiogenic growth factor, which is known to signal via two receptors, VEGFR-3 (Flt4)

[0296] The VEGFC polynucleotide may comprise the VEGFC open reading frame (ORF) sequence of SEQ ID NO: 36. The VEGFC polynucleotide may comprise a nucleic acid sequence which has at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% sequence identity to the VEGFC ORF sequence of SEQ ID NO: 36. The variant sequence may encode a VEGFC polypeptide that has retained the capability to bind and activate VEGFR-2 and VEGFR-3.

Exemplary VEGFC polynucleotide

(SEQ ID NO: 36)

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ATGCACTTGCTGGGCTTCTTCTCTGTGGCGTGTCTCTGCTCGCCGCTGCCTGCTCCCGGTCTCTG
CGAGGCGCCCGCCGCGCCGCGCTTCGAGTCCGGACTCGACCTCTCGGACGCGGAGCCCGACGCGG
GCGAGGCCACGGCTTATGCAAGCAAAGATCTGGAGGAGCAGTTACGGTCTGTGTCCAGTGTAGATGAA
CTCATGACTGTACTCTACCCAGAATATTGAAAATGTACAAGTGTACGCTAAGGAAAGGAGGCTGGCA
ACATAACAGAGAACAGGCCAACCTCAACTCAAGGACAGAAGAGACTATAAAATTTGCTGCAGCAGCAT
ATAATACAGAGATCTTGAAAAGTATTGATAATGAGTGGAGAAAGACTCAATGCATGCCACGGGAGGTG
TGTATAGATGTGGGAAGGAGTTTGGAGTCGCGACAAACACCTTCTTTAACTCCATGTGTGTCCGT
CTACAGATGTGGGGTGTCTGCAATAGTGAGGGGCTGCAGTGCATGAACACCAGCAGCAGGTACCTCA
GCAAGACGTTATTTGAAATTACAGTGCCTCTCTCTCAAGGCCCAAACAGTAACAATCAGTTTTTGCC
AATCACACTTCTGCGGATGCATGTCTAACTGGATGTTTACAGACAAGTTCATTCCATTATTAGACG
TTCCCTGCCAGCAACACTACCACAGTGTGAGGAGCGAACAAGACCTGCCCCACCAATTACATGTGGA
ATAATCACATCTGCAGATGCCTGGCTCAGGAAGATTTTATGTTTTCTCGGATGTGGAGATGACTCA
ACAGATGGATTCATGACATCTGTGGACCAACAAGAGCTGGATGAAGAGACCTGTGAGTGTGTCTG
CAGAGCGGGGCTTCGGCTGCGCAGTGTGGACCCACAAAGAACTAGACAGAAATCATGCCAGTGTG
TCTGTAAAAACAACCTTCCCCAGCAATGTGGGGCAACCGAGAATTTGATGAAAAACATGCCAG
TGTGTATGTAAAGAACCTGCCCCAGAAATCAACCCCTAAATCCTGGAAAATGTGCCTGTGAATGTAC
AGAAAGTCCACAGAAATGCTTGTGTTAAAGGAAAGAGTTCCACCACCAACATGCAGCTGTTACAGAC
GGCCATGTACGAACCGCCAGAGGCTTGTGAGCCAGGATTTTCATATAGTGAAGAAGTGTGTCGTTGT
GTCCCTTCATATTGAAAAGACCACAAATGAGC
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and VEGFR-2 (Flk4). VEGFC is produced by cells in a prepropeptide form, which dimerises before being cleaved into a tetramer.

[0294] The VEGFC transgene may comprises a polynucleotide encoding any form of VEGFC, such as the prepropeptide form, the tetramer form, the intermediate form, or fully processed mature VEGFC.

[0295] If desired, polynucleotides encoding different forms of VEGFC polypeptides may be used in any combination. Preferably the VEGFC transgene comprises a polynucleotide encoding one or more polypeptides having VEGFC biological activity, i.e., peptides that can bind to and activate VEGFR-2 and/or VEGFR-3. More preferably, the VEGFC transgene comprises a polynucleotide encoding a polypeptide comprising the VEGFC homology domain and having VEGFC biological activity, i.e., a polypeptide that can bind to and activate VEGFR-2 and/or VEGFR-3. Further details of suitable VEGFC polynucleotides and polypeptides include those described in WO 2015/022447 and US 2014/0087002.

[0297] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the VEGF (C) transgene.

[0298] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the VEGF (C) transgene.

Complement Proteins and Complement Inhibitors

[0299] The protein-coding sequence may encode a complement protein, or a fragment and/or variant thereof.

[0300] As used herein, a “complement protein” is a protein which is part of the complement system. The complement system, also known as complement cascade, is a central part of the innate immunity that serves as a first line of defence against foreign and altered host cells. The complement system is composed of plasma proteins produced mainly by the liver or membrane proteins expressed on cell surface. Complement operates in plasma, in tissues,

or within cells. Complement proteins collaborate as a cascade to opsonize pathogens and induce a series of inflammatory responses helping immune cells to fight infection and maintain homeostasis (Merle, N. S., et al., 2015. *Frontiers in immunology*, 6, 262).

[0301] There are three pathways of complement activation: the classical, the alternative, and the lectin pathways. The three complement pathways differ in their mechanisms of target recognition but converge in the activation of the central component C3. After this activation, C5 is cleaved, and the assembly of the membrane attack complex (MAC) is initiated. The enzymatic cleavage of C3 and C5 leads to the production and release of anaphylotoxins C3a and C5a.

[0302] Suitably, the complement protein is selected from the list consisting of CFI, CFH, FHL-1, C1INH, C4BP, MASP2, C3, C5aR1, C5, C5a, CD55, CD35, CD46, CD59, vitronectin, and clusterin, or fragments and/or variants thereof.

[0303] The protein-coding sequence may encode an inhibitor of the complement system, or a fragment and/or variant thereof.

[0304] As used herein, an “inhibitor of the complement system” or “complement inhibitor” is a protein which prevents activation of the complement system. Complement is tightly controlled by these inhibitors, which naturally protect self cells and tissues from unwanted complement activation. Complement inhibitors can regulate complement activation in different stages of the classical, lectin, and alternative pathways. Suitably, the complement inhibitor is a naturally-occurring complement inhibitor, or a fragment and/or variant thereof. Preferably, the inhibitor of the complement system is an inhibitor of the complement system in humans.

[0305] Complement inhibitors are grouped into two categories: soluble inhibitors and membrane-bound inhibitors. Preferably, the inhibitor of the complement system is a soluble complement inhibitor. Soluble complement inhibitors include C1 inhibitor (C1INH), complement factor I (CFI), complement factor H (CFH), complement factor H-like protein 1 (FHL-1), C4 binding protein (C4BP), clusterin and vitronectin. Membrane-bound regulators include CD46, CD55, CD59, CD35 and CUB and Sushi multiple domain 1 (CSMD1).

[0306] The inhibitor of the complement system may be selected from: CFI, CFH, FHL-1, C1INH, C4BP, CD46,

CD55, CD59, CD35, vitronectin, clusterin, and CSMD1, or fragments and/or variants thereof.

[0307] Preferably, the inhibitor of the complement system is selected from: CFI, CFH, and FHL-1, or fragments and/or variants thereof.

CFI

[0308] The protein-coding sequence may encode CFI, or a fragment and/or variant thereof.

[0309] Complement factor I (CFI) is a trypsin-like serine protease that inhibits the complement system by cleaving three peptide bonds in the alpha-chain of C3b and two bonds in the alpha-chain of C4b thereby inactivating these proteins.

[0310] CFI is a glycoprotein heterodimer consisting of a disulfide linked heavy chain and light chain. The heavy chain has four domains: an FI membrane attack complex (FIMAC) domain, CD5 domain, and low density lipoprotein receptor 1 and 2 (LDLr1 and LDLr2) domains. The heavy chain plays an inhibitory role in maintaining the enzyme inactive until it meets the complex formed by the substrate (either C3b or C4b) and a cofactor protein (Factor H, C4b-binding protein, complement receptor 1, and membrane cofactor protein). Upon binding of the enzyme to the substrate: cofactor complex, the heavy: light chain interface is disrupted, and the enzyme activated by allosteric. The light chain contains only the serine protease domain. This domain contains the catalytic triad His-362, Asp-411, and Ser-507, which is responsible for specific cleavage of C3b and C4b.

[0311] The CFI or a fragment and/or variant thereof may be capable of cleaving C3b into iC3b and/or may be capable of cleaving iC3b into C3d,g.

[0312] The fragment and/or variant of CFI may retain at least 50%, 60%, 70%, 80%, 90%, 95% or 100% of the C3b-inactivating and iC3b-degradation activity of native CFI. The C3b-inactivating and iC3b-degradation activity of the fragment and/or variant of CFI and native CFI, may be determined using any suitable method known to those of skill in the art. For example, using a proteolytic assay.

[0313] Preferably, the CFI is a human CFI. An example human CFI is the CFI having the UniProtKB accession number P05156.

[0314] Suitably, the CFI may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 49, or a variant which is at least 70% identical to SEQ ID NO: 49.

Illustrative CFI polypeptide sequence (SEQ ID NO: 49):

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MKLLHVFLFLFCFHLRFCKVYTSQEDLVEKKCLAKKYTHLSCDKVFCQPWQRCIEGTCVCKLPYQCP

KNGTAVCATNRRSFPTYCQOKSLLECLHPGTKFLNNGTCTAEGKFSVSLKHGNTDSEGIVEVKLVQDK

TMFICKSSWSMREANVACLDLGFQQGADTQRRFKLSDLINSTELHVVHCRGLETSLAECTFTKRRTM

GYQDFADVVCYTQKADSPMDFFQCVNGKYISQMKACDGINDCGDQSDCLCKACQKGPHCKSGVCI

PSQYQCNGEVDCTIGEDVGCAGFASVTQEETEILTADMDAERRRIKSLPKLSCGVKNRMHIRRKRI

VGGKRAQLGLDLPWQVAIKDASGITCGGIYIGGCWILTAACHLRASKTHRYQIWTTVVDWIHPDLKRIV

IEYVDRIIFHENYNAGTYQNDIALIEMKKDGNKKDCELPSPACVPWSPYLFQPNDTICIVSGWGREG

DNERVFSLQWGEVKLISNCSKFYGNRFYEKEMECAGTYDGSIDACKGDSGGPLVCMANNVTYVWGVV

SWGECGKPEFPVYTKVANYFDWISYHVGRPFISQYNV

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[0315] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 49.

[0316] An example nucleotide sequence encoding CFI is NM_000204.5. Suitably, a protein-coding sequence encoding CFI may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 50, or a variant which is at least 70% identical to SEQ ID NO: 50.

95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding CFI, or a fragment and/or variant thereof.

[0319] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding CFI, or a fragment and/or variant thereof.

Illustrative CFI polynucleotide sequence (SEQ ID NO: 50):

ATGAAGCTTCTTCATGTTTTCCTGTTATTCTGTGCTTCCACTTAAGGTTTGCAGGTCACCTTATAC
ATCTCAAGAGGATCTGGTGAGAAAAAGTGCTTAGCAAAAAATATACTCACCTCTCCTGCGATAAAG
TCTTCTGCCAGCCATGGCAGAGATGCATTGAGGGCACCTGTGTTTGTAACTACCGTATCAGTGCCCA
AAGAATGGCACTGCAGTGTGTGCAACTAACAGGAGAAGCTTCCCAACATACTGTCAACAAAAGAGTTT
GGAATGTCTTCATCCAGGACAAAGTTTAAATAACGGAACATGCACAGCCGAAGGAAAGTTTAGTG
TTTCCTTGAAGCATGGAATACAGATTAGAGGGAATAGTTGAAGTAAACTTGTGGACCAAGATAAG
ACAAATGTTTATATGCAAAAGCAGCTGGAGCATGAGGGAAGCCAACGTGGCCTGCCTTGACCTTGGGTT
TCAACAAGGTGCTGATACTCAAAGAAGGTTTAAAGTTGTCTGATCTCTATAAATTCACCTGAATGTG
TACATGTGCATTGCCGAGGATTAGAGACCAGTTTGGCTGAATGTACTTTTACTAAGAGAAGAACTATG
GGTTACCAGGATTTCTGCTGATGTGGTTTGTATACACAGAAAGCAGATTCTCCAATGGATGACTTCTT
TCAGTGTGTGAATGGGAAATACATTTCTCAGATGAAAGCCTGTGATGGTATCAATGATTGTGGAGACC
AAAGTGATGAAGTGTGTGTAAGCATGCCAAGGCAAAGGCTTCCATTGCAAAATCGGGTGTGTCATT
CCAAGCCAGTATCAATGCAATGGTGGGAGTGCATTACAGGGGAAGATGAAGTTGGCTGTGCAGG
CTTTCATCTGTGACTCAAGAAGAAACAGAAATTTTGACTGCTGACATGGATGCAGAAAGAAGACGGA
TAAATCATTATTACCTAACTATCTTGTGGAGTTAAACACAGATGCACATTGCAAGGAAACGAATT
GTGGGAGGAAAGCGAGCACAACTGGGAGACCTCCCATGGCAGGTGGCAATTAAGGATGCCAGTGAAT
CACCTGTGGGGAAATTTATATTGGTGGCTGTGGATTCTGACTGCTGCACATTGTCTCAGAGCCAGTA
AAACTCATCGTTACCAATATGGACAACAGTAGTAGACTGGATACACCCGACCTTAAACGTATAGTA
ATTGAATACGTGGATAGAAATTTTCCATGAAAACACAATGCAGGCACTTACCAAAATGACATCGC
TTTGATTGAAATGAAAAAGACGGAACAAAAAGATTGTGAGCTGCCTCGTTCATCCCTGCCTGTG
TCCCCTGGTCTCCTTACCTATTTCAACCTAATGATACATGCATCGTTTCTGGCTGGGACGAGAAAAA
GATAACGAAAGAGTCTTTTCACTTCAGTGGGTGAAGTTAACTAATAAGCAACTGCTCTAAGTTTGA
CGAAATCGTTTCTATGAAAAAGAAATGGAATGTGCAGGTACATATGATGGTTCATCGATGCCGTGTA
AAGGGGACTCTGGAGGCCCTTAGTCTGTATGGATGCCAACATGTGACTTATGCTGGGGTGTGTG
AGTTGGGGGAAAACTGTGAAACAGAGTTCCAGGTGTTTACACCAAGTGGCCAAATTTTGA
CTGGATTAGCTACCATGTAGGAAGGCCTTTTATTTCTCAGTACAATGTATAA

[0317] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 50.

[0318] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least

CFH

[0320] The protein-coding sequence may encode CFH, or a fragment and/or variant thereof.

[0321] Complement factor H (CFH) regulates complement activation on self cells and surfaces. CFH competes for binding of complement factor B (CFB) to C3b, acts as a cofactor for CFI-catalysed proteolytic cleavage of C3b, and accelerates the irreversible dissociation of C3bBb and

C3b2Bb into their separate components. Thus, CFH not only inhibits formation of the convertases but it also shortens the lifespan of any convertase complex that forms.

[0322] CFH is a large (155 kDa) soluble glycoprotein. CFH is composed from a total of 20 domains, each containing approximately 60 amino acid residues and termed complement control protein modules (CCPs) or short consensus repeats that are joined by short linkers consisting of 3-8 residues. The CCP modules are numbered from 1-20 (from the N-terminus of the protein): CCPs 1-4 and CCPs 19-20 engage with C3b while CCPs 7 and CCPs 19-20 bind to GAGs and sialic acid.

[0323] The CFH or a fragment and/or variant thereof may be capable of binding C3b and/or C3d; and/or acting as a

cofactor for the CFI-catalysed proteolytic cleavage of C3b; and/or increasing the irreversible dissociation of C3bBb and C3b2Bb into their separate components. The fragment and/or variant of CFH may retain at least 50%, 60%, 70%, 80%, 90%, 95% or 100% of the activity of native CFH. The activity of the fragment and/or variant of CFH and native CFH may be determined using any suitable method known to those of skill in the art.

[0324] Preferably, the CFH is a human CFH. An example human CFH is the CFH having the UniProtKB accession number P08603.

[0325] Suitably, the CFH may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 51, or a variant which is at least 70% identical to SEQ ID NO: 51.

Illustrative CFH polypeptide sequence (SEQ ID NO: 51):
 MRLAKIICLMLWAI CVAEDCNELPPRRNTEILTGSDQTYPEGTQAIYKCRPGYRSLGNVIMVCRK
 GEWVALNPLRKQKRPCGHPGDTTPFGTFTLTGGNVFEYGVKAVYTCNEG YQLLGEIN YRECDTDGWTN
 DIPICEVVKCLPVTAPENGKIVSSAMEPDREYHFGQAVRFVCNSGYKIEGDEEMHCSDDGFSKEKPK
 CVEISCKSPDVINGSPISQKIIYKENERFQYKCNMGYEYSERGDAVCTESGWRPLPSCEEKSCDNPYI
 PNGDYSPLRIKHRIGDEITYQCRNGFYPATRGNTAKCTSTGWIPAPRCTLKPDCYDPDIKHGGLYHENM
 RRPYFPVAVGKYYSYCDHFETPSGSYWDHIHCTQDGWSPAVPCLRKCYFPYLENGYNQNYGRKFVQ
 GKSIDVACHPGYALPKAQTTVTCTMENGWSPTRCIRVKTCSKSSIDIENGFISESQYTYALKEKAKYQ
 CKLGYVTADGETSGSITCGKDGWSAQPTCIKSCDIPVFMNARTKNDFTWFKLNDTLDYEC HDGYESNT
 GSTTGSIVCGYNGWSDLPICYERECELPKIDVHLVPDRKDKQYKVGEVLKFSCKPGFTIVGPNSVQCY
 HPGLSPDLPICKEQVQSCGPPPELLNGNVKEKTKEEYGHSEVVEYYCNPRFLMKGPNKIQCV DGEWTT
 LPVCIVEESTCGDIPLEHGWAQLSSPPYYYGDSVEFNCSSESFTMIGHRSITCIHGVWTLQPQCVAID
 KKKCKSSNLIILEEHLKKNKEFDHNSNIRYRCRGKEGWIHTVCINGRWDPEVNC SMAQIQLCPPPPQ
 IPNSHNMTTTLNRYDGEKVSVLQENYLIQEGEEITCKDGRWQSIPLCVEKIPCSQPPQIEHG TINSS
 RSSQESYAHGKLSYTCGGFRISEENETTCYMGKWSPPQCEGLPCKSPPEISHGVVAHMSDSYQYG
 EEVTYKCFEGFGIDGPAIAKCLGKEKWSHPPSCIKTDCLSLPSFENAIPMGKKDVYKAGEQVTTYTCAT
 YYKMDGASNVTCINSRWTRPTCRDTS CVNPPTVQNAYIVSRQMSKYPGSEVRVYQCRSPYEMPGDEE
 VMCLNGNWTEPPQCKDSTGKCGPPPIDNGDITSFPLSVYAPASSVEYQC ONLYQLEGNKRITCRNGQ
 WSEPPKCLHPCVISREIMENYNIALRWTAQKLYSRTGESVEFVCKRGYRLSSRSHTLRLITCWDGKLE
 YPTCAKR

[0326] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 51.

[0327] An example nucleotide sequence encoding CFH is NM_000186.4. Suitably, a protein-coding sequence encoding CFH may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 52, or a variant which is at least 70% identical to SEQ ID NO: 52.

Illustrative CFH polynucleotide sequence (SEQ ID NO: 52):
ATGAGACTTCTAGCAAAGATTATTGCTTATGTTATGGGCTATTGTGTAGCAGAAGATTGCAATGA
ACTTCCTCCAAGAAGAAATACAGAAATCTGACAGGTTCTGGTCTGACCAAACATATCCAGAAGGCA
CCCAGGCTATCTATAAATGCCGCCCTGGATATAGATCTCTTGAAAATGTAATAATGGTATGCAGGAAG
GGAGAATGGGTGCTCTTAATCCATTAAAGGAAATGTCAGAAAAGGCCCTGTGGACATCCTGGAGATAC
TCCTTTTGGTACTTTTACCCTTACAGGAGGAAATGTGTTTGAATATGGTGTAAAGCTGTGTATACAT
GTAATGAGGGGTATCAATTGCTAGGTGAGATTAATTACCGTGAATGTGACACAGATGGATGGACCAAT
GATATTCCTATATGTGAAGTTGTGAAGTGTTTACCAAGTGACAGCACCAGAGAATGGAAAAATGTGAG
TAGTGCAATGGAACAGATCGGGAATACCATTTTGGACAAGCAGTACGGTTTGTATGTAACCTAGGCT
ACAAGATTGAAGGAGATGAAGAAATGCATTGTTTCCAGACGATGGTTTTTGGAGTAAAGAGAAACCAAAG
TGTGTGGAATTTTATGCAATCCCGATGTTATAAATGGATCTCTATATCTCAGAAGATTATTTA
TAAGGAGAATGAACGATTTCAATATAAATGTAACATGGGTTATGAATACAGTGAAGAGGAGATGCTG
TATGCACTGAATCTGGATGGCGTCCGTTGCCTTCATGTGAAGAAAAATCATGTGATAATCCTTATATT
CCAAATGGTGACTACTCACCTTTAAGGATTAAACACAGAACTGGAGATGAAATCACGTACCAGTGTAG
AAATGGTTTTTATCCTGCAACCCGGGAAATACAGCAAAATGCACAAGTACTGGCTGGATACCTGCTC
CGAGATGTACCTTGAAACCTTGTGATTATCCAGACATTAAACATGGAGGTCTATATCATGAGAATATG
CGTAGACCATACTTTCCAGTAGCTGTAGGAAATATTACTCTTACTGTGATGAACATTTTGGAGAC
TCCGTCAGGAAGTTACTGGGATCACATTCATTGCACACAAGATGGATGGTCCGACGAGTACCATGCC
TCAGAAAATGTTATTTTCTTATTTGGAAAATGGATATAATCAAAATCATGGAAGAAAGTTTGTACAG
GGTAAATCTATAGACGTTGCCATCCTGGCTACGCTCTTCCAAAAGCGCAGACCACAGTTACATG
TATGGAGAATGGCTGCTCTACTCCAGATGCATCCGTGTCAAACATGTTCCAAATCAAGTATAG
ATATTGAGAATGGGTTTATTCTGAATCTCAGTATACATATGCCCTTAAAGAAAAGCGAAATATCAA
TGCAAACTAGGATATGTAACAGCAGATGGTGAAACATCAGGATCAATTACATGTGGGAAAGATGGATG
GTCAGCTCAACCCACGTGCATTAATCTTGTGATATCCAGTATTTATGAATGCCAGAACTAAAAATG
ACTTCACATGGTTTAAAGCTGAATGACACATTGGACTATGAATGCCATGATGGTTATGAAAGCAATACT
GGAAGCACCACTGGTTCCATAGTGTGTGGTTACAATGGTTGGTCTGATTTACCCATATGTTATGAAAG
AGAAATGCGAACTTCCTAAAATAGATGTACACTTAGTTCTGTATCGAAGAAAGACCAAGTATAAAGTTG
GAGAGGTGTTGAAATCTCCTGCAACCAGGATTACAAATAGTTGGACCTAATCCGTTTCACTGCTAC
CACTTTGGATTGTCTCCTGACCTCCCAATATGTAAGAGCAAGTACAATCATGTGGTCCACCTCCTGA
ACTCCTCAATGGGAATGTTAAGGAAAAAACGAAAGAAGATATGGACACAGTGAAGTGGTGAATATT
ATTGCAATCTAGATTTCTAATGAAGGGACCTAATAAAATCAATGTGTTGATGGAGAGTGGACAACCT
TTACCACTGTGTATTGTGGAGGAGAGTACCTGTGGAGATATACCTGAACCTGAACATGGCTGGGCCCA
GCTTTCTTCCCTCCTTATTACTATGGAGATTCAAGTGAATTCATTTGCTCAGAATCATTTACAATGA
TTGGACACAGATCAATTACGTGATTTCATGGAGTATGGACCAACTTCCCAGTGTGTGGCAATAGAT
AAACTTAAGAAGTGCAATCATCAAATTTAATTATCTTGGAGAACATTTAAAAACAGAAGGAATT

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CGATCATAATTCTAACATAAGGTACAGATGTAGAGGAAAAGAAGGATGGATACACAGTCTGCATAA
 ATGGAAGATGGGATCCAGAAGTGAAGTCTCAATGGCACAATAACAATTATGCCACCTCCACCTCAG
 ATTCCCAATTCTCACAATATGACAACCCACTGAATTATCGGGATGGAGAAAAGTATCTGTTCTTTG
 CCAAGAAAATTATCTAATTCAGGAAGGAGAAGAAATTACATGCAAAGATGGAAGATGGCAGTCAATAC
 CACTCTGTGTTGAAAAAATTCATGTTTCAACCCACTCAGATAGAACACGGAACCATTAAATTCATCC
 AGGTCTTCACAAGAAAGTTATGCACATGGGACTAAATTGAGTTACTTGTGAGGGTGGTTTCAGGAT
 ATCTGAAGAAAATGAAACAACATGCTACATGGGAAAATGGAGTTCTCCACCTCAGTGTGAAGGCCTTC
 CTTGTAAATCTCCACCTGAGATTTCTCATGGTGTGTAGCTCACATGTCAGACAGTTATCAGTATGGA
 GAAGAAGTTACGTACAATGTTTTGAAGGTTTGAATTGATGGGCCTGCAATTGCAAAATGCTTAGG
 AGAAAAATGGTCTCACCTCCATCATGCATAAAACAGATTGTCTCAGTTTACCTAGCTTTGAAAATG
 CCATACCCATGGGAGAGAAGAAGGATGTGTATAAGGCGGGTGAAGCAAGTGACTTACACTTGTGCAACA
 TATTACAAAATGGATGGAGCCAGTAATGTAACATGCATTAAATAGCAGATGGACAGGAAGGCCAACATG
 CAGAGACACCTCCTGTGTGAATCCGCCACAGTACAAAATGCTTATATAGTGTGAGACAGATGAGTA
 AATATCCATCTGGTGAAGAGTACGTTTATCAATGTAGGAGCCCTTATGAAATGTTTGGGGATGAAGAA
 GTGATGTGTTTAAATGGAAGTGGACGGAACACCTCAATGCAAAGATTCTACAGGAAAATGTGGGCC
 CCCTCCACCTATTGACAATGGGGACATTACTTCATTCCCGTTGTCTAGTATATGCTCCAGCTTCATCAG
 TTGAGTACCAATGCCAGAAGTTGTATCAACTTGAGGGTAACAAGCAATAACATGTAGAAATGGACAA
 TGGTCAGAACCCAAAATGCTTACATCCGTGTGTAATATCCGAGAAATTATGGAAAATTATAACAT
 AGCATTAAAGGTGGACAGCCAAACAGAAGCTTTATTCGAGAACAGGTGAATCAGTTGAATTTGTGTGTA
 AACGGGGATATCGTCTTTCATCACGTTCTCACACATTGCGAACAACATGTTGGGATGGGAACTGGAG
 TATCCAACCTTGTGCAAAAAGATAG

[0328] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 52.

[0329] The CFH fragment may be a splice variant. For example, complement factor H-like protein 1 (FHL-1) is a CFH gene splice variant, which is almost identical to the N-terminal 7 domains of CFH (CCPs 1-7).

[0330] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding CFH, or a fragment and/or variant thereof.

[0331] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding CFH, or a fragment and/or variant thereof.

FHL-1

[0332] The protein-coding sequence may encode FHL-1, or a fragment and/or variant thereof.

[0333] FHL-1 or a fragment and/or variant thereof may be capable of binding C3b and/or C3d. The fragment and/or variant of FHL-1 may retain at least 50%, 60%, 70%, 80%, 90%, 95% or 100% of the activity of native FHL-1. The activity of the fragment and/or variant of FHL-1 and native FHL-1 may be determined using any suitable method known to those of skill in the art. Preferably, the FHL-1 is a human FHL-1. An example human FHL-1 is the FHL-1 having the NCBI Reference Sequence: NP_001014975.1.

[0334] Suitably, the FHL-1 may comprise or consist of the polypeptide sequence shown as SEQ ID NO: 53, or a variant which is at least 70% identical to SEQ ID NO: 53.

Illustrative FHL-1 polypeptide sequence (SEQ ID NO: 53):

MRLLAKII CLMLWAICVAEDCNELPPRNTTEILTGSWSDQTYPEGTQAIYKCRPGYRSLGNVIMVCRK

GEWVALNPLRKQKRPCGHPGDTPTFGTFTLTGGNVFEYGVKAVYTCNEGYQLLGEINRECDTDGWTN

DIPICEVVKCLPVTAPENGKIVSSAMEPDREYHFGQAVRFVCNSGYKIEGDEEMHCSDDGFWSKEKPK

CVEISCKSPDVINGSPISQKIIYKENERFQYKCNMGYEYSERGDAVCTESGWRPLPSCEEKSCDNPYI

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PNGDYSPLRIKHRIGDEITYQCRNGFYPATRGNTAKCTSTGWIAPRCTLKPCDYPDIKHGGLYHENM
RRPYFPVAVGKYYSYYCDEHFETPSGSYWDHIHCTQDGWSPAVPCLRKCYFPYLENGYNQNHGRKFVQ
GKSIDVACHPGYALPKAQTTVTCMENGWSPTPRCIRVSFTL

[0335] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 53.

[0336] An example nucleotide sequence encoding FHL-1 is NM_001014975.2. Suitably, a protein-coding sequence encoding FHL-1 may comprise or consist of the polynucleotide sequence shown as SEQ ID NO: 54, or a variant which is at least 70% identical to SEQ ID NO: 54.

[0339] In some embodiments, a promoter consisting of the nucleotide sequence of SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding FHL-1, or a fragment and/or variant thereof.

Other Regulatory Elements

[0340] In addition to the promoter of the invention, the polynucleotide may comprise one or more further regulatory

Illustrative FHL-1 polynucleotide sequence (SEQ ID NO: 54):
ATGAGACTTCTAGCAAAGATTATTTGCCTTATGTTATGGGCTATTGTGTAGCAGAAGATTGCAATGA
ACTTCTCTCCAAGAAGAAATACAGAAATCTGACAGGTTCTGGTCTGACCAAACATATCCAGAAGGCA
CCCAGGCTATCTATAAATGCCGCCCTGGATATAGATCTCTTGAAATGTAATAATGGTATGCAGGAAG
GGAGAATGGGTTGGTCTTAATCCATTAAGGAAATGTGAGAAAAGGCCCTGTGGACATCCTGGAGATAC
TCCTTTTGGTACTTTTACCCTTACAGGAGGAAATGTGTTTGAATATGGTGTAAAGCTGTGTATACAT
GTAATGAGGGGTATCAATTGCTAGGTGAGATTAATTACCGTGAATGTGACACAGATGGATGGACCAAT
GATATTCCTATATGTGAAGTTGTGAAGTGTTTACCAGTGACAGCACCAGAGAATGGAAAAATGTCAG
TAGTGCAATGGAACAGATCGGAATACCATTTTGGACAAGCAGTACGGTTTGTATGTAACCTCAGGCT
ACAAGATTGAAGGAGATGAAGAAATGCATTGTTTCAGACGATGGTTTTTGGAGTAAAGAGAAACCAAAG
TGTGTGAAAATTTTCATGCAAAATCCCAGATGTTATAAATGGATCTCCTATATCTCAGAAGATTATTTA
TAAGGAGAATGAACGATTTCAATATAAATGTAACATGGGTTATGAATACAGTGAAGAGGAGATGCTG
TATGCACTGAATCTGGATGGCGTCCGTTGCCTTCATGTGAAGAAAAATCATGTGATAATCCTTATATT
CCAAATGGTGACTACTCACCTTTAAGGATTAAACACAGAACTGGAGATGAAATCACGTACCAGTGTAG
AAATGGTTTTTATCTGCAACCCGGGAAATACAGCAAAATGCACAAGTACTGGCTGGATACCTGCTC
CGAGATGTACCTTGAAACCTTGTGATTATCCAGACATTAAACATGGAGGTCTATATCATGAGAATATG
CGTAGACCATACTTTCCAGTAGCTGTAGGAAAATATTACTCCTATTACTGTGATGAACATTTTGAGAC
TCCGTCAGGAAGTTACTGGGATCACATTTCATGCACACAAGATGGATGGTCGCCAGCAGTACCATGCC
TCAGAAAATGTTATTTTCTTATTTGGAAAATGGATATAATCAAAATCATGGAAGAAAGTTTGTACAG
GGTAAATCTATAGACGTTGCCCTGCCATCCTGGCTACGCTCTTCCAAAAGCGCAGACCACAGTTACATG
TATGGAGAATGGCTGGTCTCTACTCCAGATGCATCCGTGTCAGCTTTACCCTCTGA

[0337] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 54.

[0338] In some embodiments, a promoter comprising or consisting of a nucleotide sequence having at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identity to SEQ ID NO: 47 is operably linked to the protein-coding sequence encoding FHL-1, or a fragment and/or variant thereof.

sequences which may act pre- or post-transcriptionally. Suitably, the protein-coding sequence may be operably linked to one or more further regulatory sequences. The one or more further regulatory sequences may facilitate expression of the protein in glomerular cells (e.g. podocytes).

[0341] "Regulatory sequences" are any sequences which facilitate expression of the polypeptides, e.g. act to increase expression of a transcript or to enhance mRNA stability. Suitable further regulatory sequences include for example enhancer elements, post-transcriptional regulatory elements and polyadenylation sites.

[0342] Suitably, the polynucleotide of the present invention does not comprise any further regulatory sequences (except the promoter of the invention).

Enhancers

[0343] The polynucleotide of the invention may comprise an enhancer. Suitably, the enhancer may be operably linked to the protein-coding sequence. The enhancer may facilitate expression of the protein in glomerular cells (e.g. podocytes). Suitably, the enhancer is a mammalian enhancer e.g. a human enhancer.

[0344] An “enhancer” is a region of DNA that can be bound by proteins (activators) to increase the likelihood that transcription of a particular gene will occur. Enhancers are cis-acting. They can be located up to 1 Mbp (1,000,000 bp) away from the gene, upstream or downstream from the start site. Any suitable enhancer may be used, the selection of which may be readily made by the skilled person.

[0345] The polynucleotide of the invention may comprise a tissue specific enhancer. As used herein, a “tissue-specific enhancer” is an enhancer which preferentially facilitates expression of a gene in specific cells or tissues. Suitably, a tissue-specific enhancer may facilitate higher expression of a gene in specific cells-types as compared to other cell-types. Higher expression may be measured for example by measuring the expression of a transgene, e.g. GFP, operably linked to the enhancer, wherein expression of the transgene correlates with the ability of the enhancer to facilitate expression of a gene. For example, a tissue-specific enhancer may be an enhancer which facilitates gene expression levels at least 10% higher, at least 20% higher, at least 30% higher, at least 40% higher, at least 50% higher, at least 100% higher, at least 200% higher, at least 300% higher, at least 400% higher, at least 500% higher, or at least 1000% higher in a specific cell-type compared to expression levels in other cell types.

[0346] Suitable tissue-specific enhancer will be well known to those of skill in the art. The enhancer may be a kidney-specific enhancer, preferably a glomerular-specific enhancer, more preferably a podocyte-specific enhancer. Suitably, the enhancer may be operably linked to the protein-coding sequence.

[0347] Suitably, the enhancer may be or may be derived from an enhancer associated with a gene with selective expression in human kidney, glomerular cells, and/or podocytes. Methods to identify the enhancer regions associated with genes will be well known to those of skill in the art.

[0348] Suitably, the podocyte-specific enhancer is a NPHS1 or a NPHS2 enhancer, or a fragment or derivative thereof. Suitably, the podocyte-specific enhancer is a NPHS1 enhancer, or a fragment or derivative thereof.

[0349] A NPHS1 enhancer has been described in Guo, G., et al., 2004. Journal of the American Society of Nephrology, 15 (11), pp. 2851-2856. A 186-bp fragment from the human NPHS1 promoter was capable of directing podocyte-specific

expression of a β -galactosidase transgene when placed in front of a heterologous minimal promoter in transgenic mice.

[0350] Suitably, the NPHS1 enhancer may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 27, or a variant which is at least 70% identical to SEQ ID NO: 27.

Exemplary NPHS1 enhancer (SEQ ID NO: 27):
 ctgctgagctgggagaccacettgatctgacttctcccatcttccagc
 ctaagccaggccctgggggtcacggaggctggggaggcaccgaggaacgc
 gcttgccatgtgctgacaggggattttatgctccagctgggccagctgg
 gaggagcctgctgggcagaggccagagctgggggctctg

[0351] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 27.

[0352] Suitably, the polynucleotide of the present invention does not comprise an enhancer.

Kozak Sequence

[0353] The polynucleotide of the invention may comprise a Kozak sequence. Suitably, the Kozak sequence may be operably linked to the protein-coding sequence. A Kozak sequence may be inserted before the start codon of the protein to improve the initiation of translation.

[0354] Suitable Kozak sequences will be well known to those of skill in the art.

[0355] Suitably, the Kozak sequence may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 28, or a variant which is at least 65% identical to SEQ ID NO: 28.

Exemplary Kozak sequence (SEQ ID NO: 28):
 GCCGCCACCAUGG

[0356] Suitably, the variant may be at least 75%, at least 85%, or at least 90% identical to SEQ ID NO: 28.

[0357] Suitably, the polynucleotide of the present invention does not comprise a Kozak sequence.

Post-Transcriptional Regulatory Elements

[0358] The polynucleotide of the invention may comprise a post-transcriptional regulatory element. Suitably, the post-transcriptional regulatory element may be operably linked to the protein-coding sequence. The post-transcriptional regulatory element may improve gene expression.

[0359] The polynucleotide may comprise a Woodchuck Hepatitis Virus Post-transcriptional Regulatory Element (WPRE). Suitably, the WPRE may be operably linked to the protein-coding sequence.

[0360] The WPRE sequence may have substitutions, deletions, or insertions within the X-antigen promoter and/or the initiation codon of the X-antigen. This may prevent the production of a functional X-antigen.

[0361] Suitably, the WPRE may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 29, or a variant which is at least 70% identical to SEQ ID NO: 29.

Exemplary WPRE (SEQ ID NO: 29):

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aatcaacctctggattacaaaattgtgaaagattgactggattcttaactatgttgcctctttac
gctatgtggatagcgtgctttaatgcctttgtatcatgctattgctcccgatggctttcattttct
cctccttgataaatcctgggtgctgtctctttatgaggagttgtggccggtgtcaggcaacgtggc
gtgggtgtgcaactgtgtttgctgacgcaacccccactgggtggggcattgccaccacctgtcagctcct
ttccgggactttcgtttccccctccctattgccacggcggaactcatcgccgctgcttgcctcct
gctggacaggggctcggtgttgggcactgacaattccgtggtgtgtcggggaaatcatcgctcttt
ccttggtgctcgctgtgttgccacctggattctgcgcggaactccttctgctacgtcccttcggc
cctcaatccagcggaacttccttcccgcggtgctgcggctctgcggcctcttccgctcttcgcc
ttcgccctcagacgagtcggatctcccttgggcgcctcccg
```

[0362] Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 29.

[0363] Suitably, the polynucleotide of the present invention does not comprise a post-transcriptional regulatory element.

Polyadenylation Signal

[0364] The polynucleotide of the invention may comprise a polyadenylation signal. Suitably, the polyadenylation signal may be operably linked to the protein-coding sequence. The polyadenylation signal may improve gene expression.

Exemplary SV40pA signal sequence (SEQ ID NO: 30):

```
aacttggttattgcagcttataatggtacaaaataagcaatagcatca
caaatctcacaataaagcatttttttcaactgcattctagtgtggttt
gtccaaactcatcaatgtatcttatcatgtctggatc
```

[0367] Suitably, the polyadenylation signal may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 31, or a variant which is at least 70% identical to SEQ ID NO: 31. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 31.

Exemplary bGH poly(A) signal sequence (SEQ ID NO: 31):

```
ctgtgccttctagtgtgccagccatctgtgtttgcccctcccccgctccttctgacccctggaaggt
gccactcccactgtcctttcctaataaaatgaggaaattgcatcgcatgtgtgagtaggtgtcattc
tattctgggggggtgggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctg
gggatgcggtgggctctatgg
```

[0365] Suitable polyadenylation signals include the early SV40 polyadenylation signal (SV40 pA), a chicken beta-globin polyadenylation signal, a bovine growth hormone polyadenylation signal (bGH), or a soluble neuropilin-1 polyadenylation signal. In some embodiments, the polyadenylation signal is an early SV40 polyadenylation signal (SV40 pA) or a chicken beta-globin polyadenylation signal. Preferably, the polyadenylation signal is an early SV40 polyadenylation signal (SV40 pA).

[0366] Suitably, the polyadenylation signal may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 30, or a variant which is at least 70% identical to SEQ ID NO: 30. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 30.

[0368] Suitably, the polyadenylation signal may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 32, or a variant which is at least 70% identical to SEQ ID NO: 32. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 32.

[0369] Exemplary soluble neuropilin-1 polyadenylation signal (SEQ ID NO: 32):

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aaataaaatcgaatg
```

[0370] Suitably, the polyadenylation signal may comprise or consist of the nucleotide sequence shown as SEQ ID NO: 42, or a variant which is at least 70% identical to SEQ ID NO: 42. Suitably, the variant may be at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to SEQ ID NO: 42.

Exemplary chicken beta-globin polyadenylation signal
(SEQ ID NO: 42)
caataaaagatctttatttttcattagatctgtgttggttttttgtgtg

Vectors

[0371] The present invention provides a vector comprising the polynucleotide of the present invention.

[0372] A vector is a tool that allows or facilitates the transfer of an entity from one environment to another. The four major types of vectors are plasmids, viral vectors, cosmids, and artificial chromosomes.

[0373] Preferably, the vector of the present invention is a viral vector. The vector of the invention is preferably an adeno-associated viral (AAV) vector, although it is contemplated that other viral vectors may be used.

[0374] The vector of the present invention may be in the form of a viral vector particle. Preferably, the viral vector of the present invention is in the form of an AAV vector particle.

[0375] Methods of preparing and modifying viral vectors and viral vector particles, such as those derived from AAV, are well known in the art. Suitable methods are described in Ayuso, E., et al., 2010. *Current gene therapy*, 10 (6), pp. 423-436; Merten, O. W., et al., 2016. *Molecular Therapy-Methods & Clinical Development*, 3, p. 16017; and Nadeau, I. and Kamen, A., 2003. *Biotechnology advances*, 20 (7-8), pp. 475-489.

[0376] The vector of the present invention may be capable of transducing kidney cells. In some embodiments, the vector of the present invention is capable of specifically transducing kidney cells.

[0377] The vector of the present invention is preferably capable of transducing glomerular cells (e.g. podocytes). In some embodiments, the vector of the present invention is capable of specifically transducing glomerular cells (e.g. podocytes). The vector of the present invention is preferably capable of transducing podocytes. In some embodiments, the vector of the present invention is capable of specifically transducing podocytes.

Adeno-Associated Viral (AAV) Vectors

[0378] The vector of the present invention may be an adeno-associated viral (AAV) vector. The vector of the present invention may be in the form of an AAV vector particle.

AAV genome

[0379] The AAV vector or AAV vector particle may comprise an AAV genome or a fragment or derivative thereof.

[0380] An AAV genome is a polynucleotide sequence, which may encode functions needed for production of an AAV particle. These functions include those operating in the replication and packaging cycle of AAV in a host cell, including encapsidation of the AAV genome into an AAV particle. Naturally occurring AAVs are replication-deficient and rely on the provision of helper functions in trans for completion of a replication and packaging cycle. Accordingly, the AAV genome of the AAV vector of the invention is typically replication-deficient.

[0381] The AAV genome may be in single-stranded form (ssAAV), either positive or negative-sense, or alternatively in double-stranded form (dsAAV). The use of a double-stranded form allows bypass of the DNA replication step in

the target cell and so can accelerate transgene expression. The maximum packaging capacity of the single-stranded form is larger than the double-stranded form. Suitably, the AAV genome is in single-stranded form.

[0382] AAVs occurring in nature may be classified according to various biological systems. The AAV genome may be from any naturally derived serotype, isolate or clade of AAV.

[0383] AAV may be referred to in terms of their serotype. A serotype corresponds to a variant subspecies of AAV which, owing to its profile of expression of capsid surface antigens, has a distinctive reactivity which can be used to distinguish it from other variant subspecies. Typically, an AAV vector particle having a particular AAV serotype does not efficiently cross-react with neutralising antibodies specific for any other AAV serotype. AAV serotypes include AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10 and AAV11. In some embodiments, the AAV vector of the invention may be an AAV3B, LK03, AAV9, or AAV8 serotype. In some embodiments, the AAV vector of the invention may be an AAV3B, LK03, or AAV9 serotype.

[0384] AAV may also be referred to in terms of clades or clones. This refers to the phylogenetic relationship of naturally derived AAVs, and typically to a phylogenetic group of AAVs which can be traced back to a common ancestor, and includes all descendants thereof. Additionally, AAVs may be referred to in terms of a specific isolate, i.e., a genetic isolate of a specific AAV found in nature. The term genetic isolate describes a population of AAVs which has undergone limited genetic mixing with other naturally occurring AAVs, thereby defining a recognisably distinct population at a genetic level.

[0385] Typically, the AAV genome of a naturally derived serotype, isolate or clade of AAV comprises at least one inverted terminal repeat sequence (ITR). An ITR sequence acts in cis to provide a functional origin of replication and allows for integration and excision of the vector from the genome of a cell. ITRs may be the only sequences required in cis next to the therapeutic gene.

[0386] The AAV genome may also comprise packaging genes, such as rep and/or cap genes which encode packaging functions for an AAV particle. A promoter may be operably linked to each of the packaging genes. Specific examples of such promoters include the p5, p19 and p40 promoters. For example, the p5 and p19 promoters are generally used to express the rep gene, while the p40 promoter is generally used to express the cap gene. The rep gene encodes one or more of the proteins Rep78, Rep68, Rep52 and Rep40 or variants thereof. The cap gene encodes one or more capsid proteins such as VP1, VP2 and VP3 or variants thereof. These proteins make up the capsid of an AAV particle, which determines the AAV serotype. VP1, VP2, and VP3 may be produced by alternate mRNA splicing (Trempe, J. P. and Carter, B. J., 1988. *Journal of virology*, 62 (9), pp. 3356-3363). Thus, VP1, VP2 and VP3 may have identical sequences, but wherein VP2 is truncated at the N-terminus relative to VP1, and VP3 is truncated at the N-terminus relative to VP2.

[0387] The AAV genome may be the full genome of a naturally occurring AAV. For example, a vector comprising a full AAV genome may be used to prepare an AAV vector or vector particle.

[0388] Preferably, the AAV genome is derivatised for the purpose of administration to patients. Such derivatisation is standard in the art and the invention encompasses the use of any known derivative of an AAV genome, and derivatives which could be generated by applying techniques known in the art. The AAV genome may be a derivative of any naturally occurring AAV. Suitably, the AAV genome is a derivative of AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, or AAV11. Suitably, the AAV genome is a derivative of AAV2.

[0389] Derivatives of an AAV genome include any truncated or modified forms of an AAV genome which allow for expression of a transgene from an AAV vector of the invention *in vivo*. Typically, it is possible to truncate the AAV genome significantly to include minimal viral sequence yet retain the above function. This is preferred for safety reasons to reduce the risk of recombination of the vector with wild-type virus, and also to avoid triggering a cellular immune response by the presence of viral gene proteins in the target cell.

[0390] Typically, a derivative will include at least one inverted terminal repeat sequence (ITR), preferably more than one ITR, such as two ITRs or more. One or more of the ITRs may be derived from AAV genomes having different serotypes, or may be a chimeric or mutant ITR. A preferred mutant ITR is one having a deletion of a trs (terminal resolution site). This deletion allows for continued replication of the genome to generate a single-stranded genome which contains both coding and complementary sequences, i.e., a self-complementary AAV (scAAV) genome. This allows for bypass of DNA replication in the target cell, and so enables accelerated transgene expression. However, the maximum packaging capacity of a scAAV is reduced. Suitably, the AAV genome is not a scAAV genome.

[0391] The AAV genome may comprise one or more ITR sequences from any naturally derived serotype, isolate or clade of AAV or a variant thereof. The AAV genome may comprise at least one, such as two, AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, or AAV11 ITRs, or variants thereof. Suitably, the AAV genome may comprise at least one, such as two, AAV2 ITRs.

[0392] The inclusion of one or more ITRs is preferred to aid concatamer formation of the AAV vector in the nucleus of a host cell, for example following the conversion of single-stranded vector DNA into double-stranded DNA by the action of host cell DNA polymerases. The formation of such episomal concatamers protects the AAV vector during the life of the host cell, thereby allowing for prolonged expression of the transgene *in vivo*.

[0393] Suitably, ITR elements will be the only sequences retained from the native AAV genome in the derivative. A derivative will preferably not include the rep and/or cap genes of the native genome and any other sequences of the native genome. This is preferred for the reasons described above, and also to reduce the possibility of integration of the vector into the host cell genome. Additionally, reducing the size of the AAV genome allows for increased flexibility in incorporating other sequence elements (such as regulatory elements) within the vector in addition to the transgene.

[0394] The following portions could therefore be removed in a derivative of the invention: one inverted terminal repeat (ITR) sequence, the replication (rep) and capsid (cap) genes. However, derivatives may additionally include one or more rep and/or cap genes or other viral sequences of an AAV genome. Naturally occurring AAV integrates with a high frequency at a specific site on human chromosome 19, and shows a negligible frequency of random integration, such that retention of an integrative capacity in the AAV vector may be tolerated in a therapeutic setting.

[0395] The invention additionally encompasses the provision of sequences of an AAV genome in a different order and configuration to that of a native AAV genome. The invention also encompasses the replacement of one or more AAV sequences or genes with sequences from another virus or with chimeric genes composed of sequences from more than one virus. Such chimeric genes may be composed of sequences from two or more related viral proteins of different viral species.

Aav Serotype and Capsid Proteins

[0396] The AAV vector particle may be encapsidated by capsid proteins. The serotype may facilitate the transduction of glomerular cells (e.g., podocytes), for example specific transduction of glomerular cells (e.g., podocytes).

[0397] The AAV vector particle may be a kidney-specific vector particle. Preferably, the AAV vector particle is a glomerular-specific (e.g., podocyte-specific) vector particle. The AAV vector particle may be encapsidated by a glomerular-specific (e.g., podocyte-specific) capsid. The AAV vector particle may comprise a glomerular-specific (e.g., podocyte-specific) capsid protein.

[0398] Suitably, the AAV vector particles may be transcapsidated forms wherein an AAV genome or derivative having an ITR of one serotype is packaged in the capsid of a different serotype. The AAV vector particle also includes mosaic forms wherein a mixture of unmodified capsid proteins from two or more different serotypes makes up the viral capsid. The AAV vector particle also includes chemically modified forms bearing ligands adsorbed to the capsid surface. For example, such ligands may include antibodies for targeting a particular cell surface receptor.

[0399] Where a derivative comprises capsid proteins i.e., VP1, VP2 and/or VP3, the derivative may be a chimeric, shuffled or capsid-modified derivative of one or more naturally occurring AAVs. In particular, the invention encompasses the provision of capsid protein sequences from different serotypes, clades, clones, or isolates of AAV within the same vector (i.e. a pseudotyped vector). The AAV vector may be in the form of a pseudotyped AAV vector particle.

[0400] Chimeric, shuffled or capsid-modified derivatives will be typically selected to provide one or more desired functionalities for the AAV vector. Thus, these derivatives may display increased efficiency of gene delivery, decreased immunogenicity (humoral or cellular), an altered tropism range and/or improved targeting of podocytes compared to an AAV vector comprising a naturally occurring AAV genome. Increased efficiency of gene delivery may be effected by improved receptor or co-receptor binding at the cell surface, improved internalisation, improved trafficking within the cell and into the nucleus, improved uncoating of the viral particle and improved conversion of a single-stranded genome to double-stranded form. Increased efficiency may also relate to an altered tropism range or

targeting of podocytes, such that the vector dose is not diluted by administration to tissues where it is not needed.

[0401] Chimeric capsid proteins include those generated by recombination between two or more capsid coding sequences of naturally occurring AAV serotypes. This may be performed for example by a marker rescue approach in which non-infectious capsid sequences of one serotype are co-transfected with capsid sequences of a different serotype, and directed selection is used to select for capsid sequences having desired properties. The capsid sequences of the different serotypes can be altered by homologous recombination within the cell to produce novel chimeric capsid proteins.

[0402] Chimeric capsid proteins also include those generated by engineering of capsid protein sequences to transfer specific capsid protein domains, surface loops or specific amino acid residues between two or more capsid proteins, for example between two or more capsid proteins of different serotypes.

[0403] Shuffled or chimeric capsid proteins may also be generated by DNA shuffling or by error-prone PCR. Hybrid AAV capsid genes can be created by randomly fragmenting the sequences of related AAV genes e.g., those encoding capsid proteins of multiple different serotypes and then subsequently reassembling the fragments in a self-priming polymerase reaction, which may also cause crossovers in regions of sequence homology. A library of hybrid AAV genes created in this way by shuffling the capsid genes of several serotypes can be screened to identify viral clones having a desired functionality. Similarly, error prone PCR may be used to randomly mutate AAV capsid genes to create a diverse library of variants which may then be selected for a desired property.

[0404] The sequences of the capsid genes may also be genetically modified to introduce specific deletions, substitutions or insertions with respect to the native wild-type sequence. In particular, capsid genes may be modified by the insertion of a sequence of an unrelated protein or peptide within an open reading frame of a capsid coding sequence, or at the N- and/or C-terminus of a capsid coding sequence. The unrelated protein or peptide may advantageously be one which acts as a ligand for a particular cell type, thereby conferring improved binding to a target cell or improving the specificity of targeting of the vector to a particular cell population. The unrelated protein may also be one which assists purification of the viral particle as part of the production process, i.e. an epitope or affinity tag. The site of insertion will typically be selected so as not to interfere with other functions of the viral particle e.g., internalisation, trafficking of the viral particle.

[0405] The capsid protein may be an artificial or mutant capsid protein. The term “artificial capsid” as used herein means that the capsid particle comprises an amino acid sequence which does not occur in nature or which comprises an amino acid sequence which has been engineered (e.g. modified) from a naturally occurring capsid amino acid sequence. In other words, the artificial capsid protein comprises a mutation or a variation in the amino acid sequence compared to the sequence of the parent capsid from which it is derived where the artificial capsid amino acid sequence and the parent capsid amino acid sequences are aligned.

[0406] The capsid protein may comprise a mutation or modification relative to the wild type capsid protein which improves the ability to transduce podocytes relative to an

unmodified or wild type viral particle. Improved ability to transduce podocytes may be measured for example by measuring the expression of a transgene, e.g., GFP, carried by the AAV vector particle, wherein expression of the transgene in podocytes correlates with the ability of the AAV vector particle to transduce podocytes.

[0407] The AAV vector particle may be an AAV3B, LK03, AAV9, or AAV8 vector particle. The present inventors have shown that AAV vector particles with AAV3B, LK03, AAV9 and AAV8 serotypes can transduce podocytes. Preferably, the AAV vector particle is an AAV3B vector particle or an LK03 vector particle. More preferably, the AAV vector particle is an AAV3B vector particle.

[0408] The AAV vector particle may comprise an AAV3B, LK03, AAV9, or AAV8 capsid protein. Preferably, the AAV vector particle comprises an AAV3B capsid protein or an LK03 capsid protein. More preferably, the AAV vector particle comprises an AAV3B capsid protein.

[0409] The AAV vector particle may comprise AAV3B, LK03, AAV9, or AAV8 capsid proteins VP1, VP2 and VP3. Preferably, the AAV vector particle comprises AAV3B or LK03 capsid proteins VP1, VP2 and VP3. More preferably, the AAV vector particle comprises AAV3B capsid proteins VP1, VP2 and VP3.

[0410] The AAV vector particle may comprise one or more AAV2 ITR sequences and AAV3B capsid proteins, LK03 capsid proteins, AAV9 capsid proteins, or AAV8 capsid proteins. Preferably, the AAV vector particle comprises one or more AAV2 ITR sequences and AAV3B or LK03 capsid proteins. More preferably, the AAV vector particle comprises one or more AAV2 ITR sequences and AAV3B capsid proteins.

[0411] The AAV vector particle may have an AAV2 genome and AAV3B capsid proteins (AAV2/3B), an AAV2 genome and LK03 capsid proteins, an AAV2 genome and AAV9 capsid proteins (AAV2/9), or an AAV2 genome and AAV8 capsid proteins (AAV2/8). Preferably, the AAV vector particle comprises an AAV2 genome and AAV3B or LK03 capsid proteins. More preferably, the AAV vector particle comprises an AAV2 genome and AAV3B capsid proteins.

[0412] The nomenclature AAVX/Y may denote a pseudotyped AAV, for example where the ITR sequences are from AAVX and flank a cassette harbouring a payload which is encapsidated into serotype AAVY (i.e. with AAVY capsid proteins).

AAV3B Serotype

[0413] The AAV vector particle may comprise an AAV3B capsid protein. Suitably, the AAV vector particle may be encapsidated by AAV3B capsid proteins.

[0414] Two distinct AAV3 isolates (AAV3A and AAV3B) have been cloned. In comparison with vectors based on other AAV serotypes, it is thought that AAV3 vectors inefficiently transduce most cell types. However, AAV3B may efficiently transduce podocytes. AAV3B has been described in Rutledge, E. A., et al., 1998. *Journal of virology*, 72 (1), pp. 309-319.

[0415] The AAV vector particle may comprise an AAV3B VP1 capsid protein, an AAV3B VP2 capsid protein, and/or an AAV3B VP3 capsid protein. Suitably, the AAV vector particle may be encapsidated by AAV3B VP1 capsid proteins, AAV3B VP2 capsid proteins, and/or AAV3B VP3 capsid proteins. Suitably, the AAV vector particle may be encapsidated by AAV3B VP1, VP2, and VP3 capsid proteins.

[0416] Suitably, the AAV3B VP1 capsid protein may comprise or consist of the amino acid sequence shown as SEQ ID NO: 33, or a variant which is at least 90% identical to SEQ ID NO: 33.

Exemplary AAV3B VP1 capsid protein (SEQ ID NO: 33):
 MAADGYLPDWLEDNLSEGIREWVALKPGVPQPKANQQHQDNRRGLVLPGYKYLPGNGLDKGEPVNEA
 DAAALEHDKAYDQQLKAGDNPYLKYNHADADEFQERLQEDTSFGGNLGRAVFOAKKRILEPLGLVEEAA
 KTAPGKKRPVDQSPQEPDSSSGVGKSGKQPKARKRLNFGQTGDSESVDPDQPLGEPPAAPTSLGSNTMA
 SGGGAPMADNNEGADGVGNSSGNWHCDSQWLGDREVITSTRTWALPTYNNHLYKQISSQSGASNDNH
 FGYSTPWGYDFNRFHCHFSRDLWQRLINNNWGFRPKKLSFKLFNIQVKEVTQNDGTTTIANNLSTV
 QVFTDSEYQLPYVLGSAHQGLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPPSQMLRTGNN
 FQFSYTFEDVPPHSSYAHSSQLDRLMNPLIDQYLYLNRQTGTTSGTTNQSRLLFSQAGPQSMQLQAR
 NWLPGPCYRQRLSKTANDNNNSNFPWTAASKYHLNGRDSLNVNPGPAMASHKDDDEKFFPMHGNLIFG
 KEGTTASNAELDNVMTDEEEIRITNPVATEQYGTVANLQSSNTAPTTRTVNDQGALPGMVWQDRDV
 YLQGPWAKIPHTDGHFHPSPMLMGFGGLKHPPQIMIKNTVPANPPTTFSPAKFASFITQYSTGQVS
 VEIEWELQKENSQRWNPEIQYTSNYNKSVDFTVDNNGVYSEPRPIGTRYLTRN

[0417] Suitably, the variant may be at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 33.

[0418] Suitably, the AAV3B VP2 and VP3 capsid proteins may be N-terminal truncations of SEQ ID NO: 33, or N-terminal truncations of a variant which is at least 90% identical, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 33.

Lk03 Serotype

[0419] The AAV vector particle may comprise an LK03 capsid protein. Suitably, the AAV vector particle may be encapsidated by LK03 capsid proteins.

[0420] The AAV-LK03 cap sequence consists of fragments from seven different wild-type serotypes (AAV1, 2,

3B, 4, 6, 8, 9) and is described in Lisowski, L., et al., 2014. Nature, 506 (7488), pp. 382-386. The present inventors have demonstrated that AAV-LK03 vectors can achieve high transduction of close to 100% in human podocytes in vitro.

[0421] The AAV vector particle may comprise an LK03 VP1 capsid protein, an LK03 VP2 capsid protein, and/or an LK03 VP3 capsid protein. Suitably, the AAV vector particle may be encapsidated by LK03 VP1 capsid proteins, LK03 VP2 capsid proteins, and/or LK03 VP3 capsid proteins. Suitably, the AAV vector particle may be encapsidated by LK03 VP1, VP2, and VP3 capsid proteins.

[0422] Suitably, the LK03 VP1 capsid protein may comprise or consist of the amino acid sequence shown as SEQ ID NO: 34, or a variant which is at least 90% identical to SEQ ID NO: 34.

Exemplary LK03 VP1 capsid protein (SEQ ID NO: 34):
 MAADGYLPDWLEDNLSEGIREWVALQPGAPKPKANQQHQDNARGLVLPGYKYLPGNGLDKGEPVNAA
 DAAALEHDKAYDQQLKAGDNPYLKYNHADADEFQERLKEDTSFGGNLGRAVFOAKKRILLEPLGLVEEAA
 KTAPGKKRPVDQSPQEPDSSSGVGKSGKQPKARKRLNFGQTGDSESVDPDQPLGEPPAAPTSLGSNTMA
 SGGGAPMADNNEGADGVGNSSGNWHCDSQWLGDREVITSTRTWALPTYNNHLYKQISSQSGASNDNH
 FGYSTPWGYDFNRFHCHFSRDLWQRLINNNWGFRPKKLSFKLFNIQVKEVTQNDGTTTIANNLSTV
 QVFTDSEYQLPYVLGSAHQGLPPFPADVFMVPQYGYLTLNNGSQAVGRSSFYCLEYFPPSQMLRTGNN
 FQFSYTFEDVPPHSSYAHSSQLDRLMNPLIDQYLYLNRQTGTTSGTTNQSRLLFSQAGPQSMQLQAR
 NWLPGPCYRQRLSKTANDNNNSNFPWTAASKYHLNGRDSLNVNPGPAMASHKDDDEKFFPMHGNLIFG
 KEGTTASNAELDNVMTDEEEIRITNPVATEQYGTVANLQSSNTAPTTRTVNDQGALPGMVWQDRDV
 YLQGPWAKIPHTDGHFHPSPMLMGFGGLKHPPQIMIKNTVPANPPTTFSPAKFASFITQYSTGQVS
 VEIEWELQKENSQRWNPEIQYTSNYNKSVDFTVDNNGVYSEPRPIGTRYLTRN

[0423] Suitably, the variant may be at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 34.

[0424] Suitably, the LK03 VP2 and VP3 capsid proteins may be N-terminal truncations of SEQ ID NO: 34, or N-terminal truncations of a variant which is at least 90% identical, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 34.

Aav9 Serotype

[0425] The AAV vector particle may comprise an AAV9 VP1 capsid protein. Suitably, the AAV vector particle may be encapsidated by AAV9 capsid proteins.

[0426] The present inventors have demonstrated that AAV9 vectors can achieve high transduction in human podocytes in vitro.

[0427] The AAV vector particle may comprise an AAV9 VP1 capsid protein, an AAV9 VP2 capsid protein, and/or an AAV9 VP3 capsid protein. Suitably, the AAV vector particle may be encapsidated by AAV9 VP1 capsid proteins, AAV9 VP2 capsid proteins, and/or AAV9 VP3 capsid proteins. Suitably, the AAV vector particle may be encapsidated by AAV9 VP1, VP2, and VP3 capsid proteins.

[0428] Suitably, the AAV9 VP1 capsid protein may comprise or consist of the amino acid sequence shown as SEQ ID NO: 35, or a variant which is at least 90% identical to SEQ ID NO: 35.

Exemplary AAV9 VP1 capsid protein (SEQ ID NO: 35):

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MAADGYLPDWLEDNLSEGIREWALKPGAPQPKANQHQDNARGLVLPGYKYLPGNGLDKGEVNA
DAAALEHDKAYDQQLKAGDNPYLKYNHADADEFQERLKEDTSFGGNLGRAVFQAKRRLLEPLGLVEEAA
KTAPGKKRPVEQSPQEPDSSAGIGKSGAQPAKKRLNFGQTGDTESVPDPQPIGEPPAAPSGVGLTMA
SGGGAPVADNNEGADGVGSSGNWHCDSSQWLGDRVITTTSTRTWALPTYNNHLYKQISNSTSGGSSNDN
AYFGYSTPWGYDFNRFCHFSRPRDQRLINNNWGFPRKRLNFKLFNIQVKEVTNNGVKTIANNLTS
TVQVFTDSYQLPYVLGSAHEGCLPPFPADVFMIPOYGYLTLNDGSQAVGRSSFYCLEYFPPSQMLRTG
NNFQFSYEFENVFPHSSYAHQSQSLDRLMNPLIDQYLYLSKTINGSGNQQTLPKFSVAGPSNMAVQGR
NYIPGPSYRQQRVSTTVTQNNNSEFAWPGASSWALNGRNSLMNPGPAMASHKEGEDRFFPLSGSLIFG
KQGTGRDNVDADKVMITNEEIKITNPVATESYGVATNHQSAQAQATGWQNGQILPGMVWQDRDV
YLQGPWAKIPHDTGNFHPSPMLGGFGMKHPPQILIKNTPVPADPPTAFNKDKLNSFITQYSTGQVS
VEIEWELQKENS KRWNPEIQYTSNYKSNVFEAVNTEGVYSEPRPIGTRYLTRNL
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[0429] Suitably, the variant may be at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 35.

[0430] Suitably, the AAV9 VP2 and VP3 capsid proteins may be N-terminal truncations of SEQ ID NO: 35, or N-terminal truncations of a variant which is at least 90% identical, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identical to SEQ ID NO: 35.

Other Viral Vectors

Retroviral and Lentiviral Vectors

[0431] The vector of the present invention may be a retroviral vector or a lentiviral vector. The vector of the present invention may be a retroviral vector particle or a lentiviral vector particle.

[0432] A retroviral vector may be derived from or may be derivable from any suitable retrovirus. A large number of different retroviruses have been identified. Examples include murine leukaemia virus (MLV), human T-cell leukaemia virus (HTLV), mouse mammary tumour virus (MMTV), Rous sarcoma virus (RSV), Fujinami sarcoma virus (FuSV), Moloney murine leukaemia virus (Mo-MLV), FBR murine osteosarcoma virus (FBR MSV), Moloney murine sarcoma virus (Mo-MSV), Abelson murine leukaemia virus (A-MLV), avian myelocytomatosis virus-29 (MC29) and avian erythroblastosis virus (AEV).

[0433] Retroviruses may be broadly divided into two categories, "simple" and "complex". Retroviruses may be even further divided into seven groups. Five of these groups represent retroviruses with oncogenic potential. The remaining two groups are the lentiviruses and the spumaviruses.

[0434] The basic structure of retrovirus and lentivirus genomes share many common features such as a 5' LTR and a 3' LTR. Between or within these are located a packaging signal to enable the genome to be packaged, a primer binding site, integration sites to enable integration into a host cell genome, and gag, pol and env genes encoding the packaging components—these are polypeptides required for the assembly of viral particles. Lentiviruses have additional features, such as rev and RRE sequences in HIV, which enable the efficient export of RNA transcripts of the integrated provirus from the nucleus to the cytoplasm of an infected target cell.

[0435] In the provirus, these genes are flanked at both ends by regions called long terminal repeats (LTRs). The LTRs are responsible for proviral integration and transcription. LTRs also serve as enhancer-promoter sequences and can control the expression of the viral genes.

[0436] The LTRs themselves are identical sequences that can be divided into three elements: U3, R and U5. U3 is derived from the sequence unique to the 3' end of the RNA. R is derived from a sequence repeated at both ends of the RNA. U5 is derived from the sequence unique to the 5' end of the RNA. The sizes of the three elements can vary considerably among different retroviruses.

[0437] In a defective retroviral vector genome gag, pol and env may be absent or not functional.

[0438] In a typical retroviral vector, at least part of one or more protein coding regions essential for replication may be

removed from the virus. This makes the viral vector replication-defective. Portions of the viral genome may also be replaced by a library encoding candidate modulating moieties operably linked to a regulatory control region and a reporter moiety in the vector genome in order to generate a vector comprising candidate modulating moieties which is capable of transducing a target host cell and/or integrating its genome into a host genome.

[0439] Lentivirus vectors are part of the larger group of retroviral vectors. In brief, lentiviruses can be divided into primate and non-primate groups. Examples of primate lentiviruses include but are not limited to human immunodeficiency virus (HIV), the causative agent of human acquired immunodeficiency syndrome (AIDS); and simian immunodeficiency virus (SIV). Examples of non-primate lentiviruses include the prototype “slow virus” visna/maedi virus (VMV), as well as the related caprine arthritis-encephalitis virus (CAEV), equine infectious anaemia virus (EIAV), and the more recently described feline immunodeficiency virus (FIV) and bovine immunodeficiency virus (BIV).

[0440] The lentivirus family differs from retroviruses in that lentiviruses have the capability to infect both dividing and non-dividing cells. In contrast, other retroviruses, such as MLV, are unable to infect non-dividing or slowly dividing cells such as those that make up, for example, muscle, brain, lung and liver tissue.

[0441] A lentiviral vector, as used herein, is a vector which comprises at least one component part derivable from a lentivirus. Preferably, that component part is involved in the biological mechanisms by which the vector infects cells, expresses genes or is replicated.

[0442] The lentiviral vector may be a “primate” vector. The lentiviral vector may be a “non-primate” vector (i.e. derived from a virus which does not primarily infect primates, especially humans). Examples of non-primate lentiviruses may be any member of the family of lentiviridae which does not naturally infect a primate.

[0443] As examples of lentivirus-based vectors, HIV-1- and HIV-2-based vectors are described below.

[0444] The HIV-1 vector contains cis-acting elements that are also found in simple retroviruses. It has been shown that sequences that extend into the gag open reading frame are important for packaging of HIV-1. Therefore, HIV-1 vectors often contain the relevant portion of gag in which the translational initiation codon has been mutated. In addition, most HIV-1 vectors also contain a portion of the env gene that includes the RRE. Rev binds to RRE, which permits the transport of full-length or singly spliced mRNAs from the nucleus to the cytoplasm. In the absence of Rev and/or RRE, full-length HIV-1 RNAs accumulate in the nucleus. Alternatively, a constitutive transport element from certain simple retroviruses such as Mason-Pfizer monkey virus can be used to relieve the requirement for Rev and RRE. Efficient transcription from the HIV-1 LTR promoter requires the viral protein Tat.

[0445] Most HIV-2-based vectors are structurally very similar to HIV-1 vectors. Similar to HIV-1-based vectors, HIV-2 vectors also require RRE for efficient transport of the full-length or singly spliced viral RNAs.

[0446] Preferably, the viral vector used in the present invention has a minimal viral genome.

[0447] By “minimal viral genome” it is to be understood that the viral vector has been manipulated so as to remove the non-essential elements and to retain the essential ele-

ments in order to provide the required functionality to infect, transduce and deliver a nucleotide sequence of interest to a target host cell. Further details of this strategy can be found in WO 1998/017815. Preferably, the plasmid vector used to produce the viral genome within a host cell/packaging cell will have sufficient lentiviral genetic information to allow packaging of an RNA genome, in the presence of packaging components, into a viral particle which is capable of infecting a target cell, but is incapable of independent replication to produce infectious viral particles within the final target cell. Preferably, the vector lacks a functional gag-pol and/or env gene and/or other genes essential for replication.

[0448] However, the plasmid vector used to produce the viral genome within a host cell/packaging cell will also include transcriptional regulatory control sequences operably linked to the lentiviral genome to direct transcription of the genome in a host cell/packaging cell. These regulatory sequences may be the natural sequences associated with the transcribed viral sequence (i.e. the 5' U3 region), or they may be a heterologous promoter, such as another viral promoter (e.g. the CMV promoter).

[0449] The vectors may be self-inactivating (SIN) vectors in which the viral enhancer and promoter sequences have been deleted. SIN vectors can be generated and transduce non-dividing cells in vivo with an efficacy similar to that of wild-type vectors. The transcriptional inactivation of the long terminal repeat (LTR) in the SIN provirus should prevent mobilisation by replication-competent virus. This should also enable the regulated expression of genes from internal promoters by eliminating any cis-acting effects of the LTR.

[0450] The vectors may be integration-defective. Integration defective lentiviral vectors (IDLVs) can be produced, for example, either by packaging the vector with catalytically inactive integrase (such as an HIV integrase bearing the D64V mutation in the catalytic site) or by modifying or deleting essential att sequences from the vector LTR, or by a combination of the above.

Adenoviral Vectors

[0451] The vector of the present invention may be an adenoviral vector. The vector of the present invention may be an adenoviral vector particle.

[0452] The adenovirus is a double-stranded, linear DNA virus that does not go through an RNA intermediate. There are over 50 different human serotypes of adenovirus divided into 6 subgroups based on the genetic sequence homology. The natural targets of adenovirus are the respiratory and gastrointestinal epithelia, generally giving rise to only mild symptoms. Serotypes 2 and 5 (with 95% sequence homology) are most commonly used in adenoviral vector systems and are normally associated with upper respiratory tract infections in the young.

[0453] Adenoviruses have been used as vectors for gene therapy and for expression of heterologous genes. The large (36 kb) genome can accommodate up to 8 kb of foreign insert DNA and is able to replicate efficiently in complementing cell lines to produce very high titres of up to 10¹². Adenovirus is thus one of the best systems to study the expression of genes in primary non-replicative cells.

[0454] The expression of viral or foreign genes from the adenovirus genome does not require a replicating cell. Adenoviral vectors enter cells by receptor mediated endocytosis. Once inside the cell, adenovirus vectors rarely

integrate into the host chromosome. Instead, they function episomally (independently from the host genome) as a linear genome in the host nucleus. Hence, the use of recombinant adenovirus alleviates the problems associated with random integration into the host genome.

Herpes Simplex Viral Vector

[0455] The vector of the present invention may be a herpes simplex viral vector. The vector of the present invention may be a herpes simplex viral vector particle.

[0456] Herpes simplex virus (HSV) is a neurotropic DNA virus with favorable properties as a gene delivery vector. HSV is highly infectious, so HSV vectors are efficient vehicles for the delivery of exogenous genetic material to cells. Viral replication is readily disrupted by null mutations in immediate early genes that in vitro can be complemented in trans enabling straightforward production of high-titre pure preparations of non-pathogenic vector. The genome is large (152 Kb) and many of the viral genes are dispensable for replication in vitro, allowing their replacement with large or multiple transgenes. Latent infection with wild-type virus results in episomal viral persistence in sensory neuronal nuclei for the duration of the host lifetime. The vectors are non-pathogenic, unable to reactivate and persist long-term. The latency active promoter complex can be exploited in vector design to achieve long-term stable transgene expression in the nervous system. HSV vectors transduce a broad range of tissues because of the wide expression pattern of the cellular receptors recognized by the virus. Increasing understanding of the processes involved in cellular entry has allowed targeting the tropism of HSV vectors.

Other Viral Vectors

[0457] Other suitable viral vectors include those described in Lundstrom, K., 2018. Diseases, 6 (2), p.42.

[0458] The vector of the present invention may be an alphaviral vector. The vector of the present invention may be an alphaviral vector particle. The vector of the present invention may be a flaviviral vector. The vector of the present invention may be a flaviviral vector particle.

[0459] Self-amplifying ssRNA viruses comprise of alphaviruses (e.g. Semliki Forest virus, Sindbis virus, Venezuelan equine encephalitis virus, and M1) and flaviviruses (e.g. Kunjin virus, West Nile virus, and Dengue virus) possessing a genome of positive polarity. Alphaviruses have been mainly applied in preclinical gene therapy studies for cancer treatment. Alphavirus vectors can be delivered in the form of naked RNA, layered plasmid DNA vectors and recombinant replication-deficient or proficient particles.

[0460] The vector of the present invention may be a rhabdoviral vector. The vector of the present invention may be a rhabdoviral vector particle. The vector of the present invention may be a measles viral vector. The vector of the present invention may be a measles viral vector particle.

[0461] Rhabdoviruses (e.g. rabies and vesicular stomatitis virus) and measles viruses carry negative strand genomes. Among rhabdoviruses, recombinant vesicular stomatitis virus (VSV) has been applied for preclinical gene therapy studies. Measles viruses (e.g. MV-Edm) have found a number of gene therapy applications.

[0462] The vector of the present invention may be a Newcastle disease viral vector. The vector of the present invention may be a Newcastle disease viral vector particle.

[0463] The ssRNA paramyxovirus Newcastle disease virus (NDV) replicates specifically in tumour cells and has therefore been frequently applied for cancer gene therapy.

[0464] The vector of the present invention may be a poxviral vector. The vector of the present invention may be a poxviral vector particle.

[0465] The characteristic feature of poxviruses is their dsDNA genome, which can generously accommodate more than 30 kb of foreign DNA. Poxviruses have found several applications as gene therapy vectors. For instance, vaccinia virus vectors have demonstrated potential for treatment of cancer. Vaccinia virus is large enveloped poxvirus that has an approximately 190 kb linear, double-stranded DNA genome. Vaccinia virus can accommodate up to approximately 25 kb of foreign DNA, which also makes it useful for the delivery of large genes. A number of attenuated vaccinia virus strains are known in the art that are suitable for gene therapy applications, for example the MVA and NYVAC strains.

[0466] The vector of the present invention may be a picornaviral vector. The vector of the present invention may be a picornaviral vector particle.

[0467] Picornaviruses are non-enveloped ssRNA viruses. Cocksackieviruses belonging to Picornaviridae, have been applied as oncolytic vectors.

Variants, Derivatives, Analogues, Homologues and Fragments

[0468] In addition to the specific proteins and nucleotides mentioned herein, the invention also encompasses variants, derivatives, homologues and fragments thereof.

[0469] In the context of the invention, a “variant” of any given sequence is a sequence in which the specific sequence of residues (whether amino acid or nucleic acid residues) has been modified in such a manner that the polypeptide or polynucleotide in question retains at least one of its endogenous functions. A variant sequence can be obtained by addition, deletion, substitution, modification, replacement and/or variation of at least one residue present in the naturally occurring polypeptide or polynucleotide. For example, a variant promoter sequence retains at least some level of the activity and specificity of the promoter sequence from which it is obtained.

[0470] The term “derivative” as used herein in relation to proteins or polypeptides of the invention includes any substitution of, variation of, modification of, replacement of, deletion of and/or addition of one (or more) amino acid residues from or to the sequence, providing that the resultant protein or polypeptide retains at least one of its endogenous functions.

[0471] Typically, amino acid substitutions may be made, for example from 1, 2 or 3, to 10 or 20 substitutions, provided that the modified sequence retains the required activity or ability. Amino acid substitutions may include the use of non-naturally occurring analogues.

[0472] Proteins used in the invention may also have deletions, insertions or substitutions of amino acid residues which produce a silent change and result in a functionally equivalent protein. Deliberate amino acid substitutions may be made on the basis of similarity in polarity, charge, solubility, hydrophobicity, hydrophilicity and/or the amphipathic nature of the residues as long as the endogenous function is retained. For example, negatively charged amino acids include aspartic acid and glutamic acid; posi-

tively charged amino acids include lysine and arginine; and amino acids with uncharged polar head groups having similar hydrophilicity values include asparagine, glutamine, serine, threonine and tyrosine.

[0473] Conservative substitutions may be made, for example according to the table below. Amino acids in the same block in the second column and preferably in the same line in the third column may be substituted for each other:

ALIPHATIC	Non-polar	G A P I L V
	Polar - uncharged	C S T M N Q
	Polar - charged	D E K R H
		F W Y
AROMATIC		

[0474] The term “homologue” as used herein means a variant having a certain homology with the wild type amino acid sequence or the wild type nucleotide sequence. The term “homology” can be equated with “identity”.

[0475] In the present context, a homologous sequence is taken to include an amino acid sequence which may be at least 50%, 55%, 65%, 75%, 85% or 90% identical, preferably at least 95%, 96% or 97% or 98% or 99% identical to the subject sequence. Typically, the homologues will comprise the same active sites etc. as the subject amino acid sequence. Although homology can also be considered in terms of similarity (i.e., amino acid residues having similar chemical properties/functions), in the context of the present invention it is preferred to express homology in terms of sequence identity.

[0476] In the present context, a homologous sequence is taken to include a nucleotide sequence which may be at least 50%, 55%, 65%, 75%, 85% or 90% identical, preferably at least 95%, 96% or 97% or 98% or 99% identical to the subject sequence. Although homology can also be considered in terms of similarity, in the context of the present invention it is preferred to express homology in terms of sequence identity.

[0477] Preferably, reference to a sequence which has a percent identity to any one of the SEQ ID NOs detailed herein refers to a sequence which has the stated percent identity over the entire length of the SEQ ID NO referred to.

[0478] Homology comparisons can be conducted by eye, or more usually, with the aid of readily available sequence comparison programs. These commercially available computer programs can calculate percent homology or identity between two or more sequences.

[0479] Percent homology may be calculated over contiguous sequences, i.e., one sequence is aligned with the other sequence and each amino acid or nucleotide in one sequence is directly compared with the corresponding amino acid or nucleotide in the other sequence, one residue at a time. This is called an “ungapped” alignment. Typically, such ungapped alignments are performed only over a relatively short number of residues.

[0480] Although this is a very simple and consistent method, it fails to take into consideration that, for example, in an otherwise identical pair of sequences, one insertion or deletion in the amino acid or nucleotide sequence may cause the following residues or codons to be put out of alignment, thus potentially resulting in a large reduction in percent homology when a global alignment is performed. Conse-

quently, most sequence comparison methods are designed to produce optimal alignments that take into consideration possible insertions and deletions without penalising unduly the overall homology score. This is achieved by inserting “gaps” in the sequence alignment to try to maximise local homology.

[0481] However, these more complex methods assign “gap penalties” to each gap that occurs in the alignment so that, for the same number of identical amino acids or nucleotides, a sequence alignment with as few gaps as possible, reflecting higher relatedness between the two compared sequences, will achieve a higher score than one with many gaps. “Affine gap costs” are typically used that charge a relatively high cost for the existence of a gap and a smaller penalty for each subsequent residue in the gap. This is the most commonly used gap scoring system. High gap penalties will of course produce optimised alignments with fewer gaps. Most alignment programs allow the gap penalties to be modified. However, it is preferred to use the default values when using such software for sequence comparisons. For example, when using the GCG Wisconsin Bestfit package the default gap penalty for amino acid sequences is -12 for a gap and -4 for each extension.

[0482] Calculation of maximum percent homology therefore firstly requires the production of an optimal alignment, taking into consideration gap penalties. A suitable computer program for carrying out such an alignment is the GCG Wisconsin Bestfit package (University of Wisconsin, USA; Devereux et al. (1984) *Nucleic Acids Research* 12:387). Examples of other software that can perform sequence comparisons include, but are not limited to, the BLAST package (see Ausubel et al. (1999) *ibid*—Ch. 18), FASTA (Atschul et al. (1990) *J. Mol. Biol.* 403:410), EMBOSS Needle (Madeira, F., et al., 2019. *Nucleic acids research*, 47 (W1), pp. W636-W641) and the GENWORKS suite of comparison tools. Both BLAST and FASTA are available for offline and online searching (see Ausubel et al. (1999) *ibid*, pages 7-58 to 7-60). However, for some applications, it is preferred to use the GCG Bestfit program. Another tool, BLAST 2 Sequences, is also available for comparing protein and nucleotide sequences (FEMS Microbiol. Lett. (1999) 174 (2): 247-50; FEMS Microbiol. Lett. (1999) 177 (1): 187-8).

[0483] Although the final percent homology can be measured in terms of identity, the alignment process itself is typically not based on an all-or-nothing pair comparison. Instead, a scaled similarity score matrix is generally used that assigns scores to each pairwise comparison based on chemical similarity or evolutionary distance. An example of such a matrix commonly used is the BLOSUM62 matrix (the default matrix for the BLAST suite of programs). GCG Wisconsin programs generally use either the public default values or a custom symbol comparison table if supplied (see the user manual for further details). For some applications, it is preferred to use the public default values for the GCG package, or in the case of other software, the default matrix, such as BLOSUM62.

[0484] Once the software has produced an optimal alignment, it is possible to calculate percent homology, preferably percent sequence identity. The software typically does this as part of the sequence comparison and generates a numerical result. The percent sequence identity may be calculated as the number of identical residues as a percentage of the total residues in the SEQ ID NO referred to.

[0485] “Fragments” are also variants and the term typically refers to a selected region of the polypeptide or polynucleotide that is of interest either functionally or, for example, in an assay. “Fragment” thus refers to an amino acid or nucleic acid sequence that is a portion of a full-length polypeptide or polynucleotide.

[0486] Such variants, derivatives, homologues and fragments may be prepared using standard recombinant DNA techniques such as site-directed mutagenesis. Where insertions are to be made, synthetic DNA encoding the insertion together with 5' and 3' flanking regions corresponding to the naturally-occurring sequence either side of the insertion site may be made. The flanking regions will contain convenient restriction sites corresponding to sites in the naturally-occurring sequence so that the sequence may be cut with the appropriate enzyme(s) and the synthetic DNA ligated into the cut. The DNA is then expressed in accordance with the invention to make the encoded protein. These methods are only illustrative of the numerous standard techniques known in the art for manipulation of DNA sequences and other known techniques may also be used.

Cells

[0487] In one aspect, the present invention provides a cell comprising the polynucleotide or the vector of the invention. The cell may be an isolated cell. The cell may be a human cell, suitably an isolated human cell.

[0488] Vectors comprising polynucleotides used in the invention may be introduced into cells using a variety of techniques known in the art, such as transfection, transduction and transformation. Suitably, the vector of the present invention is introduced into the cell by transfection or transduction.

[0489] The cell may be any cell type known in the prior art.

[0490] Suitably, the cell may be a producer cell. The term “producer cell” includes a cell that produces viral particles, after transient transfection, stable transfection or vector transduction of all the elements necessary to produce the viral particles or any cell engineered to stably comprise the elements necessary to produce the viral particles. Suitable producer cells will be well known to those of skill in the art. Suitable producer cell lines include HEK 293 (e.g., HEK 293T), HeLa, and A549 cell lines.

[0491] Suitably, the cell may be a packaging cell. The term “packaging cell” includes a cell which contains some or all of the elements necessary for packaging an infectious recombinant virus. The packaging cell may lack a recombinant viral vector genome. Typically, such packaging cells contain one or more vectors which are capable of expressing viral structural proteins. Cells comprising only some of the elements required for the production of enveloped viral particles are useful as intermediate reagents in the generation of viral particle producer cell lines, through subsequent steps of transient transfection, transduction or stable integration of each additional required element. These intermediate reagents are encompassed by the term “packaging cell”. Suitable packaging cells will be well known to those of skill in the art.

[0492] Suitably, the cell may be a kidney cell or glomerular cell, for example a podocyte. Suitably, the cell may be an immortalized kidney cell or glomerular cell, for example an immortalized podocyte. Suitable podocyte cell lines will be well known to those of skill in the art, for example CIHP-1.

Methods to generate immortalized podocytes will be well known to those of skill in the art. Suitable methods are described in Ni, L., et al., 2012. *Nephrology*, 17 (6), pp. 525-531.

PHARMACEUTICAL COMPOSITION

[0493] In one aspect, the present invention provides pharmaceutical composition comprising the polynucleotide or vector of the invention or the cell of the invention.

[0494] A pharmaceutical composition is a composition that comprises or consists of a therapeutically effective amount of a pharmaceutically active agent i.e. the vector. It preferably includes a pharmaceutically acceptable carrier, diluent or excipient (including combinations thereof).

[0495] By “pharmaceutically acceptable” is included that the formulation is sterile and pyrogen free. The carrier, diluent, and/or excipient must be “acceptable” in the sense of being compatible with the vector and not deleterious to the recipients thereof. Typically, the carriers, diluents, and excipients will be saline or infusion media which will be sterile and pyrogen free, however, other acceptable carriers, diluents, and excipients may be used.

[0496] Acceptable carriers, diluents, and excipients for therapeutic use are well known in the pharmaceutical art. The choice of pharmaceutical carrier, excipient or diluent can be selected with regard to the intended route of administration and standard pharmaceutical practice. The pharmaceutical compositions may comprise as—or in addition to—the carrier, excipient or diluent any suitable binder(s), lubricant(s), suspending agent(s), coating agent(s) or solubilising agent(s).

[0497] Examples of pharmaceutically acceptable carriers include, for example, water, salt solutions, alcohol, silicone, waxes, petroleum jelly, vegetable oils, polyethylene glycols, propylene glycol, liposomes, sugars, gelatin, lactose, amylose, magnesium stearate, talc, surfactants, silicic acid, viscous paraffin, perfume oil, fatty acid monoglycerides and diglycerides, petroethral fatty acid esters, hydroxymethylcellulose, polyvinylpyrrolidone, and the like.

[0498] The vector, cell, or pharmaceutical composition according to the present invention may be administered in a manner appropriate for treating and/or preventing the diseases described herein. The quantity and frequency of administration will be determined by such factors as the condition of the subject, and the type and severity of the subject's disease, although appropriate dosages may be determined by clinical trials. The pharmaceutical composition may be formulated accordingly.

[0499] The vector, cell or pharmaceutical composition according to the present invention may be administered parenterally, for example, intravenously, or by infusion techniques. The vector, cell or pharmaceutical composition may be administered in the form of a sterile aqueous solution which may contain other substances, for example, enough salts or glucose to make the solution isotonic with blood. The aqueous solution may be suitably buffered (preferably to a pH of from 3 to 9). The pharmaceutical composition may be formulated accordingly. The preparation of suitable parenteral formulations under sterile conditions is readily accomplished by standard pharmaceutical techniques well-known to those skilled in the art.

[0500] The vector, cell or pharmaceutical composition according to the present invention may be administered systemically, for example by intravenous injection.

[0501] The vector, cell or pharmaceutical composition according to the present invention may be administered locally, for example by targeting administration to the kidney. Suitably, the vector, cell or pharmaceutical composition may be administered by injection into the renal artery or by ureteral or subcapsular injection.

[0502] The pharmaceutical compositions may comprise vectors or cells of the invention in infusion media, for example sterile isotonic solution. The pharmaceutical composition may be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

[0503] The vector, cell or pharmaceutical composition may be administered in a single or in multiple doses. Particularly, the vector, cell or pharmaceutical composition may be administered in a single, one off dose. The pharmaceutical composition may be formulated accordingly.

[0504] The vector, cell or pharmaceutical composition may be administered at varying doses (e.g. measured in vector genomes (vg) per kg). The physician in any event will determine the actual dosage which will be most suitable for any individual subject and it will vary with the age, weight and response of the particular subject. Typically, however, for the AAV vectors of the invention, doses of 10^{10} to 10^{14} vg/kg, or 10^{11} to 10^{13} vg/kg may be administered.

[0505] The pharmaceutical composition may further comprise one or more other therapeutic agents.

[0506] The invention further includes the use of kits comprising the vector, cells and/or pharmaceutical composition of the present invention. Preferably said kits are for use in the methods and used as described herein, e.g., the therapeutic methods as described herein. Preferably said kits comprise instructions for use of the kit components.

Methods for Treating and/or Preventing Disease

[0507] In one aspect, the present invention provides the vector, cell or pharmaceutical composition according to the present invention for use as a medicament.

[0508] In one aspect, the present invention provides use of the vector, cell or pharmaceutical composition according to the present invention in the manufacture of a medicament.

[0509] In one aspect, the present invention provides a method of administering the vector, cell or pharmaceutical composition according to the present invention to a subject in need thereof.

Glomerular Diseases

[0510] The vector, cell or pharmaceutical composition according to the present invention may be used to treat glomerular diseases in a subject. Suitably, the subject is a human subject.

[0511] In one aspect, the present invention provides the vector, cell or pharmaceutical composition according to the present invention for use in preventing or treating a glomerular disease.

[0512] In a related aspect, the present invention provides use of the vector, cell or pharmaceutical composition according to the present invention for the manufacture of a medicament for preventing or treating a glomerular disease.

[0513] In a related aspect, the present invention provides a method of preventing or treating a glomerular disease comprising administering the vector, cell or pharmaceutical composition according to the present invention to a subject in need thereof.

[0514] Glomerular diseases can be classified as either nephrotic or nephritic. Nephrotic syndrome often involves

factors that affect the integrity of podocyte podocyte or podocyte GBM interactions. By contrast, factors involved in the etiology of nephritic syndrome can vary, but may include circulating platelets and white blood cells, the GBM, resident glomerular endothelial cells, and mesangial cells. (Chiang, C. K. and Inagi, R., 2010. *Nature Reviews Nephrology*, 6 (9), p.539).

[0515] Suitably, the glomerular disease is a genetic glomerular disease, i.e. a glomerular disease which is inherited. Genetic glomerular diseases include podocyte-associated genetic glomerular diseases, such as nephrotic syndrome, and GBM-associated glomerular diseases, such as Alport Syndrome.

[0516] Suitably, the glomerular disease is a podocyte-associated genetic glomerular disease. Podocyte-associated genetic glomerular diseases include Congenital nephrotic syndrome of the Finnish type, Congenital nephrotic syndrome type 2, Familial nephrotic syndrome type 3, Frasier syndrome and Denys-Drash syndrome, Schimke immunosseous dysplasia, Nephrotic syndrome caused by mutations in CD2AP, Nephrotic syndrome caused by mutations in actinin-4, Nephrotic syndrome caused by mutations in TRPC6, and Epstein and Fechtner syndrome. Suitably, the glomerular disease is nephrotic syndrome.

[0517] Suitably, the glomerular disease is a GBM-associated genetic glomerular disease. Podocyte-associated genetic glomerular diseases include X-linked Alport syndrome, Autosomal recessive Alport syndrome, Autosomal dominant Alport syndrome, Thin basement membrane diseases, Pierson syndrome, and Nail-patella syndrome. Suitably, the glomerular disease is Alport syndrome (AS). AS is also known as familial nephritis, hereditary nephritis, thin basement membrane disease and thin basement membrane nephropathy.

EXAMPLES

[0518] The invention will now be further described by way of Examples, which are meant to serve to assist one of ordinary skill in the art in carrying out the invention and are not intended in any way to limit the scope of the invention.

Example 1—Design, Construction, and Testing of Minimal Nephtrin Promoters

Design of Minimal Nephtrin Promoters

[0519] A human NPHS1 promoter has been described in Moeller et al. 2002 *J Am Soc Nephrol*, 13 (6): 1561-7 and Wong M A et al. 2000 *Am J Physiol Renal Physiol*, 279 (6): F1027-32. This NPHS1 promoter is a 1.2 kb fragment and appears to be podocyte-specific. This is referred to hereafter as a “FL” nephtrin promoter and is shown in FIG. 1A.

[0520] The FL nephtrin promoter was initially cut to 822 bp (819 bp excluding start codon) by deleting the N-terminal sequence. This is referred to hereafter as a “midi” nephtrin promoter and is shown in FIG. 1B.

[0521] The midi nephtrin promoter was further cut to 268 bp (265 bp excluding start codon) by removing putative general transcription domains from the central region. This is referred to hereafter as a “mini” nephtrin promoter and is shown in FIG. 1C.

Construction of Vector Constructs

Midi Nephhrin Promoter

[0522] pACE_hNPHS1 promoter was used as a template to introduce BamHI and ClaI restriction sites as shown in FIG. 2A. Fragments were then gel extracted and digested with ClaI and BamHI for 1 h at 37° C. prior to ligation into pLenti GFP Blast vector. Ligations were further transformed into stable competent *E. Coli* cells, DNA was extracted and sequenced (Midi Promoter). The final lentiviral vector is shown in FIG. 2B.

Mini Nephhrin Promoter

[0523] pACE_hNPHS1 promoter was used to PCR the overhangs (OHs) as shown in FIG. 3A. Two sections of the promoter containing OHs were gel extracted for the NEBuilder HiFi Assembly reaction into pLenti GFP Blast vector. The ligation reaction was then cleaned using the DNA clean up kit, prior to its transformation into stable competent *E. Coli* cells. DNA was extracted and sequenced. The final lentiviral vector is shown in FIG. 3B.

Testing Vector Constructs

[0524] The minimal nephhrin promoters were used to express GFP in in vitro cell models to check efficacy and podocyte-specificity.

[0525] pLenti GFP Blast Nephhrin Promoter constructs (Full Length, Midi and Mini) were used to transfect HEK293T cells for 48 h to make virus, which was further used to create human conditionally immortalized podocytes stably expressing either GFP-tagged FL NPHS1, midi or mini promoters.

[0526] Conditionally immortalised human podocytes (ciPodocytes) were transfected with the lentiviral vectors to determine whether the minimal promoters were able to drive GFP expression. Both the midi and mini nephhrin promoters were shown to drive GFP expression. FIG. 4 shows a representative fluorescence microscopy image showing GFP expression from the mini nephhrin promoter. FIG. 5 shows a representative western blot showing GFP expression from the mini nephhrin promoter. These results show that a mini nephhrin promoter is able to drive transgene expression in podocytes.

[0527] The lentiviral vectors were also used to transduce human glomerular cells. ciPodocytes and glomerular endothelial cells were transduced with a lentivirus comprising GFP coupled to the mini nephhrin reporter. FIG. 6A-C shows FACS analysis displaying median GFP fluorescence (AFU) of all live singlets of conditionally immortalised human podocytes (LY) and glomerular endothelial cells (GEnC) using a Novocyte Analyser. Untransduced cells (Cell Control) were compared with those transduced with lentivirus constructs harbouring a GFP expression cassette controlled by the full length human nephhrin promoter (hNPHS1.GFP) or the mini human nephhrin promoter (265. GFP). All cells were differentiated for 10 days, trypsinised (100 uL) and diluted in PBS, 2% FBS, 1:1000 DRAQ7 (150 uL). Data and error bars represent 3 technical repeats (100 uL, >2500 cells)+SEM. These results show podocyte specificity for a minimal nephhrin promoter when compared to glomerular endothelial cells.

Example 2—Design, Construction, and Testing of Minimal Nephhrin Promoter Coupled to COL4A3, COL4A4, and COL4A5

Design and Construction of AAV Constructs

[0528] The following AAV transfer plasmids comprising COL4A3, COL4A4, and COL4A5 coupled to the mini nephhrin promoter (“265”) were designed and constructed:

[0529] pAAV.265.Col4a3.3flag.sv40, an AAV plasmid comprising COL4A3 coupled to mini nephhrin promoter (see FIG. 7A).

[0530] pAAV.265.Col4a4.3flag.sv40, an AAV plasmid comprising COL4A4 coupled to mini nephhrin promoter (see FIG. 7B)

[0531] pAAV.265.Col4a5.3flag.sv40, an AAV plasmid comprising COL4A5 coupled to mini nephhrin promoter (see FIG. 7C)

[0532] A SmaI digestion was performed to confirm the identity of the plasmids (see FIGS. 7D-E). This shows successful cloning of COL4 a3, a4 and a5 into AAV, with the 265 bp mini nephhrin promoter and a SV40 polyA tail.

Testing of AAV Constructs

[0533] The following AAV viral vectors were prepared using standard methods:

[0534] AAV.COL4A3.nephhrin265.Sv40 with LK03 serotype.

[0535] AAV.COL4A5.nephhrin265.Sv40 with LK03 serotype

[0536] AAV.COL4A5.nephhrin265.Sv40 with 2/9 serotype

[0537] FIG. 8A shows immunoprecipitation experiments of the full-length FLAG-tagged Col4a3 (LK03) or Col4a5 (LK03) in human differentiated ciPodocytes pulled down with the anti-FLAG antibody. Anti-FLAG antibody precipitated both Col4a3 and Col4a5. Human-FLAG IgG was used as a control.

[0538] FIG. 8B shows western blots of protein lysates showing the expression levels of Col4a3 (LK03 capsid serotype), Col4a5 (LK03) and Col4a5 (2/9 capsid serotype) in human or mouse differentiated ciPodocytes. Non-infected human and mouse ciPodocytes were used as controls.

[0539] FIG. 8C shows confocal images showing immunofluorescence staining of transduced Col4a5 in Human wild-type CiPodocytes/Col4a5 3xFlag AAV CiPodocytes with F-Actin. Col4a5 is present at the cytosolic level in the human differentiated podocytes infected with Col4a5 3xFlag AAV virus in comparison to the wild-type counterpart.

[0540] These results show that we have successfully transduced human podocytes with COL4a3 and COL4a5 full length coupled to a mini nephhrin promoter and that a mini nephhrin promoter drives expression of the full length COL4a3 or COL4a5 in the human podocytes.

Example 3—Expression of CFH, CFI and CFHL1 in HEK Cells

Materials and Methods 60-80% confluent 293T Human Embryonic Kidney cells grown in DMEM supplemented with 10% FBS were triple transfected with pHelper (HGT11), one of the two pAAV Rep-Cap (LK03 or AAV9) and one of the three ITR-expression plasmids containing 1) CFH under the 265 bp minimal nephhrin promoter of SEQ ID

NO: 47 (pAAV-265-CFH), 2) CFI under the full-length minimal nephrin promoter of SEQ ID NO: 14 (pAAV-FL-CFI) or 3) CFHL1 under the full-length minimal nephrin promoter of SEQ ID NO: 14 (pAAV-FL-CFHL1). All constructs were tagged with MYC and FLAG.

[0541] Transfection was carried out on a 150 mm culture dish in serum-free media in the presence of polyethylenimine (PEI). Media was changed to DMEM with FBS the following day after transfection. On Day 4 post-transfection, media and cells were collected and processed separately. Media was frozen and stored at -80°C . Cells were lysed with RIPA buffer supplemented with proteinase inhibitors and stored at -80°C .

[0542] Protein concentration in the cell lysates was measured using Pierce BCA protein assay and 10 μg of total protein from each sample was loaded onto a 4-15% polyacrylamide Tris-Glycine gel. A total of 2.6 μl of media from each sample was loaded onto the gel. Protein was transferred to a nitrocellulose membrane using the iBlot2 dry blotting system. The following primary antibodies were used for protein detection: anti-Factor H (Abcam, cat. ab124769), anti-Factor I (Abcam, ab278524), anti-MYC-tag (CST, cat. 2276S), anti-FLAG-tag (CST, cat. 14793S) and anti-GAPDH (Millipore, cat. MAB374).

Results

[0543] Factor H was expressed in both the cell lysates and the media from 293T HEK cells transfected with the CFH expression plasmid (FIG. 9, lanes 2, 3, 8, 9). Expression was detected using a Factor H-specific antibody. Factor H was not detected in the cell lysates or media of untransfected cells or cells transfected with the CFI or CFHL1 expression plasmids (FIG. 9, lanes 1, 4-7, 10-13). These results demonstrate that a 265 bp minimal nephrin promoter allows expression of the transgene in target cells.

Example 4—Transduction of Factor H Mutated Podocytes with AAV2/9 265-CFH or Transfection with Plasmid Encoding CFH

Transduction of Factor H Mutated Podocytes with AAV2/9 265-CFH

[0544] Conditionally immortalised human podocytes with a mutation in endogenous Factor H (Muehlig et al, 2020) grown in RPMI supplemented with 10% FBS and 1% ITS were seeded in 6-well culture plates and grown at 33°C until 70-80% confluency. Cells were incubated with AAV2/9 containing a CFH transgene under the control of the 265 bp minimal nephrin promoter (SEQ ID NO: 47). Cells incubated without the virus were used as a non-transduced (NT) control. On the same day following viral transduction, cells were transferred to a non-permissive temperature of 37°C to allow for cell differentiation and transgene expression. Media was changed twice on the subsequent days following transduction. On Day 10 post-transduction, cell media was collected and the concentration of human Factor H was measured by ELISA using an anti-Factor H antibody (Abcam, cat. ab252359).

[0545] Human podocytes with a mutation in Factor H were transduced with AAV2/9 virus containing a CFH transgene under the control of the 265 bp minimal nephrin promoter demonstrated higher concentrations of human Factor H in the culture media than untransduced cells (FIGS. 10A and 10B). This demonstrates that a Factor H transgene

delivered by AAV and under the control of a 265 bp minimal nephrin promoter can be expressed in human podocytes.

Transfection of Factor H Mutated Podocytes with Plasmid Encoding CFH

[0546] Conditionally immortalised human podocytes with a mutation in endogenous Factor H (Muehlig et al, 2020) grown in RPMI supplemented with 10% FBS and 1% ITS were seeded in 6-well culture plates and grown at 33°C until 70-80% confluency. For plasmid transfection, cells were incubated with 1.5 μg of expression plasmid in serum-free media in the presence of polyethylenimine. Cells where no plasmid was added were used as a non-transfected (NT) control. The media was changed the following day to media containing FBS. On Day 3 post-transfection, media was collected and analysed by ELISA (Abcam, cat. ab252359).

[0547] Podocytes transfected with a plasmid expressing the CFH transgene under the control of a 265 bp minimal nephrin promoter demonstrated higher concentrations of human Factor H than the non-transfected control (FIG. 10C).

Example 5—Expression of CFH in the Kidney in WT Mice

Materials and Methods

[0548] 100 μl of AAV2/9 gene therapy product (pAAV.NPHS1 (265).hCFH.WPRE.bGH) or saline was administered to wild-type C57BL6 mice by IV tail vein injection. AAV expressed tagged wild-type human CFH transgene was under control of the 265 bp minimal nephrin promoter (SEQ ID NO: 47). AAV was harvested and purified by ultracentrifugation 3 days later and titrated ($\sim 1.5 \times 10^{13}/\text{ml}$) in PBS. All animals completed the study on Day 21 and culled. Kidneys were snap frozen in liquid nitrogen and used for RNA extraction by RNeasy Micro Kit (Qiagen Cat. No./ID: 74004) as per the manufacturers' protocol. RNA was then converted into cDNA using High-Capacity RNA-to-cDNA™ Kit (4387406) prior to qPCR analysis.

[0549] Quantitative qPCR was performed on DNA samples from kidneys of pAAV_CFH injected mice. Standard curve qPCR with SYBR green reagents and passive ROX method was used to detect ITR presence in the kidney DNA samples. Viral genomes per pg of DNA were calculated using a standard curve of known quantities of an ITR amplicon. Finally, viral genomes per cell were calculated based on the assumption that diploid mouse cells have 6 pg of DNA.

[0550] Immunofluorescence staining was performed on the frozen kidney sections 5 microns thick using an anti-nephrin antibody (PROGEN) and an anti-CFH antibody (ab124767).

[0551] Tissue was embedded in OCT compound (VWR, cat. number 361603E) and snap frozen in liquid nitrogen. 10 μm sections were cut using a cryostat (Thermo, Cryostat NX270). Tissue sections were fixed with 4% Paraformaldehyde for 20 minutes at room temperature, permeabilized with 0.3% Triton-X for 15 minutes at room temperature and blocked with 5% BSA for 30 minutes at room temperature.

[0552] Primary antibodies (Rabbit monoclonal [EPR6226] to Factor H, Abcam cat. number ab124769 and Nephrin (NPHS1) (1243-1256) Guinea Pig Polyclonal Antibody, Origene cat. number BP5030) were diluted in 5% BSA. (1:100 for anti-factor H and 1:300 for anti-nephrin). Incubation was done at room temperature for 1 hour.

[0553] Secondary antibodies (Goat anti-Rabbit IgG (H+L) Secondary Antibody, Alexa Fluor Plus 488, Invitrogen, cat. number A32731 and Goat anti-Guinea Pig IgG (H+L) Alexa Fluor 568, Invitrogen, cat. number A-11075) were diluted 1:500 in PBS and incubation was done 30 minutes at room temperature.

[0554] Slides were mounted in Fluoromount-G, slide mounting medium (Southern Biotech, cat. number 0100-01) and imaged on a Leica DM750 Fluorescence Microscope at 20× magnification.

Results

[0555] Injection of wild type mice with AAV containing human CFH under the 265 bp minimal nephrin promoter leads to infection of the kidney by the AAV and greater expression of the virus and human CFH in the kidney compared to wild type mice injected with saline control (FIG. 11A-C). Immunofluorescence staining demonstrates the co-localization of nephrin and CFH in the mouse glomerulus (FIG. 11D).

Example 7—Expression of GFP Using 265 bp v 818 bp v FL Minimal Nephrin Promoters (Plasmid Transfection) in Kidney Cells

[0556] Plasmids were constructed comprising eGFP under the control of: (i) the full-length (FL) minimal nephrin promoter of SEQ ID NO: 14 (PS0281); (ii) the 818 bp minimal nephrin promoter of SEQ ID NO: 59 (PS0301); or (iii) the 265 bp minimal nephrin promoter of SEQ ID NO: 47 (P20282).

[0557] Podocytes, glomerular endothelial cells (GENCs), and proximal tube epithelial cells (PTECs) were seeded at a density of 1.5e5 cells per well of a six well plate and allowed to attach overnight. For HEK293T cells, 5e5 cells per well were used. The following day 3 ug of plasmid DNA in 150 ul of DMEM was mixed with 9 ul PEI (also in 150 ul DMEM), incubated for 15 minutes at room temperature, and analysed 48 hours later by flow cytometry for eGFP expression. Non transfected cells were used as a negative control for gating of negative and positive populations.

[0558] The results in FIG. 12 show that all three promoters were capable of driving eGFP expression in all cell types tested. These results demonstrate that minimizing the full length nephrin promoter to either 818 bp or 265 bp still allows expression of the transgene in target cells, despite removing large regions of the nephrin promoter sequence.

Example 8—Expression of GFP Using 265 bp v FL Minimal Nephrin Promoters (AAV Transduction) in Kidney Cells

Materials and Methods

[0559] AAV Production: 60-80% confluent 293T Human Embryonic Kidney cells grown in DMEM supplemented with 10% FBS were triple transfected with pHelper (HGT11, PS0150), a Rep-Cap plasmid (LK03, PS0240) and one of the two ITR-expression plasmids containing the Full Length (FL) minimal nephrin promoter or the 265 bp minimal nephrin promoter driving eGFP expression (PS0281 and PS0282, respectively). Transfection was carried out on a 150 mm culture dish in the presence of polyethylenimine (PEI). Media was changed to DMEM with FBS the following day after transfection. On Day 4 post-transfection, media and

cells were collected and processed separately. PEG was added to filtered supernatants to precipitate viral particles which were subsequently centrifuged at 1500 g for 20 minutes. Cell pellets were resuspended in TD buffer and frozen at -80 degrees, followed by thawing at 37 degrees and vortexed. Freeze-thaw cycles were repeated 5 times. Pellets from the PEG treated supernatant were then added to the lysed cells and Benzonase and Sodium deoxycholate was further added and incubated at 37 degrees for 30 minutes. Samples were then centrifuged and the supernatant filtered and subsequently purified via iodixanol gradient ultracentrifugation.

[0560] AAV QC: AAV samples were titered via qPCR to determine the number of viral genomes per ml of viral supernatant. Primers designed to target the eGFP sequence were used. SDS-PAGE analysis was performed on samples to determine purity. Alkaline gel electrophoresis was used to demonstrate incorporation of full transgene cassettes into the viral particles.

[0561] AAV Transduction: Cells were seeded into wells of a 6-well plate at a density of 1.5e5 cells/well and allowed to adhere overnight. The following day viral particles were added a multiplicity of infection (MOI) of 5e5 particles per cell. Cells were placed at 37 degrees to initiate differentiation and cultured for 10 days. After 10 days cells were analysed for eGFP expression by flow cytometry.

Results

[0562] The results in FIG. 13 show that both promoters were capable of driving eGFP expression in all cell types tested. The 265 bp minimal nephrin promoter showed higher eGFP expression levels in each of the cells tested.

[0563] These data again demonstrate that a 265 bp minimal nephrin promoter is capable of driving eGFP expression in target cells. Additionally, that this promoter may be incorporated into an AAV vector for use in gene therapy. The use of a 265 bp minimal nephrin promoter, in place of the full length minimal nephrin promoter, may allow the incorporation of more genetic material into a transgene cassette for AAV purposes.

Example 9—Expression of Human Podocin Using 265 bp v FL Minimal Nephrin Promoters (Plasmid Transfection) in HEK

[0564] Plasmids were constructed comprising HA-tagged podocin under the control of: (i) the full-length (FL) minimal nephrin promoter of SEQ ID NO: 14 (FL.NPSH1-Podocin-HA); or (ii) the 265 bp minimal nephrin promoter of SEQ ID NO: 47 (265.hNPSH1-Podocin-HA).

[0565] 293T Human Embryonic Kidney cells grown in DMEM supplemented with 10% FBS, 1% Penicillin/Streptomycin and 1% sodium pyruvate were seeded at a density of 5e5 in 6-well culture plates. Transfection was carried out the following day when the cells reached 70-80% confluency. Transfection reagent mix was prepared by complexing 3 ug of expression plasmid with polyethylenimine (PEI) at ratio of 1:3 in serum-free media. Non-transfected control received a transfection mix containing PEI at the same ratio without any expression plasmid. Cells were incubated with the transfection mix at 37° C., 5% CO2 until the following day when the media was changed to complete media. Cells

were harvested on Day 2 post transfection and analysed by FACs for expression of Podocin by staining for HA-tag with anti-HA antibody.

[0566] The results in FIG. 14A show that both promoters were capable of driving human Podocin-HA expression in HEK cells. This demonstrates that minimizing the full length minimal nephrin promoter to 265 bp still allows expression of the transgene in target cells, despite removing the majority of the promoter sequence.

Example 10—Expression of Human Podocin Using
265 bp v FL Minimal Nephrin Promoters (AAV
Transduction) in Human Podocytes

Materials and Method

[0567] AAV Production: 60-80% confluent 293T Human Embryonic Kidney cells grown in DMEM supplemented with 10% FBS, 1% Penicillin/Streptomycin and 1% sodium pyruvate were triple transfected with pHelper (HGT11, PS0150), pAAV Rep-Cap (LK03, PS0240) and one of the two ITR-expression plasmids containing the Full Length (FL) minimal nephrin promoter or the 265 bp minimal nephrin promoter driving Podocin-HA expression (FL-NPSH1-Podocin-HA and 265.hNPSH1-Podocin-HA, respectively). Transfection was carried out on a 150 mm culture dish in the presence of polyethylenimine (PEI) at a ratio of 1:2. Media was changed to DMEM with FBS the following day after transfection. On Day 4 post-transfection, media and cells were collected and processed separately. PEG was added to filtered supernatants to precipitate viral particles which were subsequently centrifuged at 1500 g for 20 minutes. Cell pellets were resuspended in TD buffer and frozen at -80 degrees, followed by thawing at 37 degrees and vortexed. Freeze-thaw cycles were repeated 5 times. Pellets from the PEG treated supernatant were then added to the lysed cells. Benzonase and Sodium deoxycholate was further added and incubated at 37 degrees for 30 minutes. Samples were then centrifuged and the supernatant filtered and subsequently purified via iodixanol gradient ultracentrifugation.

[0568] AAV QC: AAV samples were titred via qPCR to determine the number of viral genomes per ml of viral supernatant. Primers designed to target the Podocin sequence were used.

[0569] AAV Transduction: Cells were seeded into wells of a 12-well plate at a density of 0.5e5 cells/well and allowed to adhere overnight. The following day viral particles were added a multiplicity of infection (MOI) of 5e5 particles per cell. Cells were placed at 37 degrees to initiate differentiation and cultured for 9 days. After 9 days cells were harvested and stained with anti-HA antibody for HA expression by flow cytometry as a measure of Podocin expression.

Results

[0570] As demonstrated with the plasmid transfection data in HEK cells (FIG. 11A), the results shown in FIG. 14B further confirm that both promoters are capable of driving expression of a transgene in human podocytes. The 265 bp minimal nephrin promoter showed higher podocin expression levels in the human podocytes.

Example 11—In Vivo Expression of GFP Using
265 bp v FL Promoters Via AAV Transduction

[0571] The 265- and FL-nephrin promoters were evaluated in vivo in WT mice (C57/Bl6). AAV particles as described in Example 8, except that AAV9 (PS0241) was used instead of LK03, were administered via tail vein injection of $\sim 1 \times 10^{13}$ AAV2/9 per mouse (2x mice per treatment group and 1x untreated mouse). Tissue harvest of kidney occurred 4 weeks post injection. qPCR analysis performed on the kidney and GFP copy number normalised to 18S (housekeeping gene). 18S rRNA is a reliable normalisation gene for real time PCR.

[0572] The results in FIG. 15 show that there was detectable GFP expression in the kidney using both the 265 bp and full length minimal nephrin promoters. The expression in the kidney was higher using the 265 bp minimal nephrin promoter.

[0573] All publications mentioned in the above specification are herein incorporated by reference. Various modifications and variations of the disclosed methods, cells, compositions and uses of the invention will be apparent to the skilled person without departing from the scope and spirit of the invention. Although the invention has been disclosed in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the disclosed modes for carrying out the invention, which are obvious to the skilled person are intended to be within the scope of the following claims.

EMBODIMENTS

[0574] Various preferred features and embodiments of the present invention will now be described with reference to the following numbered paragraphs (paras).

[0575] 1. A promoter comprising (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 and wherein the promoter has a length of about 1.1 kb or less.

[0576] 2. A promoter which comprises or consists of a nucleotide sequence having at least 70% identity to a nucleotide sequence according to SEQ ID NO: 1 but wherein:

[0577] (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 100 to 430; and/or

[0578] (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061.

[0579] 3. The promoter according to para 2, wherein the promoter comprises (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 and/or wherein the promoter has a length of about 1.1 kb or less.

[0580] 4. The promoter according to any preceding para, wherein the promoter is a podocyte-specific promoter.

[0581] 5. The promoter according to any preceding para, wherein the promoter further comprises (ii) a nucleotide sequence having at least 70% identity to SEQ ID NO: 5, a nucleotide sequence having at least 70% identity to SEQ ID NO: 6, and/or a nucleotide sequence having at least 70% identity to SEQ ID NO: 7.

[0582] 6. The promoter according to any preceding para, wherein the promoter further comprises (iii) a nucleotide sequence having at least 70% identity to SEQ ID NO: 8, or one or more fragments thereof.

[0583] 7. The promoter according to any preceding para, wherein the promoter comprises from 5' to 3': (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4; (iii) optionally a nucleotide sequence having at least 70% identity to SEQ ID NO: 8, or one or more fragments thereof; and (ii) a nucleotide sequence having at least 70% identity to SEQ ID NO: 5, a nucleotide sequence having at least 70% identity to SEQ ID NO: 6, and/or a nucleotide sequence having at least 70% identity to SEQ ID NO: 7.

[0584] 8. The promoter according to any preceding para, wherein the promoter has a length of about 1.0 kb or less, about 0.9 kb or less, about 0.8 kb or less, about 0.7 kb or less, about 0.6 kb or less, about 0.5 kb or less, about 0.4 kb or less, or about 0.3 kb or less.

[0585] 9. The promoter according to any preceding para, wherein the promoter has a length of from 0.265 kb to 1.0 kb, from 0.265 kb to 0.9 kb, from 0.265 kb to 0.8 kb, from 0.265 kb to 0.7 kb, from 0.265 kb to 0.6 kb, from 0.265 kb to 0.5 kb, from 0.265 kb to 0.4 kb, or from 0.265 kb to 0.3 kb.

[0586] 10. The promoter according to any preceding para, wherein the promoter comprises:

[0587] (a) a retinoic acid receptor binding site;

[0588] (b) a WT1 binding site;

[0589] (c) an enhancer box;

[0590] (d) a transcription factor binding region; and/or

[0591] (e) a transcription initiation site.

[0592] 11. The promoter according to para 10 wherein the retinoic acid receptor binding site comprises or consists of a nucleotide sequence shown as SEQ ID NO: 10 or a nucleotide sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 10.

[0593] 12. The promoter according to para 10 or 11 wherein the WT1 binding site comprises or consists of a nucleotide sequence shown as SEQ ID NO: 11 or a nucleotide sequence having one, two or three substitutions, deletions, or insertions compared to SEQ ID NO: 11.

[0594] 13. The promoter according to any one of paras 10 to 12 wherein the enhancer box comprises or consists of a nucleotide sequence shown as SEQ ID NO: 12 or a nucleotide sequence having one or two substitutions, deletions, or insertions compared to SEQ ID NO: 12.

[0595] 14. The promoter according to any one of paras 1, or 3 to 13 wherein (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 comprises one or more of: (a) a retinoic acid receptor binding site; (b) a WT1 binding site; and (c) an enhancer box.

[0596] 15. The promoter according to any one of paras 1, or 3 to 14 wherein one or more of the following nucleotide sequences is present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4:

[0597] (a) GGGGTCA at a position corresponding to position 7 to position 13 of SEQ ID NO: 4;

[0598] (b) CGGAGGCTGGGGAGGCA at a position corresponding to position 14 to position 30 of SEQ ID NO: 4; and

[0599] (c) ATGTG at a position corresponding to position 49 to position 53 of SEQ ID NO: 4.

[0600] 16. The promoter according to any one of paras 10 to 15 wherein the transcription factor binding region comprises or consists of a nucleotide sequence shown as SEQ ID NO: 13 or a nucleotide sequence having one, two, three, four or five substitutions, deletions, or insertions compared to SEQ ID NO: 13.

[0601] 17. The promoter according to any one of paras 10 to 16, wherein the transcription initiation site comprises or consists of an "AG" dinucleotide.

[0602] 18. The promoter according to any one of paras 10 to 17, wherein the transcription factor binding site is operably linked to the transcription initiation site, optionally wherein the transcription factor binding site is directly upstream of the transcription initiation site.

[0603] 19. The promoter according to any preceding para, wherein the promoter comprises or consists of a nucleotide sequence having at least 70% identity to SEQ ID NO: 2 or SEQ ID NO: 3.

[0604] 20. A promoter which consists of a nucleotide sequence having at least 70% identity to SEQ ID NO: 2.

[0605] 21. A promoter which consists of a nucleotide sequence having at least 70% identity to SEQ ID NO: 3.

[0606] 22. A polynucleotide comprising the promoter according to any one of paras 1 to 21.

[0607] 23. The polynucleotide according to para 22, wherein the promoter is operably linked to a protein coding sequence.

[0608] 24. The polynucleotide according to para 23, wherein the protein coding sequence is operably linked to one or more further regulatory elements, such as a post-transcriptional regulatory element and/or a polyadenylation sequence.

[0609] 25. A vector comprising the polynucleotide according to any one of paras 22 to 24.

[0610] 26. The vector according to para 25, wherein the vector is capable of transducing podocytes, optionally wherein the vector is capable of specifically transducing podocytes.

[0611] 27. The vector according to para 25 or 26, wherein the vector is a viral vector, such as an adeno-associated virus (AAV) vector, a lentiviral vector, a retroviral vector, an adenoviral vector, an adeno-associated viral vector, a herpes simplex viral vector, an alphaviral vector, a flaviviral vector, a rhabdoviral vector, a measles viral vector, a Newcastle disease viral vector, a poxviral vector, and a picornaviral vector, preferably wherein the vector is an AAV vector.

[0612] 28. The vector according to para 27, wherein the viral vector is in the form of a viral vector particle, preferably wherein the viral vector is in the form of an AAV vector particle.

[0613] 29. The vector according to any one of paras 25 to 28, wherein the vector is the form of an AAV vector particle encapsidated by AAV3B, LK03, or AAV9 capsid proteins.

[0614] 30. A cell comprising the polynucleotide according to any one of paras 22 to 24, or the vector according to any one of paras 25 to 29.

[0615] 31. A pharmaceutical composition comprising the polynucleotide according to any one of paras 22 to 24, the vector according to any one of paras 25 to 29, or the cell according to para 30.

[0616] 32. A polynucleotide according to any one of paras 22 to 24, vector according to any one of paras 25 to 29, or cell according to para 30 for use in medicine.

[0617] 33. Use of the promoter according to any one of paras 1 to 21 to drive expression of a coding sequence.

[0618] 34. The use according to para 32, wherein the expression is podocyte-specific.

 SEQUENCE LISTING

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<210> SEQ ID NO 1

<211> LENGTH: 1249

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary nephrin promoter

<400> SEQUENCE: 1

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ctgaggctcag gagttcgaga ccagcctggc caacatgatg aaaccccgctc tctagtaaaa    120
atacaaaaat tagccaggca tgggtgtata tacctgtagt accagctact tgggagacag      180
aggtgggaga attacttgaa cctgggaggt tcaagccatg ggaggtggaa gttgcagtga    240
gccgagatgc cactgcactc cagcctgagc aacagagcaa gactatctca agaaaaaaaa    300
gaaagaaaaga aagggacttg ccaaggtcat gtatcagggc aaggaagagc tggggggccca    360
gttggtgtct cccctgtgta gctgggagac caccttgatc tgacttctcc catcttccca    420
gcctaagcca ggccttgggg tcacggaggc tggggaggca ccgaggaacg cgcttggcat    480
gtgctgacag ggaattttat gctccagctg ggcagctggg gaggagcctg ctgggcagag    540
gccagagctg ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa    600
gctgggtcag agagcagggc tgactctgcc agtgcttcca tcagcctcat cgctctccta    660
ggctcctggc ctgctggact ctgggctgca ggtccttctt gaaaggctgt gtagtagtgag    720
acaaggagca ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga    780
gagacagaga gagaggaaga gacagagaca aaaggagaga gaacggctta gacaaggaga    840
gaaagatgga aagataaaga gactgggctc agtggctcac gcctgtaatc ccaacacttg    900
gggaggccaa ggtgggagga tggtctgaag gaaagagtct gagatcaacc tggccaacat    960
agtgagaccc cgtctctaaa aaaaaaaaaa aaaaaaaaaa gaaaaaagaa aaaaaagttt 1020
ttttaagag acagagaaag agactcagag attgagactg agagcaagac agagagagac    1080
actcacaggg aagaggggaa gaggaacacg agaaaggag gagagtaacg gaaagagata    1140
aaaaagaaaa gcaggtggca gagacacaga gagagggacc cagagaaagc cagacagacg    1200
caggtggctg gcagcgggcg ctgtgggggt cacagtaggg ggacctgtc      1249

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<210> SEQ ID NO 2

<211> LENGTH: 819

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary minimal nephrin promoter

<400> SEQUENCE: 2

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ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgcttggcat gtgctgacag      60
ggaattttat gctccagctg ggcagctggg gaggagcctg ctgggcagag gccagagctg    120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag    180
agagcagggc tgactctgcc agtgcttcca tcagcctcat cgctctccta ggctcctggc    240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gtagtagtgag acaaggagca    300
ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga gagacagaga    360

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gagaggaaga gacagagaca aaaggagaga gaacggctta gacaaggaga gaaagatgga	420
aagataaaga gactgggagc agtgggtcac gcctgtaac ccaacacttg gggaggccaa	480
ggtgggagga tggcttgaag gaaagagtct gagatcaacc tggccaacat agtgagaccc	540
cgtctctaaa aaaaaaaaaa aaaaaaaaaa gaaaaaagaa aaaaaagttt ttttaagag	600
acagagaaag agactcagag attgagactg agagcaagac agagagagac actcacaggg	660
aagaggggaa gaggaaaacg agaaaggag gagagtaacg gaaagagata aaaaagaaaa	720
gcaggtggca gagacacaga gagagggacc cagagaaagc cagacagacg caggtggctg	780
gcagcgggag ctgtgggggt cacagtaggg ggacctgtc	819

<210> SEQ ID NO 3
 <211> LENGTH: 265
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary minimal nephrin promoter

<400> SEQUENCE: 3

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag	60
ggaattttat gctccaggag caagacagag agagacactc acaggaaga ggggaagagg	120
aaaacgagaa agggaggaga gtaacggaaa gagataaaaa agaaaagcag gtggcagaga	180
cacagagaga gggaccaga gaaagccaga cagacgcagg tggctggcag cgggcgctgt	240
gggggtcaca gtagggggac ctgtc	265

<210> SEQ ID NO 4
 <211> LENGTH: 77
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary proximal promoter region

<400> SEQUENCE: 4

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag	60
ggaattttat gctccag	77

<210> SEQ ID NO 5
 <211> LENGTH: 32
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary core promoter region

<400> SEQUENCE: 5

gagcaagaca gagagagaca ctcacaggga ag	32
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<210> SEQ ID NO 6
 <211> LENGTH: 156
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary 5'UTR

<400> SEQUENCE: 6

aggggaagag gaaaacgaga aaggaggag agtaacggaa agagataaaa aagaaaagca	60
ggtggcagag acacagagag agggaccag agaaagccag acagacgcag gtggctggca	120

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gcggggcgctg tgggggtcac agtaggggga cctgtc 156

<210> SEQ ID NO 7
 <211> LENGTH: 188
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary core promoter region and 5'UTR

<400> SEQUENCE: 7

gagcaagaca gagagagaca ctcacagggg agaggggaag aggaaaacga gaaagggagg 60
 agagtaacgg aaagagataa aaaagaaaag caggtggcag agacacagag agagggagccc 120
 agagaaaagg agacagacgc aggtggctgg cagcggggcg tgtgggggtc acagtagggg 180
 gacctgtc 188

<210> SEQ ID NO 8
 <211> LENGTH: 554
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary optional promoter region

<400> SEQUENCE: 8

ctggggccagc tgggaggagc ctgctgggca gaggccagag ctgggggctc tggaaggtag 60
 ctggggggagg ttgactgtg agaatgagct caagctgggt cagagagcag ggctgactct 120
 gccagtgcct gcatcagcct catcgtctc ctaggctcct ggctgtctgg actctgggct 180
 gcaggtcctt cttgaaaggc tgtgagtagt gagacaagga gcaggagtga ggggtggcag 240
 gagagaagat agagattgag agagagagag agagagacag agagagagga agagacagag 300
 acaaaaggag agagaacggc ttagacaagg agagaaagat ggaaagataa agagactggg 360
 cgcagtggct cacgctgta atcccaacac ttggggaggc caaggtggga ggatggcttg 420
 aaggaaagag tctgagatca acctggccaa catagtgaga ccccgctctc aaaaaaaaaa 480
 aagaaaaaaaa aaagaaaaaa gaaaaaaaaa tttttttaa gagacagaga aagagactca 540
 gagattgaga ctga 554

<210> SEQ ID NO 9
 <211> LENGTH: 430
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary optional upstream promoter region

<400> SEQUENCE: 9

cctgcagggc ccactagtct gtaatcccag cattttggga ggctgaggca gatggatcac 60
 ctgaggtcag gagttcgaga ccagcctggc caacatgatg aaaccccgct tctagtaaaa 120
 atacaaaaat tagccaggca tgggtctata tacctgtagt accagctact tgggagacag 180
 aggtgggaga attacttgaa cctgggaggt tcaagccatg ggaggtggaa gttgcagtga 240
 gccgagatgc cactgcactc cagcctgagc aacagagcaa gactatctca agaaaaaaaa 300
 gaaagaaaga aagggaacttg ccaaggtcat gtatcagggc aaggaagagc tgggggcccc 360
 gctggtgct cccctgtga gctgggagac caccttgatc tgacttctcc catcttccca 420
 gcctaagcca 430

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<210> SEQ ID NO 10
 <211> LENGTH: 7
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary RAR binding site

 <400> SEQUENCE: 10

 ggggtca 7

<210> SEQ ID NO 11
 <211> LENGTH: 17
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary WT1 binding site

 <400> SEQUENCE: 11

 cggaggctgg ggaggca 17

<210> SEQ ID NO 12
 <211> LENGTH: 5
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary enhancer box

 <400> SEQUENCE: 12

 atgtg 5

<210> SEQ ID NO 13
 <211> LENGTH: 30
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary transcription factor binding region

 <400> SEQUENCE: 13

 gagcaagaca gagagagaca ctcacaggga 30

<210> SEQ ID NO 14
 <211> LENGTH: 1192
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary variant nephrin promoter

 <400> SEQUENCE: 14

 cacctgaggt caggagttcg agaccagcgt ggccaacatg atgaaacccc gtctctagta 60
 aaaatacaaa aattagccag gcatggtgct atatacctgt agcaccagct acttgggaga 120
 cagaggtggg agaattactt gaacctggga ggttcaagcc atgggagggtg gaagttgcag 180
 tgagccgaga tgccactgca ctccagcctg agcaacagag caagactatc tcaagaaaag 240
 aaagaaagaa agaaagagac ttgccaaggt catgtatcag ggcaaggag agctggggggc 300
 ccagctggct gctcccctgc tgagctggga gaccaccttg atctgacttc tcccatcttc 360
 ccagcctaag ccaggccctg gggtcacgga ggctggggag gcaccgagga acgcgcctgg 420
 catgtgctga caggggattt tatgtccag ctgggccagc tgggaggagc ctgctgggca 480
 gaggccagag ctggggggctc tggaaggtag ctgggggagg ttgcactgtg agaatgagct 540

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caagctgggt cagagagcag ggctgactct gccagtgcct gcatcagcct catcgctctc	600
ctaggctcct ggctgtctgg actctgggct gcaggtcctt cttgaaaggc tgtgagtagt	660
gagacaagga gcaggagtga ggggtggcag gagagaagat agagattgag agagagagag	720
agagagagac agagagagag gaagagacag agacaaaagg agagagaacg gcttagacaa	780
ggagagaaaag atggaagat aaagagactg ggcgagtggt ctcacgcctg taatcccaac	840
acttggggag gccaaagtgg gaggatggct tgaaggaaag agtctgagat caacctggcc	900
aacatagtga gaccccgctc ttaaaaaaaaa aagaaaaaaaa aaagaaaaaaaa gaaaaaaaaag	960
tttttttaaa gagacagaga aagagactca gagattgaga ctgagagcaa gacagagaga	1020
gatactcaca ggaagagagg gaagaggaaa acgagaaagg gaggagagta acggaagag	1080
ataaaaaaga aaagcaggtg gcagagacac acagagaggg acccagagaa agccagacag	1140
acgcaggtgg ctggcagcgg gcgctgtggg ggtcacagta gggggacctg tg	1192

<210> SEQ ID NO 15
 <211> LENGTH: 819
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant
 <400> SEQUENCE: 15

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag	60
gggattttat gctccagctg ggccagctgg gagagacctg ctgggcagag gccagagctg	120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag	180
agagcagggc tgactctgcc agtgctgca tcagcctcat cgctctccta ggctcctggc	240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gagtagtgag acaaggagca	300
ggagtggagg gtggcaggag agaagataga gattgagaga gagagagaga gagagacaga	360
gagagaggaa gagacagaga caaaaggaga gagaacggct tagacaagga gagaaagatg	420
gaaagataaa gagactgggc gcagtggctc acgcctgtaa tccaacact tggggaggcc	480
aagtgaggag gatggcttga aggaagagt ctgagatcaa cctggccaac atagtgagac	540
cccgctctta aaaaaaaaaa aaaaaaaaaa gaaaaagaa aaaaagtgtt ttttaagag	600
acagagaaa agactcagag attgagactg agagcaagac agagagagat actcacagg	660
aagaggggaa gaggaaaacg agaaaggag gagagtaacg gaaagagata aaaaagaaaa	720
gcaggtggca gagacacaca gagagggacc cagagaaacg cagacagacg caggtggctg	780
gcagcgggag ctgtgggggt cacagtaggg ggacctgtg	819

<210> SEQ ID NO 16
 <211> LENGTH: 265
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant
 <400> SEQUENCE: 16

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag	60
gggattttat gctccaggag caagacagag agagatactc acaggaaga ggggaagagg	120
aaaacgagaa agggaggaga gtaacggaaa gagataaaaa agaaaagcag gtggcagaga	180

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cacagagaga gggaccacaga gaaagccaga cagacgcagg tggttgccag cgggcgctgt	240
gggggtcaca gtagggggac ctgtc	265

<210> SEQ ID NO 17
 <211> LENGTH: 77
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary variant proximal promoter region

<400> SEQUENCE: 17	
ggccctgggg tcacggaggc tggggaggca cggaggaacg cgctggcat gtgctgacag	60
gggattttat gctccag	77

<210> SEQ ID NO 18
 <211> LENGTH: 32
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary variant core promoter region

<400> SEQUENCE: 18	
gagcaagaca gagagagata ctcacaggga ag	32

<210> SEQ ID NO 19
 <211> LENGTH: 156
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary variant 5'UTR

<400> SEQUENCE: 19	
aggggaagag gaaaacgaga aaggaggag agtaacggaa agagataaaa aagaaaagca	60
ggtggcagag acacacagag agggaccag agaaagccag acagacgcag gtggctggca	120
gcgggcgctg tgggggtcac agtagggga cctgtg	156

<210> SEQ ID NO 20
 <211> LENGTH: 188
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary variant core promoter region and 5'UTR

<400> SEQUENCE: 20	
gagcaagaca gagagagata ctcacaggga agaggggaag aggaaaacga gaaaggagg	60
agagtaacgg aaagagataa aaaagaaaag caggtggcag agacacacag agagggaccc	120
agagaaagcc agacagacgc aggtggctgg cagcgggcgc tgtgggggtc acagtagggg	180
gacctgtg	188

<210> SEQ ID NO 21
 <211> LENGTH: 1670
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary COL4A3 amino acid sequence

<400> SEQUENCE: 21	
Met Ser Ala Arg Thr Ala Pro Arg Pro Gln Val Leu Leu Leu Pro Leu	

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1	5	10	15
Leu Leu Val	Leu Leu Ala Ala Ala	Pro Ala Ala Ser Lys Gly Cys Val	
	20	25	30
Cys Lys Asp	Lys Gly Gln Cys Phe Cys Asp Gly Ala Lys Gly Glu Lys		
	35	40	45
Gly Glu Lys	Gly Phe Pro Gly Pro Pro Gly Ser Pro Gly Gln Lys Gly		
	50	55	60
Phe Thr Gly	Pro Glu Gly Leu Pro Gly Pro Gln Gly Pro Lys Gly Phe		
	65	70	75
Pro Gly Leu	Pro Gly Leu Thr Gly Ser Lys Gly Val Arg Gly Ile Ser		
	85	90	95
Gly Leu Pro	Gly Phe Ser Gly Ser Pro Gly Leu Pro Gly Thr Pro Gly		
	100	105	110
Asn Thr Gly	Pro Tyr Gly Leu Val Gly Val Pro Gly Cys Ser Gly Ser		
	115	120	125
Lys Gly Glu	Gln Gly Phe Pro Gly Leu Pro Gly Thr Leu Gly Tyr Pro		
	130	135	140
Gly Ile Pro	Gly Ala Ala Gly Leu Lys Gly Gln Lys Gly Ala Pro Ala		
	145	150	155
Lys Glu Glu	Asp Ile Glu Leu Asp Ala Lys Gly Asp Pro Gly Leu Pro		
	165	170	175
Gly Ala Pro	Gly Pro Gln Gly Leu Pro Gly Pro Pro Gly Phe Pro Gly		
	180	185	190
Pro Val Gly	Pro Pro Gly Pro Pro Gly Phe Phe Gly Phe Pro Gly Ala		
	195	200	205
Met Gly Pro	Arg Gly Pro Lys Gly His Met Gly Glu Arg Val Ile Gly		
	210	215	220
His Lys Gly	Glu Arg Gly Val Lys Gly Leu Thr Gly Pro Pro Gly Pro		
	225	230	235
Pro Gly Thr	Val Ile Val Thr Leu Thr Gly Pro Asp Asn Arg Thr Asp		
	245	250	255
Leu Lys Gly	Glu Lys Gly Asp Lys Gly Ala Met Gly Glu Pro Gly Pro		
	260	265	270
Pro Gly Pro	Ser Gly Leu Pro Gly Glu Ser Tyr Gly Ser Glu Lys Gly		
	275	280	285
Ala Pro Gly	Asp Pro Gly Leu Gln Gly Lys Pro Gly Lys Asp Gly Val		
	290	295	300
Pro Gly Phe	Pro Gly Ser Glu Gly Val Lys Gly Asn Arg Gly Phe Pro		
	305	310	315
Gly Leu Met	Gly Glu Asp Gly Ile Lys Gly Gln Lys Gly Asp Ile Gly		
	325	330	335
Pro Pro Gly	Phe Arg Gly Pro Thr Glu Tyr Tyr Asp Thr Tyr Gln Glu		
	340	345	350
Lys Gly Asp	Glu Gly Thr Pro Gly Pro Pro Gly Pro Arg Gly Ala Arg		
	355	360	365
Gly Pro Gln	Gly Pro Ser Gly Pro Pro Gly Val Pro Gly Ser Pro Gly		
	370	375	380
Ser Ser Arg	Pro Gly Leu Arg Gly Ala Pro Gly Trp Pro Gly Leu Lys		
	385	390	395
Gly Ser Lys	Gly Glu Arg Gly Arg Pro Gly Lys Asp Ala Met Gly Thr		
	405	410	415

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Pro Gly Ser	Pro Gly Cys	Ala Gly Ser	Pro Gly Leu	Pro Gly Ser	Pro
420		425	430		
Gly Pro	Pro Gly Pro	Pro Gly Asp	Ile Val Phe	Arg Lys Gly	Pro Pro
435		440	445		
Gly Asp His	Gly Leu Pro	Gly Tyr Leu	Gly Ser Pro	Gly Ile Pro	Gly
450		455	460		
Val Asp Gly	Pro Lys Gly	Glu Pro Gly	Leu Leu Cys	Thr Gln Cys	Pro
465		470	475		480
Tyr Ile Pro	Gly Pro Pro	Gly Leu Pro	Gly Leu Pro	Gly Leu His	Gly
	485		490		495
Val Lys Gly	Ile Pro Gly	Arg Gln Gly	Ala Ala Gly	Leu Lys Gly	Ser
	500		505		510
Pro Gly Ser	Pro Gly Asn	Thr Gly Leu	Pro Gly Phe	Pro Gly Phe	Pro
	515		520		525
Gly Ala Gln	Gly Asp Pro	Gly Leu Lys	Gly Glu Lys	Gly Glu Thr	Leu
	530		535		540
Gln Pro Glu	Gly Gln Val	Gly Val Pro	Gly Asp Pro	Gly Leu Arg	Gly
	545		550		555
Gln Pro Gly	Arg Lys Gly	Leu Asp Gly	Ile Pro Gly	Thr Pro Gly	Val
	565		570		575
Lys Gly Leu	Pro Gly Pro	Lys Gly Glu	Leu Ala Leu	Ser Gly Glu	Lys
	580		585		590
Gly Asp Gln	Gly Pro Pro	Gly Asp Pro	Gly Ser Pro	Gly Ser Pro	Gly
	595		600		605
Pro Ala Gly	Pro Ala Gly	Pro Pro Gly	Tyr Gly Pro	Gln Gly Glu	Pro
	610		615		620
Gly Leu Gln	Gly Thr Gln	Gly Val Pro	Gly Ala Pro	Gly Pro Pro	Gly
	625		630		635
Glu Ala Gly	Pro Arg Gly	Glu Leu Ser	Val Ser Thr	Pro Val Pro	Gly
	645		650		655
Pro Pro Gly	Pro Pro Gly	Pro Pro Gly	His Pro Gly	Pro Gln Gly	Pro
	660		665		670
Pro Gly Ile	Pro Gly Ser	Leu Gly Lys	Cys Gly Asp	Pro Gly Leu	Pro
	675		680		685
Gly Pro Asp	Gly Glu Pro	Gly Ile Pro	Gly Ile Gly	Phe Pro Gly	Pro
	690		695		700
Pro Gly Pro	Lys Gly Asp	Gln Gly Phe	Pro Gly Thr	Lys Gly Ser	Leu
	705		710		715
Gly Cys Pro	Gly Lys Met	Gly Glu Pro	Gly Leu Pro	Gly Lys Pro	Gly
	725		730		735
Leu Pro Gly	Ala Lys Gly	Glu Pro Ala	Val Ala Met	Pro Gly Gly	Pro
	740		745		750
Gly Thr Pro	Gly Phe Pro	Gly Glu Arg	Gly Asn Ser	Gly Glu His	Gly
	755		760		765
Glu Ile Gly	Leu Pro Gly	Leu Pro Gly	Leu Pro Gly	Thr Pro Gly	Asn
	770		775		780
Glu Gly Leu	Asp Gly Pro	Arg Gly Asp	Pro Gly Gln	Pro Gly Pro	Pro
	785		790		795
Gly Glu Gln	Gly Pro Pro	Gly Arg Cys	Ile Glu Gly	Pro Arg Gly	Ala
	805		810		815

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Gln	Gly	Leu	Pro	Gly	Leu	Asn	Gly	Leu	Lys	Gly	Gln	Gln	Gly	Arg	Arg	820	825	830	
Gly	Lys	Thr	Gly	Pro	Lys	Gly	Asp	Pro	Gly	Ile	Pro	Gly	Leu	Asp	Arg	835	840	845	
Ser	Gly	Phe	Pro	Gly	Glu	Thr	Gly	Ser	Pro	Gly	Ile	Pro	Gly	His	Gln	850	855	860	
Gly	Glu	Met	Gly	Pro	Leu	Gly	Gln	Arg	Gly	Tyr	Pro	Gly	Asn	Pro	Gly	865	870	875	880
Ile	Leu	Gly	Pro	Pro	Gly	Glu	Asp	Gly	Val	Ile	Gly	Met	Met	Gly	Phe	885	890	895	
Pro	Gly	Ala	Ile	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Asn	Pro	Gly	Thr	Pro	900	905	910	
Gly	Gln	Arg	Gly	Ser	Pro	Gly	Ile	Pro	Gly	Val	Lys	Gly	Gln	Arg	Gly	915	920	925	
Thr	Pro	Gly	Ala	Lys	Gly	Glu	Gln	Gly	Asp	Lys	Gly	Asn	Pro	Gly	Pro	930	935	940	
Ser	Glu	Ile	Ser	His	Val	Ile	Gly	Asp	Lys	Gly	Glu	Pro	Gly	Leu	Lys	945	950	955	960
Gly	Phe	Ala	Gly	Asn	Pro	Gly	Glu	Lys	Gly	Asn	Arg	Gly	Val	Pro	Gly	965	970	975	
Met	Pro	Gly	Leu	Lys	Gly	Leu	Lys	Gly	Leu	Pro	Gly	Pro	Ala	Gly	Pro	980	985	990	
Pro	Gly	Pro	Arg	Gly	Asp	Leu	Gly	Ser	Thr	Gly	Asn	Pro	Gly	Glu	Pro	995	1000	1005	
Gly	Leu	Arg	Gly	Ile	Pro	Gly	Ser	Met	Gly	Asn	Met	Gly	Met	Pro		1010	1015	1020	
Gly	Ser	Lys	Gly	Lys	Arg	Gly	Thr	Leu	Gly	Phe	Pro	Gly	Arg	Ala		1025	1030	1035	
Gly	Arg	Pro	Gly	Leu	Pro	Gly	Ile	His	Gly	Leu	Gln	Gly	Asp	Lys		1040	1045	1050	
Gly	Glu	Pro	Gly	Tyr	Ser	Glu	Gly	Thr	Arg	Pro	Gly	Pro	Pro	Gly		1055	1060	1065	
Pro	Thr	Gly	Asp	Pro	Gly	Leu	Pro	Gly	Asp	Met	Gly	Lys	Lys	Gly		1070	1075	1080	
Glu	Met	Gly	Gln	Pro	Gly	Pro	Pro	Gly	His	Leu	Gly	Pro	Ala	Gly		1085	1090	1095	
Pro	Glu	Gly	Ala	Pro	Gly	Ser	Pro	Gly	Ser	Pro	Gly	Leu	Pro	Gly		1100	1105	1110	
Lys	Pro	Gly	Pro	His	Gly	Asp	Leu	Gly	Phe	Lys	Gly	Ile	Lys	Gly		1115	1120	1125	
Leu	Leu	Gly	Pro	Pro	Gly	Ile	Arg	Gly	Pro	Pro	Gly	Leu	Pro	Gly		1130	1135	1140	
Phe	Pro	Gly	Ser	Pro	Gly	Pro	Met	Gly	Ile	Arg	Gly	Asp	Gln	Gly		1145	1150	1155	
Arg	Asp	Gly	Ile	Pro	Gly	Pro	Ala	Gly	Glu	Lys	Gly	Glu	Thr	Gly		1160	1165	1170	
Leu	Leu	Arg	Ala	Pro	Pro	Gly	Pro	Arg	Gly	Asn	Pro	Gly	Ala	Gln		1175	1180	1185	
Gly	Ala	Lys	Gly	Asp	Arg	Gly	Ala	Pro	Gly	Phe	Pro	Gly	Leu	Pro		1190	1195	1200	
Gly	Arg	Lys	Gly	Ala	Met	Gly	Asp	Ala	Gly	Pro	Arg	Gly	Pro	Thr					

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1205	1210	1215
Gly Ile Glu Gly Phe Pro Gly Pro Pro Gly Leu Pro Gly Ala Ile		
1220	1225	1230
Ile Pro Gly Gln Thr Gly Asn Arg Gly Pro Pro Gly Ser Arg Gly		
1235	1240	1245
Ser Pro Gly Ala Pro Gly Pro Pro Gly Pro Pro Gly Ser His Val		
1250	1255	1260
Ile Gly Ile Lys Gly Asp Lys Gly Ser Met Gly His Pro Gly Pro		
1265	1270	1275
Lys Gly Pro Pro Gly Thr Ala Gly Asp Met Gly Pro Pro Gly Arg		
1280	1285	1290
Leu Gly Ala Pro Gly Thr Pro Gly Leu Pro Gly Pro Arg Gly Asp		
1295	1300	1305
Pro Gly Phe Gln Gly Phe Pro Gly Val Lys Gly Glu Lys Gly Asn		
1310	1315	1320
Pro Gly Phe Leu Gly Ser Ile Gly Pro Pro Gly Pro Ile Gly Pro		
1325	1330	1335
Lys Gly Pro Pro Gly Val Arg Gly Asp Pro Gly Thr Leu Lys Ile		
1340	1345	1350
Ile Ser Leu Pro Gly Ser Pro Gly Pro Pro Gly Thr Pro Gly Glu		
1355	1360	1365
Pro Gly Met Gln Gly Glu Pro Gly Pro Pro Gly Pro Pro Gly Asn		
1370	1375	1380
Leu Gly Pro Cys Gly Pro Arg Gly Lys Pro Gly Lys Asp Gly Lys		
1385	1390	1395
Pro Gly Thr Pro Gly Pro Ala Gly Glu Lys Gly Asn Lys Gly Ser		
1400	1405	1410
Lys Gly Glu Pro Gly Pro Ala Gly Ser Asp Gly Leu Pro Gly Leu		
1415	1420	1425
Lys Gly Lys Arg Gly Asp Ser Gly Ser Pro Ala Thr Trp Thr Thr		
1430	1435	1440
Arg Gly Phe Val Phe Thr Arg His Ser Gln Thr Thr Ala Ile Pro		
1445	1450	1455
Ser Cys Pro Glu Gly Thr Val Pro Leu Tyr Ser Gly Phe Ser Phe		
1460	1465	1470
Leu Phe Val Gln Gly Asn Gln Arg Ala His Gly Gln Asp Leu Gly		
1475	1480	1485
Thr Leu Gly Ser Cys Leu Gln Arg Phe Thr Thr Met Pro Phe Leu		
1490	1495	1500
Phe Cys Asn Val Asn Asp Val Cys Asn Phe Ala Ser Arg Asn Asp		
1505	1510	1515
Tyr Ser Tyr Trp Leu Ser Thr Pro Ala Leu Met Pro Met Asn Met		
1520	1525	1530
Ala Pro Ile Thr Gly Arg Ala Leu Glu Pro Tyr Ile Ser Arg Cys		
1535	1540	1545
Thr Val Cys Glu Gly Pro Ala Ile Ala Ile Ala Val His Ser Gln		
1550	1555	1560
Thr Thr Asp Ile Pro Pro Cys Pro His Gly Trp Ile Ser Leu Trp		
1565	1570	1575
Lys Gly Phe Ser Phe Ile Met Phe Thr Ser Ala Gly Ser Glu Gly		
1580	1585	1590

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Thr Gly Gln Ala Leu Ala Ser Pro Gly Ser Cys Leu Glu Glu Phe
 1595 1600 1605
 Arg Ala Ser Pro Phe Leu Glu Cys His Gly Arg Gly Thr Cys Asn
 1610 1615 1620
 Tyr Tyr Ser Asn Ser Tyr Ser Phe Trp Leu Ala Ser Leu Asn Pro
 1625 1630 1635
 Glu Arg Met Phe Arg Lys Pro Ile Pro Ser Thr Val Lys Ala Gly
 1640 1645 1650
 Glu Leu Glu Lys Ile Ile Ser Arg Cys Gln Val Cys Met Lys Lys
 1655 1660 1665
 Arg His
 1670

<210> SEQ ID NO 22
 <211> LENGTH: 1690
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary COL4A4 amino acid sequence

<400> SEQUENCE: 22

Met Trp Ser Leu His Ile Val Leu Met Arg Cys Ser Phe Arg Leu Thr
 1 5 10 15
 Lys Ser Leu Ala Thr Gly Pro Trp Ser Leu Ile Leu Ile Leu Phe Ser
 20 25 30
 Val Gln Tyr Val Tyr Gly Ser Gly Lys Lys Tyr Ile Gly Pro Cys Gly
 35 40 45
 Gly Arg Asp Cys Ser Val Cys His Cys Val Pro Glu Lys Gly Ser Arg
 50 55 60
 Gly Pro Pro Gly Pro Pro Gly Pro Gln Gly Pro Ile Gly Pro Leu Gly
 65 70 75 80
 Ala Pro Gly Pro Ile Gly Leu Ser Gly Glu Lys Gly Met Arg Gly Asp
 85 90 95
 Arg Gly Pro Pro Gly Ala Ala Gly Asp Lys Gly Asp Lys Gly Pro Thr
 100 105 110
 Gly Val Pro Gly Phe Pro Gly Leu Asp Gly Ile Pro Gly His Pro Gly
 115 120 125
 Pro Pro Gly Pro Arg Gly Lys Pro Gly Met Ser Gly His Asn Gly Ser
 130 135 140
 Arg Gly Asp Pro Gly Phe Pro Gly Gly Arg Gly Ala Leu Gly Pro Gly
 145 150 155 160
 Gly Pro Leu Gly His Pro Gly Glu Lys Gly Glu Lys Gly Asn Ser Val
 165 170 175
 Phe Ile Leu Gly Ala Val Lys Gly Ile Gln Gly Asp Arg Gly Asp Pro
 180 185 190
 Gly Leu Pro Gly Leu Pro Gly Ser Trp Gly Ala Gly Gly Pro Ala Gly
 195 200 205
 Pro Thr Gly Tyr Pro Gly Glu Pro Gly Leu Val Gly Pro Pro Gly Gln
 210 215 220
 Pro Gly Arg Pro Gly Leu Lys Gly Asn Pro Gly Val Gly Val Lys Gly
 225 230 235 240
 Gln Met Gly Asp Pro Gly Glu Val Gly Gln Gln Gly Ser Pro Gly Pro
 245 250 255

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Thr Leu Leu Val Glu Pro Pro Asp Phe Cys Leu Tyr Lys Gly Glu Lys
 260 265 270
 Gly Ile Lys Gly Ile Pro Gly Met Val Gly Leu Pro Gly Pro Pro Gly
 275 280 285
 Arg Lys Gly Glu Ser Gly Ile Gly Ala Lys Gly Glu Lys Gly Ile Pro
 290 295 300
 Gly Phe Pro Gly Pro Arg Gly Asp Pro Gly Ser Tyr Gly Ser Pro Gly
 305 310 315 320
 Phe Pro Gly Leu Lys Gly Glu Leu Gly Leu Val Gly Asp Pro Gly Leu
 325 330 335
 Phe Gly Leu Ile Gly Pro Lys Gly Asp Pro Gly Asn Arg Gly His Pro
 340 345 350
 Gly Pro Pro Gly Val Leu Val Thr Pro Pro Leu Pro Leu Lys Gly Pro
 355 360 365
 Pro Gly Asp Pro Gly Phe Pro Gly Arg Tyr Gly Glu Thr Gly Asp Val
 370 375 380
 Gly Pro Pro Gly Pro Pro Gly Leu Leu Gly Arg Pro Gly Glu Ala Cys
 385 390 395 400
 Ala Gly Met Ile Gly Pro Pro Gly Pro Gln Gly Phe Pro Gly Leu Pro
 405 410 415
 Gly Leu Pro Gly Glu Ala Gly Ile Pro Gly Arg Pro Asp Ser Ala Pro
 420 425 430
 Gly Lys Pro Gly Lys Pro Gly Ser Pro Gly Leu Pro Gly Ala Pro Gly
 435 440 445
 Leu Gln Gly Leu Pro Gly Ser Ser Val Ile Tyr Cys Ser Val Gly Asn
 450 455 460
 Pro Gly Pro Gln Gly Ile Lys Gly Lys Val Gly Pro Pro Gly Gly Arg
 465 470 475 480
 Gly Pro Lys Gly Glu Lys Gly Asn Glu Gly Leu Cys Ala Cys Glu Pro
 485 490 495
 Gly Pro Met Gly Pro Pro Gly Pro Pro Gly Leu Pro Gly Arg Gln Gly
 500 505 510
 Ser Lys Gly Asp Leu Gly Leu Pro Gly Trp Leu Gly Thr Lys Gly Asp
 515 520 525
 Pro Gly Pro Pro Gly Ala Glu Gly Pro Pro Gly Leu Pro Gly Lys His
 530 535 540
 Gly Ala Ser Gly Pro Pro Gly Asn Lys Gly Ala Lys Gly Asp Met Val
 545 550 555 560
 Val Ser Arg Val Lys Gly His Lys Gly Glu Arg Gly Pro Asp Gly Pro
 565 570 575
 Pro Gly Phe Pro Gly Gln Pro Gly Ser His Gly Arg Asp Gly His Ala
 580 585 590
 Gly Glu Lys Gly Asp Pro Gly Pro Pro Gly Asp His Glu Asp Ala Thr
 595 600 605
 Pro Gly Gly Lys Gly Phe Pro Gly Pro Leu Gly Pro Pro Gly Lys Ala
 610 615 620
 Gly Pro Val Gly Pro Pro Gly Leu Gly Phe Pro Gly Pro Pro Gly Glu
 625 630 635 640
 Arg Gly His Pro Gly Val Pro Gly His Pro Gly Val Arg Gly Pro Asp
 645 650 655

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Gly	Leu	Lys	Gly	Gln	Lys	Gly	Asp	Thr	Ile	Ser	Cys	Asn	Val	Thr	Tyr
			660					665					670		
Pro	Gly	Arg	His	Gly	Pro	Pro	Gly	Phe	Asp	Gly	Pro	Pro	Gly	Pro	Lys
		675					680					685			
Gly	Phe	Pro	Gly	Pro	Gln	Gly	Ala	Pro	Gly	Leu	Ser	Gly	Ser	Asp	Gly
	690					695					700				
His	Lys	Gly	Arg	Pro	Gly	Thr	Pro	Gly	Thr	Ala	Glu	Ile	Pro	Gly	Pro
705					710					715					720
Pro	Gly	Phe	Arg	Gly	Asp	Met	Gly	Asp	Pro	Gly	Phe	Gly	Gly	Glu	Lys
			725						730					735	
Gly	Ser	Ser	Pro	Val	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Ser	Pro	Gly	Val
			740					745					750		
Asn	Gly	Gln	Lys	Gly	Ile	Pro	Gly	Asp	Pro	Ala	Phe	Gly	His	Leu	Gly
		755					760					765			
Pro	Pro	Gly	Lys	Arg	Gly	Leu	Ser	Gly	Val	Pro	Gly	Ile	Lys	Gly	Pro
	770					775					780				
Arg	Gly	Asp	Pro	Gly	Cys	Pro	Gly	Ala	Glu	Gly	Pro	Ala	Gly	Ile	Pro
785					790					795					800
Gly	Phe	Leu	Gly	Leu	Lys	Gly	Pro	Lys	Gly	Arg	Glu	Gly	His	Ala	Gly
			805						810					815	
Phe	Pro	Gly	Val	Pro	Gly	Pro	Pro	Gly	His	Ser	Cys	Glu	Arg	Gly	Ala
			820					825					830		
Pro	Gly	Ile	Pro	Gly	Gln	Pro	Gly	Leu	Pro	Gly	Tyr	Pro	Gly	Ser	Pro
		835					840					845			
Gly	Ala	Pro	Gly	Gly	Lys	Gly	Gln	Pro	Gly	Asp	Val	Gly	Pro	Pro	Gly
	850					855					860				
Pro	Ala	Gly	Met	Lys	Gly	Leu	Pro	Gly	Leu	Pro	Gly	Arg	Pro	Gly	Ala
865					870					875					880
His	Gly	Pro	Pro	Gly	Leu	Pro	Gly	Ile	Pro	Gly	Pro	Phe	Gly	Asp	Asp
			885						890					895	
Gly	Leu	Pro	Gly	Pro	Pro	Gly	Pro	Lys	Gly	Pro	Arg	Gly	Leu	Pro	Gly
			900					905					910		
Phe	Pro	Gly	Phe	Pro	Gly	Glu	Arg	Gly	Lys	Pro	Gly	Ala	Glu	Gly	Cys
		915					920					925			
Pro	Gly	Ala	Lys	Gly	Glu	Pro	Gly	Glu	Lys	Gly	Met	Ser	Gly	Leu	Pro
	930					935					940				
Gly	Asp	Arg	Gly	Leu	Arg	Gly	Ala	Lys	Gly	Ala	Ile	Gly	Pro	Pro	Gly
945					950					955					960
Asp	Glu	Gly	Glu	Met	Ala	Ile	Ile	Ser	Gln	Lys	Gly	Thr	Pro	Gly	Glu
			965						970					975	
Pro	Gly	Pro	Pro	Gly	Asp	Asp	Gly	Phe	Pro	Gly	Glu	Arg	Gly	Asp	Lys
			980					985					990		
Gly	Thr	Pro	Gly	Met	Gln	Gly	Arg	Arg	Gly	Glu	Pro	Gly	Arg	Tyr	Gly
		995					1000						1005		
Pro	Pro	Gly	Phe	His	Arg	Gly	Glu	Pro	Gly	Glu	Lys	Gly	Gln	Pro	
	1010					1015						1020			
Gly	Pro	Pro	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Ser	Thr	Gly	Leu	Arg	
	1025					1030						1035			
Gly	Phe	Ile	Gly	Phe	Pro	Gly	Leu	Pro	Gly	Asp	Gln	Gly	Glu	Pro	
	1040					1045					1050				
Gly	Ser	Pro	Gly	Pro	Pro	Gly	Phe	Ser	Gly	Ile	Asp	Gly	Ala	Arg	

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1055	1060	1065
Gly Pro Lys Gly Asn Lys Gly Asp Pro Ala Ser His Phe Gly Pro		
1070	1075	1080
Pro Gly Pro Lys Gly Glu Pro Gly Ser Pro Gly Cys Pro Gly His		
1085	1090	1095
Phe Gly Ala Ser Gly Glu Gln Gly Leu Pro Gly Ile Gln Gly Pro		
1100	1105	1110
Arg Gly Ser Pro Gly Arg Pro Gly Pro Pro Gly Ser Ser Gly Pro		
1115	1120	1125
Pro Gly Cys Pro Gly Asp His Gly Met Pro Gly Leu Arg Gly Gln		
1130	1135	1140
Pro Gly Glu Met Gly Asp Pro Gly Pro Arg Gly Leu Gln Gly Asp		
1145	1150	1155
Pro Gly Ile Pro Gly Pro Pro Gly Ile Lys Gly Pro Ser Gly Ser		
1160	1165	1170
Pro Gly Leu Asn Gly Leu His Gly Leu Lys Gly Gln Lys Gly Thr		
1175	1180	1185
Lys Gly Ala Ser Gly Leu His Asp Val Gly Pro Pro Gly Pro Val		
1190	1195	1200
Gly Ile Pro Gly Leu Lys Gly Glu Arg Gly Asp Pro Gly Ser Pro		
1205	1210	1215
Gly Ile Ser Pro Pro Gly Pro Arg Gly Lys Lys Gly Pro Pro Gly		
1220	1225	1230
Pro Pro Gly Ser Ser Gly Pro Pro Gly Pro Ala Gly Ala Thr Gly		
1235	1240	1245
Arg Ala Pro Lys Asp Ile Pro Asp Pro Gly Pro Pro Gly Asp Gln		
1250	1255	1260
Gly Pro Pro Gly Pro Asp Gly Pro Arg Gly Ala Pro Gly Pro Pro		
1265	1270	1275
Gly Leu Pro Gly Ser Val Asp Leu Leu Arg Gly Glu Pro Gly Asp		
1280	1285	1290
Cys Gly Leu Pro Gly Pro Pro Gly Pro Pro Gly Pro Pro Gly Pro		
1295	1300	1305
Pro Gly Tyr Lys Gly Phe Pro Gly Cys Asp Gly Lys Asp Gly Gln		
1310	1315	1320
Lys Gly Pro Val Gly Phe Pro Gly Pro Gln Gly Pro His Gly Phe		
1325	1330	1335
Pro Gly Pro Pro Gly Glu Lys Gly Leu Pro Gly Pro Pro Gly Arg		
1340	1345	1350
Lys Gly Pro Thr Gly Leu Pro Gly Pro Arg Gly Glu Pro Gly Pro		
1355	1360	1365
Pro Ala Asp Val Asp Asp Cys Pro Arg Ile Pro Gly Leu Pro Gly		
1370	1375	1380
Ala Pro Gly Met Arg Gly Pro Glu Gly Ala Met Gly Leu Pro Gly		
1385	1390	1395
Met Arg Gly Pro Ser Gly Pro Gly Cys Lys Gly Glu Pro Gly Leu		
1400	1405	1410
Asp Gly Arg Arg Gly Val Asp Gly Val Pro Gly Ser Pro Gly Pro		
1415	1420	1425
Pro Gly Arg Lys Gly Asp Thr Gly Glu Asp Gly Tyr Pro Gly Gly		
1430	1435	1440

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<210> SEQ ID NO 23
<211> LENGTH: 1685
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary COL4A5 amino acid sequence

<400> SEQUENCE: 23

Met Lys Leu Arg Gly Val Ser Leu Ala Ala Gly Leu Phe Leu Leu Ala
1             5             10             15

Leu Ser Leu Trp Gly Gln Pro Ala Glu Ala Ala Ala Cys Tyr Gly Cys
                20             25             30

Ser Pro Gly Ser Lys Cys Asp Cys Ser Gly Ile Lys Gly Glu Lys Gly
        35             40             45

Glu Arg Gly Phe Pro Gly Leu Glu Gly His Pro Gly Leu Pro Gly Phe
        50             55             60

Pro Gly Pro Glu Gly Pro Pro Gly Pro Arg Gly Gln Lys Gly Asp Asp
65             70             75             80

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Gly 85	Ile	Pro	Gly	Pro	Gly	Pro	Gly	Pro	Lys	Gly	Ile	Arg	Gly	Pro	Pro	Gly
Leu 100	Pro	Gly	Phe	Pro	Gly	Thr	Pro	Gly	Leu	Pro	Gly	Met	Pro	Gly	His	
Asp 115	Gly	Ala	Pro	Gly	Pro	Gln	Gly	Ile	Pro	Gly	Cys	Asn	Gly	Thr	Lys	
Gly 130	Glu	Arg	Gly	Phe	Pro	Gly	Ser	Pro	Gly	Phe	Pro	Gly	Leu	Gln	Gly	
Pro 145	Pro	Gly	Pro	Pro	Gly	Ile	Pro	Gly	Met	Lys	Gly	Glu	Pro	Gly	Ser	
Ile 165	Ile	Met	Ser	Ser	Leu	Pro	Gly	Pro	Lys	Gly	Asn	Pro	Gly	Tyr	Pro	
Gly 180	Pro	Pro	Gly	Ile	Gln	Gly	Leu	Pro	Gly	Pro	Thr	Gly	Ile	Pro	Gly	
Pro 195	Ile	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Leu	Met	Gly	Pro	Pro	Gly	Pro	
Pro 210	Gly	Leu	Pro	Gly	Pro	Lys	Gly	Asn	Met	Gly	Leu	Asn	Phe	Gln	Gly	
Pro 225	Lys	Gly	Glu	Lys	Gly	Glu	Gln	Gly	Leu	Gln	Gly	Pro	Pro	Gly	Pro	
Pro 245	Gly	Gln	Ile	Ser	Glu	Gln	Lys	Arg	Pro	Ile	Asp	Val	Glu	Phe	Gln	
Lys 260	Gly	Asp	Gln	Gly	Leu	Pro	Gly	Asp	Arg	Gly	Pro	Pro	Gly	Pro	Pro	
Gly 275	Ile	Arg	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Gly	Glu	Lys	Gly	Glu	Lys	
Gly 290	Glu	Gln	Gly	Glu	Pro	Gly	Lys	Arg	Gly	Lys	Pro	Gly	Lys	Asp	Gly	
Glu 305	Asn	Gly	Gln	Pro	Gly	Ile	Pro	Gly	Leu	Pro	Gly	Asp	Pro	Gly	Tyr	
Pro 325	Gly	Glu	Pro	Gly	Arg	Asp	Gly	Glu	Lys	Gly	Gln	Lys	Gly	Asp	Thr	
Gly 340	Pro	Pro	Gly	Pro	Pro	Gly	Leu	Val	Ile	Pro	Arg	Pro	Gly	Thr	Gly	
Ile 355	Thr	Ile	Gly	Glu	Lys	Gly	Asn	Ile	Gly	Leu	Pro	Gly	Leu	Pro	Gly	
Glu 370	Lys	Gly	Glu	Arg	Gly	Phe	Pro	Gly	Ile	Gln	Gly	Pro	Pro	Gly	Leu	
Pro 385	Gly	Pro	Pro	Gly	Ala	Ala	Val	Met	Gly	Pro	Pro	Gly	Pro	Pro	Gly	
Phe 405	Pro	Gly	Glu	Arg	Gly	Gln	Lys	Gly	Asp	Glu	Gly	Pro	Pro	Gly	Ile	
Ser 420	Ile	Pro	Gly	Pro	Pro	Gly	Leu	Asp	Gly	Gln	Pro	Gly	Ala	Pro	Gly	
Leu 435	Pro	Gly	Pro	Pro	Gly	Pro	Ala	Gly	Pro	His	Ile	Pro	Pro	Ser	Asp	
Glu 450	Ile	Cys	Glu	Pro	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Ser	Pro	Gly	Asp	
Lys 465	Gly	Leu	Gln	Gly	Glu	Gln	Gly	Val	Lys	Gly	Asp	Lys	Gly	Asp	Thr	

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Cys	Phe	Asn	Cys	Ile	Gly	Thr	Gly	Ile	Ser	Gly	Pro	Pro	Gly	Gln	Pro	485	490	495
Gly	Leu	Pro	Gly	Leu	Pro	Gly	Pro	Pro	Gly	Ser	Leu	Gly	Phe	Pro	Gly	500	505	510
Gln	Lys	Gly	Glu	Lys	Gly	Gln	Ala	Gly	Ala	Thr	Gly	Pro	Lys	Gly	Leu	515	520	525
Pro	Gly	Ile	Pro	Gly	Ala	Pro	Gly	Ala	Pro	Gly	Phe	Pro	Gly	Ser	Lys	530	535	540
Gly	Glu	Pro	Gly	Asp	Ile	Leu	Thr	Phe	Pro	Gly	Met	Lys	Gly	Asp	Lys	545	550	555
Gly	Glu	Leu	Gly	Ser	Pro	Gly	Ala	Pro	Gly	Leu	Pro	Gly	Leu	Pro	Gly	565	570	575
Thr	Pro	Gly	Gln	Asp	Gly	Leu	Pro	Gly	Leu	Pro	Gly	Pro	Lys	Gly	Glu	580	585	590
Pro	Gly	Gly	Ile	Thr	Phe	Lys	Gly	Glu	Arg	Gly	Pro	Pro	Gly	Asn	Pro	595	600	605
Gly	Leu	Pro	Gly	Leu	Pro	Gly	Asn	Ile	Gly	Pro	Met	Gly	Pro	Pro	Gly	610	615	620
Phe	Gly	Pro	Pro	Gly	Pro	Val	Gly	Glu	Lys	Gly	Ile	Gln	Gly	Val	Ala	625	630	635
Gly	Asn	Pro	Gly	Gln	Pro	Gly	Ile	Pro	Gly	Pro	Lys	Gly	Asp	Pro	Gly	645	650	655
Gln	Thr	Ile	Thr	Gln	Pro	Gly	Lys	Pro	Gly	Leu	Pro	Gly	Asn	Pro	Gly	660	665	670
Arg	Asp	Gly	Asp	Val	Gly	Leu	Pro	Gly	Asp	Pro	Gly	Leu	Pro	Gly	Gln	675	680	685
Pro	Gly	Leu	Pro	Gly	Ile	Pro	Gly	Ser	Lys	Gly	Glu	Pro	Gly	Ile	Pro	690	695	700
Gly	Ile	Gly	Leu	Pro	Gly	Pro	Pro	Gly	Pro	Lys	Gly	Phe	Pro	Gly	Ile	705	710	715
Pro	Gly	Pro	Pro	Gly	Ala	Pro	Gly	Thr	Pro	Gly	Arg	Ile	Gly	Leu	Glu	725	730	735
Gly	Pro	Pro	Gly	Pro	Pro	Gly	Phe	Pro	Gly	Pro	Lys	Gly	Glu	Pro	Gly	740	745	750
Phe	Ala	Leu	Pro	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Leu	Pro	Gly	Phe	Lys	755	760	765
Gly	Ala	Leu	Gly	Pro	Lys	Gly	Asp	Arg	Gly	Phe	Pro	Gly	Pro	Pro	Gly	770	775	780
Pro	Pro	Gly	Arg	Thr	Gly	Leu	Asp	Gly	Leu	Pro	Gly	Pro	Lys	Gly	Asp	785	790	795
Val	Gly	Pro	Asn	Gly	Gln	Pro	Gly	Pro	Met	Gly	Pro	Pro	Gly	Leu	Pro	805	810	815
Gly	Ile	Gly	Val	Gln	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Ile	Pro	Gly	Pro	820	825	830
Ile	Gly	Gln	Pro	Gly	Leu	His	Gly	Ile	Pro	Gly	Glu	Lys	Gly	Asp	Pro	835	840	845
Gly	Pro	Pro	Gly	Leu	Asp	Val	Pro	Gly	Pro	Pro	Gly	Glu	Arg	Gly	Ser	850	855	860
Pro	Gly	Ile	Pro	Gly	Ala	Pro	Gly	Pro	Ile	Gly	Pro	Pro	Gly	Ser	Pro	865	870	875
Gly	Leu	Pro	Gly	Lys	Ala	Gly	Ala	Ser	Gly	Phe	Pro	Gly	Thr	Lys	Gly			

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885						890						895			
Glu	Met	Gly	Met	Met	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Pro	Leu	Gly	Ile
900						905						910			
Pro	Gly	Arg	Ser	Gly	Val	Pro	Gly	Leu	Lys	Gly	Asp	Asp	Gly	Leu	Gln
915						920						925			
Gly	Gln	Pro	Gly	Leu	Pro	Gly	Pro	Thr	Gly	Glu	Lys	Gly	Ser	Lys	Gly
930						935						940			
Glu	Pro	Gly	Leu	Pro	Gly	Pro	Pro	Gly	Pro	Met	Asp	Pro	Asn	Leu	Leu
945						950						955			
Gly	Ser	Lys	Gly	Glu	Lys	Gly	Glu	Pro	Gly	Leu	Pro	Gly	Ile	Pro	Gly
965						970						975			
Val	Ser	Gly	Pro	Lys	Gly	Tyr	Gln	Gly	Leu	Pro	Gly	Asp	Pro	Gly	Gln
980						985						990			
Pro	Gly	Leu	Ser	Gly	Gln	Pro	Gly	Leu	Pro	Gly	Pro	Pro	Gly	Pro	Lys
995						1000						1005			
Gly	Asn	Pro	Gly	Leu	Pro	Gly	Gln	Pro	Gly	Leu	Ile	Gly	Pro	Pro	
1010						1015						1020			
Gly	Leu	Lys	Gly	Thr	Ile	Gly	Asp	Met	Gly	Phe	Pro	Gly	Pro	Gln	
1025						1030						1035			
Gly	Val	Glu	Gly	Pro	Pro	Gly	Pro	Ser	Gly	Val	Pro	Gly	Gln	Pro	
1040						1045						1050			
Gly	Ser	Pro	Gly	Leu	Pro	Gly	Gln	Lys	Gly	Asp	Lys	Gly	Asp	Pro	
1055						1060						1065			
Gly	Ile	Ser	Ser	Ile	Gly	Leu	Pro	Gly	Leu	Pro	Gly	Pro	Lys	Gly	
1070						1075						1080			
Glu	Pro	Gly	Leu	Pro	Gly	Tyr	Pro	Gly	Asn	Pro	Gly	Ile	Lys	Gly	
1085						1090						1095			
Ser	Val	Gly	Asp	Pro	Gly	Leu	Pro	Gly	Leu	Pro	Gly	Thr	Pro	Gly	
1100						1105						1110			
Ala	Lys	Gly	Gln	Pro	Gly	Leu	Pro	Gly	Phe	Pro	Gly	Thr	Pro	Gly	
1115						1120						1125			
Pro	Pro	Gly	Pro	Lys	Gly	Ile	Ser	Gly	Pro	Pro	Gly	Asn	Pro	Gly	
1130						1135						1140			
Leu	Pro	Gly	Glu	Pro	Gly	Pro	Val	Gly	Gly	Gly	Gly	His	Pro	Gly	
1145						1150						1155			
Gln	Pro	Gly	Pro	Pro	Gly	Glu	Lys	Gly	Lys	Pro	Gly	Gln	Asp	Gly	
1160						1165						1170			
Ile	Pro	Gly	Pro	Ala	Gly	Gln	Lys	Gly	Glu	Pro	Gly	Gln	Pro	Gly	
1175						1180						1185			
Phe	Gly	Asn	Pro	Gly	Pro	Pro	Gly	Leu	Pro	Gly	Leu	Ser	Gly	Gln	
1190						1195						1200			
Lys	Gly	Asp	Gly	Gly	Leu	Pro	Gly	Ile	Pro	Gly	Asn	Pro	Gly	Leu	
1205						1210						1215			
Pro	Gly	Pro	Lys	Gly	Glu	Pro	Gly	Phe	His	Gly	Phe	Pro	Gly	Val	
1220						1225						1230			
Gln	Gly	Pro	Pro	Gly	Pro	Pro	Gly	Ser	Pro	Gly	Pro	Ala	Leu	Glu	
1235						1240						1245			
Gly	Pro	Lys	Gly	Asn	Pro	Gly	Pro	Gln	Gly	Pro	Pro	Gly	Arg	Pro	
1250						1255						1260			
Gly	Leu	Pro	Gly	Pro	Glu	Gly	Pro	Pro	Gly	Leu	Pro	Gly	Asn	Gly	
1265						1270						1275			

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Gly Ile	Lys Gly Glu Lys	Gly Asn Pro Gly Gln Pro	Gly Leu Pro
1280		1285	1290
Gly Leu	Pro Gly Leu Lys	Gly Asp Gln Gly Pro Pro	Gly Leu Gln
1295		1300	1305
Gly Asn	Pro Gly Arg Pro	Gly Leu Asn Gly Met Lys	Gly Asp Pro
1310		1315	1320
Gly Leu	Pro Gly Val Pro	Gly Phe Pro Gly Met Lys	Gly Pro Ser
1325		1330	1335
Gly Val	Pro Gly Ser Ala	Gly Pro Glu Gly Glu Pro	Gly Leu Ile
1340		1345	1350
Gly Pro	Pro Gly Pro Pro	Gly Leu Pro Gly Pro Ser	Gly Gln Ser
1355		1360	1365
Ile Ile	Ile Lys Gly Asp Ala	Gly Pro Pro Gly Ile	Pro Gly Gln
1370		1375	1380
Pro Gly	Leu Lys Gly Leu Pro	Gly Pro Gln Gly Pro	Gln Gly Leu
1385		1390	1395
Pro Gly	Pro Thr Gly Pro	Pro Gly Asp Pro Gly Arg	Asn Gly Leu
1400		1405	1410
Pro Gly	Phe Asp Gly Ala	Gly Arg Lys Gly Asp	Pro Gly Leu
1415		1420	1425
Pro Gly	Gln Pro Gly Thr	Arg Gly Leu Asp Gly Pro	Pro Gly Pro
1430		1435	1440
Asp Gly	Leu Gln Gly Pro	Pro Gly Pro Pro Gly Thr	Ser Ser Val
1445		1450	1455
Ala His	Gly Phe Leu Ile Thr	Arg His Ser Gln Thr	Thr Asp Ala
1460		1465	1470
Pro Gln	Cys Pro Gln Gly Thr	Leu Gln Val Tyr Glu	Gly Phe Ser
1475		1480	1485
Leu Leu	Tyr Val Gln Gly	Asn Lys Arg Ala His	Gly Gln Asp Leu
1490		1495	1500
Gly Thr	Ala Gly Ser Cys Leu	Arg Arg Phe Ser Thr	Met Pro Phe
1505		1510	1515
Met Phe	Cys Asn Ile Asn Asn	Val Cys Asn Phe Ala	Ser Arg Asn
1520		1525	1530
Asp Tyr	Ser Tyr Trp Leu Ser	Thr Pro Glu Pro Met	Pro Met Ser
1535		1540	1545
Met Gln	Pro Leu Lys Gly	Gln Ser Ile Gln Pro Phe	Ile Ser Arg
1550		1555	1560
Cys Ala	Val Cys Glu Ala Pro	Ala Val Val Ile Ala	Val His Ser
1565		1570	1575
Gln Thr	Ile Gln Ile Pro His	Cys Pro Gln Gly Trp	Asp Ser Leu
1580		1585	1590
Trp Ile	Gly Tyr Ser Phe Met	Met His Thr Ser Ala	Gly Ala Glu
1595		1600	1605
Gly Ser	Gly Gln Ala Leu Ala	Ser Pro Gly Ser Cys	Leu Glu Glu
1610		1615	1620
Phe Arg	Ser Ala Pro Phe Ile	Glu Cys His Gly Arg	Gly Thr Cys
1625		1630	1635
Asn Tyr	Tyr Ala Asn Ser Tyr	Ser Phe Trp Leu Ala	Thr Val Asp
1640		1645	1650

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Val	Ser	Asp	Met	Phe	Ser	Lys	Pro	Gln	Ser	Glu	Thr	Leu	Lys	Ala
1655						1660					1665			

Gly	Asp	Leu	Arg	Thr	Arg	Ile	Ser	Arg	Cys	Gln	Val	Cys	Met	Lys
1670						1675					1680			

Arg	Thr
1685	

<210> SEQ ID NO 24

<211> LENGTH: 5013

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary COL4A3 transgene sequence

<400> SEQUENCE: 24

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<210> SEQ ID NO 25

<211> LENGTH: 5073

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary COL4A4 transgene sequence

<400> SEQUENCE: 25

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<210> SEQ ID NO 26

<211> LENGTH: 5058

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary COL4A5 transgene sequence

<400> SEQUENCE: 26

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gaaagaggtc cccctgggaa cccagggtta ccaggcctcc cagggaatat agggcctatg	1860
ggtccccctg gtttcggccc tccaggccca gtaggtgaaa aaggcataca aggtgtggca	1920
ggaaatccag gccagccagg aataccaggc ctaaaagggg atccaggtea gactataacc	1980
cagccgggga agcctggctt gcctggtaac ccaggcagag atggtgatgt aggtcttcca	2040
ggtgacctg gacttccagg gcaaccaggc ttgccaggga tacctggtag caaaggagaa	2100
ccaggtatcc ctggaattgg gcttctctgga ccacctggtc ccaaaggctt tctggaatt	2160
ccaggacctc caggagcacc tgggacacct ggaagaattg gtctagaagg cctctctggg	2220
ccaccggct ttccaggacc aaagggtgaa ccaggatttg cattacctgg gccacctggg	2280
ccaccaggac ttccagggtt caaaggagca cttggtccaa aaggtagatg tggtttccca	2340
ggacctccgg gtctccagg acgcactggc ttagatgggc tccctggacc aaaaggtag	2400

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gttgaccacaa atggacaacc tggaccaatg ggacctcctg ggctgccagg aataggtgtt	2460
cagggaccac caggaccacc agggattcct gggccaatag gtcaacctgg ttacatgga	2520
ataccaggag agaaggggga tccaggacct cctggacttg atgttcagg acccccagg	2580
gaaagaggca gtccagggat ccccgagca cctggctcta taggacctcc aggatcacca	2640
gggttccag gaaaagcagg tgctctgga ttccaggta ccaaaggta aatgggtatg	2700
atgggacctc caggcccacc aggacctttg ggaattcctg gcaggagtgg tgtacctggt	2760
cttaagggtg atgatggctt gcagggtcag ccaggacttc ctggccctac aggagaaaaa	2820
ggtagtaaa gagagcctgg ccttcaggc cctcctggac caatggatcc aaatcttctg	2880
ggctcaaaag gagagaagg ggaacctggc ttaccaggta tacctggagt ttcagggcc	2940
aaaggttatc aggggttgc tggagacca gggcaacctg gactgagtgg acaacctgga	3000
ttaccaggac caccaggctc caaaggtaac cctggtctcc ctggacagcc aggtcttata	3060
ggacctcctg gacttaaaag aaccatcggg gatatgggtt ttccagggcc tcagggtgtg	3120
gaagggctc ctggaccttc tggagtctc ggacaacctg gctcccagg attacctgga	3180
cagaaaggcg acaaaggta tcttggtatt tcaagcattg gtcttcagg tcttctggt	3240
ccaaagggtg agcctgtct gcttggtatc ccagggaacc ctggtatcaa aggttctgtg	3300
ggagatcctg gtttgcctg attacaggga acccctggag caaaaggaca accaggcctt	3360
cctggattcc caggaacccc aggcctcct ggaccaaag gtattagtgg ccctcctggg	3420
aaccccgcc ttccaggaga acctggtcct gtaggtggtg gaggtcatcc tgggcaacca	3480
gggctccag gcgaaaaagg caaacccgt caagatggtt ttctggacc agctggacag	3540
aagggtgaac caggtcaacc aggtttgga aaccaggac cccctggact tccaggactt	3600
tctggccaaa aggggtgagg aggattacct gggattccag gaaatcctg ccttccagg	3660
ccaaaggcg aaccaggctt tcacggttc cctggtgtg agggtcctcc aggcctcct	3720
ggttctccg gtccagctc ggaaggacct aaaggcaacc ctgggcccc aggtcctcct	3780
gggagaccag gtctaccagg tccagaagg cctccaggtc tccctggaaa tggaggtatt	3840
aaaggagaga agggaaatc aggccaacct gggctacctg gcttgctgg ttgaaagga	3900
gatcaaggac caccaggact ccagggtaat cctggccggc cgggtctcaa tggaatgaaa	3960
ggagatcctg gtctcctcgt tgttcaggga ttccaggca tgaaggacc cagtggagta	4020
cctggatcag ctggccctga gggggaaccg ggacttattg gtctccagg tctcctgga	4080
ttacctggtc cttcaggaca gattatcata attaaaggag atgctggtc tccaggaatc	4140
cctggccagc ctgggctaaa ggtctacca ggacccaag gacctcaagg cttaccagg	4200
ccaactggcc ctccaggaga tcttggaagc aatggactcc ctggcttga tgggtcagga	4260
gggcgcaaa gagaccagg tctgccagga cagccaggta cccgtggtt ggatggtccc	4320
cctggtccag atggattgca aggtccccc ggtcccctg gaacctcctc tgtgcacat	4380
ggatttctta ttacacgcca cagccagaca acggatgcac cacaatgccc acaggaaca	4440
cttcaggctc atgaaggctt ttctctcctg tatgtacaag gaaataaaa agcccacggt	4500
caagacttgg ggacggctg cagctgctt cgtcgttta gtaccatgcc ttcatgttc	4560
tgcaacatca ataattgtt caactttgct tcaagaaatg actattctta ctggctctc	4620
acccagagc catgccaat gagcatgcaa cccctaaagg gccagagcat ccagccattc	4680

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attagtcgat gtgcagtatg tgaagctcca gctgtgggtga tgcagttca cagtcagacg 4740
atccagattc cccattgtcc tcagggatgg gattctctgt ggattgggta ttccttcacg 4800
atgcatacaa gtgcaggggc agaaggtcca ggtcaagccc tagcctcccc tggttcctgc 4860
ttggaagagt ttcgttcagc tcccttcacg gaatgtcatg ggaggggtac ctgtaactac 4920
tatgccaaact cctacagctt ttggctggca actgtagatg tgtcagacat gttcagtaaa 4980
cctcagtcag aaacgctgaa agcaggagac ttgaggacac gaattagccg atgtcaagtg 5040
tgcatgaaga ggacataa 5058

```

```

<210> SEQ ID NO 27
<211> LENGTH: 186
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary NPHS1 enhancer

```

```

<400> SEQUENCE: 27

```

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ctgctgagct gggagaccac cttgatctga cttctcccat cttccagcc taagccaggc 60
cctgggggtca cggaggctgg ggaggcaccg aggaacgcgc ctggcatgtg ctgacagggg 120
attttatgct ccagctgggc cagctgggag gagcctgctg ggcagaggcc agagctgggg 180
gctctg 186

```

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<210> SEQ ID NO 28
<211> LENGTH: 13
<212> TYPE: DNA
<213> ORGANISM: Exemplary Kozak sequence

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<400> SEQUENCE: 28

```

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gccgccacca ugg 13

```

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<210> SEQ ID NO 29
<211> LENGTH: 589
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary WPRE

```

```

<400> SEQUENCE: 29

```

```

aatcaacctc tggattacaa aatttgtgaa agattgactg gtattcttaa ctatgttgct 60
ccttttacgc tatgtggata cgtgcttta atgcctttgt atcatgctat tgettcccg 120
atggccttca ttttctctc cttgtataaa tcctgggtgc tgtctcttta tgaggagt 180
tggcccggtg tcaggcaacg tggcgtgggtg tgcactgtgt ttgctgacgc aacccccact 240
ggttggggca ttgccaccac ctgtcagctc ctttcgggga ctttcgcttt cccctccct 300
attgccacgg cggaactcat cgcgcctgc cttgcccgt gctggacagg ggctcggtg 360
ttgggcactg acaattccgt ggtgtgtcg gggaaatcat cgtcctttcc ttggctgctc 420
gcctgtgttg ccacctggat tctgcgcggg acgtccttct gctacgtccc ttcggccctc 480
aatccagcgg accttccttc ccgcggcctg ctgcgggctc tgccggcctct tccgcgtctt 540
cgcccttcgc ctcagacgag tcggatctcc ctttgggcgg cctcccccgc 589

```

```

<210> SEQ ID NO 30
<211> LENGTH: 135

```

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<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary SV40pA signal sequence

<400> SEQUENCE: 30

aacttggttta ttgcagctta taatgggttac aaataaagca atagcatcac aaatttcaca      60
aataaagcat ttttttcact gcattctagt tgtggtttgt ccaaactcat caatgtatct      120
tatcatgtct ggatc                                          135

<210> SEQ ID NO 31
<211> LENGTH: 225
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary bGH poly(A) signal sequence

<400> SEQUENCE: 31

ctgtgccttc tagttgccag ccattctgttg ttgcccctc cccgtgcct tccttgacct      60
tggaagggtgc cactcccact gtcctttcct aataaaatga ggaaattgca tcgcattgtc      120
tgagtaggtg tcattctatt ctgggggggtg ggggtggggca ggacagcaag ggggaggatt      180
gggaagacaa tagcaggcat gctgggggatg cgggtgggctc tatgg                                          225

<210> SEQ ID NO 32
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary soluble neuropilin-1 polyadenylation
signal

<400> SEQUENCE: 32

aaataaaata cgaaatg                                          17

<210> SEQ ID NO 33
<211> LENGTH: 735
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary AAV3B VP1 capsid protein

<400> SEQUENCE: 33

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
1          5          10          15

Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Val Pro Gln Pro
20          25          30

Lys Ala Asn Gln Gln His Gln Asp Asn Arg Arg Gly Leu Val Leu Pro
35          40          45

Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
50          55          60

Val Asn Glu Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
65          70          75          80

Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
85          90          95

Asp Ala Glu Phe Gln Glu Arg Leu Gln Glu Asp Thr Ser Phe Gly Gly
100         105         110

Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Ile Leu Glu Pro

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115	120	125
Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg		
130	135	140
Pro Val Asp Gln Ser Pro Gln Glu Pro Asp Ser Ser Ser Gly Val Gly		
145	150	155
Lys Ser Gly Lys Gln Pro Ala Arg Lys Arg Leu Asn Phe Gly Gln Thr		
165	170	175
Gly Asp Ser Glu Ser Val Pro Asp Pro Gln Pro Leu Gly Glu Pro Pro		
180	185	190
Ala Ala Pro Thr Ser Leu Gly Ser Asn Thr Met Ala Ser Gly Gly Gly		
195	200	205
Ala Pro Met Ala Asp Asn Asn Glu Gly Ala Asp Gly Val Gly Asn Ser		
210	215	220
Ser Gly Asn Trp His Cys Asp Ser Gln Trp Leu Gly Asp Arg Val Ile		
225	230	235
Thr Thr Ser Thr Arg Thr Trp Ala Leu Pro Thr Tyr Asn Asn His Leu		
245	250	255
Tyr Lys Gln Ile Ser Ser Gln Ser Gly Ala Ser Asn Asp Asn His Tyr		
260	265	270
Phe Gly Tyr Ser Thr Pro Trp Gly Tyr Phe Asp Phe Asn Arg Phe His		
275	280	285
Cys His Phe Ser Pro Arg Asp Trp Gln Arg Leu Ile Asn Asn Asn Trp		
290	295	300
Gly Phe Arg Pro Lys Lys Leu Ser Phe Lys Leu Phe Asn Ile Gln Val		
305	310	315
Lys Glu Val Thr Gln Asn Asp Gly Thr Thr Thr Ile Ala Asn Asn Leu		
325	330	335
Thr Ser Thr Val Gln Val Phe Thr Asp Ser Glu Tyr Gln Leu Pro Tyr		
340	345	350
Val Leu Gly Ser Ala His Gln Gly Cys Leu Pro Pro Phe Pro Ala Asp		
355	360	365
Val Phe Met Val Pro Gln Tyr Gly Tyr Leu Thr Leu Asn Asn Gly Ser		
370	375	380
Gln Ala Val Gly Arg Ser Ser Phe Tyr Cys Leu Glu Tyr Phe Pro Ser		
385	390	395
Gln Met Leu Arg Thr Gly Asn Asn Phe Gln Phe Ser Tyr Thr Phe Glu		
405	410	415
Asp Val Pro Phe His Ser Ser Tyr Ala His Ser Gln Ser Leu Asp Arg		
420	425	430
Leu Met Asn Pro Leu Ile Asp Gln Tyr Leu Tyr Tyr Leu Asn Arg Thr		
435	440	445
Gln Gly Thr Thr Ser Gly Thr Thr Asn Gln Ser Arg Leu Leu Phe Ser		
450	455	460
Gln Ala Gly Pro Gln Ser Met Ser Leu Gln Ala Arg Asn Trp Leu Pro		
465	470	475
Gly Pro Cys Tyr Arg Gln Gln Arg Leu Ser Lys Thr Ala Asn Asp Asn		
485	490	495
Asn Asn Ser Asn Phe Pro Trp Thr Ala Ala Ser Lys Tyr His Leu Asn		
500	505	510
Gly Arg Asp Ser Leu Val Asn Pro Gly Pro Ala Met Ala Ser His Lys		
515	520	525

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Asp Asp Glu Glu Lys Phe Phe Pro Met His Gly Asn Leu Ile Phe Gly
 530 535 540
 Lys Glu Gly Thr Thr Ala Ser Asn Ala Glu Leu Asp Asn Val Met Ile
 545 550 555 560
 Thr Asp Glu Glu Glu Ile Arg Thr Thr Asn Pro Val Ala Thr Glu Gln
 565 570 575
 Tyr Gly Thr Val Ala Asn Asn Leu Gln Ser Ser Asn Thr Ala Pro Thr
 580 585 590
 Thr Arg Thr Val Asn Asp Gln Gly Ala Leu Pro Gly Met Val Trp Gln
 595 600 605
 Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
 610 615 620
 Thr Asp Gly His Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Leu
 625 630 635 640
 Lys His Pro Pro Pro Gln Ile Met Ile Lys Asn Thr Pro Val Pro Ala
 645 650 655
 Asn Pro Pro Thr Thr Phe Ser Pro Ala Lys Phe Ala Ser Phe Ile Thr
 660 665 670
 Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
 675 680 685
 Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
 690 695 700
 Tyr Asn Lys Ser Val Asn Val Asp Phe Thr Val Asp Thr Asn Gly Val
 705 710 715 720
 Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn
 725 730 735

<210> SEQ ID NO 34

<211> LENGTH: 736

<212> TYPE: PRT

<213> ORGANISM: Exemplary LK03 VP1 capsid protein

<400> SEQUENCE: 34

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Gln Pro Gly Ala Pro Lys Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Asp Gln Ser Pro Gln Glu Pro Asp Ser Ser Ser Gly Val Gly

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145					150						155					160
Lys	Ser	Gly	Lys	Gln	Pro	Ala	Arg	Lys	Arg	Leu	Asn	Phe	Gly	Gln	Thr	
				165					170					175		
Gly	Asp	Ser	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Leu	Gly	Glu	Pro	Pro	
			180					185					190			
Ala	Ala	Pro	Thr	Ser	Leu	Gly	Ser	Asn	Thr	Met	Ala	Ser	Gly	Gly	Gly	
		195					200					205				
Ala	Pro	Met	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Asn	Ser	
	210					215					220					
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225					230					235					240	
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250					255		
Tyr	Lys	Gln	Ile	Ser	Ser	Gln	Ser	Gly	Ala	Ser	Asn	Asp	Asn	His	Tyr	
		260						265					270			
Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	Phe	His	
	275						280					285				
Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	Asn	Trp	
	290					295					300					
Gly	Phe	Arg	Pro	Lys	Lys	Leu	Ser	Phe	Lys	Leu	Phe	Asn	Ile	Gln	Val	
305					310					315					320	
Lys	Glu	Val	Thr	Gln	Asn	Asp	Gly	Thr	Thr	Thr	Ile	Ala	Asn	Asn	Leu	
				325					330					335		
Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Glu	Tyr	Gln	Leu	Pro	Tyr	
			340					345					350			
Val	Leu	Gly	Ser	Ala	His	Gln	Gly	Cys	Leu	Pro	Pro	Phe	Pro	Ala	Asp	
	355						360					365				
Val	Phe	Met	Val	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asn	Gly	Ser	
	370					375					380					
Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	Pro	Ser	
385					390					395					400	
Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Thr	Phe	Glu	
				405					410					415		
Asp	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	Asp	Arg	
			420					425					430			
Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Asn	Arg	Thr	
	435						440					445				
Gln	Gly	Thr	Thr	Ser	Gly	Thr	Thr	Asn	Gln	Ser	Arg	Leu	Leu	Phe	Ser	
	450					455					460					
Gln	Ala	Gly	Pro	Gln	Ser	Met	Ser	Leu	Gln	Ala	Arg	Asn	Trp	Leu	Pro	
465					470					475					480	
Gly	Pro	Cys	Tyr	Arg	Gln	Gln	Arg	Leu	Ser	Lys	Thr	Ala	Asn	Asp	Asn	
				485					490					495		
Asn	Asn	Ser	Asn	Phe	Pro	Trp	Thr	Ala	Ala	Ser	Lys	Tyr	His	Leu	Asn	
			500					505					510			
Gly	Arg	Asp	Ser	Leu	Val	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
		515					520					525				
Asp	Asp	Glu	Glu	Lys	Phe	Phe	Pro	Met	His	Gly	Asn	Leu	Ile	Phe	Gly	
	530					535					540					
Lys	Glu	Gly	Thr	Thr	Ala	Ser	Asn	Ala	Glu	Leu	Asp	Asn	Val	Met	Ile	
545					550				555					560		

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Thr Asp Glu Glu Glu Ile Arg Thr Thr Asn Pro Val Ala Thr Glu Gln
 565 570 575
 Tyr Gly Thr Val Ala Asn Asn Leu Gln Ser Ser Asn Thr Ala Pro Thr
 580 585 590
 Thr Arg Thr Val Asn Asp Gln Gly Ala Leu Pro Gly Met Val Trp Gln
 595 600 605
 Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
 610 615 620
 Thr Asp Gly His Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Leu
 625 630 635 640
 Lys His Pro Pro Pro Gln Ile Met Ile Lys Asn Thr Pro Val Pro Ala
 645 650 655
 Asn Pro Pro Thr Thr Phe Ser Pro Ala Lys Phe Ala Ser Phe Ile Thr
 660 665 670
 Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
 675 680 685
 Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
 690 695 700
 Tyr Asn Lys Ser Val Asn Val Asp Phe Thr Val Asp Thr Asn Gly Val
 705 710 715 720
 Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Pro Leu
 725 730 735

<210> SEQ ID NO 35
 <211> LENGTH: 736
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary AAV9 VP1 capsid protein

<400> SEQUENCE: 35

Met Ala Ala Asp Gly Tyr Leu Pro Asp Trp Leu Glu Asp Asn Leu Ser
 1 5 10 15
 Glu Gly Ile Arg Glu Trp Trp Ala Leu Lys Pro Gly Ala Pro Gln Pro
 20 25 30
 Lys Ala Asn Gln Gln His Gln Asp Asn Ala Arg Gly Leu Val Leu Pro
 35 40 45
 Gly Tyr Lys Tyr Leu Gly Pro Gly Asn Gly Leu Asp Lys Gly Glu Pro
 50 55 60
 Val Asn Ala Ala Asp Ala Ala Ala Leu Glu His Asp Lys Ala Tyr Asp
 65 70 75 80
 Gln Gln Leu Lys Ala Gly Asp Asn Pro Tyr Leu Lys Tyr Asn His Ala
 85 90 95
 Asp Ala Glu Phe Gln Glu Arg Leu Lys Glu Asp Thr Ser Phe Gly Gly
 100 105 110
 Asn Leu Gly Arg Ala Val Phe Gln Ala Lys Lys Arg Leu Leu Glu Pro
 115 120 125
 Leu Gly Leu Val Glu Glu Ala Ala Lys Thr Ala Pro Gly Lys Lys Arg
 130 135 140
 Pro Val Glu Gln Ser Pro Gln Glu Pro Asp Ser Ser Ala Gly Ile Gly
 145 150 155 160
 Lys Ser Gly Ala Gln Pro Ala Lys Lys Arg Leu Asn Phe Gly Gln Thr
 165 170 175

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Gly	Asp	Thr	Glu	Ser	Val	Pro	Asp	Pro	Gln	Pro	Ile	Gly	Glu	Pro	Pro	
	180							185					190			
Ala	Ala	Pro	Ser	Gly	Val	Gly	Ser	Leu	Thr	Met	Ala	Ser	Gly	Gly	Gly	
	195						200					205				
Ala	Pro	Val	Ala	Asp	Asn	Asn	Glu	Gly	Ala	Asp	Gly	Val	Gly	Ser	Ser	
	210				215						220					
Ser	Gly	Asn	Trp	His	Cys	Asp	Ser	Gln	Trp	Leu	Gly	Asp	Arg	Val	Ile	
225				230						235					240	
Thr	Thr	Ser	Thr	Arg	Thr	Trp	Ala	Leu	Pro	Thr	Tyr	Asn	Asn	His	Leu	
				245					250					255		
Tyr	Lys	Gln	Ile	Ser	Asn	Ser	Thr	Ser	Gly	Gly	Ser	Ser	Asn	Asp	Asn	
		260						265					270			
Ala	Tyr	Phe	Gly	Tyr	Ser	Thr	Pro	Trp	Gly	Tyr	Phe	Asp	Phe	Asn	Arg	
	275						280					285				
Phe	His	Cys	His	Phe	Ser	Pro	Arg	Asp	Trp	Gln	Arg	Leu	Ile	Asn	Asn	
	290					295					300					
Asn	Trp	Gly	Phe	Arg	Pro	Lys	Arg	Leu	Asn	Phe	Lys	Leu	Phe	Asn	Ile	
305					310					315					320	
Gln	Val	Lys	Glu	Val	Thr	Asp	Asn	Asn	Gly	Val	Lys	Thr	Ile	Ala	Asn	
			325						330					335		
Asn	Leu	Thr	Ser	Thr	Val	Gln	Val	Phe	Thr	Asp	Ser	Asp	Tyr	Gln	Leu	
		340						345					350			
Pro	Tyr	Val	Leu	Gly	Ser	Ala	His	Glu	Gly	Cys	Leu	Pro	Pro	Phe	Pro	
		355				360						365				
Ala	Asp	Val	Phe	Met	Ile	Pro	Gln	Tyr	Gly	Tyr	Leu	Thr	Leu	Asn	Asp	
	370					375					380					
Gly	Ser	Gln	Ala	Val	Gly	Arg	Ser	Ser	Phe	Tyr	Cys	Leu	Glu	Tyr	Phe	
385				390						395					400	
Pro	Ser	Gln	Met	Leu	Arg	Thr	Gly	Asn	Asn	Phe	Gln	Phe	Ser	Tyr	Glu	
			405					410						415		
Phe	Glu	Asn	Val	Pro	Phe	His	Ser	Ser	Tyr	Ala	His	Ser	Gln	Ser	Leu	
		420					425						430			
Asp	Arg	Leu	Met	Asn	Pro	Leu	Ile	Asp	Gln	Tyr	Leu	Tyr	Tyr	Leu	Ser	
		435				440					445					
Lys	Thr	Ile	Asn	Gly	Ser	Gly	Gln	Asn	Gln	Gln	Thr	Leu	Lys	Phe	Ser	
	450					455					460					
Val	Ala	Gly	Pro	Ser	Asn	Met	Ala	Val	Gln	Gly	Arg	Asn	Tyr	Ile	Pro	
465					470					475				480		
Gly	Pro	Ser	Tyr	Arg	Gln	Gln	Arg	Val	Ser	Thr	Thr	Val	Thr	Gln	Asn	
			485					490						495		
Asn	Asn	Ser	Glu	Phe	Ala	Trp	Pro	Gly	Ala	Ser	Ser	Trp	Ala	Leu	Asn	
			500					505					510			
Gly	Arg	Asn	Ser	Leu	Met	Asn	Pro	Gly	Pro	Ala	Met	Ala	Ser	His	Lys	
		515					520					525				
Glu	Gly	Glu	Asp	Arg	Phe	Phe	Pro	Leu	Ser	Gly	Ser	Leu	Ile	Phe	Gly	
	530					535					540					
Lys	Gln	Gly	Thr	Gly	Arg	Asp	Asn	Val	Asp	Ala	Asp	Lys	Val	Met	Ile	
545					550					555					560	
Thr	Asn	Glu	Glu	Glu	Ile	Lys	Thr	Thr	Asn	Pro	Val	Ala	Thr	Glu	Ser	
			565					570						575		

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Tyr Gly Gln Val Ala Thr Asn His Gln Ser Ala Gln Ala Gln Ala Gln
 580 585 590
 Thr Gly Trp Val Gln Asn Gln Gly Ile Leu Pro Gly Met Val Trp Gln
 595 600 605
 Asp Arg Asp Val Tyr Leu Gln Gly Pro Ile Trp Ala Lys Ile Pro His
 610 615 620
 Thr Asp Gly Asn Phe His Pro Ser Pro Leu Met Gly Gly Phe Gly Met
 625 630 635 640
 Lys His Pro Pro Pro Gln Ile Leu Ile Lys Asn Thr Pro Val Pro Ala
 645 650 655
 Asp Pro Pro Thr Ala Phe Asn Lys Asp Lys Leu Asn Ser Phe Ile Thr
 660 665 670
 Gln Tyr Ser Thr Gly Gln Val Ser Val Glu Ile Glu Trp Glu Leu Gln
 675 680 685
 Lys Glu Asn Ser Lys Arg Trp Asn Pro Glu Ile Gln Tyr Thr Ser Asn
 690 695 700
 Tyr Tyr Lys Ser Asn Asn Val Glu Phe Ala Val Asn Thr Glu Gly Val
 705 710 715 720
 Tyr Ser Glu Pro Arg Pro Ile Gly Thr Arg Tyr Leu Thr Arg Asn Leu
 725 730 735

<210> SEQ ID NO 36

<211> LENGTH: 1257

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary VEGFC polynucleotide

<400> SEQUENCE: 36

```

atgcacttgc tgggcttctt ctctgtggcg tgttctctgc tcgccgctgc gctgctcccg      60
ggtcctcgcg aggcgcccgc cgccgcccgc gccttcgagt ccggactcga cctctcggac      120
gcgagccccg acgcgggcga ggccacggct tatgcaagca aagatctgga ggagcagtta      180
cggctctgtg ccagtgtaga tgaactcatg actgtactct acccagaata ttggaaaatg      240
tacaagtgtc agctaaggaa aggaggttgg caacataaca gagaacaggc caacctcaac      300
tcaaggacag aagagactat aaaatttgct gcagcacatt ataatacaga gatcttgaaa      360
agtattgata atgagtggag aaagactcaa tgcatgccac gggaggtgtg tatagatgtg      420
gggaaggagt ttggagtgcg gacaaacacc ttctttaaac ctccatgtgt gtccgtctac      480
agatgtgggg gttgtgcaa tagtgagggg ctgcagtgcg tgaacaccag cagcagctac      540
ctcagcaaga cgttatttga aattacagtg cctctctctc aaggccccaa accagtaaca      600
atcagttttg ccaatcacac ttcttgccga tgcattgtct aactggatgt ttacagacaa      660
gttcattcca ttattagacg ttccctgccg gcaacactac cacagtgtca ggcagcgaac      720
aagacctgcc ccaccaatta catgtggaat aatcacatct gcagatgcct ggctcaggaa      780
gattttatgt tttcctcgga tgetggagat gactcaacag atggattcca tgacatctgt      840
ggaccaaaca aggagctgga tgaagagacc tgtcagtgtg tctgcagagc ggggcttcgg      900
cctgccagct gtggacccca caaagaacta gacagaaact catgccagtg tgtctgtaaa      960
aacaactctt tccccagcca atgtggggcc aaccgagaat ttgatgaaaa cacatgccag     1020
tgtgtatgta aaagaacctg ccccgaaaat caaccctaa atcctggaaa atgtgcctgt     1080

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gaatgtacag aaagtcacaca gaaatgcttg ttaaaaggaa agaagttcca ccaccaaaaca	1140
tgcagctgtt acagacggcc atgtacgaac cgccagaagg cttgtgagcc aggattttca	1200
tatagtgaag aagtgtgtcg ttgtgtccct tcatattgga aaagaccaca aatgagc	1257

<210> SEQ ID NO 37
<211> LENGTH: 6
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: transcription factor binding region

<400> SEQUENCE: 37

tacgat	6
--------	---

<210> SEQ ID NO 38
<211> LENGTH: 6
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: transcription factor binding region

<400> SEQUENCE: 38

tataat	6
--------	---

<210> SEQ ID NO 39
<211> LENGTH: 6
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: transcription factor binding region

<400> SEQUENCE: 39

gatact	6
--------	---

<210> SEQ ID NO 40
<211> LENGTH: 6
<212> TYPE: DNA
<213> ORGANISM: transcription factor binding region

<400> SEQUENCE: 40

tatgat	6
--------	---

<210> SEQ ID NO 41
<211> LENGTH: 6
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: transcription factor binding region

<400> SEQUENCE: 41

tatggt	6
--------	---

<210> SEQ ID NO 42
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary chicken beta-globin polyadenylation signal

<400> SEQUENCE: 42

caataaaaga tctttatctt cattagatct gtgtgttggt tttttgtgtg	50
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<210> SEQ ID NO 43
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary transcription factor binding region

<400> SEQUENCE: 43

gagcaagaca gagagagata ctcacagggga 30

<210> SEQ ID NO 44
<211> LENGTH: 1192
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary variant nephrin promoter

<400> SEQUENCE: 44

cacctgaggt caggagtctg agaccagcgt ggccaacatg atgaaacccc gtctctagta 60
aaaatacaaaa aattagccag gcatggtgct atatacctgt agcaccagct acttggggaga 120
cagaggtggg agaattactt gaacctggga ggttcaagcc atgggaggtg gaagttgcag 180
tgagccgaga tgccactgca ctcacgcctg agcaacagag caagactatc tcaagaaaag 240
aaagaaagaa agaagagac ttgccaaagt catgtatcag ggcaagggaag agctggggggc 300
ccagctggct gctcccctgc tgagctggga gaccaccttg atctgacttc tcccattctc 360
ccagcctaag ccaggccctg gggtcacgga ggctggggag gcaccgagga acgcgcctgg 420
catgtgctga cagggaatct tatgtccag ctggggccagc tgggaggagc ctgctgggca 480
gaggccagag ctgggggctc tggaaggtac ctgggggagg ttgcactgtg agaatgagct 540
caagctgggt cagagagcag ggctgactct gccagtgcct gcacacagct catcgctctc 600
ctaggctcct ggctgctgg actctgggct gcaggtcctt cttgaaaggc tgtgagtagt 660
gagacaagga gcaggagtga ggggtggcag gagagaagat agagattgag agagagagag 720
agagagagac agagagagag gaagagacag agacaaaagg agagagaacg gcttagacaa 780
ggagagaaaag atggaaaagat aaagagactg ggcgcagtgg ctcacgcctg taatcccaac 840
acttggggag gccaaaggtg gaggatggct tgaaggaaaag agtctgagat caacctggcc 900
aacatagtga gacccctct ctaaaaaaaaa aagaaaaaaaa aaagaaaaaaaa gaaaaaaaaag 960
tttttttaaa gagacagaga aagagactca gagattgaga ctgagagcaa gacagagaga 1020
gacactcaca gggaaagagg gaagaggaaa acgagaaagg gaggagagta acggaagag 1080
ataaaaaaga aaagcagggtg gcagagacac agagagaggg acccagagaa agccagacag 1140
acgcaggtgg ctggcagcgg gcgctgtggg ggtcacagta gggggacctg tc 1192

<210> SEQ ID NO 45
<211> LENGTH: 1192
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary variant nephrin promoter

<400> SEQUENCE: 45

cacctgaggt caggagtctg agaccagcgt ggccaacatg atgaaacccc gtctctagta 60
aaaatacaaaa aattagccag gcatggtgct atatacctgt agcaccagct acttggggaga 120

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cagagggtggg agaattactt gaacctggga ggttcaagcc atgggagggtg gaagttgcag	180
tgagccgaga tgccactgca ctccagcctg agcaacagag caagactatc tcaagaaaag	240
aaagaaagaa agaagagac ttgccaaagt catgtatcag ggcaaggaaag agctggggggc	300
ccagctggct gctcccctgc tgagctggga gaccaccttg atctgacttc tcccatcttc	360
ccagcctaag ccaggccctg gggtcacgga ggctggggag gcaccgagga acgcgcctgg	420
catgtgctga caggggattt tatgtctcag ctgggccagc tgggaggagc ctgctgggca	480
gaggccagag ctgggggctc tggaaggtag ctgggggagg ttgcaactgtg agaatgagct	540
caagctgggt cagagagcag ggctgactct gccagtgcct gcatcagcct catcgctctc	600
ctaggctcct ggctgtgtgg actctgggct gcaggctcct cttgaaaggc tgtgagtagt	660
gagacaagga gcaggagtga ggggtggcag gagagaagat agagattgag agagagagag	720
agagagagac agagagagag gaagagacag agacaaaagg agagagaacg gcttagacaa	780
ggagagaaaag atgaaaagat aaagagactg ggcgcagtgg ctcacgcctg taatcccaac	840
acttggggag gccaaagtgg gaggatggct tgaaggaaag agtctgagat caacctggcc	900
aacatagtga gaccccgctc ctaaaaaaaaa aagaaaaaaaa aaagaaaaaaaa gaaaaaaaaag	960
tttttttaaa gagacagaga aagagactca gagattgaga ctgagagcaa gacagagaga	1020
gatactcaca ggggaagagg gaagagggaaa acgagaaaagg gaggagagta acggaaaagag	1080
ataaaaaaga aaagcagggtg gcagagacac agagagaggg acccagagaa agccagacag	1140
acgcagggtg ctggcagcgg gcgctgtggg ggtcacagta gggggacctg tc	1192

<210> SEQ ID NO 46

<211> LENGTH: 819

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant

<400> SEQUENCE: 46

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag	60
gggattttat gctccagctg ggccagctgg gaggagcctg ctgggcagag gccagagctg	120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag	180
agagcagggc tgactctgcc agtgcttgca tcagcctcat cgctctccta ggctcctggc	240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gagtgtgag acaaggagca	300
ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga gagacagaga	360
gagaggaaga gacagagaca aaaggagaga gaacggctta gacaaggaga gaaagatgga	420
aagataaaga gactggggcg agtggctcac gcctgtaatc ccaacacttg gggaggccaa	480
ggtgggagga tggcttgaag gaaagagtct gagatcaacc tggccaacat agtgagaccc	540
cgtctctaaa aaaaaaaaaa aaaaaaaaaa gaaaaaagaa aaaaaagttt ttttaaagag	600
acagagaaaag agactcagag attgagactg agagcaagac agagagagat actcacaggg	660
aagaggggaa gaggaaaacg agaaaggag gagagtaacg gaaagagata aaaaagaaaa	720
gcaggtggca gagacacaga gagagggacc cagagaaagc cagacagacg caggtggctg	780
gcagcgggcy ctgtgggggt cacagtaggg ggacctgtc	819

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<210> SEQ ID NO 47
 <211> LENGTH: 265
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant

<400> SEQUENCE: 47

```

ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgcttggcat gtgctgacag      60
gggattttat gctccaggag caagacagag agagatactc acaggaaga ggggaagagg      120
aaaacgagaa agggaggaga gtaacggaaa gagataaaaa agaaaagcag gtggcagaga      180
cacacagaga gggaccaga gaaagccaga cagacgcagg tggctggcag cgggcgctgt      240
gggggtcaca gtagggggac ctgtg                                           265

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<210> SEQ ID NO 48

<400> SEQUENCE: 48

000

<210> SEQ ID NO 49
 <211> LENGTH: 583
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Illustrative CFI polypeptide sequence

<400> SEQUENCE: 49

```

Met Lys Leu Leu His Val Phe Leu Leu Phe Leu Cys Phe His Leu Arg
1             5             10             15
Phe Cys Lys Val Thr Tyr Thr Ser Gln Glu Asp Leu Val Glu Lys Lys
20             25             30
Cys Leu Ala Lys Lys Tyr Thr His Leu Ser Cys Asp Lys Val Phe Cys
35             40             45
Gln Pro Trp Gln Arg Cys Ile Glu Gly Thr Cys Val Cys Lys Leu Pro
50             55             60
Tyr Gln Cys Pro Lys Asn Gly Thr Ala Val Cys Ala Thr Asn Arg Arg
65             70             75             80
Ser Phe Pro Thr Tyr Cys Gln Gln Lys Ser Leu Glu Cys Leu His Pro
85             90             95
Gly Thr Lys Phe Leu Asn Asn Gly Thr Cys Thr Ala Glu Gly Lys Phe
100            105            110
Ser Val Ser Leu Lys His Gly Asn Thr Asp Ser Glu Gly Ile Val Glu
115            120            125
Val Lys Leu Val Asp Gln Asp Lys Thr Met Phe Ile Cys Lys Ser Ser
130            135            140
Trp Ser Met Arg Glu Ala Asn Val Ala Cys Leu Asp Leu Gly Phe Gln
145            150            155            160
Gln Gly Ala Asp Thr Gln Arg Arg Phe Lys Leu Ser Asp Leu Ser Ile
165            170            175
Asn Ser Thr Glu Cys Leu His Val His Cys Arg Gly Leu Glu Thr Ser
180            185            190
Leu Ala Glu Cys Thr Phe Thr Lys Arg Arg Thr Met Gly Tyr Gln Asp
195            200            205
Phe Ala Asp Val Val Cys Tyr Thr Gln Lys Ala Asp Ser Pro Met Asp

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210	215	220
Asp Phe Phe Gln Cys Val Asn Gly Lys Tyr Ile Ser Gln Met Lys Ala		
225	230	235 240
Cys Asp Gly Ile Asn Asp Cys Gly Asp Gln Ser Asp Glu Leu Cys Cys		
	245	250 255
Lys Ala Cys Gln Gly Lys Gly Phe His Cys Lys Ser Gly Val Cys Ile		
	260	265 270
Pro Ser Gln Tyr Gln Cys Asn Gly Glu Val Asp Cys Ile Thr Gly Glu		
	275	280 285
Asp Glu Val Gly Cys Ala Gly Phe Ala Ser Val Thr Gln Glu Glu Thr		
	290	295 300
Glu Ile Leu Thr Ala Asp Met Asp Ala Glu Arg Arg Arg Ile Lys Ser		
305	310	315 320
Leu Leu Pro Lys Leu Ser Cys Gly Val Lys Asn Arg Met His Ile Arg		
	325	330 335
Arg Lys Arg Ile Val Gly Gly Lys Arg Ala Gln Leu Gly Asp Leu Pro		
	340	345 350
Trp Gln Val Ala Ile Lys Asp Ala Ser Gly Ile Thr Cys Gly Gly Ile		
	355	360 365
Tyr Ile Gly Gly Cys Trp Ile Leu Thr Ala Ala His Cys Leu Arg Ala		
	370	375 380
Ser Lys Thr His Arg Tyr Gln Ile Trp Thr Thr Val Val Asp Trp Ile		
385	390	395 400
His Pro Asp Leu Lys Arg Ile Val Ile Glu Tyr Val Asp Arg Ile Ile		
	405	410 415
Phe His Glu Asn Tyr Asn Ala Gly Thr Tyr Gln Asn Asp Ile Ala Leu		
	420	425 430
Ile Glu Met Lys Lys Asp Gly Asn Lys Lys Asp Cys Glu Leu Pro Arg		
	435	440 445
Ser Ile Pro Ala Cys Val Pro Trp Ser Pro Tyr Leu Phe Gln Pro Asn		
	450	455 460
Asp Thr Cys Ile Val Ser Gly Trp Gly Arg Glu Lys Asp Asn Glu Arg		
465	470	475 480
Val Phe Ser Leu Gln Trp Gly Glu Val Lys Leu Ile Ser Asn Cys Ser		
	485	490 495
Lys Phe Tyr Gly Asn Arg Phe Tyr Glu Lys Glu Met Glu Cys Ala Gly		
	500	505 510
Thr Tyr Asp Gly Ser Ile Asp Ala Cys Lys Gly Asp Ser Gly Gly Pro		
	515	520 525
Leu Val Cys Met Asp Ala Asn Asn Val Thr Tyr Val Trp Gly Val Val		
	530	535 540
Ser Trp Gly Glu Asn Cys Gly Lys Pro Glu Phe Pro Gly Val Tyr Thr		
545	550	555 560
Lys Val Ala Asn Tyr Phe Asp Trp Ile Ser Tyr His Val Gly Arg Pro		
	565	570 575
Phe Ile Ser Gln Tyr Asn Val		
	580	

<210> SEQ ID NO 50

<211> LENGTH: 1752

<212> TYPE: DNA

<213> ORGANISM: Unknown

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Illustrative CFI polynucleotide sequence

<400> SEQUENCE: 50

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atgaagcttc ttcattgttt cctgttattt ctgtgcttcc acttaagggt ttgcaagggt    60
acttatacat ctcaagagga tctgggtggag aaaaagtgtc tagcaaaaaa atatactcac    120
ctctcctgcg ataaagtctt ctgccagcca tggcagagat gcattgaggg cacctgtgtt    180
tgtaaaactac cgtatcagtg cccaaagaat ggcaactgcag tgtgtgcaac taacaggaga    240
agcttcccaa catactgtca acaaaagagt ttggaatgtc ttcattccagg gacaaagttt    300
ttaaataacg gaacatgcac agccgaagga aagtttagtg ttcccttgaa gcatggaaat    360
acagattcag agggaatagt tgaagtaaaa cttgtggacc aagataagac aatgttcata    420
tgcaaaagca gctggagcat gaggggaagcc aacgtggcct gccttgacct tgggtttcaa    480
caaggtgctg atactcaaag aaggtttaag ttgtctgac tctctataaa ttccactgaa    540
tgtctacatg tgcattgccg aggattagag accagtttgg ctgaatgtac ttttactaag    600
agaagaacta tgggttacca ggatttcgct gatgtggttt gttatacaca gaaagcagat    660
tctccaatgg atgacttctt tcagtgtgtg aatgggaaat acattttctca gatgaaagcc    720
tgtgatggta tcaatgattg tggagaccaa agtgatgaac tgtgttgtaa agcatgccaa    780
ggcaaaggct tccattgcaa atcgggtgtt tgcattccaa gccagtatca atgcaatggt    840
gaggtggact gcattacagg ggaagatgaa gttggctgtg caggctttgc atctgtgact    900
caagaagaaa cagaaatttt gactgctgac atggatgcag aaagaagacg gataaaatca    960
ttattaccta aactatcttg tggagttaaa aacagaatgc acattcgaag gaaacgaatt   1020
gtgggaggaa agcagacaca actgggagac ctcccatggc aggtggcaat taaggatgcc   1080
agtgaatca cctgtggggg aatttatatt ggtggctgtt ggattctgac tgcctcacat   1140
tgtctcagag ccagtaaaac tcatcggtac caaatatgga caacagtagt agactggata   1200
caccctgacc ttaaactgat agtaattgaa tacgtggata gaattatttt ccatgaaaac   1260
tacaatgcag gcacttacca aaatgacatc gctttgattg aaatgaaaaa agacggaaac   1320
aaaaaagatt gtgagctgcc tcgttccatc cctgcctgtg tccctgggtc tccttaccta   1380
ttccaaccta atgatacatg catcgtttct ggctggggac gagaaaaaga taacgaaaga   1440
gtcttttcac ttcagtgggg tgaagttaaa ctaataagca actgctctaa gttttacgga   1500
aatcgtttct atgaaaaaga aatggaatgt gcaggtacat atgatggttc catcgatgcc   1560
tgtaaagggg actctggagg ccccttagtc tgtatggatg ccaacaatgt gacttatgtc   1620
tggggtgttg tgagttgggg ggaaaactgt ggaaaaccag agttcccagg tgtttacacc   1680
aaagtggcca attattttga ctggattagc taccatgtag gaaggccttt tatttctcag   1740
tacaatgtat aa                                     1752

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<210> SEQ ID NO 51

<211> LENGTH: 1231

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Illustrative CFH polypeptide sequence

<400> SEQUENCE: 51

Met Arg Leu Leu Ala Lys Ile Ile Cys Leu Met Leu Trp Ala Ile Cys

-continued

1	5	10	15
Val Ala Glu Asp Cys Asn Glu Leu Pro Pro Arg Arg Asn Thr Glu Ile	20	25	30
Leu Thr Gly Ser Trp Ser Asp Gln Thr Tyr Pro Glu Gly Thr Gln Ala	35	40	45
Ile Tyr Lys Cys Arg Pro Gly Tyr Arg Ser Leu Gly Asn Val Ile Met	50	55	60
Val Cys Arg Lys Gly Glu Trp Val Ala Leu Asn Pro Leu Arg Lys Cys	65	70	75
Gln Lys Arg Pro Cys Gly His Pro Gly Asp Thr Pro Phe Gly Thr Phe	85	90	95
Thr Leu Thr Gly Gly Asn Val Phe Glu Tyr Gly Val Lys Ala Val Tyr	100	105	110
Thr Cys Asn Glu Gly Tyr Gln Leu Leu Gly Glu Ile Asn Tyr Arg Glu	115	120	125
Cys Asp Thr Asp Gly Trp Thr Asn Asp Ile Pro Ile Cys Glu Val Val	130	135	140
Lys Cys Leu Pro Val Thr Ala Pro Glu Asn Gly Lys Ile Val Ser Ser	145	150	155
Ala Met Glu Pro Asp Arg Glu Tyr His Phe Gly Gln Ala Val Arg Phe	165	170	175
Val Cys Asn Ser Gly Tyr Lys Ile Glu Gly Asp Glu Glu Met His Cys	180	185	190
Ser Asp Asp Gly Phe Trp Ser Lys Glu Lys Pro Lys Cys Val Glu Ile	195	200	205
Ser Cys Lys Ser Pro Asp Val Ile Asn Gly Ser Pro Ile Ser Gln Lys	210	215	220
Ile Ile Tyr Lys Glu Asn Glu Arg Phe Gln Tyr Lys Cys Asn Met Gly	225	230	235
Tyr Glu Tyr Ser Glu Arg Gly Asp Ala Val Cys Thr Glu Ser Gly Trp	245	250	255
Arg Pro Leu Pro Ser Cys Glu Glu Lys Ser Cys Asp Asn Pro Tyr Ile	260	265	270
Pro Asn Gly Asp Tyr Ser Pro Leu Arg Ile Lys His Arg Thr Gly Asp	275	280	285
Glu Ile Thr Tyr Gln Cys Arg Asn Gly Phe Tyr Pro Ala Thr Arg Gly	290	295	300
Asn Thr Ala Lys Cys Thr Ser Thr Gly Trp Ile Pro Ala Pro Arg Cys	305	310	315
Thr Leu Lys Pro Cys Asp Tyr Pro Asp Ile Lys His Gly Gly Leu Tyr	325	330	335
His Glu Asn Met Arg Arg Pro Tyr Phe Pro Val Ala Val Gly Lys Tyr	340	345	350
Tyr Ser Tyr Tyr Cys Asp Glu His Phe Glu Thr Pro Ser Gly Ser Tyr	355	360	365
Trp Asp His Ile His Cys Thr Gln Asp Gly Trp Ser Pro Ala Val Pro	370	375	380
Cys Leu Arg Lys Cys Tyr Phe Pro Tyr Leu Glu Asn Gly Tyr Asn Gln	385	390	395
Asn Tyr Gly Arg Lys Phe Val Gln Gly Lys Ser Ile Asp Val Ala Cys	405	410	415

His	Pro	Gly	Tyr	Ala	Leu	Pro	Lys	Ala	Gln	Thr	Thr	Val	Thr	Cys	Met
			420						425			430			
Glu	Asn	Gly	Trp	Ser	Pro	Thr	Pro	Arg	Cys	Ile	Arg	Val	Lys	Thr	Cys
			435						440			445			
Ser	Lys	Ser	Ser	Ile	Asp	Ile	Glu	Asn	Gly	Phe	Ile	Ser	Glu	Ser	Gln
			450						455			460			
Tyr	Thr	Tyr	Ala	Leu	Lys	Glu	Lys	Ala	Lys	Tyr	Gln	Cys	Lys	Leu	Gly
			465						470			475			
Tyr	Val	Thr	Ala	Asp	Gly	Glu	Thr	Ser	Gly	Ser	Ile	Thr	Cys	Gly	Lys
			485						490			495			
Asp	Gly	Trp	Ser	Ala	Gln	Pro	Thr	Cys	Ile	Lys	Ser	Cys	Asp	Ile	Pro
			500						505			510			
Val	Phe	Met	Asn	Ala	Arg	Thr	Lys	Asn	Asp	Phe	Thr	Trp	Phe	Lys	Leu
			515						520			525			
Asn	Asp	Thr	Leu	Asp	Tyr	Glu	Cys	His	Asp	Gly	Tyr	Glu	Ser	Asn	Thr
			530						535			540			
Gly	Ser	Thr	Thr	Gly	Ser	Ile	Val	Cys	Gly	Tyr	Asn	Gly	Trp	Ser	Asp
			545						550			555			
Leu	Pro	Ile	Cys	Tyr	Glu	Arg	Glu	Cys	Glu	Leu	Pro	Lys	Ile	Asp	Val
			565						570			575			
His	Leu	Val	Pro	Asp	Arg	Lys	Lys	Asp	Gln	Tyr	Lys	Val	Gly	Glu	Val
			580						585			590			
Leu	Lys	Phe	Ser	Cys	Lys	Pro	Gly	Phe	Thr	Ile	Val	Gly	Pro	Asn	Ser
			595						600			605			
Val	Gln	Cys	Tyr	His	Phe	Gly	Leu	Ser	Pro	Asp	Leu	Pro	Ile	Cys	Lys
			610						615			620			
Glu	Gln	Val	Gln	Ser	Cys	Gly	Pro	Pro	Pro	Glu	Leu	Leu	Asn	Gly	Asn
			625						630			635			
Val	Lys	Glu	Lys	Thr	Lys	Glu	Glu	Tyr	Gly	His	Ser	Glu	Val	Val	Glu
			645						650			655			
Tyr	Tyr	Cys	Asn	Pro	Arg	Phe	Leu	Met	Lys	Gly	Pro	Asn	Lys	Ile	Gln
			660						665			670			
Cys	Val	Asp	Gly	Glu	Trp	Thr	Thr	Leu	Pro	Val	Cys	Ile	Val	Glu	Glu
			675						680			685			
Ser	Thr	Cys	Gly	Asp	Ile	Pro	Glu	Leu	Glu	His	Gly	Trp	Ala	Gln	Leu
			690						695			700			
Ser	Ser	Pro	Pro	Tyr	Tyr	Tyr	Gly	Asp	Ser	Val	Glu	Phe	Asn	Cys	Ser
			705						710			715			
Glu	Ser	Phe	Thr	Met	Ile	Gly	His	Arg	Ser	Ile	Thr	Cys	Ile	His	Gly
			725						730			735			
Val	Trp	Thr	Gln	Leu	Pro	Gln	Cys	Val	Ala	Ile	Asp	Lys	Leu	Lys	Lys
			740						745			750			
Cys	Lys	Ser	Ser	Asn	Leu	Ile	Ile	Leu	Glu	Glu	His	Leu	Lys	Asn	Lys
			755						760			765			
Lys	Glu	Phe	Asp	His	Asn	Ser	Asn	Ile	Arg	Tyr	Arg	Cys	Arg	Gly	Lys
			770						775			780			
Glu	Gly	Trp	Ile	His	Thr	Val	Cys	Ile	Asn	Gly	Arg	Trp	Asp	Pro	Glu
			785						790			795			
Val	Asn	Cys	Ser	Met	Ala	Gln	Ile	Gln	Leu	Cys	Pro	Pro	Pro	Pro	Gln
			805						810			815			

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Ile	Pro	Asn	Ser	His	Asn	Met	Thr	Thr	Thr	Leu	Asn	Tyr	Arg	Asp	Gly	820	825	830
Glu	Lys	Val	Ser	Val	Leu	Cys	Gln	Glu	Asn	Tyr	Leu	Ile	Gln	Glu	Gly	835	840	845
Glu	Glu	Ile	Thr	Cys	Lys	Asp	Gly	Arg	Trp	Gln	Ser	Ile	Pro	Leu	Cys	850	855	860
Val	Glu	Lys	Ile	Pro	Cys	Ser	Gln	Pro	Pro	Gln	Ile	Glu	His	Gly	Thr	865	870	875
Ile	Asn	Ser	Ser	Arg	Ser	Ser	Gln	Glu	Ser	Tyr	Ala	His	Gly	Thr	Lys	885	890	895
Leu	Ser	Tyr	Thr	Cys	Glu	Gly	Gly	Phe	Arg	Ile	Ser	Glu	Glu	Asn	Glu	900	905	910
Thr	Thr	Cys	Tyr	Met	Gly	Lys	Trp	Ser	Ser	Pro	Pro	Gln	Cys	Glu	Gly	915	920	925
Leu	Pro	Cys	Lys	Ser	Pro	Pro	Glu	Ile	Ser	His	Gly	Val	Val	Ala	His	930	935	940
Met	Ser	Asp	Ser	Tyr	Gln	Tyr	Gly	Glu	Glu	Val	Thr	Tyr	Lys	Cys	Phe	945	950	955
Glu	Gly	Phe	Gly	Ile	Asp	Gly	Pro	Ala	Ile	Ala	Lys	Cys	Leu	Gly	Glu	965	970	975
Lys	Trp	Ser	His	Pro	Pro	Ser	Cys	Ile	Lys	Thr	Asp	Cys	Leu	Ser	Leu	980	985	990
Pro	Ser	Phe	Glu	Asn	Ala	Ile	Pro	Met	Gly	Glu	Lys	Lys	Asp	Val	Tyr	995	1000	1005
Lys	Ala	Gly	Glu	Gln	Val	Thr	Tyr	Thr	Cys	Ala	Thr	Tyr	Tyr	Lys		1010	1015	1020
Met	Asp	Gly	Ala	Ser	Asn	Val	Thr	Cys	Ile	Asn	Ser	Arg	Trp	Thr		1025	1030	1035
Gly	Arg	Pro	Thr	Cys	Arg	Asp	Thr	Ser	Cys	Val	Asn	Pro	Pro	Thr		1040	1045	1050
Val	Gln	Asn	Ala	Tyr	Ile	Val	Ser	Arg	Gln	Met	Ser	Lys	Tyr	Pro		1055	1060	1065
Ser	Gly	Glu	Arg	Val	Arg	Tyr	Gln	Cys	Arg	Ser	Pro	Tyr	Glu	Met		1070	1075	1080
Phe	Gly	Asp	Glu	Glu	Val	Met	Cys	Leu	Asn	Gly	Asn	Trp	Thr	Glu		1085	1090	1095
Pro	Pro	Gln	Cys	Lys	Asp	Ser	Thr	Gly	Lys	Cys	Gly	Pro	Pro	Pro		1100	1105	1110
Pro	Ile	Asp	Asn	Gly	Asp	Ile	Thr	Ser	Phe	Pro	Leu	Ser	Val	Tyr		1115	1120	1125
Ala	Pro	Ala	Ser	Ser	Val	Glu	Tyr	Gln	Cys	Gln	Asn	Leu	Tyr	Gln		1130	1135	1140
Leu	Glu	Gly	Asn	Lys	Arg	Ile	Thr	Cys	Arg	Asn	Gly	Gln	Trp	Ser		1145	1150	1155
Glu	Pro	Pro	Lys	Cys	Leu	His	Pro	Cys	Val	Ile	Ser	Arg	Glu	Ile		1160	1165	1170
Met	Glu	Asn	Tyr	Asn	Ile	Ala	Leu	Arg	Trp	Thr	Ala	Lys	Gln	Lys		1175	1180	1185
Leu	Tyr	Ser	Arg	Thr	Gly	Glu	Ser	Val	Glu	Phe	Val	Cys	Lys	Arg		1190	1195	1200
Gly	Tyr	Arg	Leu	Ser	Ser	Arg	Ser	His	Thr	Leu	Arg	Thr	Thr	Cys				

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1205	1210	1215	
Trp Asp Gly Lys Leu Glu Tyr Pro Thr Cys Ala Lys Arg			
1220	1225	1230	
<210> SEQ ID NO 52 <211> LENGTH: 3696 <212> TYPE: DNA <213> ORGANISM: Unknown <220> FEATURE: <223> OTHER INFORMATION: Illustrative CFH polynucleotide sequence <400> SEQUENCE: 52			
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acatatccag aaggcaccca ggctatctat aaatgccgcc ctggatatag atctcttgga		180	
aatgtaataa tggatgcag gaaggagaa tgggtgtctc ttaatccatt aaggaaatgt		240	
cagaaaaggc cctgtggaca tcttgagat actccttttg gtacttttac ccttacagga		300	
ggaaatgtgt ttgaatatgg tgtaaaagct gtgtatacat gtaatgagg gtatcaattg		360	
ctaggtgaga ttaattaccg tgaatgtgac acagatggat ggaccaatga tttcctata		420	
tgtgaagttg tgaagtgttt accagtgcgc gcaccagaga atggaaaaat tgtcagtagt		480	
gcaatggaac cagatcgga ataccatttt ggacaagcag tacggtttgt atgtaactca		540	
ggctacaaga ttgaaggaga tgaagaaatg cattgttcag acgatggttt ttggagtaaa		600	
gagaaaccaa agtgtgtgga aatttcattgc aaatccccag atgttataaa tggatctcct		660	
atatctcaga agattattta taaggagaat gaacgatttc aatataaatg taacatgggt		720	
tatgaataca gtgaagagg agatgctgta tgcactgaat ctggatggcg tccgttgccct		780	
tcatgtgaag aaaaatcatg tgataatcct tatattccaa atggtgacta ctacacctta		840	
aggattaaac acagaactgg agatgaaatc acgtaccagt gtagaaatgg tttttatcct		900	
gcaaccggcg gaaatacagc aaaatgcaca agtactggct ggatacctgc tccgagatgt		960	
accttgaaac cttgtgatta tccagacatt aaacatggag gtctatatca tgagaatatg		1020	
cgtagaccat actttccagt agctgtagga aaatattact cctattactg tgatgaacat		1080	
tttgagactc cgtcaggaag ttactgggat cacattcatt gcacacaaga tggatggctc		1140	
ccagcagtac catgcctcag aaaatgttat ttctcttatt tggaaaatgg atataatcaa		1200	
aatcatggaa gaaagtgtgt acagggtaaa tctatagacg ttgcctgcc aacctggctac		1260	
gctcttccaa aagcgcagac cacagttaca tgtatggaga atggctggct tcctactccc		1320	
agatgcattc gtgtcaaac atgttccaaa tcaagtatag atattgagaa tgggtttatt		1380	
tctgaatctc agtatacata tgccttaaaa gaaaaagcga aatatcaatg caaactagga		1440	
tatgtaacag cagatggatg aacatcagga tcaattacat gtgggaaaga tggatggatc		1500	
gctcaaccca cgtgcattaa atcttggat atcccagat ttatgaatgc cagaactaaa		1560	
aatgacttca catggtttta gctgaatgac acattggact atgaatgcca tgatggttat		1620	
gaaagcaata ctggaagcac cactgggtcc atagtgtgtg gttacaatgg ttggtctgat		1680	
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gatcgcaaga aagaccagta taaagtggga gaggtgttga aattctcctg caaaccagga		1800	
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gaatcattta caatgattgg acacagatca attacgtgta ttcatggagt atggacccaa 2220
cttccccagt gtgtggcaat agataaactt aagaagtgca aatcatcaaa tttaattata 2280
cttgaggaaac atttaaaaaa caagaaggaa ttcgatcata attctaacat aaggtaacaga 2340
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gaaaattatc taattcagga aggagaagaa attacatgca aagatggaag atggcagtca 2580
ataccactct gtgttgaaaa aattccatgt tcacaaccac ctcagataga acacggaacc 2640
attaattcat ccaggtcttc acaagaaagt tatgcacatg ggactaaatt gagttatact 2700
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tcgagacaga tgagtaaata tccatctggt gagagagtac gttatcaatg taggagccct 3240
tatgaaatgt ttggggatga agaagtgatg tgtttaaatg gaaactggac ggaaccacct 3300
caatgcaaag attctacagg aaaatgtggg ccccctccac ctattgacaa tggggacatt 3360
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ttgtatcaac ttgagggtaa caagcgaata acatgtagaa atggacaatg gtcagaacca 3480
ccaaaatgct tacatccgtg tgtaatatcc cgagaaatta tggaaaatta taacatagca 3540
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<210> SEQ ID NO 53

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Illustrative FHL-1 polypeptide sequence

<400> SEQUENCE: 53

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Val	Ala	Glu	Asp	Cys	Asn	Glu	Leu	Pro	Pro	Arg	Arg	Asn	Thr	Glu	Ile
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Leu	Thr	Gly	Ser	Trp	Ser	Asp	Gln	Thr	Tyr	Pro	Glu	Gly	Thr	Gln	Ala
		35					40					45			
Ile	Tyr	Lys	Cys	Arg	Pro	Gly	Tyr	Arg	Ser	Leu	Gly	Asn	Val	Ile	Met
		50				55					60				
Val	Cys	Arg	Lys	Gly	Glu	Trp	Val	Ala	Leu	Asn	Pro	Leu	Arg	Lys	Cys
65					70					75					80
Gln	Lys	Arg	Pro	Cys	Gly	His	Pro	Gly	Asp	Thr	Pro	Phe	Gly	Thr	Phe
				85					90					95	
Thr	Leu	Thr	Gly	Gly	Asn	Val	Phe	Glu	Tyr	Gly	Val	Lys	Ala	Val	Tyr
			100					105					110		
Thr	Cys	Asn	Glu	Gly	Tyr	Gln	Leu	Leu	Gly	Glu	Ile	Asn	Tyr	Arg	Glu
		115					120					125			
Cys	Asp	Thr	Asp	Gly	Trp	Thr	Asn	Asp	Ile	Pro	Ile	Cys	Glu	Val	Val
		130				135					140				
Lys	Cys	Leu	Pro	Val	Thr	Ala	Pro	Glu	Asn	Gly	Lys	Ile	Val	Ser	Ser
145					150					155					160
Ala	Met	Glu	Pro	Asp	Arg	Glu	Tyr	His	Phe	Gly	Gln	Ala	Val	Arg	Phe
				165					170					175	
Val	Cys	Asn	Ser	Gly	Tyr	Lys	Ile	Glu	Gly	Asp	Glu	Glu	Met	His	Cys
			180					185					190		
Ser	Asp	Asp	Gly	Phe	Trp	Ser	Lys	Glu	Lys	Pro	Lys	Cys	Val	Glu	Ile
		195					200					205			
Ser	Cys	Lys	Ser	Pro	Asp	Val	Ile	Asn	Gly	Ser	Pro	Ile	Ser	Gln	Lys
		210				215					220				
Ile	Ile	Tyr	Lys	Glu	Asn	Glu	Arg	Phe	Gln	Tyr	Lys	Cys	Asn	Met	Gly
225				230						235					240
Tyr	Glu	Tyr	Ser	Glu	Arg	Gly	Asp	Ala	Val	Cys	Thr	Glu	Ser	Gly	Trp
				245					250					255	
Arg	Pro	Leu	Pro	Ser	Cys	Glu	Glu	Lys	Ser	Cys	Asp	Asn	Pro	Tyr	Ile
			260					265					270		
Pro	Asn	Gly	Asp	Tyr	Ser	Pro	Leu	Arg	Ile	Lys	His	Arg	Thr	Gly	Asp
		275					280					285			
Glu	Ile	Thr	Tyr	Gln	Cys	Arg	Asn	Gly	Phe	Tyr	Pro	Ala	Thr	Arg	Gly
		290				295					300				
Asn	Thr	Ala	Lys	Cys	Thr	Ser	Thr	Gly	Trp	Ile	Pro	Ala	Pro	Arg	Cys
305				310						315					320
Thr	Leu	Lys	Pro	Cys	Asp	Tyr	Pro	Asp	Ile	Lys	His	Gly	Gly	Leu	Tyr
				325					330					335	
His	Glu	Asn	Met	Arg	Arg	Pro	Tyr	Phe	Pro	Val	Ala	Val	Gly	Lys	Tyr
		340						345					350		
Tyr	Ser	Tyr	Tyr	Cys	Asp	Glu	His	Phe	Glu	Thr	Pro	Ser	Gly	Ser	Tyr
		355					360					365			
Trp	Asp	His	Ile	His	Cys	Thr	Gln	Asp	Gly						

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420				425				430								
Glu	Asn	Gly	Trp	Ser	Pro	Thr	Pro	Arg	Cys	Ile	Arg	Val	Ser	Phe	Thr	
435				440				445								
Leu																
<210> SEQ ID NO 54																
<211> LENGTH: 1350																
<212> TYPE: DNA																
<213> ORGANISM: Unknown																
<220> FEATURE:																
<223> OTHER INFORMATION: Illustrative FHL-1 polynucleotide sequence																
<400> SEQUENCE: 54																
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acatatccag aaggcaccca ggctatctat aaatgccgcc ctggatatag atctcttgga															180	
aatgtaataa tggatgacag gaaggagaaa tgggttgctc ttaatccatt aaggaaatgt															240	
cagaaaaggc cctgtggaca tcttgagat actccttttg gtacttttac ccttacagga															300	
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gcaatggaac cagatcggga ataccatctt ggacaagcag tacggtttgt atgtaactca															540	
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gagaaaccaa agtgtgtgga aatttcattg aaatccccag atgttataaa tggatctcct															660	
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tatgaataga gtgaagagg agatgtgtga tgcaactgaat ctggatggcg tccgttgcc															780	
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aggattaaac acagaactgg agatgaaatc acgtaccagt gtagaaatgg tttttatcct															900	
gcaacccggg gaaatacagc aaaatgcaca agtactggct ggatacctgc tccgagatgt															960	
accttgaaac cttgtgatta tccagacatt aaacatggag gtctatatca tgagaatatg															1020	
cgtagaccat actttccagt agctgttaga aaatattact cctattactg tgatgaacat															1080	
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ccagcagtag catgctcag aaaaatgttat tttccttatt tggaaaatgg atataatcaa															1200	
aatcatggaa gaaagtgtgt acagggtaaa tctatagacg ttgcctgcca tcctggctac															1260	
gctcttccaa aagcgcagac cacagttaca tgtatggaga atggctggct tcctactccc															1320	
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<210> SEQ ID NO 55
 <211> LENGTH: 383
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary NPHS2 amino acid sequence
 <400> SEQUENCE: 55

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Gly	Arg	Thr	Pro	His	Lys	Glu	Asn	Lys	Arg	Ala	Lys	Ala	Glu	Arg	Ser	20	25	30
Gly	Gly	Gly	Arg	Gly	Arg	Gln	Glu	Ala	Gly	Pro	Glu	Pro	Ser	Gly	Ser	35	40	45
Gly	Arg	Ala	Gly	Thr	Pro	Gly	Glu	Pro	Arg	Ala	Pro	Ala	Ala	Thr	Val	50	55	60
Val	Asp	Val	Asp	Glu	Val	Arg	Gly	Ser	Gly	Glu	Glu	Gly	Thr	Glu	Val	65	70	80
Val	Ala	Leu	Leu	Glu	Ser	Glu	Arg	Pro	Glu	Glu	Gly	Thr	Lys	Ser	Ser	85	90	95
Gly	Leu	Gly	Ala	Cys	Glu	Trp	Leu	Leu	Val	Leu	Ile	Ser	Leu	Leu	Phe	100	105	110
Ile	Ile	Met	Thr	Phe	Pro	Phe	Ser	Ile	Trp	Phe	Cys	Val	Lys	Val	Val	115	120	125
Gln	Glu	Tyr	Glu	Arg	Val	Ile	Ile	Phe	Arg	Leu	Gly	His	Leu	Leu	Pro	130	135	140
Gly	Arg	Ala	Lys	Gly	Pro	Gly	Leu	Phe	Phe	Phe	Leu	Pro	Cys	Leu	Asp	145	150	155
Thr	Tyr	His	Lys	Val	Asp	Leu	Arg	Leu	Gln	Thr	Leu	Glu	Ile	Pro	Phe	165	170	175
His	Glu	Ile	Val	Thr	Lys	Asp	Met	Phe	Ile	Met	Glu	Ile	Asp	Ala	Ile	180	185	190
Cys	Tyr	Tyr	Arg	Met	Glu	Asn	Ala	Ser	Leu	Leu	Leu	Ser	Ser	Leu	Ala	195	200	205
His	Val	Ser	Lys	Ala	Val	Gln	Phe	Leu	Val	Gln	Thr	Thr	Met	Lys	Arg	210	215	220
Leu	Leu	Ala	His	Arg	Ser	Leu	Thr	Glu	Ile	Leu	Leu	Glu	Arg	Lys	Ser	225	230	235
Ile	Ala	Gln	Asp	Ala	Lys	Val	Ala	Leu	Asp	Ser	Val	Thr	Cys	Ile	Trp	245	250	255
Gly	Ile	Lys	Val	Glu	Arg	Ile	Glu	Ile	Lys	Asp	Val	Arg	Leu	Pro	Ala	260	265	270
Gly	Leu	Gln	His	Ser	Leu	Ala	Val	Glu	Ala	Glu	Ala	Gln	Arg	Gln	Ala	275	280	285
Lys	Val	Arg	Met	Ile	Ala	Ala	Glu	Ala	Glu	Lys	Ala	Ala	Ser	Glu	Ser	290	295	300
Leu	Arg	Met	Ala	Ala	Glu	Ile	Leu	Ser	Gly	Thr	Pro	Ala	Ala	Val	Gln	305	310	315
Leu	Arg	Tyr	Leu	His	Thr	Leu	Gln	Ser	Leu	Ser	Thr	Glu	Lys	Pro	Ser	325	330	335
Thr	Val	Val	Leu	Pro	Leu	Pro	Phe	Asp	Leu	Leu	Asn	Cys	Leu	Ser	Ser	340	345	350
Pro	Ser	Asn	Arg	Thr	Gln	Gly	Ser	Leu	Pro	Phe	Pro	Ser	Pro	Ser	Lys	355	360	365
Pro	Val	Glu	Pro	Leu	Asn	Pro	Lys	Lys	Lys	Asp	Ser	Pro	Met	Leu		370	375	380

<210> SEQ ID NO 56

<211> LENGTH: 1149

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary NPHS2 transgene sequence

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<400> SEQUENCE: 56

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gctgggcccc agccgtcggg ctccggacgg gcggggaccc cgggggagcc ccgagcgccc    180
gccgccacgg tggtagacgt ggatgaggtc cgaggctccg gcgagggagg caccgaggtg    240
gtggcgctgt tggagagcga gcggcccagg gaaggtacca aatcctccgg cttagggggc    300
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atctggttct gcgtaaagg tgtacaagag tatgaaagag taattatatt ccgactggga    420
catctgcttc ctggaagagc caaaggccct ggtcttttct tttttttgcc ctgcctggat    480
acctaccaca aggttgacct tcgtctccaa actctggaga taccttttca tgagatcgtg    540
accaaagaca tggtttataat ggagatagat gccatttgc actaccgaat ggaaaatgcc    600
tctcttctcc taagcagtct tgcctcatgta tctaaagctg tgcaattcct tgtgcaaacc    660
actatgaagc gtctcctagc acatcgatcc ctactgaaa ttcttctaga gaggaagagc    720
atcgcccaag atgcaaagg tgccttggat tcagtgcact gtatttgggg aatcaaagtg    780
gagagaatag aaattaaaga tgtgaggttg ccagctgggc ttcagcactc actggctgtg    840
gaggctgaag gcgaaagaca agccaaagtg cggatgattg ctgcagaagc ggaaaaggct    900
gcttctgagt ccctgaggat ggcagctgag attctgtcag gcacccctgc tgctgttcag    960
cttcgatacc tccacaccct tcagctctctg tccacagaga agccttccac tgtggtttta   1020
cctttgccat ttgaactact gaattgctg tcttctccca gcaacagaac tcaggaagc   1080
ctccccttcc caagtcttc caaacctgtt gagccactaa atcctaaaaa gaaagactct   1140
cccatgtta                                     1149

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<210> SEQ ID NO 57

<211> LENGTH: 1152

<212> TYPE: DNA

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Exemplary NPHS2 transgene sequence

<400> SEQUENCE: 57

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atggagagga gggcgcgagg ctctccagg ggtcccgcg ggcgagggcg caggactccg    60
cacaaggaga acaagagggc aaaggccgag aggagcggcg gaggccgcg gcgccaggag    120
gctgggcccc agccgtcggg ctccggacgg gcggggaccc cgggggagcc ccgagcgccc    180
gccgccacgg tggtagacgt ggatgaggtc cgaggctccg gcgagggagg caccgaggtg    240
gtggcgctgt tggagagcga gcggcccagg gaaggtacca aatcctccgg cttagggggc    300
tgtgagtggc ttctgtctct ctttccctg ctcttcacca tcatgacctt ccttttttcc    360
atctggttct gcgtaaagg tgtacaagag tatgaaagag taattatatt ccgactggga    420
catctgcttc ctggaagagc caaaggccct ggtcttttct tttttttgcc ctgcctggat    480
acctaccaca aggttgacct tcgtctccaa actctggaga taccttttca tgagatcgtg    540
accaaagaca tggtttataat ggagatagat gccatttgc actaccgaat ggaaaatgcc    600
tctcttctcc taagcagtct tgcctcatgta tctaaagctg tgcaattcct tgtgcaaacc    660
actatgaagc gtctcctagc acatcgatcc ctactgaaa ttcttctaga gaggaagagc    720

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atgcccgaag atgcaaaggt tgccttgat tcagtgacct gtatttggg aatcaaagt 780
gagagaatag aaattaaaga tgtgaggttg ccagctgggc ttcagcactc actggctgtg 840
gaggctgaag cgcaaagaca agccaaagt cggtatgattg ctgcagaagc gaaaaggct 900
gcttctgagt ccttgaggat ggcagctgag attctgtcag gcacctctgc tgcggttcag 960
cttcgatacc tccacacctc tcagtctctg tccacagaga agccttccac tgtggtttta 1020
cctttgccat ttgacctact gaattgcctg tcttctccca gcaacagaac tcaggaagc 1080
ctccctctcc caagtccttc caaacctgtt gagccactaa atcctaaaaa gaaagactct 1140
cccatgttat ag 1152

```

```

<210> SEQ ID NO 58
<211> LENGTH: 818
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary minimal nephrin promoter

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<400> SEQUENCE: 58

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ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag 60
ggaattttat gctccagctg ggccagctgg gaggagcctg ctgggcagag gccagagctg 120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag 180
agagcagggc tgactctgcc agtgctgca tcagcctcat cgctctccta ggctcctggc 240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gtagtgtag acaaggagca 300
ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga gagacagaga 360
gagaggaaga gacagagaca aaaggagaga gaacggctta gacaaggaga gaaagatgga 420
aagataaaga gactgggcgc agtggtcac gcctgtaatc ccaacacttg gggaggccaa 480
ggtggggagga tggcttgaag gaaagagtct gagatcaacc tggccaacat agtgagacct 540
cgtctctaaa aaaaaaaaaa aaaaaaaaaa gaaaaagaa aaaaaagttt ttttaagag 600
acagagaaag agactcagag attgagactg agagcaagac agagagagac actcacaggg 660
aagaggggaa gaggaaaacg agaaaggag gagagtaacy gaaagagata aaaaagaaaa 720
gcaggtggca gagacacaga gagagggacc cagagaaagc cagacagacg caggtggctg 780
gcagcgggcg ctgtgggggt cacagtaggg ggacctgt 818

```

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<210> SEQ ID NO 59
<211> LENGTH: 818
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant

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<400> SEQUENCE: 59

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ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag 60
gggattttat gctccagctg ggccagctgg gaggagcctg ctgggcagag gccagagctg 120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag 180
agagcagggc tgactctgcc agtgctgca tcagcctcat cgctctccta ggctcctggc 240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gtagtgtag acaaggagca 300
ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga gagagacaga 360

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gagagaggaa gagacagaga caaaaggaga gagaacggct tagacaagga gagaaagatg 420
gaaagataaa gagactgggc gcagtggctc acgcctgtaa tccaacact tggggaggcc 480
aaggtgggag gatggcttga aggaaagagt ctgagatcaa cctggccaac atagtgagac 540
cccgctctcta aaaaaaaaaa aaaaaaaaaa gaaaaaagaa aaaaaagttt ttttaagag 600
acagagaaaag agactcagag attgagactg agagcaagac agagagagat actcacaggg 660
aagaggggaa gaggaaaacg agaaaggag gagagtaacg gaaagagata aaaaagaaaa 720
gcaggtggca gagacacaca gagagggacc cagagaaagc cagacagacg caggtggctg 780
gcagcgggcg ctgtgggggt cacagtaggg ggacctgt 818

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<210> SEQ ID NO 60
<211> LENGTH: 818
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary minimal nephrin promoter variant

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<400> SEQUENCE: 60

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ggccctgggg tcacggaggc tggggaggca ccgaggaacg cgctggcat gtgctgacag 60
gggattttat gctccagctg ggcacgctgg gaggagcctg ctgggcagag gccagagctg 120
ggggctcttg aaggtacctg ggggaggttg cactgtgaga atgagctcaa gctgggtcag 180
agagcagggc tgactctgcc agtgctgca tcagcctcat cgctctccta ggctcctggc 240
ctgctggact ctgggctgca ggtccttctt gaaaggctgt gagtgtgag acaaggagca 300
ggagtgaggg gtggcaggag agaagataga gattgagaga gagagagaga gagacagaga 360
gagaggaaga gacagagaca aaaggagaga gaacggctta gacaaggaga gaaagatgga 420
aagataaaga gactgggcgc agtggctcac gcctgtaatc ccaacacttg gggaggccaa 480
ggtggggagga tggcttgaag gaaagagtct gagatcaacc tggccaacat agtgagacct 540
cgtctctaaa aaaaaaaaaa aaaaaaaaaa gaaaaaagaa aaaaaagttt ttttaagag 600
acagagaaaag agactcagag attgagactg agagcaagac agagagagat actcacaggg 660
aagaggggaa gaggaaaacg agaaaggag gagagtaacg gaaagagata aaaaagaaaa 720
gcaggtggca gagacacaga gagagggacc cagagaaagc cagacagacg caggtggctg 780
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<210> SEQ ID NO 61
<211> LENGTH: 7
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary DNA sequence

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<400> SEQUENCE: 61

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ggggtca

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7

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<210> SEQ ID NO 62
<211> LENGTH: 17
<212> TYPE: DNA
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Exemplary DNA sequence

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<400> SEQUENCE: 62

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-continued

cgagggtgg ggaggca

17

<210> SEQ ID NO 63
 <211> LENGTH: 5
 <212> TYPE: DNA
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Exemplary DNA sequence

<400> SEQUENCE: 63

atgtg

5

1. A promoter comprising (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17 and wherein the promoter has a length of about 1.1 kb or less.

2. A promoter which comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 but wherein:

- (i) position 1 to position n1 of SEQ ID NO: 1 is deleted, wherein n1 is an integer from 100 to 430; and/or
- (ii) position n2 to position n3 of SEQ ID NO: 1 is deleted, wherein $n3 \geq n2$, n2 is an integer from 508 to 1061, and n3 is an integer from 508 to 1061;

or a nucleotide sequence with at least 70% identity thereto.

3. The promoter according to claim 2, wherein:

- (iii) the position corresponding to position 493 of SEQ ID NO: 1 is G, the position corresponding to position 1080 of SEQ ID NO: 1 is T, the position corresponding to position 1169 of SEQ ID NO: 1 is C, and the position corresponding to position 1249 of SEQ ID NO: 1 is G.

4. The promoter according to claim 2, wherein the promoter comprises (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17 and/or wherein the promoter has a length of about 1.1 kb or less.

5. The promoter according to claim 1, wherein the promoter is a kidney-specific promoter.

6-7. (canceled)

8. The promoter according to claim 1, wherein the promoter comprises from 5' to 3': (i) a nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17; (iii) optionally a nucleotide sequence having at least 70% identity to SEQ ID NO: 8, or one or more fragments thereof; and (ii) a nucleotide sequence having at least 70% identity to SEQ ID NO: 5 or 18, a nucleotide sequence having at least 70% identity to SEQ ID NO: 6 or 19, and/or a nucleotide sequence having at least 70% identity to SEQ ID NO: 7 or 20.

9. The promoter according to claim 1, wherein the promoter has a length of 0.7 kb or less.

10. The promoter according to claim 1, wherein the promoter has a length of from 0.265 kb to 0.5 kb.

11. The promoter according to claim 1, wherein the promoter comprises:

- (a) a retinoic acid receptor binding site;
- (b) a WT1 binding site;
- (c) an enhancer box;
- (d) a transcription factor binding region; and/or
- (e) a transcription initiation site.

12-15. (canceled)

16. The promoter according to claim 1 wherein one or more of the following nucleotide sequences is present in (i) the nucleotide sequence having at least 70% identity to SEQ ID NO: 4 or 17:

- (a) GGGGTCA at a position corresponding to position 7 to position 13 of SEQ ID NO: 4 or 17;
- (b) CGGAGGCTGGGGAGGCA at a position corresponding to position 14 to position 30 of SEQ ID NO: 4 or 17; and
- (c) ATGTG at a position corresponding to position 49 to position 53 of SEQ ID NO: 4 or 17.

17-28. (canceled)

29. A promoter which consists of a nucleotide sequence having at least 70% identity to SEQ ID NO: 47.

30. The promoter according to claim 29, wherein the promoter consists of a nucleotide sequence having at least 90% identity to SEQ ID NO: 47.

31. The promoter according to claim 29, wherein the promoter consists of a nucleotide sequence having at least 95% identity to SEQ ID NO: 47.

32. The promoter according to claim 29, wherein the promoter consists of a nucleotide sequence having at least 98% identity to SEQ ID NO: 47.

33. The promoter according to claim 29, wherein the promoter consists of a nucleotide sequence having at least 99% identity to SEQ ID NO: 47.

34. The promoter according to claim 29, wherein the promoter consists of the nucleotide sequence of SEQ ID NO: 47.

35. A polynucleotide comprising the promoter according to claim 1.

36. The polynucleotide according to claim 35, wherein the promoter is operably linked to a protein coding sequence.

37. The polynucleotide according to claim 35, wherein the promoter is operably linked to a protein coding sequence encoding NPHS2 or a fragment and/or variant thereof; a COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof; or CFI, CFH or FHL-1, or a fragment and/or variant thereof.

38-41. (canceled)

42. A vector comprising the polynucleotide according to claim 35.

43. (canceled)

44. The vector according to claim 42, wherein the vector is a viral vector.

45. The vector according to claim 44, wherein the viral vector is in the form of a viral vector particle.

46. The vector according to claim **42**, wherein the vector is the form of an AAV vector particle encapsidated by AAV3B, LK03, or AAV9 capsid proteins.

47. A cell comprising the polynucleotide according to claim **35**.

48. A pharmaceutical composition comprising the polynucleotide according to claim **35**.

49. A method of preventing or treating glomerular disease comprising administering a polynucleotide according to claim **35** to a subject in need thereof.

50-51. (canceled)

52. A polynucleotide comprising the promoter according to claim **29**.

53. The polynucleotide according to claim **52**, wherein the promoter is operably linked to a protein coding sequence.

54. The polynucleotide according to claim **52**, wherein the promoter is operably linked to a protein coding sequence encoding NPHS2 or a fragment and/or variant thereof; a

COL4A3, COL4A4 or COL4A5 polypeptide, or a fragment or derivative thereof; or CFI, CFH or FHL-1, or a fragment and/or variant thereof.

55. A vector comprising the polynucleotide according to claim **52**.

56. The vector according to claim **55**, wherein the vector is a viral vector.

57. The vector according to claim **56**, wherein the viral vector is in the form of a viral vector particle.

58. The vector according to claim **55**, wherein the vector is the form of an AAV vector particle encapsidated by AAV3B, LK03, or AAV9 capsid proteins.

59. A cell comprising the polynucleotide according to claim **52**.

60. A pharmaceutical composition comprising the polynucleotide according to claim **52**.

61. A method of preventing or treating glomerular disease comprising administering a polynucleotide according to claim **52** to a subject in need thereof.

* * * * *